EduSpark

NCERT based · Rationalised syllabus

CBSE Class 10

SCIENCE

KHATAM IN 99 PAGES

Session 2026-27



FREE e-book

Chemistry | Biology | Physics

By Nitesh Bhaiya

Handwritten-style notes · All NCERT activities · Self Practice in every chapter

Eduspark · Free for students, not for sale

Puri Science, Ek Notebook Mein

Most asked: 13 chapters | notes + Most Important Questions with answers | last-night revision

1. Chemistry

Ch	Chapter	Page
1	Chemical Reactions and Equations	3
2	Acids, Bases and Salts	9
3	Metals and Non-metals	16
4	Carbon and its Compounds	23

2. Biology

Ch	Chapter	Page
5	Life Processes	31
6	Control and Coordination	40
7	How do Organisms Reproduce?	47
8	Heredity	54
13	Our Environment	89

3. Physics

Ch	Chapter	Page
9	Light: Reflection and Refraction	60
10	The Human Eye and the Colourful World	69
11	Electricity	74
12	Magnetic Effects of Electric Current	82

4. Revision

Section	Page
Last-night revision: formulas, reactions, one-liners, quick check	94
Topper answer sheets (real CBSE copies)	99

Nitesh Bhaiya ki baat



Is notebook mein poori NCERT hai, exam wali language mein. Har chapter ke end mein board ke Most Important Questions answers ke saath. Roz ek chapter + uske questions, aur board exam bas ek aur test lagega!

Nitesh Bhaiya | Eduspark

5. Board Paper Pattern (Theory 80 + Internal 20)

Section	Type of questions	Marks
A	20 MCQ / Assertion-Reason x 1	20
B	6 very short answer x 2	12
C	7 short answer x 3	21
D	3 long answer x 5	15
E	3 case-based x 4	12
Total	**39 questions, 3 hours**	**80**

6. Unit-wise Marks

Unit	Marks
I. Chemical Substances: Nature and Behaviour (Ch 1-4)	25
II. World of Living (Ch 5-8)	25
III. Natural Phenomena (Ch 9-10)	12
IV. Effects of Current (Ch 11-12)	13
V. Natural Resources (Ch 13)	5
Total	**80**

Is book ko kaise padhein

1. Pehle notes padho: yellow = must learn, red = keyword.
2. Har box ka kaam: **Yaad karo** = trick, **Exam Alert** = marks katne wali galti, **Quick fact** = 1-mark wali line.
3. Phir chapter ke **Most Important Questions** khud likh ke solve karo, answer green mein match karo.
4. Exam se pehle wali raat: page 94 se revision.

Not in board exam (2026-27)

Periodic Classification of Elements, Sources of Energy, Sustainable Management of Natural Resources, astigmatism, electric motor, electromagnetic induction and generator (removed in rationalised NCERT).

Exam hall tips

- → 15 min reading time mein choice wale questions pehle tick kar lo, phir likhna shuru karo.
- → Diagram pencil se, labels seedhi lines se; equation balanced with states.
- → Numericals: Given, Formula, Solution, unit ke saath boxed answer.

Last 7 days plan

Day	Kya karna hai
1-2	Chemistry Ch 1-4 + reactions list (p94)
3-4	Biology Ch 5-8, 13 + all diagrams
5-6	Physics Ch 9-12 + numericals, ray diagrams
7	Revision pages + one full sample paper in 3 hours

Diagrams jo board mein baar-baar aate hain

Digestive system, heart (double circulation), nephron, stomata, neuron, reflex arc, human brain, LS of flower, pollen tube, ray diagrams (mirror + lens), prism, circuit with ammeter-voltmeter, solenoid field lines, domestic circuit. **Pencil se, labels seedhi line se.**

Marks ka formula

Notes padho + MIQ khud likho + diagram practice = 80/80.
Ek baar padha, ek baar likha, toh exam mein yaad aayega!

Chemical Reactions and Equations



Most asked: balancing with state symbols, types of reactions with examples, redox (oxidised/reduced), corrosion vs rancidity

Milk turning into curd, food getting cooked, iron rusting, even breathing: in each case **new substances** are formed. That is a chemical change.

1. Chemical Reaction

Chemical reaction: a process in which one or more substances (reactants) change into new substances (products) with new properties. Atoms are only rearranged; no atom is created or destroyed.

How to know a reaction has taken place

- → change in **state**
- → change in **colour**
- → **evolution of a gas**
- → change in **temperature**
- → formation of a **precipitate**

Meme samjho



Sodium (catches fire in water) + chlorine (poisonous gas) milke ban gaye... khaane ka namak!

Asli baat: reaction ke baad product ki properties reactants se bilkul alag hoti hain. Yahi chemical change hai.

Paheli-Bujho



Bujho: Chemical reaction matlab lab mein chemicals ka jhagda?

Paheli: Nahi bhondu! Jab purane substances badal kar naye substances bante hain, wahi chemical reaction hai.

Bujho: Toh doodh se dahi bhi reaction hai?

Paheli: Bilkul! Naya substance bana, toh chemical change.

Activity 1.1: Burning a magnesium ribbon

Done: Clean a Mg ribbon with sandpaper and burn it with tongs over a watch-glass. **Observed:** It burns with a **dazzling white flame** and leaves a white powder. **Conclusion:** Mg combines with oxygen to form magnesium oxide (MgO).

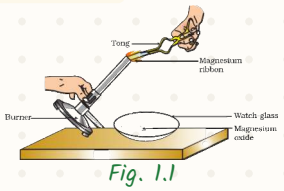


Fig. 1.1

$$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$$

Activity 1.2: Lead nitrate + potassium iodide

Done: Mix lead nitrate solution with potassium iodide solution. **Observed:** A **yellow precipitate** forms. **Conclusion:** Lead iodide (PbI_2) is formed.

$$Pb(NO_3)_2(aq) + 2KI(aq) \longrightarrow PbI_2(s) + 2KNO_3(aq)$$

Activity 1.3: Zinc + dilute acid

Done: Add dilute HCl or H_2SO_4 to zinc granules in a conical flask. **Observed:** Bubbles of gas form and the flask becomes **warm**. **Conclusion:** Hydrogen gas is evolved and heat is released.

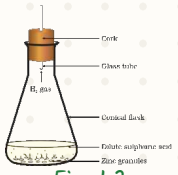


Fig. 1.2

$$Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$$

Quick fact

Mg ribbon is rubbed with sandpaper to remove the **protective layer of MgO** on its surface, so that it burns easily in air.

2. Chemical Equations

Word equation: Magnesium + Oxygen ⟶ Magnesium oxide

Chemical equation: a reaction written with symbols and formulae. Reactants on the left (LHS), products on the right (RHS), arrow pointing towards the products; "+" between substances on the same side.

Skeletal equation: an **unbalanced** equation, e.g. $Mg + O_2 \longrightarrow MgO$.

Key terms

- → **Law of Conservation of Mass:** mass can neither be created nor destroyed in a chemical reaction.
- → **Balanced equation:** number of atoms of each element is equal on both sides.
- → **States:** (s) solid, (l) liquid, (g) gas, (aq) aqueous solution.
- → **Conditions** (heat, pressure, catalyst, light) are written above/below the arrow.

3. Balancing a Chemical Equation

NCERT example, iron + steam (hit-and-trial method):

$$Fe + H_2O \longrightarrow Fe_3O_4 + H_2 \text{ (skeletal)}$$

1. List the atoms of each element on both sides.
2. Start with the compound having the **maximum atoms** (Fe_3O_4): balance O by writing **4**H_2O.
3. Balance H: write **4**H_2.
4. Balance Fe: write **3**Fe. Check every element once more.
5. Add the physical states.

$$3Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2(g)$$

Equations with conditions:

$$CO(g) + 2H_2(g) \xrightarrow{340\ atm} CH_3OH(l)$$

$$6CO_2(aq) + 12H_2O(l) \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6(aq) + 6O_2(aq) + 6H_2O(l)$$

Paheli-Bujho



Paheli: Bujho, equation balance karna zaroori kyun hai?
Bujho: Taaki dono side barabar dikhe, page sundar lagta hai!
Paheli: Bhondu! Law of Conservation of Mass: atoms na bante hain, na mitte hain, bas rearrange hote hain.

Exam Alert

- → **Subscript kabhi mat badlo** (H_2O ko H_2O_2 mat banao); sirf aage coefficient lagao.
- → "Write balanced equation **with state symbols**" likha ho toh (s), (l), (g), (aq) ke bina marks katenge.

4. Types of Chemical Reactions

Yaad karo: 5 types

"Chhota Dost Dhoondhe Double Rasgulla": Combination, Decomposition, Displacement, Double displacement, Redox.

(a) Combination reaction

Two or more substances combine to form a single product. A + B ⟶ AB

Activity 1.4: Quick lime + water

Done: Slowly add water to calcium oxide (quick lime) in a beaker. **Observed:** A hissing sound; the beaker becomes **very hot**.
Conclusion: Slaked lime, $Ca(OH)_2$, forms with release of a large amount of heat.

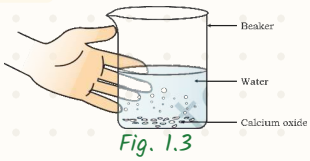


Fig. 1.3

$$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(aq) + Heat$$

Whitewashing: a solution of slaked lime reacts slowly with CO_2 of air and forms a thin layer of $CaCO_3$, which gives a **shiny finish** after 2-3 days.

$$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(l)$$

More examples: $C(s) + O_2(g) \longrightarrow CO_2(g)$; $2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$

Exothermic and endothermic reactions

Exothermic: heat is **released** along with products, e.g. burning of natural gas, respiration, decomposition of vegetable matter into compost.

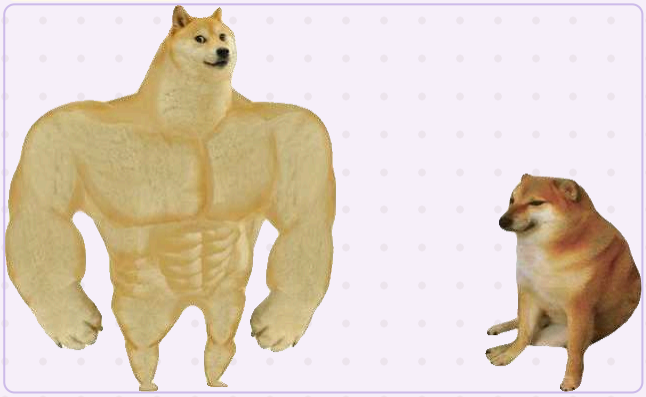
$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g) + Energy$$

$$C_6H_{12}O_6(aq) + 6O_2(aq) \longrightarrow 6CO_2(aq) + 6H_2O(l) + Energy$$

Endothermic: heat is **absorbed**, e.g. most decomposition reactions, photosynthesis. Mixing barium hydroxide with ammonium chloride makes the test tube **cold**:

$$Ba(OH)_2 + 2NH_4Cl \longrightarrow BaCl_2 + 2NH_3 + 2H_2O$$

Meme samjho



Exothermic: "lo, garmi le lo!" Endothermic: "garmi do, tabhi kaam karunga."
Asli baat: respiration exothermic hai (energy deti hai), isliye hum zinda hain; decomposition ko zyadatar energy chahiye.

(b) Decomposition reaction

A single reactant breaks down into two or more simpler products. AB ⟶ A + B

Activity 1.5: Heating ferrous sulphate crystals

Done: Heat green $FeSO_4 \cdot 7H_2O$ crystals in a dry boiling tube.
Observed: Crystals lose water and turn white, then brown, with a **smell of burning sulphur**.
Conclusion: $FeSO_4$ decomposes into ferric oxide, SO_2 and SO_3.

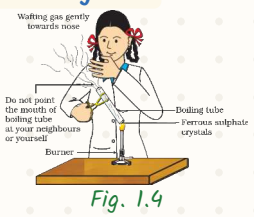


Fig. 1.4

$$2FeSO_4(s) \xrightarrow{Heat} Fe_2O_3(s) + SO_2(g) + SO_3(g)$$

1. Thermal decomposition (by heat). Limestone gives quick lime, used to make cement:

$$CaCO_3(s) \xrightarrow{Heat} CaO(s) + CO_2(g)$$

Activity 1.6: Heating lead nitrate

Done: Heat lead nitrate powder in a boiling tube. **Observed:** **Brown fumes** are given off. **Conclusion:** The brown fumes are nitrogen dioxide (NO_2).

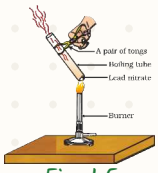


Fig. 1.5

$$2Pb(NO_3)_2(s) \xrightarrow{Heat} 2PbO(s) + 4NO_2(g) + O_2(g)$$

2. Electrolytic decomposition (by electricity):

Activity 1.7: Electrolysis of water

Done: Pass current through water (a few drops of dilute H_2SO_4 added) using carbon electrodes. **Observed:** Gas at the **cathode** (H_2) is **double** the volume of gas at the anode (O_2). H_2 burns with a pop; O_2 makes a candle burn brighter. **Conclusion:** Water decomposes into H_2 and O_2 in the ratio **2 : 1** by volume.

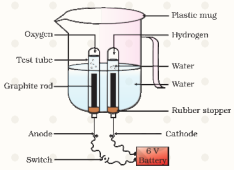


Fig. 1.6

$$2H_2O(l) \xrightarrow{Electricity} 2H_2(g) + O_2(g)$$

3. Photolytic decomposition (by light), used in **black and white photography**:

Activity 1.8: Silver chloride in sunlight

Done: Keep white silver chloride in a china dish in sunlight. **Observed:** White silver chloride turns **grey**. **Conclusion:** AgCl decomposes into silver and chlorine.

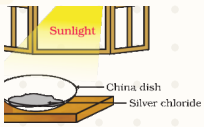


Fig. 1.7

$$2AgCl(s) \xrightarrow{Sunlight} 2Ag(s) + Cl_2(g)$$

$$2AgBr(s) \xrightarrow{Sunlight} 2Ag(s) + Br_2(g)$$

(c) Displacement reaction

A **more reactive** element displaces a **less reactive** element from its compound. A + BC ⟶ AC + B

Activity 1.9: Iron nails in copper sulphate

Done: Dip clean iron nails in $CuSO_4$ solution for about 20 minutes. **Observed:** Nails become **brownish**; the **blue colour fades** (turns light green). **Conclusion:** Iron displaces copper from copper sulphate.

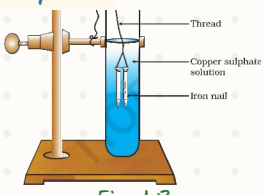


Fig. 1.8

$$Fe(s) + CuSO_4(aq) \longrightarrow FeSO_4(aq) + Cu(s)$$

$$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$$

$$Pb(s) + CuCl_2(aq) \longrightarrow PbCl_2(aq) + Cu(s)$$

$$Cu(s) + 2AgNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$ (silver refining)

Meme samjho

Fe + CuSO4 → FeSO4 + Cu
Zn + CuSO4 → ZnSO4 + Cu
Mg + CuSO4 → MgSO4 + Cu



Iron ne $CuSO_4$ mein entry maari aur copper ko bahar nikaal diya. Zyada reactive = boss!

Asli baat: zinc, lead aur iron copper se zyada reactive hain, isliye copper ko displace kar dete hain.

(d) Double displacement reaction

Two compounds exchange their ions. AB + CD ⟶ AD + CB

Activity 1.10: Sodium sulphate + barium chloride

Done: Mix sodium sulphate solution with barium chloride solution. **Observed:** A **white**, insoluble substance forms. **Conclusion:** Barium sulphate precipitate forms by exchange of ions.

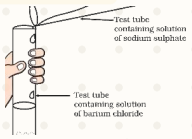


Fig. 1.9

$$Na_2SO_4(aq) + BaCl_2(aq) \longrightarrow BaSO_4(s)\downarrow + 2NaCl(aq)$$

Precipitation reaction: any reaction that produces an **insoluble solid** (precipitate), shown by ↓; an evolved gas is shown by ↑. Activities 1.2 and 1.10 are precipitation reactions.

(e) Oxidation and reduction (redox)

Activity 1.11: Heating copper powder

Done: Heat copper powder in a china dish. **Observed:** The surface becomes coated with a **black** substance. **Conclusion:** Copper combines with oxygen to form black copper(II) oxide; passing hydrogen over hot CuO turns it brown again.

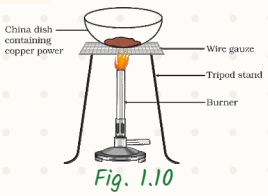


Fig. 1.10

$$2Cu + O_2 \xrightarrow{Heat} 2CuO$$

$$CuO + H_2 \xrightarrow{Heat} Cu + H_2O$$

Oxidation	Reduction
gain of oxygen	loss of oxygen
loss of hydrogen	gain of hydrogen
loss of electrons	gain of electrons

Redox reaction: one reactant is **oxidised** while the other is **reduced**. The substance that gets reduced is the oxidising agent; the one that gets oxidised is the reducing agent.

Reaction	Oxidised	Reduced
$CuO + H_2 \longrightarrow Cu + H_2O$	H_2	CuO
$ZnO + C \longrightarrow Zn + CO$	C	ZnO
$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$	HCl	MnO_2

Yaad karo: OIL RIG

Oxidation Is Loss, Reduction Is Gain (of electrons).
Oxidising agent khud reduce hota hai, reducing agent khud oxidise.

5. Oxidation in Daily Life

Corrosion: a metal is attacked by substances around it like moisture and acids and slowly gets eaten up. Iron gets a **reddish-brown** coating (rust), silver turns **black**, copper gets a **green** coating.

It damages car bodies, bridges, iron railings and ships. Rusting needs both air and moisture; prevented by painting, oiling, greasing, galvanising.

$$4Fe + 3O_2 + 2xH_2O \longrightarrow 2Fe_2O_3.xH_2O \text{ (rust)}$$

Rancidity: fats and oils get **oxidised** and their **smell and taste change**. Prevented by adding **antioxidants**, airtight containers, refrigeration, and flushing chips packets with **nitrogen**.

Paheli-Bujho



Bujho: Chips ke packet mein aadhi hawa! Company ne loot liya yaar.
Paheli: Woh hawa nahi, nitrogen gas hai. Oxygen andar nahi aayegi, toh chips oxidise hoke rancid nahi honge.
Bujho: Matlab hawa bhi science hai... samajh gaya!

NCERT in-text: quick answers

- **Why do we apply paint on iron articles?** Paint keeps air and moisture away, so iron does not rust.
- **Why are chips packets flushed with nitrogen?** To prevent oxidation of the oil/fat (rancidity).
- **Oil and fat containing food items are flushed with nitrogen. Why?** Same reason: nitrogen is unreactive and keeps oxygen out.
- **Why does the colour of $CuSO_4$ solution change on dipping an iron nail?** Iron displaces copper; blue $CuSO_4$ becomes light green $FeSO_4$.

Reaction	Colour change
$Pb(NO_3)_2 + KI$	yellow ppt (PbI_2)
$Na_2SO_4 + BaCl_2$	white ppt ($BaSO_4$)
heating $FeSO_4$	green $\longrightarrow$ brown (Fe_2O_3)
heating $Pb(NO_3)_2$	brown fumes (NO_2)
$Fe + CuSO_4$	blue $\longrightarrow$ light green
$AgCl$ in sunlight	white $\longrightarrow$ grey
heating Cu	brown $\longrightarrow$ black (CuO)

Type (general form)	Example
Combination $A + B \longrightarrow AB$	$CaO + H_2O \longrightarrow Ca(OH)_2$
Decomposition $AB \longrightarrow A + B$	$CaCO_3 \xrightarrow{\text{Heat}} CaO + CO_2$
Displacement $A + BC \longrightarrow AC + B$	$Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$
Double displacement $AB + CD \longrightarrow AD + CB$	$Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2NaCl$
Redox oxidation + reduction	$CuO + H_2 \longrightarrow Cu + H_2O$

Exam Alert

- Ek reaction ek se zyada type ki ho sakti hai: $Fe_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Fe$ = **displacement + redox** (thermite reaction, used to join railway tracks).
- Neutralisation (acid + base $\longrightarrow$ salt + water) bhi double displacement hai: $NaOH + HCl \longrightarrow NaCl + H_2O$.

1 minute recap

Reaction → Equation → Balance (mass conserved) → 5 types → Redox → Corrosion + rancidity

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM Board fav Which of the following is a combination reaction?

(a) $2KClO_3 \longrightarrow 2KCl + 3O_2$
(b) $CaO + H_2O \longrightarrow Ca(OH)_2$
(c) $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$
(d) $NaOH + HCl \longrightarrow NaCl + H_2O$

Ans: (b) $CaO + H_2O \longrightarrow Ca(OH)_2$: two reactants form a single product.

Q2. IM In the electrolysis of water, the ratio of volumes of gases collected at the cathode and anode is:

(a) 1 : 2 (b) 2 : 1 (c) 1 : 1 (d) 1 : 8

Ans: (b) 2 : 1 (hydrogen at the cathode, oxygen at the anode).

Q3. 1M Board fav $Fe_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Fe$. This reaction is an example of:
(a) combination (b) double displacement
(c) decomposition (d) displacement
Ans: (d) displacement reaction: Al displaces Fe from its oxide (it is also a redox reaction).

Q4. 1M What happens when dilute HCl is added to iron filings?
(a) H_2 gas and iron chloride are produced
(b) Cl_2 gas and iron hydroxide are produced
(c) no reaction (d) iron salt and water are produced
Ans: (a) Hydrogen gas and iron chloride are produced: $Fe + 2HCl \longrightarrow FeCl_2 + H_2$.

Q5. 1M Name the gas evolved and the colour of the residue when lead nitrate is heated.
Ans: Brown fumes of nitrogen dioxide (NO_2); O_2 is also released and the residue is lead(II) oxide, PbO (yellow).

Q6. 1M Board fav **Assertion (A):** Decomposition of vegetable matter into compost is exothermic. **Reason (R):** Heat is released when organic matter breaks down.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q7. 1M Why is silver chloride stored in dark coloured bottles?
Ans: In sunlight it decomposes into silver and chlorine ($2AgCl \longrightarrow 2Ag + Cl_2$); dark bottles cut off light.

Q8. 1M Which of the following gases can be used for storage of fresh samples of an oil for a long time?
(a) CO_2 or O_2 (b) N_2 or O_2
(c) CO_2 or He (d) He or N_2
Ans: (d) Nitrogen or helium: they are unreactive, so the oil is not oxidised.

Q9. 1M Board fav The following reaction is an example of: $4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$. (i) displacement (ii) combination (iii) redox (iv) neutralisation
(a) (i) and (iv) (b) (ii) and (iii) (c) (i) and (iii) (d) (iii) and (iv)
Ans: (c) (i) and (iii): oxygen displaces nitrogen from NH_3, and NH_3 is oxidised while O_2 is reduced.

Q10. 1M **Assertion (A):** In a reaction of copper with oxygen, copper serves as a reducing agent. **Reason (R):** The substance which gains oxygen in a chemical reaction acts as a reducing agent.
Ans: (a) Both A and R are true, and R is the correct explanation of A (Cu gains oxygen, gets oxidised, so it is the reducing agent).

Q11. 1M A white salt on heating decomposes to give brown fumes and a yellow residue. Name the salt.
Ans: Lead nitrate, $Pb(NO_3)_2$; brown fumes = NO_2, yellow residue = PbO.

Short answer (2–3 marks)

Q12. 2M Why should a magnesium ribbon be cleaned before burning in air? Write the balanced equation.
Ans: Mg ribbon has a protective layer of MgO on it that stops burning; rubbing with sandpaper removes it. $2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$.

Q13. 3M Board fav Write balanced equations with state symbols: (a) barium chloride + aluminium sulphate ⟶ barium sulphate + aluminium chloride (b) sodium + water ⟶ sodium hydroxide + hydrogen.
Ans: (a) $3BaCl_2(aq) + Al_2(SO_4)_3(aq) \longrightarrow 3BaSO_4(s) + 2AlCl_3(aq)$
(b) $2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$

Q14. 2M Board fav Why does the colour of copper sulphate solution change when an iron nail is dipped in it? Name the type of reaction.
Ans: Iron is more reactive than copper, so it displaces copper: $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$. Blue $CuSO_4$ changes to light green $FeSO_4$ and brown copper coats the nail. Displacement reaction.

Q15. 3M Board fav Identify the substances oxidised and reduced: (a) $ZnO + C \longrightarrow Zn + CO$ (b) $4Na + O_2 \longrightarrow 2Na_2O$ (c) $CuO + H_2 \longrightarrow Cu + H_2O$.
Ans: (a) C oxidised, ZnO reduced. (b) Na oxidised, O_2 reduced. (c) H_2 oxidised, CuO reduced.

Q16. 2M Why are decomposition reactions called the opposite of combination reactions? Write one equation for each.
Ans: In combination two or more reactants form one product; in decomposition one reactant gives two or more products. $CaO + H_2O \longrightarrow Ca(OH)_2$; $CaCO_3 \xrightarrow{Heat} CaO + CO_2$.

Q17. 2M Board fav What is rancidity? Write two ways to prevent it.
Ans: Oxidation of fats and oils in food that changes their smell and taste. Prevention: add antioxidants; keep food in airtight containers; flush packets with nitrogen; refrigerate.

Q18. 2M Board fav A shiny brown element X on heating in air turns black. Name X and the black compound. Write the equation.
Ans: X = copper (Cu); black compound = copper(II) oxide (CuO). $2Cu + O_2 \xrightarrow{Heat} 2CuO$.

Q19. 2M What is a precipitation reaction? Give an example.
Ans: A reaction in which an insoluble solid (precipitate) is formed. $Na_2SO_4(aq) + BaCl_2(aq) \longrightarrow BaSO_4(s) + 2NaCl(aq)$ (white ppt).

Q20. 2M Why is respiration considered an exothermic reaction?

Ans: Glucose combines with oxygen in our cells to give CO_2 and water, and energy is released: $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$ + energy.

Q21. 3M Board fav What is observed when a solution of potassium iodide is added to lead nitrate solution? Name the type of reaction and write the balanced equation.

Ans: A yellow precipitate of lead iodide forms. Double displacement (precipitation) reaction. $Pb(NO_3)_2(aq) + 2KI(aq) \longrightarrow PbI_2(s) + 2KNO_3(aq)$.

Q22. 3M Board fav Write one equation each for decomposition reactions where energy is supplied in the form of heat, light and electricity.

Ans: Heat: $2Pb(NO_3)_2 \xrightarrow{Heat} 2PbO + 4NO_2 + O_2$. Light: $2AgBr \xrightarrow{Sunlight} 2Ag + Br_2$. Electricity: $2H_2O \xrightarrow{Electricity} 2H_2 + O_2$.

Q23. 3M What is the difference between displacement and double displacement reactions? Write equations.

Ans: Displacement: a more reactive element replaces a less reactive one from its compound, $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$. Double displacement: two compounds exchange ions, $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2NaCl$.

Q24. 2M In the refining of silver, silver is recovered from silver nitrate solution using copper metal. Write the reaction involved.

Ans: $Cu(s) + 2AgNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$. Copper is more reactive than silver and displaces it.

Q25. 2M Board fav A solution of a substance X is used for whitewashing. (a) Name X and write its formula. (b) Write the reaction of X with water.

Ans: (a) X = calcium oxide (quick lime), CaO. (b) $CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(aq)$ + heat. The slaked lime formed reacts with CO_2 of air to form a shiny $CaCO_3$ layer.

Q26. 2M Why is hydrogen peroxide kept in coloured bottles? Write the equation of its decomposition.

Ans: It decomposes slowly in light into water and oxygen; coloured bottles cut off light. $2H_2O_2(l) \longrightarrow 2H_2O(l) + O_2(g)$.

Q27. 2M Why do silver articles become black and copper vessels lose their shine after some time in air?

Ans: Silver reacts with sulphur compounds in air to form a black coating of silver sulphide; copper reacts with moist CO_2 in air to form a green coating of basic copper carbonate. Both are corrosion.

Q28. 2M Name the reducing agent in: (a) $CuO + H_2 \longrightarrow Cu + H_2O$ (b) $ZnO + C \longrightarrow Zn + CO$. Give a reason.

Ans: (a) H_2 (b) C: each of them gains oxygen (gets oxidised) and removes oxygen from the other substance.

Long answer (5 marks)

Q29. 5M Board fav (a) Define a decomposition reaction. Explain thermal, electrolytic and photolytic decomposition with one balanced equation each. (b) Why is the amount of gas collected in one test tube double that in the other in electrolysis of water?

Ans: (a) A single reactant breaks into two or more simpler products. Thermal: $CaCO_3 \xrightarrow{Heat} CaO + CO_2$. Electrolytic: $2H_2O \xrightarrow{Electricity} 2H_2 + O_2$. Photolytic: $2AgCl \xrightarrow{Sunlight} 2Ag + Cl_2$. (b) Water ($H_2O$) has 2 H atoms for every O atom, so hydrogen gas formed is twice oxygen by volume.

Q30. 5M (a) What is corrosion? Name the coating formed on iron, copper and silver. (b) Write two conditions needed for rusting and two methods to prevent it. (c) State one harmful effect of corrosion.

Ans: (a) Slow eating up of a metal by moisture, air, acids around it. Iron: reddish-brown rust; copper: green coating; silver: black coating. (b) Air (oxygen) and moisture; prevented by painting, oiling/greasing, galvanising. (c) Weakens car bodies, bridges, railings and ships; huge money spent on replacing iron.

Q31. 5M Board fav (a) What is a balanced chemical equation? Why should chemical equations be balanced? (b) Balance and add state symbols: (i) $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$ (ii) $NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O$ (iii) $Mg + HCl \longrightarrow MgCl_2 + H_2$

Ans: (a) An equation with equal numbers of atoms of each element on both sides; needed because of the law of conservation of mass. (b) (i) $3Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2(g)$ (ii) $2NaOH(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$ (iii) $Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$

Case-based (4 marks)

Q32. 4M Board fav Riya added zinc granules to dilute sulphuric acid in a conical flask. Bubbles formed and the flask became warm. (a) Name the gas and give a test for it. (b) Write the balanced equation. (c) Is the reaction exothermic or endothermic? (d) Name the type of reaction.

Ans: (a) Hydrogen; it burns with a pop sound near a burning matchstick. (b) $Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$. (c) Exothermic (flask warms up). (d) Displacement reaction.

Q33. 4M Board fav Read: When a green crystalline substance X is heated in a dry boiling tube, it first loses water and then gives off gases with a smell of burning sulphur, leaving a brown solid. (a) Identify X. (b) Write the balanced equation. (c) Name the type of reaction. (d) Name the brown residue.

Ans: (a) X = ferrous sulphate crystals, $FeSO_4.7H_2O$. (b) $2FeSO_4(s) \xrightarrow{Heat} Fe_2O_3(s) + SO_2(g) + SO_3(g)$. (c) Thermal decomposition. (d) Ferric oxide, Fe_2O_3.

ACIDS, BASES AND SALTS



Most asked: pH scale and its importance, chlor-alkali process, bleaching powder / baking soda / washing soda / PoP (preparation + uses), reactions of acids

Lemon and vinegar taste sour because they contain **acids**; baking soda tastes bitter and feels soapy because it is a **base**. Acid + base gives a **salt**.

1. Acids, Bases and Indicators

Acids: sour in taste, turn blue litmus red, give H^+ ions in water. e.g. HCl, H_2SO_4, HNO_3, CH_3COOH.

Bases: bitter in taste, soapy to touch, turn red litmus blue, give OH^- ions in water. e.g. NaOH, KOH, $Ca(OH)_2$, $Mg(OH)_2$, NH_4OH.

Indicators: substances that show whether a substance is acidic or basic by a change in **colour or smell**.

- → **Natural:** litmus (purple dye from **lichen**; purple in neutral solution), turmeric, red cabbage, petals of hydrangea, petunia and geranium.
- → **Synthetic:** methyl orange, phenolphthalein.
- → **Olfactory:** change their smell in acid or base, e.g. onion, vanilla essence, clove oil.

Key terms

- → **Acid:** gives H^+ (H_3O^+) ions in water.
- → **Base:** gives OH^- ions in water; **alkali** = water-soluble base.
- → **Salt:** ionic compound formed by neutralisation of an acid by a base.
- → **Neutralisation:** acid + base ⟶ salt + water (exothermic).
- → **Indicator:** shows acid/base by colour or smell change.

Yaad karo: litmus

"BRA" = Blue litmus Red in Acid. Base mein ulta: red litmus blue. Neutral mein litmus purple hi rehta hai.

Activity 2.1: Testing acids and bases with indicators

Done: Put a drop each of HCl, H_2SO_4, HNO_3, CH_3COOH, NaOH, $Ca(OH)_2$, KOH, $Mg(OH)_2$ and NH_4OH on red and blue litmus, and test with phenolphthalein and methyl orange. **Observed:** Acids and bases change the indicators to different colours. **Conclusion:** Indicators can identify acids and bases (table below).

Indicator	In acid	In base
Blue litmus	Red	No change
Red litmus	No change	Blue
Phenolphthalein	Colourless	Pink
Methyl orange	Red	Yellow
Turmeric	Yellow	Reddish-brown

Activity 2.2: Olfactory indicators

Done: Keep cloth strips with chopped onion overnight; put dilute HCl on one strip and dilute NaOH on another. Repeat with vanilla essence and clove oil. **Observed:** The smell of onion and vanilla **disappears in base** but stays in acid. **Conclusion:** Onion and vanilla are olfactory indicators; clove oil is not.

Paheli-Bujho



Bujho: Paheli, mera pyaaz wala kapda NaOH mein daala toh smell hi gayab! Pyaaz chori ho gaya kya?

Paheli: Bhondu! Pyaaz **olfactory indicator** hai. Base mein uski smell chali jaati hai, acid mein rehti hai.

Bujho: Aur clove oil?

Paheli: Uski smell dono mein rehti hai, isliye woh indicator nahi hai.

2. Chemical Properties of Acids and Bases

(a) Acid + Metal ⟶ Salt + Hydrogen gas

Activity 2.3: Zinc + dilute sulphuric acid

Done: Add zinc granules to dilute H_2SO_4; pass the gas through soap solution and bring a burning candle near a gas-filled bubble. **Observed:** Bubbles form on zinc; the gas burns with a **pop sound**. **Conclusion:** Hydrogen gas is evolved.

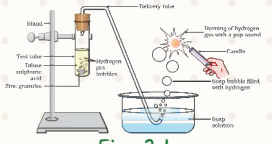


Fig. 2.1

$$Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$$

$$Mg + 2HCl \longrightarrow MgCl_2 + H_2; \quad 2Al + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2$$

Quick fact

Curd and sour substances are **never kept in brass or copper vessels**: their acids react with the metal and form poisonous compounds.

(b) Base + Metal ⟶ Salt + Hydrogen (some metals only)

Activity 2.4: Zinc + sodium hydroxide

Done: Warm zinc granules with NaOH solution and test the gas with a burning splinter. **Observed:** The gas burns with a pop sound. **Conclusion:** Hydrogen is formed along with **sodium zincate**.

$$2NaOH(aq) + Zn(s) \longrightarrow Na_2ZnO_2(s) + H_2(g)$$

(c) Acid + Metal carbonate / hydrogencarbonate ⟶ Salt + CO_2 + Water

Activity 2.5: Na_2CO_3 and $NaHCO_3$ + HCl

Done: Add dilute HCl to Na_2CO_3 (tube A) and $NaHCO_3$ (tube B); pass the gas through lime water. **Observed:** Brisk effervescence; lime water turns **milky**. **Conclusion:** Carbon dioxide gas is evolved.

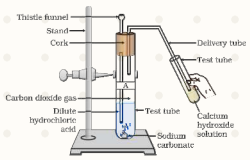


Fig. 2.2

$$Na_2CO_3(s) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$$

$$NaHCO_3(s) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$$

Lime water test: CO_2 turns lime water milky; excess CO_2 makes it clear again.

$$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(l)$$ (milky)

$$CaCO_3(s) + H_2O(l) + CO_2(g) \longrightarrow Ca(HCO_3)_2(aq)$$ (soluble)

Limestone, chalk and marble are different forms of calcium carbonate: $CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$.

Exam Alert

→ Gas test yaad karo: H_2 = jalti tili pe **pop sound**; CO_2 = lime water **milky**; O_2 = bujhti tili phir se jal uthti hai.

→ Equation mein states (s), (aq), (g) likhoge toh poore marks.

(d) Acid + Base ⟶ Salt + Water (Neutralisation)

Activity 2.6: Neutralisation with phenolphthalein

Done: Add 2 drops of phenolphthalein to dilute NaOH, then add dilute HCl drop by drop; then add a few drops of NaOH again. **Observed:** Pink colour **disappears** on adding acid and **comes back** on adding more base. **Conclusion:** Acid and base neutralise each other's effect.

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

(e) Metal oxide + Acid ⟶ Salt + Water

Activity 2.7: Copper oxide + dilute HCl

Done: Add dilute HCl slowly to copper oxide in a beaker while stirring. **Observed:** Copper oxide dissolves; the solution turns **blue-green**. **Conclusion:** Copper(II) chloride is formed. Metal oxides are **basic oxides**.

$$CuO(s) + 2HCl(aq) \longrightarrow CuCl_2(aq) + H_2O(l)$$

(f) Non-metal oxide + Base ⟶ Salt + Water

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

This shows that non-metallic oxides are acidic in nature.

Meme samjho



Baking soda pe nimbu nichoda aur seedha FIZZZ! Seagull bhi chilla utha.

Asli baat: acid + hydrogencarbonate gives salt + water + CO_2; the fizz is CO_2, which turns lime water milky.

Reaction	General equation	Example
acid + metal	salt + H_2	$Zn + H_2SO_4$
base + metal	salt + H_2 (Zn, Al)	$Zn + 2NaOH$
acid + carbonate	salt + CO_2 + H_2O	$Na_2CO_3 + 2HCl$
acid + base	salt + H_2O	$NaOH + HCl$
acid + metal oxide	salt + H_2O	$CuO + 2HCl$
base + non-metal oxide	salt + H_2O	$Ca(OH)_2 + CO_2$

3. What Do All Acids and Bases Have in Common?

Activity 2.8: Do acids conduct electricity?

Done: Fix two nails on a cork in a beaker, connect them to a 6 V battery through a bulb, and pour in dilute HCl, then H_2SO_4, then glucose and alcohol solutions. **Observed:** The bulb glows with acids but **not** with glucose or alcohol. **Conclusion:** Acids give H^+ ions in solution, which carry current. Glucose and alcohol do not ionise.

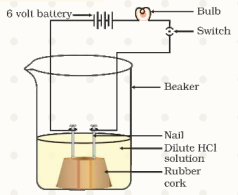


Fig. 2.3

Distilled water does not conduct (no ions); **rain water** conducts because dissolved acids give ions.

Activity 2.9: Dry HCl gas and litmus

Done: Add conc. H_2SO_4 to solid NaCl and test the gas with **dry** and then **wet** blue litmus. In humid weather pass the gas through a **guard tube** of calcium chloride to dry it. **Observed:** Dry litmus does not change; wet litmus turns red. **Conclusion:** HCl shows acidic behaviour only in water, because H^+ ions form only in water.

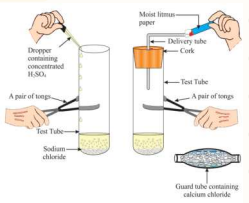


Fig. 2.4

$$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$$ (H^+ exists as hydronium ion)

$$NaOH(s) \xrightarrow{H_2O} Na^+(aq) + OH^-(aq)$$

Alkalis: bases that are soluble in water, e.g. NaOH, KOH, $Ca(OH)_2$. All alkalis are bases, but not all bases are alkalis.

Basic solutions also contain H^+ ions, but far **fewer than OH^-** ions; acidic solutions have more H^+ than OH^-.

Quick fact

The atmosphere of **Venus** is made of thick white and yellowish clouds of **sulphuric acid**. Life as we know it cannot survive there.

Activity 2.10: Dissolving an acid or a base in water

Done: Add a few drops of conc. H_2SO_4 to water and touch the beaker; repeat with NaOH pellets. **Observed:** The beaker becomes hot in both cases. **Conclusion:** Dissolving an acid or base in water is **highly exothermic**.

Dilution: mixing an acid or base with water decreases the concentration of H_3O^+ / OH^- ions per unit volume.

Meme samjho



Dry HCl gas ka attitude: "litmus red karunga". Par bina paani ke kuch nahi hota!

Asli baat: H^+ ions form only in water, so dry HCl does not change dry litmus; wet litmus turns red.

Exam Alert

→ Hamesha **acid ko paani mein** dheere-dheere daalo, paani ko acid mein kabhi nahi: mixture ubal ke uchhal sakta hai aur glass toot sakta hai.

4. How Strong are Acid or Base Solutions? (pH)

Universal indicator: a mixture of indicators that shows **different colours** at different concentrations of H^+ ions.

pH scale: measures the H^+ ion concentration, from **0** (very acidic) to **14** (very alkaline). "p" stands for *potenz* (German: power).

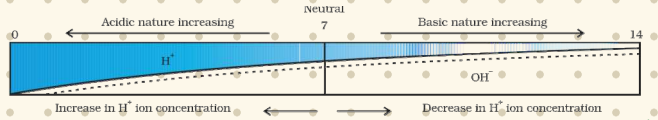


Fig. 2.6 Variation of pH with the change in concentration of $H^+(aq)$ and $OH^-(aq)$ ions

✓ **pH < 7:** acidic **pH = 7:** neutral **pH > 7:** basic

✓ Higher H^+ concentration = lower pH.

Quick fact

Approximate pH: gastric juice **1.2**, lemon juice **2.2**, pure water **7**, blood **7.4**, milk of magnesia **10**, sodium hydroxide solution **14**.

Activity 2.11: pH of common substances

Done: Test the pH of saliva, lemon juice, carrot juice, coffee, soda solution, dilute HCl and dilute NaOH with pH paper. **Observed:** Each solution gives a different colour on pH paper. **Conclusion:** Lemon juice and HCl are acidic; soda solution and NaOH are basic; pure water is neutral.

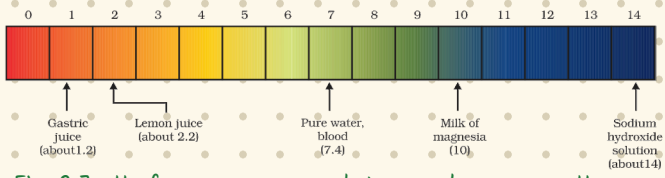


Fig. 2.7 pH of some common substances shown on a pH paper (colours are only a rough guide)

Strong	Weak
give **more** H^+ / OH^- ions	give **fewer** ions
HCl, H_2SO_4, NaOH, KOH	CH_3COOH, NH_4OH
strong electrolyte	weak electrolyte

Exam Alert

→ **Strong vs concentrated** alag hain: strong = zyada ions deta hai; concentrated = paani kam hai.

→ pH 1 wala acid pH 4 wale se zyada strong; pH 13 wala base pH 9 se zyada strong.

5. Importance of pH in Everyday Life

1. **Plants and animals:** our body works within a narrow pH range of **7.0 to 7.8**. Rain with pH below **5.6** is **acid rain**; it makes river water acidic and harms aquatic life.
2. **Soil:** plants need a specific pH range for healthy growth. Too acidic soil is treated with **quick lime, slaked lime or chalk**.
3. **Digestive system:** the stomach makes HCl for digestion. Excess acid causes **acidity** (pain, irritation); an **antacid** (mild base) like milk of magnesia, $Mg(OH)_2$, neutralises it.
4. **Tooth decay:** starts when the pH in the mouth falls **below 5.5**. Bacteria turn leftover sugar into acids that corrode tooth enamel (calcium hydroxyapatite, the hardest substance in the body). Toothpastes are basic and neutralise the acid.
5. **Self-defence by animals and plants:** a bee sting injects **methanoic acid**; rubbing baking soda (mild base) gives relief. Nettle leaves also inject methanoic acid; rubbing a **dock leaf** is the traditional remedy.

Activity 2.12: pH of soil

Done: Shake 2 g soil with 5 mL water, filter, and test the filtrate with universal indicator paper. **Observed:** The colour shows the soil's pH. **Conclusion:** Each soil has its own pH; plants grow well only in a particular range.

Natural source	Acid	Natural source	Acid
Vinegar	Acetic	Tamarind	Tartaric
Orange, lemon	Citric	Tomato	Oxalic
Sour milk (curd)	Lactic	Ant / nettle sting	Methanoic

Meme samjho



Shaadi ka khaana, phir pet mein jalan? Krabs bole: antacid lo!
Asli baat: excess HCl in the stomach causes acidity; an antacid (mild base like $Mg(OH)_2$) neutralises it.

6. More About Salts

Family of salts: salts with the same positive or negative radical, e.g. NaCl and Na_2SO_4 (**sodium** salts); NaCl and KCl (**chloride** salts).

Activity 2.13: Formulae of salts

Done: Write the formulae of K_2SO_4, Na_2SO_4, $CaSO_4$, $MgSO_4$, $CuSO_4$, NaCl, $NaNO_3$, Na_2CO_3 and NH_4Cl and name the acid and base for each. **Observed:** Each salt is made of a positive radical from a base and a negative radical from an acid. **Conclusion:** Salts can be grouped into families.

Activity 2.14: pH of salts

Done: Dissolve NaCl, KNO_3, $AlCl_3$, $ZnSO_4$, $CuSO_4$, CH_3COONa, Na_2CO_3 and $NaHCO_3$ in water and test with pH paper. **Observed:** Some solutions are neutral, some acidic, some basic. **Conclusion:** The pH of a salt depends on the acid and base it comes from.

Salt of	pH	Examples
strong acid + strong base	= 7 (neutral)	NaCl, KNO_3, Na_2SO_4
strong acid + weak base	< 7 (acidic)	NH_4Cl, $CuSO_4$, $AlCl_3$
weak acid + strong base	> 7 (basic)	Na_2CO_3, CH_3COONa, $NaHCO_3$

Exam Alert

→ Salt ka pH poocha ho toh pehle socho: kaunsa acid + kaunsa base? Strong + strong = 7; jo strong hai uska "nature" jeetta hai.
→ Na_2CO_3 ka pH 7 se zyada (basic), NH_4Cl ka pH 7 se kam (acidic): yeh MCQ mein bahut aata hai.

7. Chemicals from Common Salt

Common salt, NaCl (from HCl + NaOH), is obtained from sea water and mined as **rock salt** (brown crystals); salt was a symbol in the **Dandi March**. It is the raw material for NaOH, bleaching powder, baking soda and washing soda.

(a) Sodium hydroxide: Chlor-alkali process

Electricity is passed through **brine** (aqueous NaCl): Cl_2 at the anode, H_2 at the cathode, NaOH solution forms near the cathode. Name: **chlor** (chlorine) + **alkali** (NaOH). NaOH is also called **caustic soda**. H_2 + Cl_2 together give **HCl** (cleaning steel, making NH_4Cl, medicines, cosmetics).

$$2NaCl(aq) + 2H_2O(l) \longrightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$$

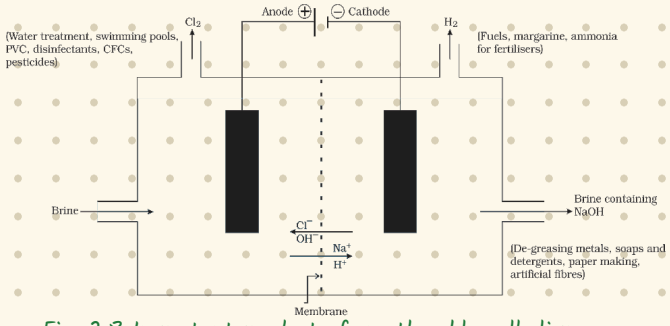


Fig. 2.8 Important products from the chlor-alkali process

(b) Bleaching powder, $Ca(ClO)_2$ (calcium hypochlorite)

Made by the action of chlorine on **dry slaked lime**.

$$2Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(ClO)_2 + CaCl_2 + 2H_2O$$

→ Bleaching cotton and linen (textiles), wood pulp (paper) and washed clothes (laundry).
→ Oxidising agent in chemical industries.
→ Disinfecting drinking water (kills germs).

(c) Baking soda, $NaHCO_3$ (sodium hydrogencarbonate)

$$NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$$

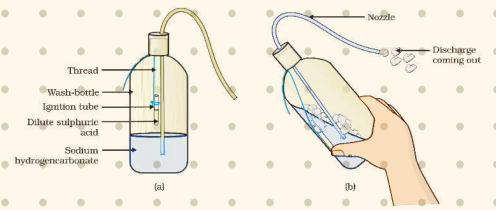
A mild, non-corrosive base. On heating (cooking):

$$2NaHCO_3 \xrightarrow{\text{Heat}} Na_2CO_3 + H_2O + CO_2$$

Baking powder = baking soda + a mild edible acid (tartaric acid). On heating or with water, CO_2 is released and makes bread or cake **soft and spongy**; the acid removes the bitter taste of Na_2CO_3.

$$NaHCO_3 + H^+ \longrightarrow CO_2 + H_2O + \text{sodium salt of acid}$$

→ Antacid (neutralises excess stomach acid).
→ Ingredient of baking powder.
→ Soda-acid **fire extinguishers**.



Soda-acid fire extinguisher: $NaHCO_3$ + H_2SO_4 gives CO_2

(d) Washing soda, $Na_2CO_3.10H_2O$

Sodium carbonate is **recrystallised** from water:

$$Na_2CO_3 + 10H_2O \longrightarrow Na_2CO_3.10H_2O$$

→ Glass, soap and paper industries; manufacture of **borax**.
→ Cleaning agent for domestic purposes.
→ Removing **permanent hardness** of water.

Yaad karo: 4 salts

"Bleach Ca ka, Baking ek Na, Washing mein 10 paani": Bleaching powder = $Ca(ClO)_2$; Baking soda = $NaHCO_3$; Washing soda = $Na_2CO_3.10H_2O$; PoP = $^1/_2\ H_2O$.

8. Are the Crystals of Salts Really Dry?

Activity 2.15: Heating copper sulphate crystals

Done: Heat a few crystals of copper sulphate in a dry boiling tube; then add 2–3 drops of water to the residue.
Observed: Blue crystals turn **white** and water drops appear; the blue colour **returns** on adding water.
Conclusion: Crystals contain **water of crystallisation**: $CuSO_4.5H_2O$.

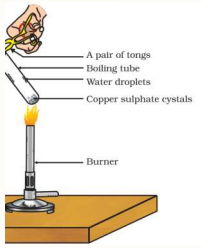


Fig. 2.9

Water of crystallisation: fixed number of water molecules in one formula unit of a salt, e.g. $CuSO_4.5H_2O$, gypsum $CaSO_4.2H_2O$, $Na_2CO_3.10H_2O$.

Plaster of Paris ($CaSO_4.^1/_2H_2O$, calcium sulphate hemihydrate): gypsum heated at **373 K**. With water it sets into hard gypsum, so it is stored in a **moisture-proof** container. $^1/_2H_2O$: **two formula units** of $CaSO_4$ share one molecule of water.

$$CaSO_4.2H_2O \xrightarrow{373\ K} CaSO_4.^1/_2H_2O + 1^1/_2H_2O$$

$$CaSO_4.^1/_2H_2O + 1^1/_2H_2O \longrightarrow CaSO_4.2H_2O \text{ (sets hard)}$$

Uses: plaster for supporting **fractured bones**, toys, decoration material, making surfaces smooth.

Paheli-Bujho



Bujho: POP ka full form Pop music hai na? Isliye doctor haddi pe lagate hain, dard pe music!
Paheli: Bhondu!
Plaster of Paris. Paani milate hi sakht gypsum ban jaata hai, isliye toota haath seedha rehta hai.

Compound	Formula	Made from	Main use
Bleaching powder	$Ca(ClO)_2$	Cl_2 + dry slaked lime	disinfecting water
Baking soda	$NaHCO_3$	NaCl + H_2O + CO_2 + NH_3	antacid, baking powder
Washing soda	$Na_2CO_3.10H_2O$	recrystallising Na_2CO_3	hard water, cleaning
Plaster of Paris	$CaSO_4.^1/_2H_2O$	gypsum at 373 K	fractured bones

1 minute recap

Indicators → Reactions of acids/bases → H^+ / OH^- in water → pH scale → pH in daily life → Salts and their pH → NaOH, bleaching powder, baking soda, washing soda → PoP

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM Board fav A solution turns red litmus blue. Its pH is likely to be:
(a) 1 (b) 4 (c) 5 (d) 10
Ans: (d) 10: it is basic (pH more than 7).

Q2. IM Which of the following is used as an antacid?
(a) NaOH (b) $Ca(OH)_2$
(c) $Mg(OH)_2$ (d) KOH
Ans: (c) Milk of magnesia, $Mg(OH)_2$: a mild base that neutralises excess HCl in the stomach.

Q3. IM Board fav The formula of Plaster of Paris is:
(a) $CaSO_4.2H_2O$ (b) $CaSO_4.^1/_2H_2O$
(c) $CaSO_4$ (d) $CaSO_4.H_2O$
Ans: (b) $CaSO_4.^1/_2H_2O$ (calcium sulphate hemihydrate).

Q4. IM Name the acid present in (a) ant sting (b) tomato.
Ans: (a) Methanoic acid (b) Oxalic acid.

Q5. IM Which gas is evolved at the anode and which at the cathode in the chlor-alkali process?
Ans: Anode: chlorine (Cl_2); cathode: hydrogen (H_2).

Q6. IM Board fav **Assertion (A):** Dry HCl gas does not change the colour of dry blue litmus paper. **Reason (R):** HCl forms H^+ ions only in the presence of water.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q7. IM Why does distilled water not conduct electricity, whereas rain water does?
Ans: Distilled water has no ions. Rain water has dissolved acids (from gases like CO_2) that give ions, so it conducts.

Q8. IM Colour of phenolphthalein in (a) acid (b) base?
Ans: (a) Colourless (b) Pink.

Q9. IM Which sodium compound is used for removing permanent hardness of water?
Ans: Washing soda, $Na_2CO_3.10H_2O$.

Q10. IM **Assertion (A):** Bleaching powder is used to disinfect drinking water. **Reason (R):** Bleaching powder releases chlorine, which kills germs.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q11. IM Why does an acid solution conduct electricity?
Ans: Because the acid gives H^+ (H_3O^+) ions in water, and these ions carry the current.

Short answer (2–3 marks)

Q12. 3M Board fav Alcohol and glucose contain hydrogen but are not acids. Describe an activity to prove this.
Ans: Activity 2.8: fix two nails on a cork in a beaker, connect to a 6 V battery and bulb. With dilute HCl or H_2SO_4 the bulb glows; with glucose or alcohol solution it does not. So glucose and alcohol do not give H^+ ions in water and are not acids.

Q13. 2M Why do HCl and HNO_3 show acidic character in aqueous solution, while their dry forms do not?
Ans: Acidic character is due to H^+ (H_3O^+) ions, which form only when the acid dissolves in water: $HCl + H_2O \longrightarrow H_3O^+ + Cl^-$. Without water no ions form.

Q14. 3M Board fav You have three test tubes with distilled water, an acidic solution and a basic solution. Using only red litmus paper, how will you identify them?
Ans: Dip red litmus in each: the one that turns it blue is the base. Put a drop of this base on another red litmus to make it blue, then dip this blue paper in the other two: the one that turns it red is the acid; the remaining one, with no change, is distilled water.

Q15. 2M Board fav Why should curd and sour substances not be kept in brass and copper vessels?
Ans: Curd and sour foods contain acids, which react with copper/brass and form poisonous, harmful compounds that spoil the food.

Q16. 2M Board fav While diluting, why should the acid be added to water and never water to the acid?
Ans: Mixing acid with water is highly exothermic. If water is added to concentrated acid, the heat can make the mixture splash out and cause burns, and the glass may break. Adding acid slowly to water spreads the heat.

Q17. 3M Board fav Five solutions A, B, C, D and E have pH 4, 1, 11, 7 and 9. Which is (a) neutral (b) strongly alkaline (c) strongly acidic (d) weakly acidic (e) weakly alkaline? Arrange them in increasing order of H^+ concentration.
Ans: (a) D (b) C (c) B (d) A (e) E. Increasing H^+: C (11) < E (9) < D (7) < A (4) < B (1).

Q18. 2M How is the concentration of H_3O^+ ions affected when a solution of an acid is diluted?
Ans: It decreases, because the same number of H_3O^+ ions is now spread over a larger volume of solution.

Q19. 3M Board fav A milkman adds a very small amount of baking soda to fresh milk. (a) Why does he shift the pH of milk from 6 to slightly alkaline? (b) Why does this milk take longer to set as curd?
Ans: (a) An alkaline medium keeps the milk from turning sour (lactic acid) quickly, so it lasts longer. (b) The lactic acid formed is first neutralised by the baking soda, so more time is needed for curd to set.

Q20. 3M Board fav What is Plaster of Paris? How is it prepared? Why is it stored in a moisture-proof container?
Ans: $CaSO_4.{}^1/_2H_2O$. Gypsum is heated at 373 K: $CaSO_4.2H_2O \xrightarrow{373\ K} CaSO_4.{}^1/_2H_2O + 1{}^1/_2H_2O$. With moisture it sets into hard gypsum ($CaSO_4.2H_2O$), so it would be useless.

Q21. 2M What is meant by water of crystallisation? Give the formulae of two salts having it.
Ans: The fixed number of water molecules present in one formula unit of a salt. e.g. $CuSO_4.5H_2O$ (blue vitriol), $Na_2CO_3.10H_2O$ (washing soda), $CaSO_4.2H_2O$ (gypsum).

Q22. 2M Why does tooth decay start when the pH of the mouth is lower than 5.5? How can it be prevented?
Ans: Bacteria break down sugar and food particles left in the mouth into acids. Below pH 5.5, the acid corrodes tooth enamel (calcium hydroxyapatite). Brushing with basic toothpaste after eating neutralises the acid.

Q23. 2M Equal lengths of Mg ribbon are put in tube A (HCl) and tube B (acetic acid) of the same concentration. In which tube is the fizzing more vigorous, and why?
Ans: Tube A. HCl is a strong acid and gives more H^+ ions than acetic acid (a weak acid), so it reacts faster with magnesium and releases hydrogen faster.

Q24. 2M Board fav A metal compound A reacts with dilute HCl to produce effervescence. The gas evolved extinguishes a burning candle. Write a balanced equation if one of the compounds formed is calcium chloride.
Ans: A is calcium carbonate; the gas is CO_2. $CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$.

Q25. 3M A white pellet B, a by-product of the chlor-alkali process, absorbs moisture and becomes sticky when kept in the open. Identify B. What type of reaction occurs when B reacts with HCl? Write the equation.
Ans: B is sodium hydroxide (NaOH). With HCl it undergoes neutralisation: $NaOH + HCl \longrightarrow NaCl + H_2O$.

Q26. 2M Give two uses each of washing soda and baking soda.
Ans: Washing soda: glass, soap and paper industries; removing permanent hardness; cleaning agent. Baking soda: antacid; baking powder; soda-acid fire extinguishers.

Q27. 2M What is a family of salts? Name two salts each of the sodium family and the chloride family.
Ans: Salts having the same positive or negative radical belong to one family. Sodium family: NaCl, Na_2SO_4. Chloride family: NaCl, KCl.

Exam Alert
Yeh 4 equations har saal aati hain: chlor-alkali, bleaching powder, baking soda heating, gypsum $\longrightarrow$ PoP. States ke saath likhna! Bleaching powder ka formula ab NCERT ke hisaab se $Ca(ClO)_2$ likho.

Long answer (5 marks)

Q28. 5M Board fav (a) Describe an activity to show that acids react with metals to produce hydrogen gas. How is the gas tested? (b) Write the equation for zinc with NaOH. (c) Why is HCl a stronger acid than acetic acid?

Ans: (a) Add zinc granules to dilute H_2SO_4 in a test tube fitted with a delivery tube; pass the gas into soap solution. Bring a burning candle near a gas-filled bubble: it burns with a pop sound, so the gas is hydrogen. $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$. (b) $2NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$. (c) HCl ionises completely and gives more H^+ ions in water; acetic acid ionises only partly.

Q29. 5M Board fav (a) What is the chlor-alkali process? Write the equation and name the products at the anode and cathode. (b) Give one use each of the three products. (c) How is bleaching powder made from one of them? Write the equation.

Ans: (a) Electrolysis of brine (aqueous NaCl) gives NaOH, Cl_2 and H_2: $2NaCl(aq) + 2H_2O(l) \longrightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$. Cl_2 at the anode, H_2 at the cathode, NaOH near the cathode. (b) H_2: fuels, margarine, ammonia; Cl_2: water treatment, PVC, pesticides; NaOH: soaps and detergents, paper, degreasing metals. (c) Cl_2 is passed over dry slaked lime: $2Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(ClO)_2 + CaCl_2 + 2H_2O$.

Q30. 5M Board fav (a) Write the chemical name and formula of baking soda. How is it prepared? (b) What happens when it is heated? (c) Why does baking powder make a cake soft and spongy, and why is tartaric acid added? (d) Give two other uses.

Ans: (a) Sodium hydrogencarbonate, $NaHCO_3$: $NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$. (b) $2NaHCO_3 \xrightarrow{Heat} Na_2CO_3 + H_2O + CO_2$. (c) On heating/with water, CO_2 is released and puffs up the dough; tartaric acid neutralises the bitter Na_2CO_3 formed. (d) Antacid; soda-acid fire extinguisher.

Q31. 5M (a) What is the pH scale? Why does pH matter in our digestive system and for plants? (b) What happens when a bee stings, and how is it treated? (c) Write the reaction of a metal carbonate with an acid and the test for the gas formed.

Ans: (a) A scale (0 to 14) to measure H^+ ion concentration; below 7 acidic, 7 neutral, above 7 basic. Excess stomach HCl causes acidity, cured by antacids like $Mg(OH)_2$; plants grow only in a particular soil pH, so acidic soil is treated with lime. (b) The sting injects methanoic acid, causing pain; rubbing a mild base like baking soda neutralises it. (c) $Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$; CO_2 turns lime water milky.

Yaad karo: formula list (MCQ mein seedha aata hai)

Baking soda $NaHCO_3$ | Washing soda $Na_2CO_3.10H_2O$ | Bleaching powder $Ca(ClO)_2$ | PoP $CaSO_4.{}^1/_2H_2O$ | Gypsum $CaSO_4.2H_2O$ | Blue vitriol $CuSO_4.5H_2O$ | Milk of magnesia $Mg(OH)_2$ | Quick lime CaO | Slaked lime $Ca(OH)_2$ | Limestone, chalk, marble $CaCO_3$ | Caustic soda NaOH | Sodium zincate Na_2ZnO_2 | Hydronium ion H_3O^+ | Common salt / rock salt NaCl | Soda ash Na_2CO_3 | Methanoic (formic) acid HCOOH | Acetic acid CH_3COOH

Case-based (4 marks)

Q32. 4M Board fav Rahul tested soil from his field: its pH was 5. His crops were not growing well. His teacher also told him that rain with pH below 5.6 had fallen in the area. (a) Is the soil acidic or basic? (b) Name two substances he can add to correct it. (c) What is acid rain? (d) How does it affect aquatic life?

Ans: (a) Acidic (pH below 7). (b) Quick lime (CaO), slaked lime ($Ca(OH)_2$) or chalk ($CaCO_3$). (c) Rain with pH less than 5.6. (d) It lowers the pH of river and lake water, which makes survival of aquatic animals difficult.

Q33. 4M Board fav A student heated a white salt X used in cooking cakes; a gas that turned lime water milky was evolved, leaving a residue Y. Y, on recrystallisation with water, gave a salt Z used to remove permanent hardness of water. (a) Identify X, Y and Z. (b) Write the equation for heating X. (c) Write the equation for forming Z. (d) What is the pH of a solution of Y: less than, equal to or more than 7?

Ans: (a) X = baking soda $NaHCO_3$; Y = sodium carbonate Na_2CO_3; Z = washing soda $Na_2CO_3.10H_2O$. (b) $2NaHCO_3 \xrightarrow{Heat} Na_2CO_3 + H_2O + CO_2$. (c) $Na_2CO_3 + 10H_2O \longrightarrow Na_2CO_3.10H_2O$. (d) More than 7 (salt of a weak acid and a strong base).

Q34. 4M Priya added a few drops of universal indicator to three colourless solutions: P turned red, Q turned green and R turned blue-violet. (a) Arrange P, Q and R in increasing order of pH. (b) Which of them could be sodium chloride solution? (c) Which solution would react with zinc to give hydrogen gas? (d) What happens when P and R are mixed?

Ans: (a) P (acidic) < Q (neutral) < R (basic). (b) Q: NaCl is a salt of a strong acid and a strong base, so its pH is 7. (c) P, the acid (R may also react if it is a strong alkali like NaOH, giving sodium zincate). (d) Neutralisation: a salt and water are formed and the pH moves towards 7.

Q35. 4M A worker adds a pale yellow powder that smells strongly of chlorine to a water tank. (a) Name the powder and give its formula. (b) Write the equation for its preparation. (c) Why is it added to water? (d) Give one other use.

Ans: (a) Bleaching powder, $Ca(ClO)_2$ (calcium hypochlorite). (b) $2Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(ClO)_2 + CaCl_2 + 2H_2O$. (c) To disinfect the water: it kills germs. (d) Bleaching cotton and linen, or as an oxidising agent in industry.

Q36. 4M A doctor used a white powder P mixed with water to set a fractured arm; the paste became hard within minutes. (a) Name P and give its formula. (b) How is P made? Give the equation. (c) Write the equation for its setting. (d) Why must P be stored in a moisture-proof container? (e) Give two other uses of P.

Ans: (a) Plaster of Paris, $CaSO_4.{}^1/_2H_2O$. (b) By heating gypsum at 373 K: $CaSO_4.2H_2O \xrightarrow{373\ K} CaSO_4.{}^1/_2H_2O + 1{}^1/_2H_2O$. (c) $CaSO_4.{}^1/_2H_2O + 1{}^1/_2H_2O \longrightarrow CaSO_4.2H_2O$. (d) Moisture would make it set into hard gypsum before use. (e) Making toys and decoration items; making surfaces smooth.

METALS AND NON-METALS



Most asked: reactivity series and displacement, ionic bond formation (NaCl, $MgCl_2$, MgO), extraction (roasting, calcination, thermit, electrolysis), electrolytic refining, corrosion and alloys

About 118 elements are known and more than 90 are **metals** (iron, copper, gold, aluminium). Oxygen, sulphur, carbon and iodine are **non-metals**.

1. Physical Properties of Metals

Activity 3.1: Lustre

Done: Rub iron, copper, aluminium and magnesium with sandpaper. **Observed:** The cleaned surface **shines**. **Conclusion:** Pure metals have **metallic lustre**.

Activity 3.2: Hardness

Done: Try to cut iron, copper, Al, Mg and sodium with a knife. **Observed:** Only **sodium** cuts easily. **Conclusion:** Metals are generally hard; Na and K are soft.

Activity 3.3: Malleability

Done: Strike iron, zinc, lead and copper with a hammer. **Observed:** They flatten into thin sheets. **Conclusion:** Metals are **malleable**; gold and silver are the most malleable.

Activity 3.4: Ductility

Done: List the metals whose wires we use daily. **Observed:** Copper, aluminium and iron wires are common. **Conclusion:** Metals are **ductile**; a wire about 2 km long can be drawn from 1 g of gold.

Activity 3.5: Conduction of heat

Done: Clamp an aluminium or copper wire, fix a pin at its free end with wax, and heat the wire near the clamp. **Observed:** The wax melts and the pin falls; the wire itself does not melt.

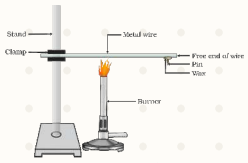


Fig. 3.1

Conclusion: Metals are good conductors of heat and have high melting points. **Silver and copper** are the best; **lead and mercury** are poor.

Activity 3.6: Conduction of electricity

Done: Place a metal between terminals A and B of a circuit with a bulb. **Observed:** The bulb glows. **Conclusion:** Metals conduct electricity; wires are made of copper and aluminium and coated with PVC.

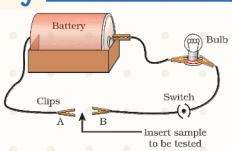


Fig. 3.2

Sonorous: metals make a ringing sound when struck, which is why school bells are made of metal.

2. Non-metals

Activity 3.7: Properties of non-metals

Done: Repeat Activities 3.1 to 3.4 and 3.6 with carbon, sulphur and iodine. **Observed:** They are dull and brittle, and do not conduct (except graphite). **Conclusion:** Non-metals are solids or gases (bromine is a liquid) with properties opposite to metals.

Exceptions (favourite MCQ)

- → **Mercury** is a liquid metal; **gallium and caesium** melt on the palm.
- → **Na, K** are soft and have low density and low melting points.
- → **Iodine** is a lustrous non-metal; **diamond** (carbon) is the hardest natural substance with a very high melting point.
- → **Graphite** (carbon) conducts electricity.

Meme samjho



Graphite dekh ke lagta hai non-metal hai, phir bhi current conduct karta hai! Pigeon confuse.

Asli baat: graphite is a non-metal (carbon) but a good conductor of electricity: an exception.

Activity 3.8: Oxides of Mg and S

Done: Burn Mg ribbon and sulphur powder; dissolve the products in water; test with litmus. **Observed:** MgO solution turns red litmus **blue**; SO_2 solution turns blue litmus **red**. **Conclusion:** Metal oxides are **basic**; non-metal oxides are **acidic**.

$$2Mg + O_2 \longrightarrow 2MgO; \quad MgO + H_2O \longrightarrow Mg(OH)_2$$

$$S + O_2 \longrightarrow SO_2; \quad SO_2 + H_2O \longrightarrow H_2SO_3$$

Non-metals do **not** displace H_2 from dilute acids, because they cannot give electrons to H^+ ions. They react with hydrogen to form **hydrides**: H_2S, NH_3, HCl.

Property	Metals	Non-metals
Lustre	shiny	dull (except iodine)
Malleable, ductile	yes	no, brittle
Conductivity	good	poor (except graphite)
State	solids (Hg liquid)	solid / gas (Br_2 liquid)
Oxides	basic	acidic or neutral (CO, H_2O)
Dilute acids	many give H_2	no H_2
Ions formed	positive (lose e^-)	negative (gain e^-)

3. Chemical Properties of Metals

(a) Metal + Oxygen ⟶ Metal oxide

$$2Cu + O_2 \longrightarrow 2CuO \text{ (black)}$$

$$4Al + 3O_2 \longrightarrow 2Al_2O_3$$

Amphoteric oxides: metal oxides that react with both acids and bases to give salt and water, e.g. Al_2O_3, ZnO.

$$Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$$

$$Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O \text{ (sodium aluminate)}$$

Most metal oxides are insoluble in water; soluble ones form **alkalis**:

$$Na_2O(s) + H_2O(l) \longrightarrow 2NaOH(aq)$$

$$K_2O(s) + H_2O(l) \longrightarrow 2KOH(aq)$$

Activity 3.9: Burning metals in air

Done: Hold samples of metals with tongs and burn them over a flame. **Observed:** Na and K catch fire easily; Mg burns with a white flame; iron filings burn with sparks; copper does not burn but turns black. **Conclusion:** Reactivity with oxygen differs.

→ **K and Na** react so vigorously that they are kept immersed in kerosene.

→ **Mg, Al, Zn, Pb** form a thin protective oxide layer that stops further oxidation.

→ **Silver and gold** do not react with oxygen even at high temperature.

(b) Metal + Water ⟶ Metal oxide / hydroxide + H_2

Activity 3.10: Metals with water

Done: Put metal pieces in cold water, then in hot water; pass steam over those that still do not react. **Observed:** Na and K react violently with cold water (the H_2 catches fire); Ca reacts less violently and floats; Mg reacts with hot water and floats; Al, Fe, Zn react only with steam; Pb, Cu, Ag, Au do not react. **Conclusion:** Reactivity of metals with water differs.

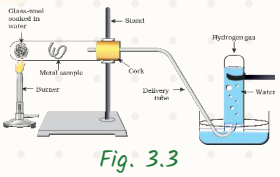


Fig. 3.3

$$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g) + heat$$

$$2K(s) + 2H_2O(l) \longrightarrow 2KOH(aq) + H_2(g) + heat$$

$$Ca(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(aq) + H_2(g)$$

$$2Al(s) + 3H_2O(g) \longrightarrow Al_2O_3(s) + 3H_2(g)$$

$$3Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2(g)$$

Exam Alert

→ Ca aur Mg paani mein **float** karte hain kyunki H_2 gas ke bubbles unki surface pe chipak jaate hain.

→ Al, Fe, Zn sirf **steam** se react karte hain aur **oxide** banate hain, hydroxide nahi.

(c) Metal + Dilute acid ⟶ Salt + Hydrogen

Activity 3.11: Metals with dilute HCl

Done: Put clean Mg, Al, Zn, Fe and Cu pieces in dilute HCl with a thermometer in each. **Observed:** Bubbles form fastest with Mg (temperature rises most), then Al, Zn, Fe; copper shows no reaction. **Conclusion:** Reactivity order: Mg > Al > Zn > Fe; Cu does not react with dilute HCl.

$$Fe(s) + 2HCl(aq) \longrightarrow FeCl_2(aq) + H_2(g)$$

H_2 is **not** evolved with nitric acid: HNO_3 is a strong **oxidising agent** that oxidises H_2 to water. Exception: Mg and Mn give H_2 with very dilute HNO_3.

Aqua regia ("royal water"): freshly mixed **conc. HCl + conc. HNO_3 in 3 : 1**; a fuming liquid that dissolves gold and platinum.

Paheli–Bujho



Bujho: Gold ki chain kisi bhi acid mein daal do, kuch nahi hoga. Gold toh invincible hai!

Paheli: Aqua regia mein daal ke dekh! Conc. HCl aur HNO_3 3:1 mein, gold bhi ghul jaayega.

Bujho: Rehne do, chain mummy ki hai...

(d) Metal + Salt solution of another metal

A more reactive metal displaces a less reactive metal from its salt solution.

Activity 3.12: Copper and iron in salt solutions

Done: Put a copper wire in iron sulphate solution and an iron nail in copper sulphate solution. **Observed:** Only the iron nail gets a **brown coating**, and the blue solution turns **green**. **Conclusion:** Iron is more reactive than copper.

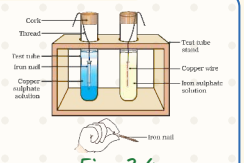


Fig. 3.4

$$Fe(s) + CuSO_4(aq) \longrightarrow FeSO_4(aq) + Cu(s)$$

$$Cu(s) + 2AgNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$

4. The Reactivity Series

Metals arranged in **decreasing order of reactivity**:

Reactivity series

K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au

Metals above hydrogen displace H_2 from dilute acids; Cu, Hg, Ag, Au (below H) do not.

Yaad karo: reactivity series

"Please Stop Calling Me A Zebra, I Like Her Calling Me Smart Guy": K, Na, Ca, Mg, Al, Zn, Fe, Pb, H, Cu, Hg, Ag, Au.

5. How do Metals and Non-metals React?

Noble gases (He, Ne, Ar) have a **completely filled outer shell**, so they are unreactive. Other elements react to attain this stable **octet** by losing, gaining or sharing electrons.

$$Na \rightarrow Na^+ + e^-$$
$$2,8,1 \quad 2,8$$
(Sodium cation)

$$Cl + e^- \rightarrow Cl^-$$
$$2,8,7 \quad 2,8,8$$
(Chloride anion)

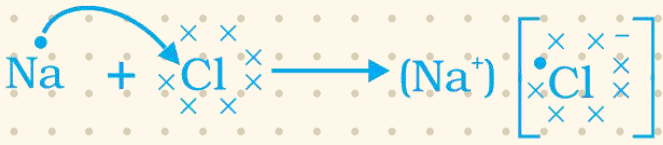


Fig. 3.5 Formation of sodium chloride

Ionic (electrovalent) compounds: formed by the transfer of electrons from a metal to a non-metal. The metal becomes a **cation**, the non-metal an **anion**; they are held by strong electrostatic forces. e.g. NaCl, $MgCl_2$, MgO.

$$Mg \longrightarrow Mg^{2+} + 2e^-$$
$$2,8,2 \quad 2,8$$
(Magnesium cation)

$$Cl + e^- \longrightarrow Cl^-$$
$$2,8,7 \quad 2,8,8$$
(Chloride anion)

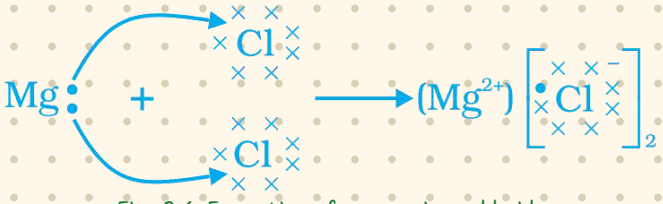
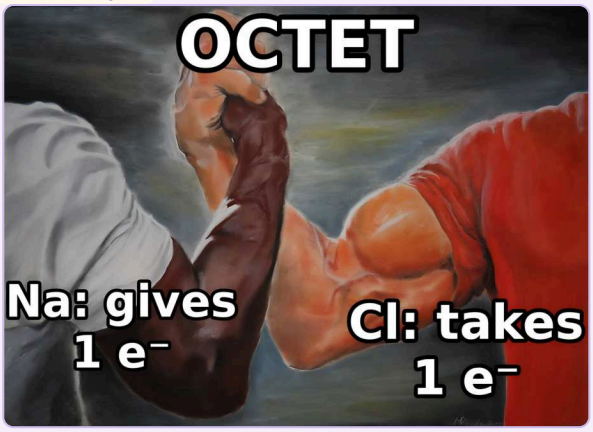


Fig. 3.6 Formation of magnesium chloride

Compound	Electron transfer
$MgCl_2$	Mg (2,8,2) gives 1 e^- to each of two Cl (2,8,7) ⟶ Mg^{2+} + $2Cl^-$
Na_2O	each of two Na (2,8,1) gives 1 e^- to O (2,6) ⟶ $2Na^+$ + O^{2-}
MgO	Mg (2,8,2) gives 2 e^- to O (2,6) ⟶ Mg^{2+} + O^{2-}

Meme samjho



Na bola "1 electron le lo", Cl bola "de do". Dono ka octet complete, deal pakki!

Asli baat: in NaCl, sodium transfers one electron to chlorine; Na^+ and Cl^- are held by electrostatic attraction.

Properties of ionic compounds

1. **Physical nature:** solid and hard (strong forces), but brittle.
2. **Melting and boiling points:** high, because much energy is needed to break the strong attraction (NaCl: m.p. 1074 K, b.p. 1686 K).
3. **Solubility:** soluble in water; insoluble in kerosene and petrol.
4. **Conduction:** conduct electricity in the **molten state and in solution** (ions move), not in the solid state (ions are fixed).

Activity 3.13: Properties of ionic compounds

Done: Heat NaCl, KI and $BaCl_2$ on a spatula; dissolve them in water, petrol and kerosene; test the solutions in a circuit. **Observed:** They melt only at high temperature, dissolve only in water, and the solution makes the bulb glow. **Conclusion:** Ionic compounds have high melting points, dissolve in water and conduct in solution.

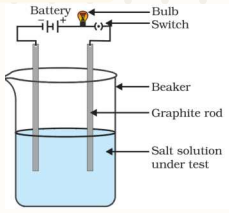


Fig. 3.8

6. Occurrence and Extraction of Metals

Minerals: naturally occurring elements or compounds in the earth's crust. **Ores:** minerals from which a metal can be extracted **profitably**.

Metals low in the series (Au, Ag, Pt, Cu) occur **free**; others occur as oxides, sulphides or carbonates. K, Na, Ca, Mg, Al are never found free.

Gangue: impurities like soil and sand in the ore, removed first by **enrichment (concentration)** of the ore.

Reactivity	Metals	Steps after concentration
high	K, Na, Ca, Mg, Al	electrolysis of molten ore ⟶ metal
middle	Zn, Fe, Pb	sulphide: roasting / carbonate: calcination ⟶ oxide ⟶ reduction with carbon ⟶ metal
low	Cu, Hg	sulphide ore heated in air ⟶ metal
all		refining (electrolytic) ⟶ pure metal

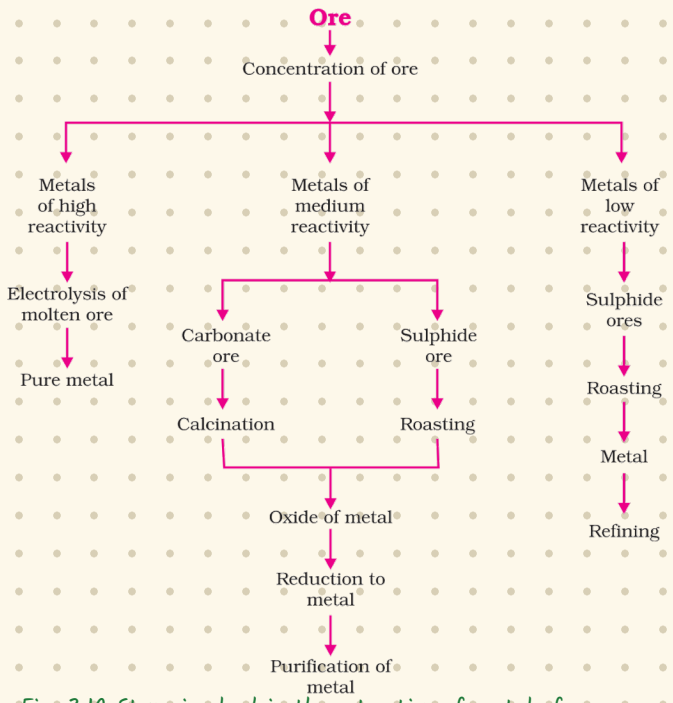


Fig. 3.10 Steps involved in the extraction of metals from ores

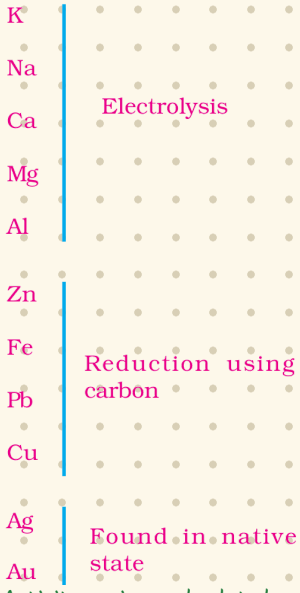


Fig. 3.9 Activity series and related metallurgy

(a) Metals low in the series: heating alone

The ore is heated in air with no other reducing agent (**self-reduction**): mercury from cinnabar (HgS), copper from Cu_2S.

$$2HgS(s) + 3O_2(g) \xrightarrow{Heat} 2HgO(s) + 2SO_2(g)$$

$$2HgO(s) \xrightarrow{Heat} 2Hg(l) + O_2(g)$$

$$2Cu_2S + 3O_2 \xrightarrow{Heat} 2Cu_2O + 2SO_2$$

$$2Cu_2O + Cu_2S \xrightarrow{Heat} 6Cu + SO_2$$

(b) Metals in the middle (Zn, Fe, Pb): oxide first

It is easier to get a metal from its oxide, so the ore is first converted to the oxide.

Roasting	Calcination
sulphide ore heated strongly in **excess air**	carbonate ore heated strongly in **limited air**
$2ZnS + 3O_2 \xrightarrow{Heat} 2ZnO + 2SO_2$	$ZnCO_3 \xrightarrow{Heat} ZnO + CO_2$

The oxide is then **reduced with carbon (coke)**:

$$ZnO(s) + C(s) \longrightarrow Zn(s) + CO(g)$$

Thermit reaction: highly reactive Al reduces metal oxides with so much heat that the metal forms in the **molten state**. Used to join **railway tracks** and cracked machine parts.

$$Fe_2O_3(s) + 2Al(s) \longrightarrow 2Fe(l) + Al_2O_3(s) + heat$$

$$3MnO_2(s) + 4Al(s) \longrightarrow 3Mn(l) + 2Al_2O_3(s) + heat$$



Fig. 3.11 Thermit process for joining railway tracks

Meme samjho



Gru ka plan: Al + Fe_2O_3 mix karo, aag lagao, pighla hua iron... aur railway track jud gaya!

Asli baat: thermit reaction: Al displaces Fe from Fe_2O_3 with huge heat; molten iron welds railway tracks.

(c) Metals at the top: electrolytic reduction

K, Na, Ca, Mg, Al have more affinity for oxygen than carbon, so they are obtained by **electrolysis of their molten chlorides** (Al from molten aluminium oxide). Metal at the cathode, chlorine at the anode.

$$Na^+ + e^- \longrightarrow Na \text{ (cathode)}$$

$$2Cl^- \longrightarrow Cl_2 + 2e^- \text{ (anode)}$$

7. Refining of Metals

Electrolytic refining (Cu, Zn, Sn, Ni, Ag, Au): impure metal = anode, a thin strip of pure metal = cathode, a solution of the metal salt (e.g. acidified $CuSO_4$) = electrolyte.

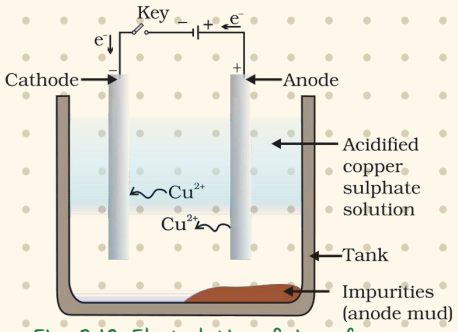


Fig. 3.12 Electrolytic refining of copper

On passing current, the impure metal of the anode **dissolves** into the electrolyte and an equal amount of pure metal from the electrolyte **deposits on the cathode**. Soluble impurities go into solution; insoluble ones settle below the anode as **anode mud**.

Exam Alert

→ **Impure = anode (+), pure = cathode (–).** Ulta likha toh poore marks gaye!

→ Anode mud mein silver, gold jaisi insoluble impurities hoti hain.

8. Corrosion

Corrosion: the slow eating away of a metal when its surface is attacked by **air, moisture or acids**. Silver turns **black** (Ag_2S), copper gets a **green** coat (basic copper carbonate), iron gets brown flaky **rust**.

Activity 3.14: Conditions for rusting

Done: Keep iron nails in three test tubes: A with water (air + water), B with boiled water and an oil layer (no air), C with anhydrous $CaCl_2$ (no moisture). Leave for a few days. **Observed:** Nails rust only in tube A. **Conclusion:** Rusting needs **both air and water**.

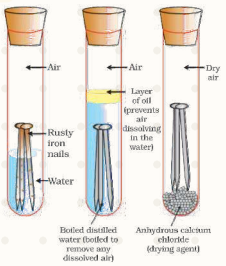


Fig. 3.13

Meme samjho



Chamakti chaandi ki payal ka mask utaara toh nikla... kaala Ag_2S coat!

Asli baat: silver reacts with sulphur in air and forms a black coating of silver sulphide (Ag_2S).

Prevention of corrosion

→ Painting, oiling, greasing; chrome plating, anodising, electroplating; making **alloys** (stainless steel).

→ **Galvanising:** coating iron or steel with **zinc**; the iron stays protected even if the zinc coat breaks.

Alloys

Alloy: a homogeneous mixture of two or more metals, or a metal and a non-metal. An alloy of mercury is an **amalgam**.

Alloy	Composition	Special point
Steel	Fe + C	hard and strong
Stainless steel	Fe + Ni + Cr	does not rust
Brass	Cu + Zn	lower conductivity than Cu
Bronze	Cu + Sn	used for statues
Solder	Pb + Sn	low m.p.; welds electric wires

Pure iron is very soft and stretches easily when hot, so it is never used pure; a little carbon (0.05%) makes it hard and strong (steel).

Pure gold (**24 carat**) is very soft, so it is alloyed with silver or copper to make **22 carat** jewellery. Alloys have lower electrical conductivity and melting point than pure metals.

Paheli-Bujho



Bujho: Mera bracelet 24 carat ka hai, ekdum pure! Ab toh roz pehnunga.

Paheli: 24 carat gold bahut soft hota hai, roz pehnoge toh mud jaayega.

Paheli: Jewellery 22 carat se banti hai: gold mein silver ya copper milaate hain. Ise **alloy** kehte hain.

Quick fact: iron pillar

The **iron pillar near the Qutub Minar** in Delhi was made about 1600 years ago by Indian iron workers. It is about 8 m high, weighs about 6 tonnes, and has **not rusted**.

1 minute recap

Physical properties → Non-metals + exceptions → Reactions with O_2, water, acids, salts → Reactivity series → Ionic bond → Extraction → Refining → Corrosion and alloys

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM Board fav Which of the following metals is a liquid at room temperature?

(a) Sodium (b) Gallium (c) Mercury (d) Zinc

Ans: (c) Mercury.

Q2. IM An element forms an oxide A_2O_3 which is acidic in nature. The element is likely to be:

(a) Na (b) Mg (c) a non-metal (d) Al

Ans: (c) a non-metal: non-metal oxides are acidic.

Q3. 1M Board fav Which of the following pairs will give a displacement reaction?
(a) NaCl solution and copper
(b) $MgCl_2$ solution and aluminium
(c) $FeSO_4$ solution and silver
(d) $AgNO_3$ solution and copper
Ans: (d) $AgNO_3$ solution and copper metal: Cu is more reactive than Ag.

Q4. 1M Name a non-metal that conducts electricity, and a metal that is a poor conductor of heat.
Ans: Graphite (carbon); lead or mercury.

Q5. 1M Board fav Why is sodium kept immersed in kerosene oil?
Ans: Sodium reacts so vigorously with oxygen and moisture in air that it catches fire; kerosene keeps it away from air and water.

Q6. 1M **Assertion (A):** Zinc oxide is amphoteric. **Reason (R):** ZnO reacts with both acids and bases to give salt and water.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q7. 1M What is aqua regia?
Ans: A freshly prepared mixture of conc. HCl and conc. HNO_3 in the ratio 3 : 1; it dissolves gold and platinum.

Q8. 1M **Assertion (A):** Hydrogen gas is not evolved when most metals react with nitric acid. **Reason (R):** Nitric acid is a strong oxidising agent.
Ans: (a) Both A and R are true, and R is the correct explanation of A (HNO_3 oxidises H_2 to water).

Q9. 1M Name two metals that are found in the free state in nature.
Ans: Gold and platinum (also silver and copper sometimes).

Q10. 1M Which gas is produced when dilute HCl is added to a reactive metal? Write the equation for zinc.
Ans: Hydrogen: $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$.

Short answer (2–3 marks)

Q11. 3M Board fav What are amphoteric oxides? Give two examples with equations.
Ans: Metal oxides that react with both acids and bases to give salt and water, e.g. Al_2O_3, ZnO. $Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$; $Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$.

Q12. 3M Board fav Write balanced equations for the reaction of (a) iron with steam (b) calcium with water (c) potassium with water.
Ans: (a) $3Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2(g)$ (b) $Ca(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(aq) + H_2(g)$ (c) $2K(s) + 2H_2O(l) \longrightarrow 2KOH(aq) + H_2(g)$ + heat.

Q13. 2M Why is copper not able to displace hydrogen from dilute acids?
Ans: Copper is less reactive than hydrogen (below H in the reactivity series), so it cannot give electrons to H^+ ions.

Q14. 3M Board fav Show the formation of Na_2O and MgO by the transfer of electrons.
Ans: Na (2,8,1): each of two Na atoms gives 1 e^- to O (2,6): $2Na \longrightarrow 2Na^+ + 2e^-$; $O + 2e^- \longrightarrow O^{2-}$ (2,8); $Na_2O = 2Na^+ O^{2-}$. Mg (2,8,2) gives 2 e^- to O: $Mg \longrightarrow Mg^{2+} + 2e^-$; $O + 2e^- \longrightarrow O^{2-}$; $MgO = Mg^{2+} O^{2-}$.

Q15. 3M Board fav Why do ionic compounds have high melting points? Why do they conduct electricity in the molten state but not as solids?
Ans: Strong electrostatic forces hold the ions, so much energy is needed to break them. In the solid the ions are fixed; on melting or dissolving they become free to move and carry current.

Q16. 3M Board fav Define roasting and calcination with one equation each.
Ans: Roasting: heating a sulphide ore strongly in excess air: $2ZnS + 3O_2 \xrightarrow{Heat} 2ZnO + 2SO_2$. Calcination: heating a carbonate ore strongly in limited air: $ZnCO_3 \xrightarrow{Heat} ZnO + CO_2$.

Q17. 2M How is mercury obtained from cinnabar? Write the equations.
Ans: Cinnabar (HgS) is heated in air: $2HgS + 3O_2 \xrightarrow{Heat} 2HgO + 2SO_2$; on further heating $2HgO \xrightarrow{Heat} 2Hg + O_2$.

Q18. 2M What is galvanisation? Why is it used?
Ans: Coating iron or steel with a thin layer of zinc. It protects iron from rusting, even if the zinc coat breaks, because zinc is more reactive.

Q19. 2M Food cans are coated with tin and not with zinc. Why?
Ans: Zinc is more reactive than tin and can react with acids in the food, forming harmful compounds; tin is less reactive.

Q20. 2M Why do we use copper and aluminium wires for electricity, and copper for hot-water tanks, not steel?
Ans: Copper and Al are good conductors of electricity. Copper conducts heat well and does not react with water even when hot, while steel (iron) rusts.

Q21. 2M Board fav Tarnished copper vessels are cleaned with lemon or tamarind juice. Why?
Ans: The green coating is basic copper carbonate; the acids in lemon or tamarind dissolve (neutralise) it and the vessel shines again.

Q22. 3M Board fav What is the thermit reaction? Give its equation and one use.
Ans: Al reduces a metal oxide with so much heat that the metal forms in the molten state: $Fe_2O_3 + 2Al \longrightarrow 2Fe + Al_2O_3$ + heat. Used to join railway tracks and cracked machine parts.

Q23. 3M Board fav Give reasons: (a) Platinum, gold and silver are used to make jewellery. (b) Sodium, potassium and lithium are stored under oil. (c) Aluminium is highly reactive, yet used for cooking utensils.

Ans: (a) They are lustrous, malleable and very unreactive, so they do not corrode. (b) They react vigorously with air and water and catch fire. (c) A protective layer of Al_2O_3 forms on its surface and stops further reaction; Al is also a good conductor of heat.

Q24. 3M Differentiate between metals and non-metals on the basis of their chemical properties (any three).

Ans: Metals form basic oxides, non-metals acidic or neutral oxides. Metals displace H_2 from dilute acids, non-metals do not. Metals lose electrons to form positive ions, non-metals gain electrons to form negative ions (and form hydrides with H_2).

Q25. 2M Why is pure gold not used for making jewellery? What is meant by 22 carat gold?

Ans: Pure (24 carat) gold is very soft and bends easily. 22 carat gold is 22 parts gold alloyed with 2 parts silver or copper, which makes it hard.

Q26. 3M You are given a hammer, a battery, a bulb, wires and a switch. How could you use them to tell metals from non-metals?

Ans: Hammer: metals flatten into sheets (malleable), non-metals break (brittle). Circuit: put the sample between the wires; the bulb glows with metals (good conductors) and not with non-metals (except graphite).

Long answer (5 marks)

Q27. 5M Board fav (a) Draw a labelled diagram of electrolytic refining of copper. (b) What are the anode, cathode and electrolyte? (c) What happens at each electrode? (d) What is anode mud?

Ans: (a) Diagram as in Fig. 3.12. (b) Anode: impure copper; cathode: thin strip of pure copper; electrolyte: acidified copper sulphate solution. (c) At the anode impure copper dissolves into the solution as Cu^{2+}; at the cathode an equal amount of pure copper deposits. (d) Insoluble impurities that settle below the anode.

Q28. 5M Board fav (a) Explain, with equations, how metals of low, middle and high reactivity are obtained from their ores. (b) Why can carbon not reduce oxides of Na, Mg, Al?

Ans: (a) Low (Hg): heating the ore alone: $2HgS + 3O_2 \longrightarrow 2HgO + 2SO_2$; $2HgO \longrightarrow 2Hg + O_2$. Middle (Zn): roasting or calcination to ZnO, then $ZnO + C \longrightarrow Zn + CO$. High (Na): electrolysis of molten NaCl: $Na^+ + e^- \longrightarrow Na$ at the cathode, $2Cl^- \longrightarrow Cl_2 + 2e^-$ at the anode. (b) These metals have more affinity for oxygen than carbon.

Q29. 5M (a) What is corrosion? Describe an activity to show that air and water are both needed for rusting. (b) Give three methods of preventing rusting. (c) What is an alloy? Give the composition of brass and solder.

Ans: (a) Slow eating away of a metal by air, moisture or acids. Iron nails in three tubes: A in water (rusts), B in boiled water under oil (no air, no rust), C with anhydrous $CaCl_2$ (no moisture, no rust). (b) Painting/oiling, galvanising, alloying (stainless steel). (c) A homogeneous mixture of metals, or a metal and a non-metal. Brass: Cu + Zn; solder: Pb + Sn.

Case-based (4 marks)

Q30. 4M Board fav Four metals A, B, C, D were added to solutions: A displaced copper from $CuSO_4$; B reacted with cold water violently; C showed no reaction with dilute HCl; D displaced A from its salt solution. (a) Which is the most reactive? (b) Which could be copper or silver? (c) Arrange A, C, D in decreasing order of reactivity. (d) What happens when C is added to dilute HCl?

Ans: (a) B (reacts violently with cold water, like Na or K). (b) C. (c) D > A > C. (d) No reaction: C is below hydrogen, so no H_2 is evolved.

Q31. 4M Riya heated three white solids NaCl, KI and $BaCl_2$ on a spatula and then dissolved them in water, kerosene and petrol. (a) Did they melt easily? Why? (b) In which solvent did they dissolve? (c) Will the solid conduct electricity? (d) Will the solution conduct? Why?

Ans: (a) No: they have high melting points because strong electrostatic forces hold their ions. (b) Only in water, not in kerosene or petrol. (c) No: in the solid the ions cannot move. (d) Yes: in solution the ions move freely and carry current.

Q32. 4M Amit noticed that the iron gate of his house had brown flaky deposits, but the galvanised iron bucket was shiny. His copper vessel had turned green and his silver chain black. (a) Name the brown deposit and the conditions for its formation. (b) Why does the galvanised bucket not rust even if scratched? (c) Name the green and black coatings. (d) Give one way to protect the gate.

Ans: (a) Rust (hydrated iron(III) oxide); it needs both air (oxygen) and moisture. (b) Zinc is more reactive than iron, so zinc corrodes first and protects iron even where the coat is broken. (c) Green: basic copper carbonate; black: silver sulphide (Ag_2S). (d) Painting (or oiling/greasing) the gate.

Yaad karo: ores aur alloys (MCQ favourite)

Ores: cinnabar HgS | zinc blende ZnS | calamine $ZnCO_3$ | copper glance Cu_2S | haematite Fe_2O_3 | bauxite $Al_2O_3.2H_2O$ | galena PbS | rock salt NaCl

Alloys: brass Cu + Zn | bronze Cu + Sn | solder Pb + Sn | stainless steel Fe + Ni + Cr | steel Fe + C | amalgam = Hg + any metal

Exam Alert

Refining mein **impure metal = anode**, pure = cathode. Reactivity series ka mnemonic aur thermit equation har saal kaam aata hai!

CARBON AND ITS COMPOUNDS



Most asked: electron dot structures, catenation and tetravalency, ethanol and ethanoic acid reactions, cleansing action of soap

Food, clothes, medicines, fuels, plastics and even our body are made of **carbon compounds**. The crust has only 0.02% carbon and air only 0.03% CO_2, yet carbon forms millions of compounds.

1. Carbon: Introduction

Organic compounds were once believed to need a "vital force" and to come only from living things. In 1828 Friedrich Wöhler prepared urea from ammonium cyanate in the laboratory, and the vital force idea was dropped.

Activity 4.1: Things around us

Done: List ten things used since morning and sort them by the material they are made of (metal, glass, carbon compounds). **Observed:** Most items (food, cloth, paper, plastic, wood, medicines) contain carbon. **Conclusion:** Carbon compounds are everywhere; all living structures are carbon based.

2. Bonding in Carbon: the Covalent Bond

Carbon (atomic number 6) has electronic configuration 2, 4. To reach a noble gas configuration it would have to:

→ gain 4 electrons to form C^{4-}: difficult, as 6 protons cannot hold 10 electrons;

→ lose 4 electrons to form C^{4+}: needs a very large amount of energy.

So carbon shares its valence electrons. **Covalent bond**: a bond formed by the sharing of electron pairs between two atoms.

Bond	Shared pairs	Examples
Single	1	H_2 (H–H), Cl_2 (Cl–Cl), CH_4
Double	2	O_2 (O=O), CO_2 (O=C=O)
Triple	3	N_2 (N≡N), C_2H_2

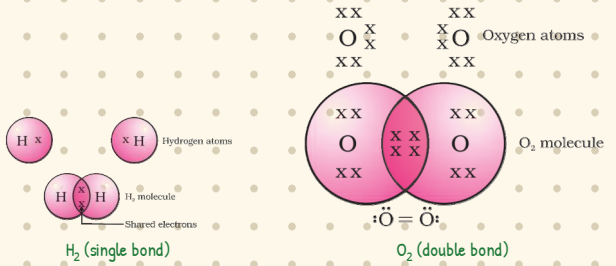


Fig. 4.1, 4.3 Electron dot structures of H_2 and O_2

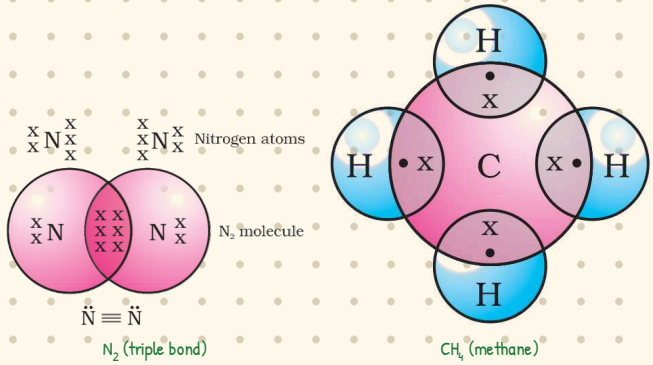


Fig. 4.4, 4.5 Electron dot structures of N_2 and CH_4

Methane (CH_4): carbon shares its 4 electrons with 4 hydrogen atoms. It is used as a fuel and is the major part of biogas and CNG.

Other dot structures: **CO_2** has two double bonds (O=C=O); **H_2O**: O shares one pair with each H (2 single bonds, 2 lone pairs on O); **NH_3**: N shares one pair with each of 3 H (1 lone pair on N); **S_8**: 8 S atoms in a crown-shaped ring, each sharing one pair with each of its 2 neighbours and keeping 2 lone pairs.

Exam Alert

→ Ionic bond = electrons ka **transfer** (metal + non-metal, jaise NaCl).

→ Covalent bond = electrons ki **sharing** (non-metals, jaise CH_4, O_2).

→ Dot structure mein har atom ke outer electrons count karo: H = 2 (duplet), baaki = 8 (octet).

Properties of covalent compounds

→ Low melting and boiling points, because the forces between molecules are weak.

→ Poor conductors of electricity, because no ions are formed.

Compound	m.p. (K)	b.p. (K)
Acetic acid (CH_3COOH)	290	391
Chloroform ($CHCl_3$)	209	334
Ethanol (CH_3CH_2OH)	156	351
Methane (CH_4)	90	111

Allotropes of carbon

Allotropes: different physical forms of the same element.

Diamond	Graphite	Fullerene
Each C bonded to 4 others; rigid 3-D structure	Each C bonded to 3 others; hexagonal layers	C_{60}, football shaped
Hardest natural substance; non-conductor	Smooth, slippery; good conductor	Named after Buckminster Fuller
Cutting/drilling tools, jewellery	Pencil leads, lubricant, electrodes	Research, medicine

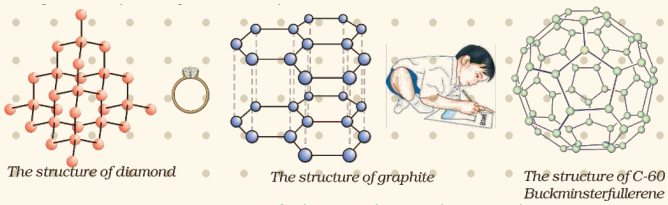


Fig. Structures of diamond, graphite and C_{60}

Quick fact

Diamond aur graphite dono pure carbon hain, phir bhi ek sabse hard aur doosra itna soft ki pencil mein chalta hai. Farak sirf **atoms ki arrangement** ka hai! Newer forms: graphene (single graphite layer) and carbon nanotubes.

Key terms

- → **Covalent bond**: bond formed by sharing electron pairs.
- → **Allotropes**: different forms of the same element (diamond, graphite, C_{60}).
- → **Catenation**: self-linking of carbon atoms into chains and rings.
- → **Isomers**: same molecular formula, different structures.
- → **Functional group**: atom or group that decides the properties of a compound.

3. Versatile Nature of Carbon

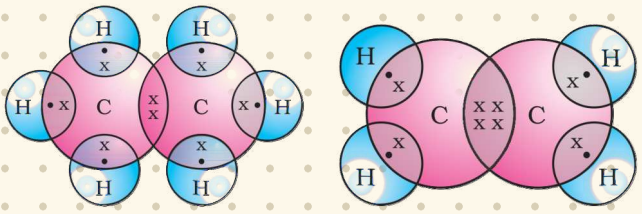
1. **Catenation**: carbon links with other carbon atoms to form long chains, branched chains and rings, because the C–C bond is very strong. Silicon forms chains of only 7–8 atoms, and these are very reactive.
2. **Tetravalency**: valency 4, so carbon bonds with four atoms of carbon or of O, H, N, S, Cl. The bonds are strong because the carbon atom is small and its nucleus holds the shared pairs tightly.

Yaad karo: Carbon ka famous hona

Catenation + Tetravalency = lakhon compounds. Carbon khud se hi lambi chains banata hai aur chaaron haath se bond karta hai. Ye do words = 2 marks pakke!

Saturated and unsaturated hydrocarbons

Hydrocarbons: compounds of carbon and hydrogen only.
Saturated (alkanes): only single C–C bonds, e.g. ethane C_2H_6.
Unsaturated: a double bond (alkenes, e.g. ethene C_2H_4) or a triple bond (alkynes, e.g. ethyne C_2H_2).



Ethane $H_3C–CH_3$ Ethene $H_2C=CH_2$

Fig. 4.6, 4.7 Electron dot structures of ethane and ethene

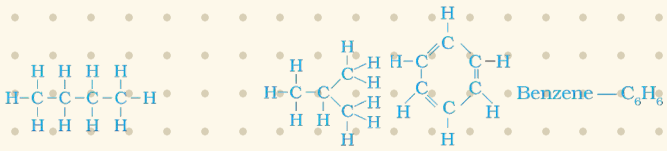
C atoms	Alkane C_nH_{2n+2}	Alkene C_nH_{2n}	Alkyne C_nH_{2n-2}
1	Methane CH_4	–	–
2	Ethane C_2H_6	Ethene C_2H_4	Ethyne C_2H_2
3	Propane C_3H_8	Propene C_3H_6	Propyne C_3H_4
4	Butane C_4H_{10}	Butene C_4H_8	Butyne C_4H_6
5	Pentane C_5H_{12}	Pentene C_5H_{10}	Pentyne C_5H_8
6	Hexane C_6H_{14}	Hexene C_6H_{12}	Hexyne C_6H_{10}

Alkanes are also called paraffins and alkenes olefins.

Chains, branches and rings

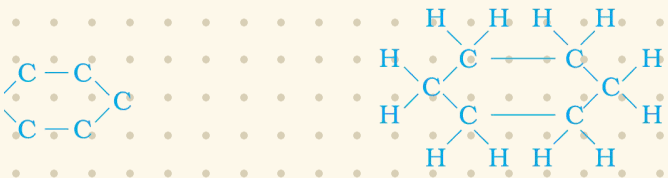
Structural isomers: compounds with the same molecular formula but different structures. Butane C_4H_{10} exists as a straight chain (n-butane) and a branched chain (isobutane).

Rings: cyclohexane C_6H_{12} is a saturated ring; benzene C_6H_6 is an unsaturated ring with alternate single and double bonds.



n-butane Benzene

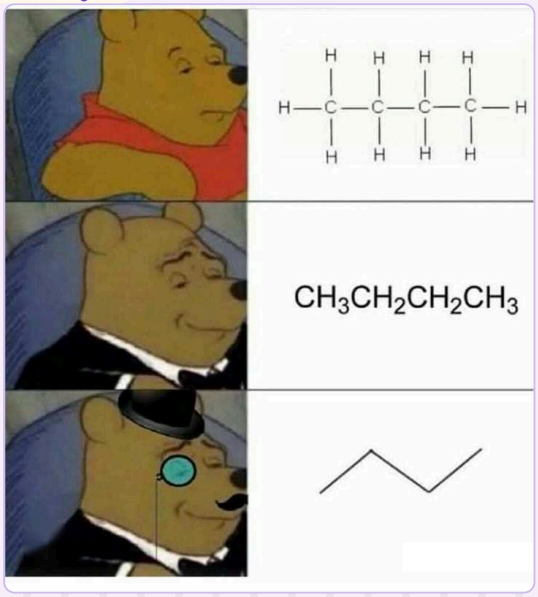
Fig. 4.8, 4.10 Butane chain and benzene ring



(a) (b)

Fig. 4.9 (a) Branched and ring skeletons (b) Cyclohexane

Meme samjho



Same butane, teen tarike se likho: full structure, condensed formula, ya bas zig-zag line. Pooh ka style badhta gaya!

Asli baat: Different ways of writing one structure; the molecular formula C_4H_{10} stays the same.

Paheli-Bujho



Bujho: Paheli, carbon ke hi itne laakhon compounds kyun? Baaki elements kya so rahe hain?

Paheli: Do superpowers: **catenation** (khud se lambi chains) aur **tetravalency** (chaar bonds).

Bujho: Matlab carbon sabse dosti kar leta hai!

Paheli: Haan, isliye carbon compounds itne zyada hain.

4. Functional Groups and Homologous Series

Heteroatom: an atom other than C and H (Cl, O, N, S) that replaces hydrogen in a hydrocarbon. The group containing it decides the properties of the compound: the functional group.

Group	Formula	Suffix / prefix	Example (3 C)
Halo	-Cl, -Br	chloro-, bromo-	Chloropropane
Alcohol	-OH	-ol	Propanol
Aldehyde	-CHO	-al	Propanal
Ketone	>C=O	-one	Propanone
Carboxylic acid	-COOH	-oic acid	Propanoic acid
Double bond	C=C	-ene	Propene
Triple bond	C≡C	-yne	Propyne

Homologous series: a series of compounds with the same functional group in which successive members differ by a $-CH_2-$ unit (14 u), e.g. CH_4, C_2H_6, C_3H_8 and CH_3OH, C_2H_5OH, C_3H_7OH.

→ Physical properties (m.p., b.p., solubility) change gradually with molecular mass.
→ Chemical properties stay similar, as they depend on the functional group.

Activity 4.2: Homologous series

Done: Calculate the difference in formulae and molecular masses of CH_3OH, C_2H_5OH, C_3H_7OH and C_4H_9OH. **Observed:** Each differs from the next by CH_2 and 14 u. **Conclusion:** They form a homologous series of alcohols.

5. Nomenclature of Carbon Compounds

1. Count the carbon atoms in the longest chain: 1 meth, 2 eth, 3 prop, 4 but, 5 pent, 6 hex. Number the chain from the end that gives the functional group the lowest number.
2. Add the functional group as a prefix (chloro-, bromo-) or a suffix (-ol, -al, -one, -oic acid, -ene, -yne).
3. If the suffix starts with a vowel, drop the final "e" of the alkane name: propane ⟶ propan**ol**, propan**one**.

Formula	Name
CH_3CH_2Cl	Chloroethane
HCHO	Methanal
CH_3COCH_3	Propanone
HCOOH	Methanoic acid
$CH_3CH_2CH_2OH$	Propanol
$CH_3CH_2CH_2COOH$	Butanoic acid

Drawing a structure from a formula

1. Join the carbon atoms with single bonds.
2. Fill the remaining valencies of each carbon with hydrogen.
3. If the formula has fewer hydrogens, put in a double or triple bond.

C_3H_8: $CH_3-CH_2-CH_3$ (propane); C_3H_6: $CH_2=CH-CH_3$ (propene); C_3H_4: $CH\equiv C-CH_3$ (propyne).

Yaad karo: carbon count

Mummy Ek Pyaali Bournvita Pilao Hamesha = Meth (1), Eth (2), Prop (3), But (4), Pent (5), Hex (6).

6. Chemical Properties of Carbon Compounds

(a) Combustion

Carbon and its compounds burn in oxygen to give CO_2, heat and light.

$$C + O_2 \longrightarrow CO_2 + \text{heat and light}$$

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + \text{heat and light}$$

$$CH_3CH_2OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O + \text{heat and light}$$

Activity 4.3: Burning carbon compounds

Done: Burn naphthalene, camphor and alcohol on a spatula; hold a metal plate above. **Observed:** Naphthalene and camphor give a sooty yellow flame and a black deposit; alcohol burns cleanly. **Conclusion:** Unsaturated compounds give a sooty flame.

Activity 4.4: Bunsen burner flame

Done: Change the air hole of a Bunsen burner. **Observed:** Less air: yellow sooty flame. More air: clean blue flame. **Conclusion:** Complete combustion gives a blue flame.

→ Saturated hydrocarbons burn with a clean blue flame; unsaturated ones give a yellow flame with black smoke.
→ Gas stoves have air holes for complete burning; a blackened vessel bottom means they are blocked.
→ **Welding** uses ethyne + pure oxygen (complete combustion, very hot flame); with air the flame is sooty and cooler.
→ A substance burns with a flame only if it vaporises: wood and candle wax give a flame; charcoal only glows.

Fossil fuels: coal formed from buried trees and plants; petroleum and natural gas from tiny sea plants and animals buried under the sea bed, under heat and pressure, millions of years ago. Burning them gives oxides of sulphur and nitrogen, which pollute the air.

(b) Oxidation

Alcohols are oxidised to carboxylic acids by **oxidising agents** like alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$. An oxidising agent adds oxygen to (or removes hydrogen from) another substance.

$$CH_3CH_2OH \xrightarrow[\text{or acidified } K_2Cr_2O_7]{\text{Alk. } KMnO_4 + \text{Heat}} CH_3COOH$$

Activity 4.5: Oxidation of ethanol

Done: Warm 3 mL ethanol in a water bath and add 5% alkaline $KMnO_4$ drop by drop. **Observed:** The purple colour disappears at first; it stays when excess is added. **Conclusion:** $KMnO_4$ is used up in oxidising ethanol to ethanoic acid; the excess stays unused.

Combustion is also oxidation: with limited oxygen carbon gives poisonous carbon monoxide; with plenty of oxygen it gives CO_2.

$$2C + O_2 \text{ (limited)} \longrightarrow 2CO \qquad C + O_2 \text{ (excess)} \longrightarrow CO_2$$

(c) Addition reaction

Unsaturated hydrocarbons add hydrogen in the presence of a **catalyst** (nickel or palladium) to give saturated hydrocarbons.

Used in the hydrogenation of vegetable oils.

$$R_2C{=}CR_2 + H_2 \xrightarrow{\text{Nickel catalyst}} R_2CH{-}CHR_2$$

Catalyst: a substance that changes the rate of a reaction without itself being affected. Vegetable oils (unsaturated) are healthier; animal fats (saturated) are less healthy and are best avoided for cooking.

(d) Substitution reaction

Saturated hydrocarbons are fairly unreactive, but in sunlight chlorine replaces their hydrogen atoms one by one.

$$CH_4 + Cl_2 \xrightarrow{\text{Sunlight}} CH_3Cl + HCl$$

Exam Alert

- → **Addition** = unsaturated mein H_2 judta hai (Ni catalyst).
- → **Substitution** = saturated mein H ki jagah Cl aata hai (sunlight).
- → Equation mein catalyst aur condition arrow ke upar zaroor likho.

7. Ethanol (Ethyl Alcohol), C_2H_5OH

A liquid at room temperature, soluble in water, a good solvent; used in tincture iodine, cough syrups and tonics. **Denatured alcohol**: industrial alcohol made unfit to drink by adding methanol, pyridine and a blue dye. **Alcohol as fuel**: ethanol from fermented sugarcane molasses is mixed with petrol; it burns cleanly in air.

Activity 4.6: Sodium and ethanol

Done: Drop a small piece of sodium into absolute ethanol (teacher's demo). **Observed:** Bubbles of a gas that burns with a pop sound. **Conclusion:** Hydrogen is evolved and sodium ethoxide is formed.

$$2Na + 2CH_3CH_2OH \longrightarrow 2CH_3CH_2O^-Na^+ + H_2$$

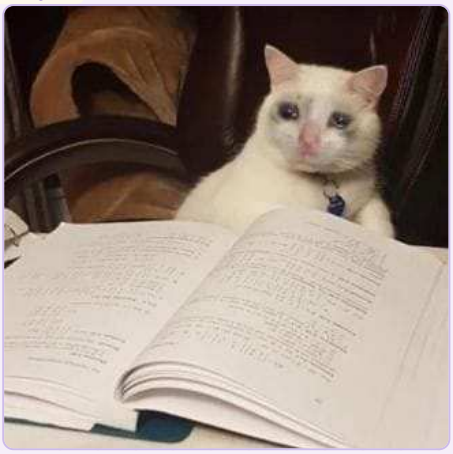
Dehydration: ethanol heated at 443 K with excess conc. H_2SO_4 gives ethene. Conc. H_2SO_4 acts as a dehydrating agent, removing water from ethanol.

$$CH_3CH_2OH \xrightarrow[443\ K]{\text{Hot conc. } H_2SO_4} CH_2{=}CH_2 + H_2O$$

Effects of alcohol

- → Small quantities of dilute ethanol cause drunkenness; even a little pure (absolute) ethanol can be lethal.
- → Large amounts slow down metabolism and depress the central nervous system: lack of coordination, mental confusion, drowsiness and lowering of normal inhibitions.
- → **Methanol** kills even in very small amounts: in the liver it is oxidised to methanal, which coagulates the protoplasm (like an egg on cooking). It also affects the optic nerve, causing blindness.

Meme samjho



Ethanol ke saare reactions ek saath yaad karne baitho toh aisi shakal ban jaati hai. Bas 4 hain: combustion, oxidation, sodium, dehydration!

Asli baat: Ethanol: burns to CO_2; oxidised to ethanoic acid; with Na gives H_2; with hot conc. H_2SO_4 gives ethene.

8. Ethanoic Acid (Acetic Acid), CH_3COOH

A 5–8% solution of ethanoic acid in water is **vinegar**, used as a preservative in pickles. Its melting point is 290 K, so it often freezes in cold climates: glacial acetic acid.

Activity 4.7: Weak acid

Done: Compare the pH of dilute acetic acid and dilute HCl with litmus and universal indicator. **Observed:** Both turn blue litmus red, but acetic acid shows a higher pH. **Conclusion:** Ethanoic acid is a weak acid; HCl is a strong acid.

(a) Esterification

Acid + alcohol, in the presence of an acid catalyst, gives a sweet-smelling **ester**, used in perfumes and as flavouring agents.

Activity 4.8: Making an ester

Done: Warm 1 mL ethanol with 1 mL glacial acetic acid and a few drops of conc. H_2SO_4 in a water bath for 5 min; pour into water. **Observed:** A sweet, fruity smell. **Conclusion:** An ester (ethyl ethanoate) is formed.

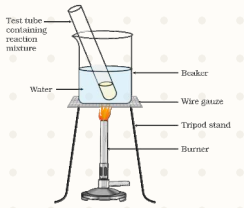


Fig. 4.11

$$CH_3COOH + CH_3CH_2OH \xrightarrow{\text{Acid}} CH_3COOCH_2CH_3 + H_2O$$

Saponification: an ester reacts with NaOH to give back the alcohol and the sodium salt of the acid. This is how soap is made.

$$CH_3COOC_2H_5 \xrightarrow{NaOH} C_2H_5OH + CH_3COONa$$

(b) Reaction with a base

$$NaOH + CH_3COOH \longrightarrow CH_3COONa + H_2O$$

(c) Reaction with carbonates and hydrogencarbonates

Activity 4.9: Acid + sodium carbonate

Done: Add dilute ethanoic acid to sodium carbonate and pass the gas through lime water; repeat with $NaHCO_3$. **Observed:** Brisk effervescence; lime water turns milky. **Conclusion:** CO_2 is evolved, with a salt and water.

$$2CH_3COOH + Na_2CO_3 \longrightarrow 2CH_3COONa + H_2O + CO_2$$

$$CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2$$

Test	Ethanol	Ethanoic acid
Smell	Pleasant (alcohol)	Vinegar-like
Litmus	No change	Blue to red
$NaHCO_3$	No reaction	Brisk effervescence (CO_2)
Sodium	H_2 gas, sodium ethoxide	H_2 gas, sodium ethanoate
Functional group	-OH	-COOH

Yaad karo: sirka

Sirka = vinegar = 5–8% ethanoic acid. Isliye mummy achaar mein daalti hain, achaar jaldi kharab nahi hota. Thand mein jam jaaye toh glacial!

Paheli-Bujho



Bujho: Paheli, perfume aur deo mein itni meethi khushboo kahan se aati hai?

Paheli: **Esters** se! Acid + alcohol, acid catalyst ke saath garam karo: fruity smell wala ester banta hai.

Bujho: Matlab mera deo bhi ek reaction ka result hai!

Quick fact

Carbon monoxide (from incomplete burning in a closed room, e.g. an angithi) is colourless, odourless and poisonous: it binds with haemoglobin and stops oxygen transport. Always keep a window open.

9. Soaps and Detergents

Activity 4.10: Soap and oil

Done: Add a drop of oil to water in tubes A and B; add soap solution to B only. Shake both. **Observed:** In A the oil separates quickly; in B it stays mixed longer. **Conclusion:** Soap helps oil mix with water.

Soap: sodium or potassium salt of a long-chain carboxylic acid. Its molecule has an ionic (hydrophilic) end that dissolves in water and a long hydrocarbon (hydrophobic) tail that dissolves in oil.

Micelle: in water, soap molecules form clusters with the tails inside (holding the oily dirt) and the ionic ends outside facing water. Micelles stay in solution as a colloid, so the dirt is washed away. They are large enough to scatter light, so soap solution looks cloudy. Micelles do not come together to precipitate, because the ionic ends repel each other.

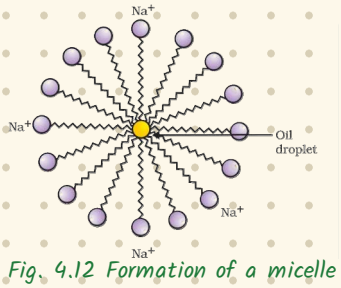


Fig. 4.12 Formation of a micelle

Yaad karo: soap molecule

Head = paani ka dost (hydrophilic), Tail = tel ka dost (hydrophobic). Beech mein phansa maila micelle ke andar band hokar paani ke saath bahar!

Cleansing action of soap

1. The hydrophobic tails stick to the oily dirt.
2. The dirt gets surrounded, with the ionic ends facing water: a micelle forms.
3. On rinsing, the micelles carry the dirt away. Rubbing or agitation helps pull the dirt out of the cloth.

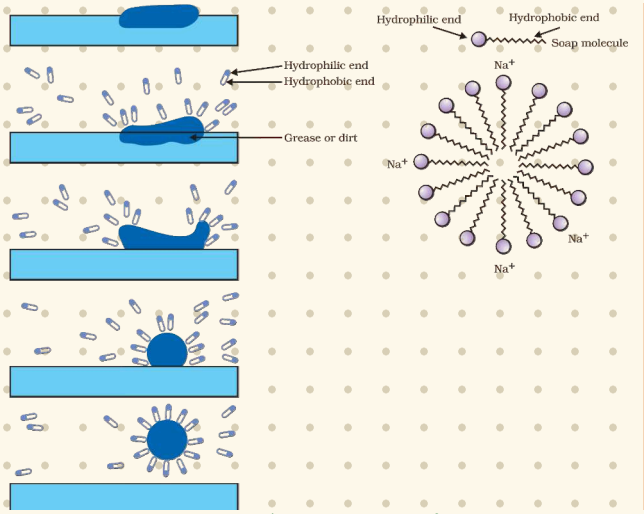


Fig. 4.13 Cleaning action of soap

Hard water: water containing salts of calcium and magnesium (hydrogencarbonates, chlorides or sulphates). Soap reacts with them to form an insoluble, curdy precipitate called scum, so a lot of soap is wasted before lather forms.

Activity 4.11: Soap in hard water

Done: Shake soap solution with distilled water and with hard water. **Observed:** Lots of foam in distilled water; a white curdy precipitate in hard water. **Conclusion:** Soap forms insoluble scum with Ca and Mg salts of hard water.

Activity 4.12: Soap vs detergent

Done: Shake hard water with soap in one tube and detergent in another. **Observed:** Detergent gives foam and no scum; soap gives scum. **Conclusion:** Detergents work well even in hard water.

Detergents: generally sodium salts of sulphonic acids or ammonium salts with chloride or bromide ions; like soaps, they have a long hydrocarbon chain. Their charged ends form no insoluble precipitate with Ca and Mg ions, so they work in hard water.

Meme samjho



Soft water mein soap full khush, hard water mein aate hi scum ban gaya aur sab barbaad!

Asli baat: Ca^{2+} and Mg^{2+} ions of hard water react with soap to form insoluble scum; detergents do not.

	Soap	Detergent
Chemical nature	Na/K salts of long-chain carboxylic acids	Sodium salts of long-chain sulphonic acids, or ammonium salts with Cl^-/Br^-
Hard water	Forms scum, wasted	Works well, no scum
Uses	Bathing, washing	Shampoos, washing clothes

All reactions at a glance

Reaction	Condition	Example
Combustion	Oxygen, heat	$CH_4 \longrightarrow CO_2 + H_2O$
Oxidation	Alk. $KMnO_4$ / acidified $K_2Cr_2O_7$	Ethanol ⟶ ethanoic acid
Addition	H_2, Ni/Pd catalyst	Oil ⟶ vanaspati ghee
Substitution	Cl_2, sunlight	$CH_4 \longrightarrow CH_3Cl$
With sodium	Room temperature	Ethanol ⟶ sodium ethoxide + H_2
Dehydration	Conc. H_2SO_4, 443 K	Ethanol ⟶ ethene
Esterification	Acid catalyst, warm	Acid + alcohol ⟶ ester
Saponification	NaOH	Ester ⟶ alcohol + salt (soap)

1 minute recap

Covalent bond → Catenation + tetravalency →
Hydrocarbons → Functional groups →
Combustion, oxidation, addition, substitution →
Ethanol, ethanoic acid → Soap and micelle

Exam Alert

→ Naming mein vowel se pehle "e" drop karo: propan**ol**, propan**one**. Double bond = -ene, triple = -yne.
→ Dot structure mein sirf outermost electrons dikhao, aur shared pair dono atoms ke beech mein.

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM Board fav How many covalent bonds are present in a molecule of ethane (C_2H_6)?
Ans: 7 covalent bonds (1 C–C + 6 C–H).

Q2. IM Write the next homologue of C_2H_5OH and its IUPAC name.
Ans: C_3H_7OH ($CH_3CH_2CH_2OH$), propanol.

Q3. IM Board fav How many electron pairs are shared between the two atoms in a molecule of N_2?
(a) one shared pair
(b) two shared pairs
(c) three shared pairs
(d) four shared pairs
Ans: (c) A triple bond: three shared pairs, with one lone pair on each N.

Q4. IM Board fav Which of the following hydrocarbons undergo addition reactions: C_2H_6, C_3H_8, C_3H_6, C_2H_2, CH_4?
Ans: C_3H_6 and C_2H_2, because they are unsaturated (C=C and C≡C).

Q5. IM Board fav Butanone is a four-carbon compound with which functional group?
(a) carboxylic acid
(b) aldehyde
(c) ketone
(d) alcohol
Ans: (c) Ketone (>C=O).

Q6. IM While cooking, the bottom of a vessel is getting blackened on the outside. What does it mean?
(a) the food is not cooked completely
(b) the fuel is not burning completely
(c) the fuel is wet
(d) the fuel is burning completely
Ans: (b) The fuel is not burning completely (air holes blocked), so soot is deposited.

Q7. IM Board fav **Assertion (A):** Soap solution looks cloudy. **Reason (R):** Micelles are large enough to scatter light.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q8. IM **Assertion (A):** Diamond conducts electricity. **Reason (R):** Each carbon in diamond is bonded to four other carbon atoms.
Ans: (d) A is false but R is true. Diamond has no free electrons, so it does not conduct.

Q9. IM Board fav Name the gas evolved when a small piece of sodium is dropped into ethanol. How is it tested?
Ans: Hydrogen; it burns with a pop sound when a burning matchstick is brought near.

Q10. IM Name the functional group present in CH_3CHO and write its IUPAC name.
Ans: Aldehyde (–CHO); ethanal.

Short answer (2-3 marks)

Q11. 3M Board fav Why does carbon form compounds mainly by covalent bonding? Give two reasons why covalent compounds are poor conductors and have low melting points.

Ans: Carbon cannot gain 4 electrons (6 protons cannot hold 10 electrons) or lose 4 (too much energy), so it shares electrons. Covalent compounds have no ions, so they do not conduct; forces between molecules are weak, so m.p. and b.p. are low.

Q12. 3M Board fav Describe the electron dot structures of (a) CO_2 (b) H_2O (c) cyclohexane.

Ans: (a) O=C=O: C shares two pairs with each O; each O keeps 2 lone pairs. (b) H–O–H: O shares one pair with each H and keeps 2 lone pairs. (c) Six carbon atoms in a ring joined by single bonds, each carbon also sharing one pair with each of 2 H atoms (C_6H_{12}).

Q13. 3M Board fav What are catenation and tetravalency? Why does carbon form a very large number of compounds?

Ans: Catenation: carbon links with carbon atoms to form long chains, branches and rings. Tetravalency: valency 4, so it bonds with four atoms of C, H, O, N, Cl, etc. Strong C–C bonds and the small size of carbon make these compounds stable.

Q14. 2M Board fav What is a homologous series? Give two characteristics.

Ans: A series of compounds with the same functional group in which successive members differ by $-CH_2-$ (14 u). Physical properties change gradually with molecular mass; chemical properties remain similar.

Q15. 2M Name the compounds: (a) CH_3CH_2Br (b) HCHO (c) CH_3COCH_3 (d) $CH_3CH_2CH_2COOH$.

Ans: (a) Bromoethane (b) Methanal (c) Propanone (d) Butanoic acid.

Q16. 3M Board fav Explain the hydrogenation of vegetable oils with an equation. Why are animal fats harmful?

Ans: Unsaturated vegetable oils add hydrogen in the presence of a nickel catalyst to form saturated fats (vanaspati ghee): $R_2C=CR_2 + H_2 \xrightarrow{Ni} R_2CH-CHR_2$. Animal fats are saturated and are harmful to health (heart).

Q17. 3M Board fav What happens when ethanol is heated with excess conc. H_2SO_4 at 443 K? Write the equation and the role of the acid.

Ans: Ethene is formed: $CH_3CH_2OH \xrightarrow[443\ K]{\text{Hot conc. } H_2SO_4} CH_2=CH_2 + H_2O$. Conc. H_2SO_4 is a dehydrating agent that removes water from ethanol.

Q18. 2M Board fav How would you distinguish between ethanol and ethanoic acid experimentally?

Ans: Add $NaHCO_3$ (or Na_2CO_3) to each: ethanoic acid gives brisk effervescence of CO_2 that turns lime water milky; ethanol shows no reaction. Ethanoic acid also turns blue litmus red.

Q19. 2M Why is the conversion of ethanol to ethanoic acid an oxidation reaction?

Ans: Oxygen is added to ethanol (and hydrogen removed): CH_3CH_2OH gains an O atom to form CH_3COOH using alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$.

Q20. 2M Board fav Why is a mixture of ethyne and air not used for welding?

Ans: Ethyne burns in air with incomplete combustion: a sooty, cooler flame. Ethyne + pure oxygen burns completely, giving the very hot flame (about 3000 °C) needed to melt metals.

Q21. 2M Why does micelle formation take place when soap is added to water? Will a micelle form in ethanol too?

Ans: The hydrophobic tails avoid water and cluster inside, while the ionic ends face water, forming micelles. In ethanol no micelles form, because the hydrocarbon tail dissolves in ethanol.

Q22. 2M Methane and ethane can both be burnt as fuels. Write the balanced equations for their complete combustion.

Ans: $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$ + heat and light; $2C_2H_6 + 7O_2 \longrightarrow 4CO_2 + 6H_2O$ + heat and light.

Q23. 2M Board fav What are structural isomers? Draw the two isomers of butane in condensed form.

Ans: Compounds with the same molecular formula but different structures. n-butane: $CH_3–CH_2–CH_2–CH_3$; isobutane (2-methylpropane): $CH_3–CH(CH_3)–CH_3$.

Q24. 2M Why is the reaction between methane and chlorine considered a substitution reaction?

Ans: The hydrogen atoms of methane are replaced one by one by chlorine atoms in sunlight: $CH_4 + Cl_2 \xrightarrow{Sunlight} CH_3Cl + HCl$. One atom is substituted for another.

Q25. 3M Board fav Why is drinking even a small amount of methanol dangerous? What is denatured alcohol?

Ans: In the liver methanol is oxidised to methanal, which coagulates protoplasm and damages the optic nerve, causing blindness and even death. Denatured alcohol is industrial ethanol made unfit to drink by adding methanol, pyridine and a blue dye.

Q26. 2M Why are carbon and its compounds used as fuels for most applications?

Ans: They burn in air to release a large amount of heat and light, have a high calorific value, burn at a controllable rate, and most are easy to store and transport.

Q27. 3M What is the difference between the molecules of soaps and detergents in terms of their structure? Why can some detergents harm rivers?

Ans: Soap: Na/K salt of a long-chain carboxylic acid, with a $-COO^-Na^+$ end. Detergent: sodium salt of a sulphonic acid, with a $-SO_3^-Na^+$ end, or an ammonium salt with Cl^-/Br^-. Some detergents are not easily broken down by microbes and pollute rivers.

Q28. 2M Give a test to check whether a sample of water is hard or soft.

Ans: Shake the water with soap solution: soft water gives plenty of lather; hard water gives little lather and a white curdy scum.

Q29. 2M Board fav What would be the electron dot structure and formula of cyclopentane? How many single bonds does it have?

Ans: Cyclopentane C_5H_{10}: five carbon atoms in a ring joined by single bonds, each carbon sharing one pair with each of 2 H atoms. Bonds: 5 C–C + 10 C–H = 15 single covalent bonds.

Q30. 3M Board fav Give reasons: (a) Pure ethanoic acid is called glacial acetic acid. (b) Carbon does not form C^{4+} ions. (c) Graphite conducts electricity but diamond does not.

Ans: (a) Its melting point is 290 K, so it freezes into ice-like crystals in cold weather. (b) Removing 4 electrons needs a very large amount of energy. (c) In graphite each C is bonded to only 3 others, leaving one free electron per atom; in diamond all 4 electrons are in bonds.

Q31. 2M An organic compound burns with a sooty flame. Is it saturated or unsaturated? Why do we avoid such fuels at home?

Ans: Unsaturated (higher carbon content, incomplete combustion). The sooty flame wastes fuel, blackens vessels and gives poisonous carbon monoxide.

Long answer (5 marks)

Q32. 5M Board fav (a) What is an ester? Write the equation for the formation of ethyl ethanoate. (b) What is saponification? (c) Write two uses of esters. (d) Give the reaction of ethanoic acid with sodium carbonate.

Ans: (a) A sweet-smelling compound formed by an acid and an alcohol: $CH_3COOH + CH_3CH_2OH \xrightarrow{Acid} CH_3COOCH_2CH_3 + H_2O$. (b) An ester reacts with NaOH to give the alcohol and the sodium salt of the acid; it is used to make soap: $CH_3COOC_2H_5 \xrightarrow{NaOH} C_2H_5OH + CH_3COONa$. (c) Perfumes, flavouring agents. (d) $2CH_3COOH + Na_2CO_3 \longrightarrow 2CH_3COONa + H_2O + CO_2$.

Q33. 5M Board fav (a) Explain the cleansing action of soap with a labelled micelle. (b) Why does soap not work well in hard water? (c) How do detergents overcome this?

Ans: (a) A soap molecule has an ionic hydrophilic head and a long hydrocarbon hydrophobic tail. The tails attach to oily dirt and the heads face water, forming a micelle with the dirt at the centre; on rinsing the micelles carry the dirt away. (b) Ca and Mg salts of hard water react with soap to form insoluble scum, wasting soap. (c) Detergents (sodium salts of sulphonic acids or ammonium salts with Cl^-/Br^-) do not form insoluble precipitates with Ca and Mg ions, so they lather in hard water.

Q34. 5M (a) Distinguish between saturated and unsaturated hydrocarbons with one example each. (b) Explain addition and substitution reactions with equations. (c) Why do unsaturated compounds burn with a sooty flame?

Ans: (a) Saturated: only C–C single bonds, e.g. ethane C_2H_6. Unsaturated: C=C or C≡C, e.g. ethene C_2H_4. (b) Addition: $R_2C{=}CR_2 + H_2 \xrightarrow{Ni} R_2CH{-}CHR_2$. Substitution: $CH_4 + Cl_2 \xrightarrow{Sunlight} CH_3Cl + HCl$. (c) They have a higher carbon content, so they do not burn completely in air and unburnt carbon forms soot.

Q35. 5M Board fav (a) What are allotropes? (b) Compare the structures of diamond and graphite and explain why diamond is hard while graphite is soft and a good conductor. (c) Give one use of each and name a third allotrope.

Ans: (a) Different physical forms of the same element with different properties. (b) In diamond each carbon is bonded to four others in a rigid three-dimensional network, so it is the hardest substance and has no free electrons to conduct. In graphite each carbon is bonded to three others in flat hexagonal layers that slide over each other (soft, slippery); the fourth electron is free, so it conducts electricity. (c) Diamond: cutting and drilling tools; graphite: pencil leads, lubricant, electrodes. Third allotrope: fullerene C_{60}.

Case-based (4 marks)

Q36. 4M Board fav A compound X (C_2H_6O) is used in cough syrups and as a fuel. On oxidation with alkaline $KMnO_4$ it gives Y, which turns blue litmus red. X and Y react in the presence of an acid to give a sweet-smelling Z. (a) Identify X, Y, Z. (b) Write the equation for X ⟶ Y. (c) Write the equation for the formation of Z. (d) Name the reaction in (c).

Ans: (a) X = ethanol (CH_3CH_2OH), Y = ethanoic acid (CH_3COOH), Z = ethyl ethanoate. (b) $CH_3CH_2OH \xrightarrow{Alk.\ KMnO_4 + Heat} CH_3COOH$. (c) $CH_3COOH + CH_3CH_2OH \xrightarrow{Acid} CH_3COOCH_2CH_3 + H_2O$. (d) Esterification.

Q37. 4M Board fav Rahul washed clothes with soap using water from a borewell. He saw very little lather and a white curdy layer on the surface. With a detergent the same water gave plenty of foam. (a) What type of water is borewell water? (b) Name the curdy layer and why it forms. (c) Why did the detergent work? (d) Give one reason why soap solution looks cloudy.

Ans: (a) Hard water (contains Ca and Mg salts). (b) Scum: soap reacts with Ca^{2+}/Mg^{2+} ions to form an insoluble precipitate. (c) The charged ends of detergents do not form insoluble precipitates with Ca and Mg ions. (d) Micelles are large enough to scatter light.

Exam Alert

Case-based mein X, Y, Z identify karo, phir har equation condition ke saath likho (acid, 443 K, Ni, sunlight).

LIFE PROCESSES



Most asked: photosynthesis and its activities, human digestion (enzymes table), aerobic vs anaerobic respiration, double circulation and heart, nephron

Even a sleeping animal is alive: inside it, molecules keep moving to **repair and maintain** the body. The processes doing this maintenance never stop.

1. What are Life Processes?

Life processes: all the processes which together perform the maintenance job of the body, even when it is not doing anything. To prevent damage and breakdown, organisms need **energy**, which comes from food taken in from outside. This food is mostly carbon-based.

- → **Nutrition:** taking in food (source of energy and raw materials) and using it.
- → **Respiration:** breaking down food in cells to release energy.
- → **Transportation:** carrying food, oxygen and wastes from one part of the body to another.
- → **Excretion:** removing harmful by-products (wastes) from the body.

In unicellular organisms the whole body surface is in contact with the surroundings, so simple diffusion is enough. In large multicellular organisms specialised tissues and organs are needed for each job.

Yaad karo: 4 life processes

"Neend Raat Tak Easy": Nutrition, Respiration, Transportation, Excretion.

NCERT in-text quick answers

- → **How do we decide something is alive?** Visible movement is not enough; **molecular movement** (repair and maintenance) inside the body is the sign of life.
- → **Outside raw materials used by organisms:** food (carbon source), water, oxygen (and CO_2 for plants), minerals.
- → **Processes essential for life:** nutrition, respiration, transportation, excretion.

2. Nutrition

Autotrophic	Heterotrophic
Makes its own food from simple inorganic substances (CO_2, water)	Takes complex food made by others and breaks it down using enzymes
e.g. green plants, some bacteria	e.g. animals, fungi; depend directly or indirectly on autotrophs

Enzymes: biological catalysts that break complex substances into simpler ones.

(a) Autotrophic nutrition: Photosynthesis

Photosynthesis: the process by which autotrophs take in CO_2 and water and convert them into carbohydrates in the presence of sunlight and chlorophyll. Extra carbohydrate is stored as **starch** in plants and as **glycogen** in our body.

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

Three events of photosynthesis

1. Absorption of light energy by **chlorophyll**.
2. Conversion of light energy to chemical energy and **splitting of water** into hydrogen and oxygen.
3. **Reduction of CO_2** to carbohydrates.

These steps need not be continuous: desert plants take up CO_2 at night (stomata open) and use it in the day.

Meme samjho



Leaf bole: dhoop, paani aur CO_2 do, khaana main khud bana loongi. Isliye toh hum autotroph hain!

Asli baat: light energy chlorophyll absorb karta hai aur chemical energy (glucose) mein badalta hai.

Site: green parts of the leaf. Mesophyll cells contain green dots called **chloroplasts**, which have chlorophyll.

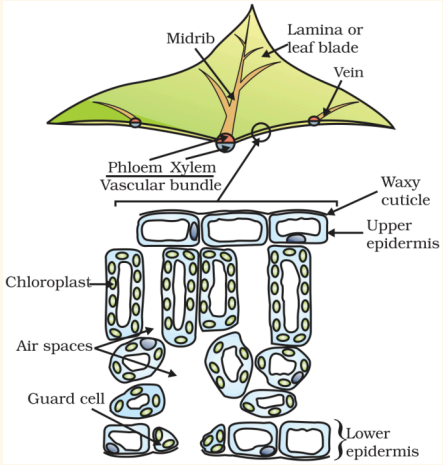


Fig. 5.1 Cross-section of a leaf

Stomata: tiny pores on the leaf surface; most of the gas exchange (CO_2 in, O_2 out) happens through them. Plants also lose a lot of water through them, so the pores close when CO_2 is not needed.

Guard cells: open and close each pore. Water flows **in** → guard cells **swell** → pore **opens**; water flows **out** → cells shrink → pore closes.

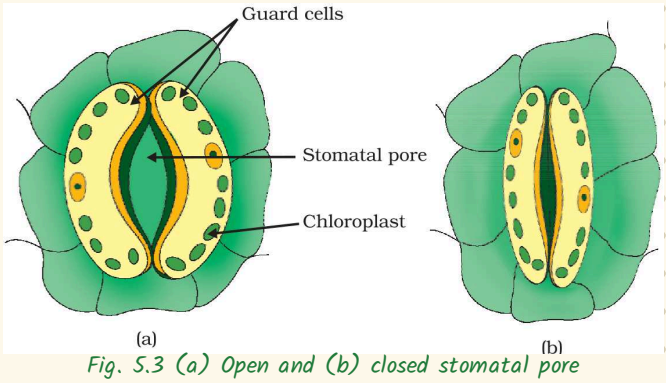


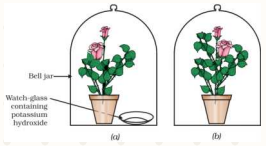
Fig. 5.3 (a) Open and (b) closed stomatal pore

Activity 5.1: Is chlorophyll needed for photosynthesis?



Done: A variegated-leaf plant (money plant/croton) is kept in the dark for 3 days (destarched), then in sunlight for 6 hours. A leaf is traced, boiled in water, then in alcohol on a water-bath, and dipped in dilute iodine. **Observed:** The leaf loses green colour in alcohol. After iodine, only the parts that were **green turn blue-black**. **Conclusion:** starch is formed only where chlorophyll is present, so **chlorophyll is essential** for photosynthesis.

Activity 5.2: Is CO_2 needed for photosynthesis?



Done: Two similar destarched potted plants are covered with air-tight bell-jars; a watch-glass of **potassium hydroxide** (absorbs CO_2) is kept beside one. After 2 hours in sunlight a leaf from each is tested for starch. **Observed:** The leaf from the jar with KOH shows **no starch**; the other turns blue-black. **Conclusion:** without CO_2 no starch is made, so **CO_2 is essential.**

Other raw materials: water is absorbed from the soil by roots. Nitrogen, phosphorus, iron and magnesium are also taken from the soil. **Nitrogen** is taken as inorganic **nitrates or nitrites**, or as organic compounds made by bacteria from atmospheric nitrogen.

Paheli-Bujho



Bujho: Paheli, plant toh din bhar dhoop mein khaana banata hai, raat ko bhookha so jaata hoga na?

Paheli: Bhondu! Extra glucose **starch** ban ke store hota hai, raat ko wahi kaam aata hai.

Paheli: Aur desert plants raat ko hi CO_2 le lete hain, taaki din mein stomata band rakh ke paani bacha sakein.

Bujho: Matlab plant ka bhi tiffin pack hota hai!

Exam Alert

→ Activity 5.1 mein leaf ko **alcohol** mein boil karte hain (chlorophyll hatane ke liye), seedha flame pe nahi: **water-bath** use karo.

→ KOH ka kaam: **CO_2 absorb** karna. Ye 2-mark ka pakka question hai.

→ Plant ko pehle dark mein rakhna = **destarching**.

(b) Heterotrophic nutrition

Type	How	Examples
Saprophytic	breaks down dead, decaying matter outside the body, then absorbs it	fungi: bread mould, yeast, mushrooms
Parasitic	feeds on a living host without killing it	*Cuscuta* (amar-bel), ticks, lice, leeches, tapeworm
Holozoic	takes in whole food and digests it inside the body	Amoeba, cow, humans

How do organisms obtain nutrition?

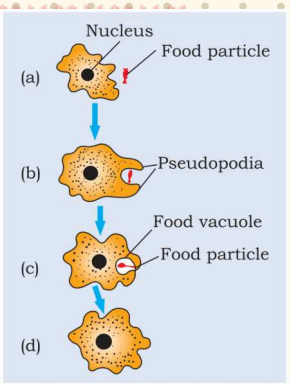


Fig. 5.5 Nutrition in Amoeba

Amoeba: takes in food using temporary finger-like extensions called **pseudopodia**, which fuse over the food particle to form a **food vacuole**. Complex substances are broken down inside it; simpler ones diffuse into the cytoplasm. Undigested material moves to the surface and is thrown out.

Paramoecium: has a definite shape, so food is taken in at a **specific spot**; it is moved there by **cilia** covering the whole cell.

(c) Nutrition in human beings

Ingestion → Digestion → Absorption → Assimilation → Egestion

The **alimentary canal** is a long tube from mouth to anus; different parts do different jobs.

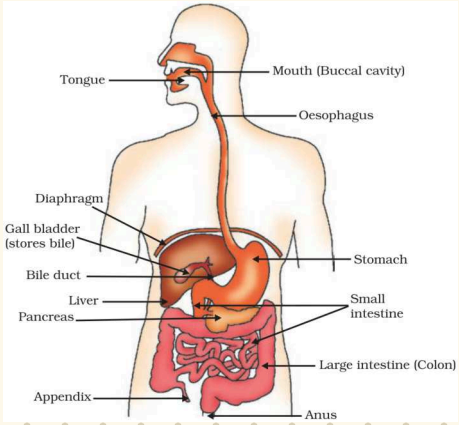


Fig. 5.6 Human alimentary canal

Mouth: teeth crush food into small particles; the tongue mixes it. **Salivary glands** secrete saliva containing the enzyme salivary amylase, which breaks **starch into simple sugar**.

Peristalsis: rhythmic contraction of the muscles lining the canal, pushing food forward in a regulated manner. Food goes from the mouth to the stomach through the **oesophagus** (food-pipe).

Activity 5.3: Action of saliva on starch

Done: Take 1 mL starch solution (1%) in test tubes A and B. Add 1 mL saliva to A only; after 20–30 min add dilute iodine to both. **Observed:** B turns **blue-black**; A shows no blue-black colour. **Conclusion:** starch in A has been digested by salivary amylase into sugar.

Stomach: its muscular walls mix food with juices from the **gastric glands**:

- **Hydrochloric acid:** makes the medium acidic so that pepsin can act; also kills germs that come with food.
- **Pepsin:** protein-digesting enzyme.
- **Mucus:** protects the inner lining of the stomach from the acid.

Exit of food from the stomach into the small intestine is regulated by a **sphincter muscle**.

Small intestine: the longest part of the canal; site of complete digestion of carbohydrates, proteins and fats. Herbivores eating grass have a longer one (to digest cellulose); carnivores have a shorter one.

- **Liver:** secretes **bile**. Food from the stomach is acidic; bile makes it **alkaline** so pancreatic enzymes can act. Bile salts break large fat globules into small ones (**emulsification**).
- **Pancreas:** pancreatic juice with **trypsin** (digests proteins) and **lipase** (breaks down emulsified fats).
- **Intestinal juice:** finally converts proteins into **amino acids**, complex carbohydrates into **glucose**, fats into **fatty acids and glycerol**.

Gland	Juice / enzyme	Acts on
Salivary	saliva: salivary amylase	starch ⟶ sugar
Gastric	HCl, pepsin, mucus	proteins (in acid)
Liver	bile (no enzyme)	emulsifies fats
Pancreas	trypsin, lipase	proteins; fats
Intestinal	intestinal juice	amino acids, glucose, fatty acids + glycerol

Villi: numerous finger-like projections of the inner lining of the small intestine. They increase the surface area for absorption and are richly supplied with blood vessels, which take absorbed food to every cell, where it is used for energy, new tissues and repair (**assimilation**).

Large intestine: unabsorbed food enters it; its walls absorb more water. The rest is removed through the **anus**, regulated by the anal sphincter.

Quick fact: Dental caries

Bacteria acting on sugars make **acids** that soften enamel. Bacteria and food stick as **plaque**, so saliva cannot neutralise the acid. Brushing after eating removes plaque.

Exam Alert

- → **Bile mein koi enzyme nahi hota:** sirf fats emulsify karta hai. MCQ trap!
- → HCl khud digest nahi karta: medium acidic banata hai aur germs maarta hai.
- → Villi ka question ho toh "**surface area badhata hai**" likhna mat bhoolna.

3. Respiration

Respiration: breakdown of food (glucose) in cells to release energy. The first step, **glucose (6-carbon) ⟶ pyruvate (3-carbon)**, happens in the cytoplasm of all organisms. After that the pathway differs:

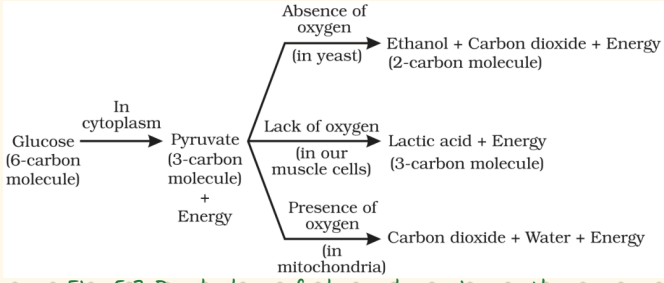


Fig. 5.8 Break-down of glucose by various pathways

$$\text{Glucose} \xrightarrow{\text{cytoplasm}} \text{Pyruvate} + \text{Energy}$$

$$\text{Pyruvate} \xrightarrow[\text{yeast}]{\text{No } O_2} \text{Ethanol} + CO_2 + \text{Energy}$$

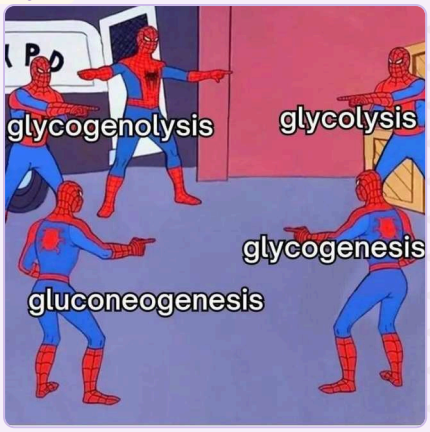
$$\text{Pyruvate} \xrightarrow[\text{muscle cells}]{\text{Lack of } O_2} \text{Lactic acid} + \text{Energy}$$

$$\text{Pyruvate} \xrightarrow[\text{mitochondria}]{O_2} CO_2 + H_2O + \text{Energy}$$

Aerobic respiration	Anaerobic respiration
in the presence of oxygen	in the absence (or lack) of oxygen
glucose broken completely into CO_2 and water	broken incompletely into ethanol + CO_2, or lactic acid
cytoplasm + mitochondria	cytoplasm only
much more energy	less energy

Muscle cramps: during sudden heavy activity (sprinting) oxygen falls short in muscle cells, so pyruvate becomes **lactic acid**; its build-up causes cramps.

Meme samjho

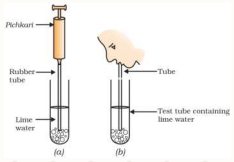


Ye sab ek doosre ko point kar rahe hain, par board mein sirf ek naam aata hai: glucose ⟶ pyruvate!

Asli baat: Class 10 mein bas itna: pehla step cytoplasm mein (glucose ⟶ pyruvate), baaki raasta oxygen decide karti hai.

Activity 5.4: CO_2 in exhaled air

Done: Air is passed through lime water with a syringe; exhaled air is blown into another tube of lime water. **Observed:** Lime water turns milky **much faster** with exhaled air. **Conclusion:** exhaled air has **more CO_2** than atmospheric air.



Activity 5.5: Products of fermentation

Done: Yeast is added to sugar solution in a test tube; the gas formed is passed into lime water. **Observed:** Lime water turns milky. **Conclusion:** yeast respires anaerobically, producing **CO_2 and ethanol.**

ATP (adenosine triphosphate): the energy currency of the cell. Energy released in respiration makes ATP from ADP and inorganic phosphate. Breaking the terminal phosphate bond using water releases **30.5 kJ/mol.** ATP powers muscle contraction, protein synthesis and nerve impulses.

Photosynthesis	Respiration
makes food (glucose)	breaks down food
in chloroplasts, only in light	in cytoplasm and mitochondria, all the time
CO_2 + water used, O_2 released	O_2 used, CO_2 + water released
light energy stored	energy released (ATP)

Respiration in plants: gases are exchanged by diffusion through **stomata** in leaves and **lenticels** in stems. At night only respiration occurs, so CO_2 is given out; in the day CO_2 from respiration is used in photosynthesis, so **O_2 release** is the major event.

Activity 5.6: Breathing in fish

Done: Count how many times a fish opens and closes its mouth and gill-slits in a minute; compare with your own breathing. **Observed:** They open and close together, **much faster** than we breathe. **Conclusion:** water has much less dissolved oxygen than air, so aquatic animals breathe faster; gills absorb the dissolved O_2.

Organs of gas exchange: unicellular organisms use the body surface, fishes use **gills**, insects use spiracles and tracheae, and land animals use **lungs**. All have a large, thin surface with a rich blood supply.

Human respiratory system

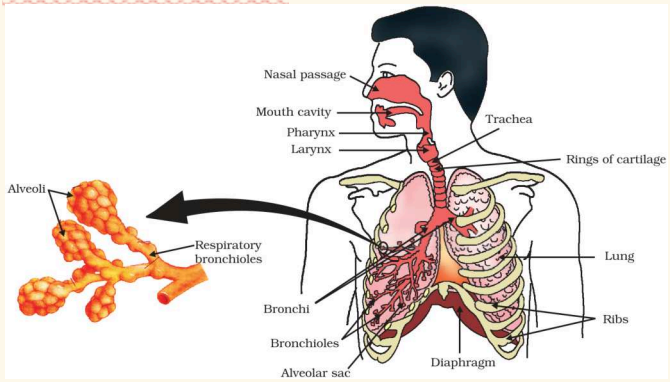


Fig. 5.9 Human respiratory system

Nostrils → Pharynx → Trachea → Bronchi → Bronchioles → Alveoli

- **Nasal passage:** fine hairs and mucus filter dust and germs.
- **Rings of cartilage** in the throat stop the air-passage from collapsing.
- **Breathing in:** ribs lift, diaphragm flattens, chest cavity enlarges and air rushes into the lungs.
- **Alveoli:** balloon-like sacs at the ends of bronchioles; spread out they would cover about **80 m^2**; rich network of blood vessels. The lungs always hold a **residual volume** of air, giving time for O_2 absorption and CO_2 release.
- **Haemoglobin:** respiratory pigment in RBCs with a very high affinity for O_2. CO_2 is more soluble in water, so it is mostly carried **dissolved in blood.**

Quick fact

If diffusion alone had to move oxygen in our body, a molecule would take about **3 years** to reach our toes from the lungs. That is why we need haemoglobin and circulation.

More to know: Smoking

- **Cilia** in the air passage sweep out germs and dust; smoking **destroys cilia**, so germs and smoke reach the lungs: infection, cough, **lung cancer.**
- Tobacco in any form (cigarette, bidi, hookah, gutkha) harms lungs, heart and liver; **gutkha causes oral cancer.**

Paheli-Bujho



Bujho: Paheli, naak mein baal aur mucus kyun? Naak ka software kharab hai kya?

Paheli: Software nahi, **security system** hai! Baal aur mucus dhool aur germs ko rokte hain.

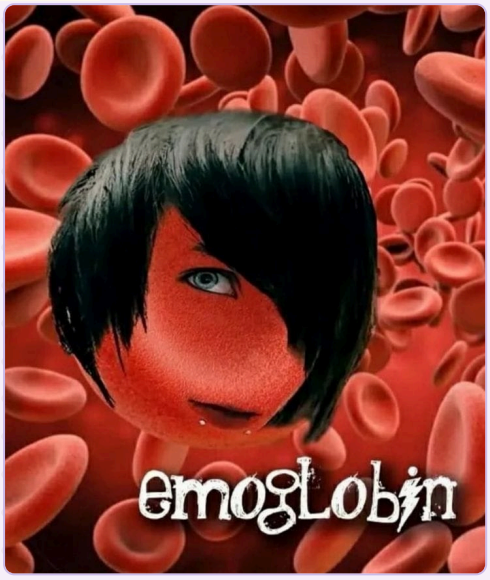
Paheli: Smoking inhi cilia ko khatam karti hai, phir germs seedha lungs tak. Isliye just say NO!

4. Transportation

(a) Blood

Blood: a fluid connective tissue. Its fluid medium, **plasma**, carries food, CO_2, salts and nitrogenous wastes in dissolved form; **red blood corpuscles** carry oxygen; **platelets** help clotting. We need a **pump** (heart), a network of **tubes** (blood vessels) and a system to **repair leaks** (platelets).

Meme samjho



Haemoglobin ka emo avatar: RBC ke andar baithi oxygen ko kabhi akela nahi chhodti!

Asli baat: haemoglobin ki oxygen se affinity bahut zyada hai; kam ho toh (anaemia) thakaan aur saans phoolti hai.

Activity 5.7: Haemoglobin survey

Done: Find the normal haemoglobin range of humans (children, men, women) and cattle from a health centre and a vet clinic. **Observed:** Levels differ with **age and sex**. **Conclusion:** less haemoglobin (anaemia) means less O_2 is carried, so the person feels **tired and breathless**.

(b) Heart

A **muscular organ** about the size of our fist, with four chambers: two upper **atria** and two lower **ventricles**. Ventricles have thicker walls as they pump blood to organs. **Valves** stop blood flowing backwards when atria or ventricles contract.

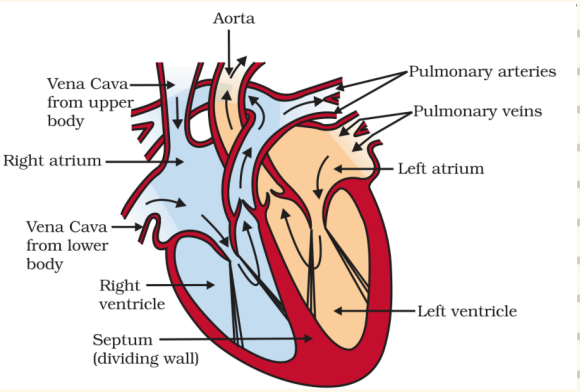


Fig. 5.10 Schematic sectional view of the human heart

→ **Oxygen-rich blood:** lungs ⟶ pulmonary vein ⟶ left atrium ⟶ left ventricle ⟶ aorta ⟶ body.

→ **Deoxygenated blood:** body ⟶ vena cava ⟶ right atrium ⟶ right ventricle ⟶ pulmonary artery ⟶ lungs.

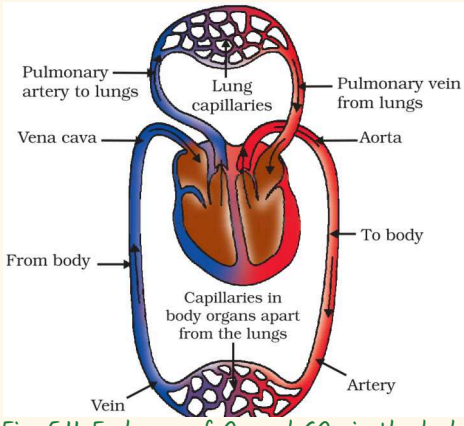


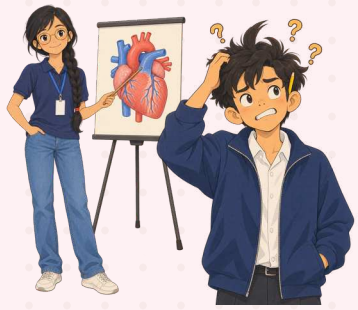
Fig. 5.11 Exchange of O_2 and CO_2 in the body

Why separate sides? The right and left sides keep oxygenated and deoxygenated blood from mixing; this highly efficient O_2 supply is needed by **birds and mammals**, which spend energy to keep a constant body temperature.

Double circulation: blood passes through the heart twice in one cycle: once to the lungs (pulmonary) and once to the body (systemic).

Animal	Heart	Circulation
Fishes	2 chambers	single: heart ⟶ gills ⟶ body
Amphibians, many reptiles	3 chambers	some mixing tolerated
Birds, mammals	4 chambers	double, no mixing

Paheli-Bujho



Bujho: Heart ke paas 4 kamre hain, ek mujhe rent pe de do!

Paheli: Chaaron full hain! Left side mein **oxygen wala** khoon, right side mein bina oxygen wala, beech mein deewar taaki mix na ho.

Bujho: Matlab khoon ek round mein heart ke do chakkar lagata hai: double circulation. Samajh gaya!

Yaad karo

"Artery = Away": artery heart se door le jaati hai, vein wapas laati hai. Exception: **pulmonary artery** mein deoxygenated, **pulmonary vein** mein oxygenated khoon!

(c) Blood vessels

Arteries	Veins
carry blood away from the heart to organs	bring blood from organs back to the heart
thick, elastic walls (high pressure)	thin walls (no high pressure)
no valves	valves keep flow in one direction

Capillaries: arteries divide into the smallest vessels, with walls **one cell thick**. Exchange of materials with cells happens here; capillaries join to form veins.

Blood pressure: force that blood exerts against the wall of a vessel. **Systolic** (ventricle contracts) about **120 mm Hg**; **diastolic** (ventricle relaxes) about **80 mm Hg**. Measured with a **sphygmomanometer**. High BP (hypertension) is caused by constriction of arterioles and can rupture an artery (internal bleeding).

Lymph (tissue fluid): some plasma, proteins and blood cells escape through pores in capillary walls into intercellular spaces. It is like plasma but **colourless with less protein**. It drains into lymphatic capillaries and vessels and then into large veins. Lymph carries digested and absorbed fat from the intestine and drains excess fluid back into the blood.

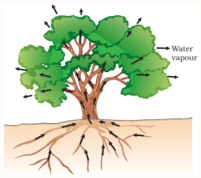
(d) Transportation in plants

Plants have low energy needs (they do not move and many cells are dead), so a slow transport system works. **Xylem** moves water and minerals from the soil; **phloem** moves products of photosynthesis from leaves to other parts.

Transport of water: xylem vessels of roots, stems and leaves are interconnected. Root cells take up ions from the soil, creating a concentration difference; water moves in from the soil and is pushed up (**root pressure**, important at night).

Activity 5.8: Transpiration from a plant

Done: Two pots, one with a plant and one with a stick of the same height; soil covered with plastic, each set covered with a plastic sheet, kept in sunlight for 30 min. **Observed:** Droplets form only on the sheet over the **plant**. **Conclusion:** plants lose water as vapour from their aerial parts (transpiration).



Transpiration: loss of water as vapour from the aerial parts of the plant. Evaporation from leaf cells creates a **suction** that pulls water up through the xylem. It helps in upward movement of water and minerals and in **temperature regulation**. In the day, transpiration pull is the main driving force.

Translocation: transport of soluble products of photosynthesis (and amino acids) through **phloem**, in **sieve tubes** with the help of **companion cells**, upward and downward to roots, fruits, seeds and growing parts. Sucrose is loaded into phloem using **ATP**; this raises osmotic pressure, water moves in, and the pressure pushes material to areas of lower pressure.

Xylem	Phloem
water and minerals	food (sucrose, amino acids)
only upward	upward and downward
no energy (physical forces)	uses energy from ATP

Exam Alert

→ **Xylem = paani, Phloem = khaana.** Xylem free mein chalta hai, phloem ATP kharch karta hai.

→ Din mein main force = **transpiration pull**; raat mein = **root pressure**.

5. Excretion

Excretion: the biological process of removing harmful nitrogenous metabolic wastes such as **urea and uric acid** from the body. Unicellular organisms use simple diffusion through the body surface; *Amoeba* and *Paramoecium* also use **contractile vacuoles** to throw out excess water.

Human excretory system

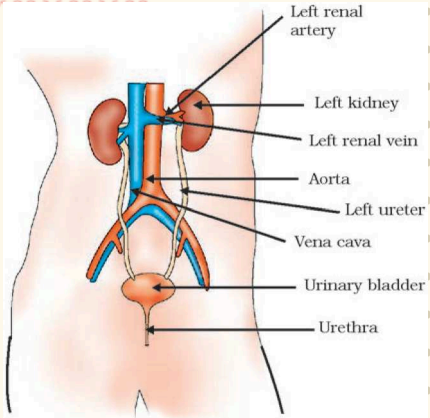


Fig. 5.13 Excretory system in human beings

A pair of **kidneys**, a pair of **ureters**, a **urinary bladder** and a **urethra**. Urine made in the kidneys passes through the ureters into the bladder, is stored there and leaves through the urethra. The bladder is muscular and under nervous control, so we can usually control the urge to urinate.

Kidneys lie in the abdomen, one on either side of the backbone. They filter urea and uric acid out of the blood (just as CO_2 is removed in the lungs) and keep the balance of water and salts.

Nephron: the basic filtration unit; each kidney has a large number of them. A cluster of thin-walled capillaries, the **glomerulus**, sits inside a cup-shaped **Bowman's capsule**, which collects the filtrate. The filtrate then flows through a long **tubule**.

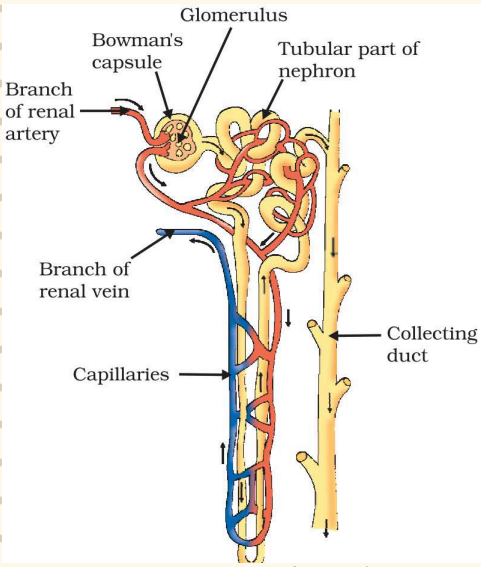
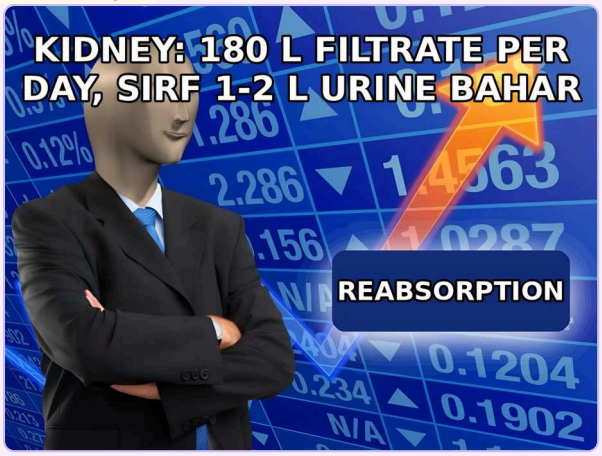


Fig. 5.14 Structure of a nephron

Re-absorption: as the filtrate flows along the tubule, useful substances such as **glucose, amino acids, salts** and a major amount of **water** are selectively re-absorbed into the surrounding capillaries. The amount of water re-absorbed depends on how much excess water there is in the body and how much dissolved waste has to be removed.

Meme samjho



Kidney roz 180 L filtrate banata hai, par bahar sirf 1–2 L urine: baaki sab wapas!

Asli baat: tubule mein glucose, amino acids, salts aur zyada tar paani ***re-absorb*** ho jaata hai.

Artificial kidney (haemodialysis): used when kidneys fail. Blood flows through long **semi-permeable tubes** in a tank of dialysing fluid (same osmotic pressure as blood, but no nitrogenous wastes); wastes diffuse out and purified blood returns to the patient. No re-absorption takes place.

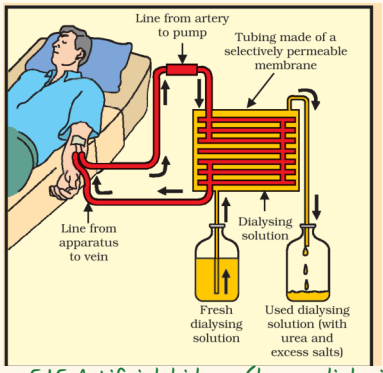


Fig. 5.15 Artificial kidney (haemodialysis)

Think it over: Organ donation

Giving an organ, with the **consent of the donor and family**, to a person whose organ has failed. Most donations happen after death or brain death; a **kidney or part of the liver** can be donated while alive. Common transplants: cornea, kidney, heart, liver, lungs, pancreas, bone marrow.

Excretion in plants

- → O_2 and CO_2 diffuse out through stomata; excess water is removed by **transpiration**.
- → Many wastes are stored in **cell vacuoles** or in leaves that fall off.
- → Others are stored as **resins and gums** in old xylem; some are excreted into the soil.

Yaad karo: urine ka raasta

"Kamla Uncle Bus Utha": Kidney ⟶ Ureter ⟶ Bladder ⟶ Urethra. Filter kidney mein, store bladder mein!

Exam Alert

- → **Glomerulus = filter, tubule = re-absorption.** Nephron ka labelled diagram 3-mark mein aata hai.
- → Dialysis mein re-absorption nahi hota: yahi artificial aur real kidney ka main difference hai.

1 minute recap

Nutrition (photosynthesis, digestion) → Respiration (glucose ⟶ pyruvate) → Transport (heart, xylem, phloem) → Excretion (nephron)

Most Important Questions

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM Board fav The opening and closing of the stomatal pore depends on:
(a) oxygen (b) temperature
(c) water in guard cells (d) CO_2 in stomata
Ans: (c) the water content of guard cells: they swell and open the pore when water flows in.

Q2. IM Board fav The breakdown of pyruvate to give carbon dioxide, water and energy takes place in:
(a) cytoplasm (b) chloroplast
(c) mitochondria (d) nucleus
Ans: (c) mitochondria.

Q3. IM Which is the correct sequence of air passage?
(a) pharynx, nostrils, trachea, alveoli
(b) nostrils, trachea, pharynx, bronchi
(c) nostrils, pharynx, trachea, bronchi, bronchioles, alveoli
(d) nostrils, bronchi, trachea, alveoli
Ans: (c) Nostrils ⟶ pharynx ⟶ trachea ⟶ bronchi ⟶ bronchioles ⟶ alveoli.

Q4. IM Name the salivary enzyme and the food it acts on.
Ans: Salivary amylase; it breaks down starch into simple sugar.

Q5. IM Why is the wall of the left ventricle thicker than that of the left atrium?
Ans: The ventricle has to pump blood with force to all body organs, while the atrium only pushes blood into the ventricle.

Q6. IM Which blood vessel carries deoxygenated blood from the heart to the lungs?
Ans: Pulmonary artery (the only artery carrying deoxygenated blood).

Q7. IM Board fav **Assertion (A):** Bile juice helps in digestion of fats. **Reason (R):** Bile contains lipase, which is an enzyme that digests fats.
Ans: (c) A is true but R is false: bile has no enzyme; its salts emulsify fats, and lipase comes from the pancreas.

Q8. IM **Assertion (A):** Fishes have single circulation. **Reason (R):** In fishes, blood goes from the heart to the gills and then directly to the rest of the body.
Ans: (a) Both A and R are true and R is the correct explanation of A.

Q9. 1M The site of complete digestion of carbohydrates, proteins and fats is the:
(a) stomach (b) mouth
(c) small intestine (d) large intestine
Ans: (c) small intestine.

Q10. 1M Why is diffusion insufficient to meet the oxygen needs of multicellular organisms like humans?
Ans: Their body is large and most cells are not in direct contact with air, so diffusion would be far too slow; special organs and a transport system are needed.

Q11. 1M In which of the following groups do all organisms show saprophytic nutrition?
(a) Cuscuta, tapeworm, lice
(b) bread mould, yeast, mushrooms
(c) Amoeba, cow, humans (d) grass, algae, moss
Ans: (b) bread mould, yeast and mushrooms (fungi).

Q12. 1M Name the cells that open and close stomata, and the tissue that carries food in plants.
Ans: Guard cells; phloem.

Short answer (2-3 marks)

Q13. 2M What are the differences between autotrophic and heterotrophic nutrition?
Ans: Autotrophic: food made from simple inorganic CO_2 and water using sunlight and chlorophyll; e.g. green plants. Heterotrophic: complex food made by others is taken in and broken down by enzymes; e.g. animals, fungi. Autotrophs are producers; heterotrophs depend on them.

Q14. 3M Board fav List the three events that occur during photosynthesis. Do they occur continuously?
Ans: (i) Absorption of light energy by chlorophyll (ii) conversion of light energy to chemical energy and splitting of water into H and O (iii) reduction of CO_2 to carbohydrates. They need not be continuous: desert plants take in CO_2 at night and use it during the day.

Q15. 3M Board fav What are the functions of HCl, pepsin and mucus in the stomach?
Ans: HCl makes the medium acidic for pepsin to act and kills germs; pepsin digests proteins; mucus protects the inner lining of the stomach from the acid.

Q16. 2M Board fav How is the small intestine designed to absorb digested food?
Ans: Its inner lining has numerous finger-like projections called villi, which increase the surface area for absorption. Villi are richly supplied with blood vessels that carry absorbed food to every cell of the body.

Q17. 3M Board fav Write the pathways of breakdown of glucose in different organisms. Why do we get muscle cramps?
Ans: Glucose $\longrightarrow$ pyruvate (cytoplasm). Then: (i) no O_2, yeast: ethanol + CO_2 + energy (ii) lack of O_2, muscle: lactic acid + energy (iii) O_2, mitochondria: CO_2 + H_2O + energy. During heavy exercise O_2 is short, pyruvate forms lactic acid, and its build-up causes cramps.

Q18. 2M How are alveoli designed to maximise the exchange of gases?
Ans: Alveoli are thin-walled balloon-like sacs giving a huge surface area (about 80 m^2), with a rich network of blood capillaries, so O_2 and CO_2 diffuse quickly between air and blood.

Q19. 2M Board fav Why is separation of oxygenated and deoxygenated blood necessary in mammals and birds?
Ans: Birds and mammals are warm-blooded and need a lot of energy to maintain a constant body temperature. Separation prevents mixing, so a highly efficient supply of oxygen reaches the body for respiration.

Q20. 3M Give three differences between arteries and veins.
Ans: Arteries carry blood away from the heart; veins bring it back to the heart. Arteries have thick elastic walls; veins have thin walls. Arteries have no valves; veins have valves to keep flow in one direction.

Q21. 3M What is translocation? Why is it essential for plants? Which tissue carries it out?
Ans: Transport of soluble products of photosynthesis (sucrose, amino acids) from leaves to other parts of the plant. It supplies food to roots, fruits, seeds and growing parts that cannot make food. It occurs in phloem (sieve tubes with companion cells) and uses energy from ATP.

Q22. 2M How do plants get rid of their excretory products?
Ans: O_2/CO_2 through stomata; excess water by transpiration; wastes stored in vacuoles or in leaves that fall off; resins and gums in old xylem; some wastes excreted into the soil.

Q23. 2M How is an artificial kidney different from a real one?
Ans: In the artificial kidney blood is filtered through semi-permeable tubes in dialysing fluid and wastes diffuse out, but there is no re-absorption. In the natural kidney, useful substances like glucose, amino acids, salts and water are re-absorbed in the tubule.

Q24. 2M What is lymph? Give two of its functions.
Ans: Lymph is the colourless fluid formed from plasma, proteins and blood cells that escape through pores in capillary walls into intercellular spaces. Functions: (i) carries digested and absorbed fat from the intestine (ii) drains excess tissue fluid back into the blood.

Q25. 3M Board fav How do guard cells regulate the opening and closing of stomata? Why do they close when CO_2 is not needed?
Ans: When water flows into guard cells they swell and curve, opening the pore; when they lose water they shrink and the pore closes. Large amounts of water are lost through open stomata, so they close when CO_2 is not required, saving water.

Q26. 2M Why do herbivores have a longer small intestine than carnivores? Name the part where water is absorbed from undigested food.
Ans: Herbivores eat grass, which contains cellulose; a longer small intestine gives more time to digest it. Carnivores eat meat, which is easier to digest, so their small intestine is shorter. Water is absorbed in the large intestine.

Long answer (5 marks)

Q27. 5M Board fav (a) Describe an activity to show that CO_2 is necessary for photosynthesis. (b) What is the role of KOH? (c) Why are the plants destarched first?

Ans: (a) Two healthy destarched potted plants are kept under separate air-tight bell-jars on glass plates, with a watch-glass of KOH beside one. Both are kept in sunlight for about 2 hours; a leaf from each is boiled in water, then in alcohol, and tested with iodine. The leaf from the jar without KOH turns blue-black; the one with KOH does not. So CO_2 is necessary. (b) KOH absorbs CO_2 from the jar. (c) Keeping them in the dark removes stored starch, so any starch found later must be newly made.

Q28. 5M Board fav (a) Draw a schematic sectional view of the human heart and label any four parts. (b) Trace the path of oxygenated and deoxygenated blood. (c) What is double circulation?

Ans: (a) Diagram as Fig. 5.10: right/left atrium, right/left ventricle, aorta, vena cava, pulmonary artery, pulmonary vein. (b) Lungs ⟶ pulmonary vein ⟶ left atrium ⟶ left ventricle ⟶ aorta ⟶ body; body ⟶ vena cava ⟶ right atrium ⟶ right ventricle ⟶ pulmonary artery ⟶ lungs. (c) Blood passes through the heart twice in one complete cycle: pulmonary (heart–lungs–heart) and systemic (heart–body–heart).

Q29. 5M Board fav (a) Draw a labelled diagram of a nephron. (b) Describe how urine is formed. (c) How is the amount of urine regulated?

Ans: (a) Diagram as Fig. 5.14: glomerulus, Bowman's capsule, tubule, collecting duct, renal artery and vein. (b) Blood enters the glomerulus; water, urea, uric acid, glucose, salts and amino acids are filtered into Bowman's capsule. In the tubule, glucose, amino acids, salts and much water are selectively re-absorbed; the remaining fluid is urine, which goes to the collecting duct, ureter and bladder. (c) The water re-absorbed depends on how much excess water and dissolved waste the body has.

Q30. 5M (a) Name the organs of the human alimentary canal in order and the glands that secrete juices into it. (b) What happens to food in the small intestine?

Ans: (a) Mouth, oesophagus, stomach, small intestine, large intestine, anus; glands: salivary glands, gastric glands, liver, pancreas, intestinal glands. (b) Bile makes food alkaline and emulsifies fats; trypsin digests proteins and lipase breaks emulsified fats; intestinal juice converts proteins to amino acids, carbohydrates to glucose and fats to fatty acids and glycerol. Villi absorb the digested food into blood.

Q31. 5M (a) Draw a labelled diagram of the human respiratory system. (b) How does air enter the lungs during breathing in? (c) Why do the lungs always hold a residual volume of air?

Ans: (a) Diagram as Fig. 5.9: nasal passage, pharynx, larynx, trachea, rings of cartilage, bronchi, bronchioles, alveolar sacs, lungs, diaphragm. (b) Ribs are lifted and the diaphragm flattens, so the chest cavity becomes larger; air is sucked into the lungs and fills the alveoli. (c) So that there is enough time for O_2 to be absorbed into blood and CO_2 to be released.

Case-based (4 marks)

Q32. 4M Board fav An athlete sprinting 400 m felt pain and cramps in his leg muscles at the end of the race. (a) Which type of respiration occurred in his muscles? (b) Name the product that caused the cramps. (c) Where in the cell does the first step of glucose breakdown happen? (d) Give one difference between this and the respiration in yeast.

Ans: (a) Anaerobic respiration (lack of oxygen). (b) Lactic acid. (c) Cytoplasm (glucose ⟶ pyruvate). (d) Muscles form lactic acid; yeast forms ethanol and CO_2.

Q33. 4M Ravi's blood test shows haemoglobin of 9 g/dL (normal 13–17 g/dL). (a) Name the condition. (b) Why does he feel tired and breathless? (c) Where is haemoglobin found? (d) How is CO_2 mainly transported in blood?

Ans: (a) Anaemia (haemoglobin deficiency). (b) Less haemoglobin carries less oxygen to cells, so less energy is released by respiration. (c) In red blood corpuscles. (d) Dissolved in plasma, since CO_2 is more soluble in water.

Q34. 4M A student covered a potted plant and a stick (in a similar pot) with plastic sheets and kept both in sunlight. After 30 minutes droplets appeared only on the sheet over the plant. (a) Name the process. (b) Through which pores does it mainly occur? (c) Give two advantages of this process to the plant. (d) Which force moves water up at night?

Ans: (a) Transpiration. (b) Stomata. (c) Creates a suction (transpiration pull) that draws water and minerals up from the roots; helps regulate temperature. (d) Root pressure.

More board favourites

Q35. 2M Board fav Why is the rate of breathing much faster in aquatic organisms than in terrestrial organisms?

Ans: The amount of oxygen dissolved in water is much lower than in air, so aquatic organisms must breathe faster to get enough oxygen.

Q36. 3M Board fav How are fats digested in our body? Where does this process take place?

Ans: In the small intestine. Bile from the liver emulsifies fats (breaks large globules into small ones); lipase from the pancreatic juice and intestinal juice then breaks them into fatty acids and glycerol.

Exam Alert: diagrams to practise

Leaf section, stomata, alimentary canal, respiratory system, heart, nephron: ye 6 diagrams har saal aate hain. Pencil se banao, labels seedhi lines se.

Control and Coordination



Most asked: neuron and synapse, reflex arc, parts of the brain, phototropism and plant hormones, endocrine glands and their hormones

A plant bends towards light; we pull our hand away from a hot object. Every such response needs **control and coordination**: nerves and hormones in animals, only hormones in plants.

1. Animals: Nervous System

Movement is a sign of life. Some movements are due to growth (a seed sprouting), others are not (running, the beating heart). In all of them the response must be controlled, and body parts must work together in a **coordinated** way.

Stimulus: any change in the environment to which an organism responds (heat, light, touch, smell). **Receptors:** specialised tips of nerve cells in the sense organs that detect stimuli. If the receptors do not work, the stimulus is not detected and there is no response (e.g. a blocked nose in a cold: no smell).

Receptor	Organ	Detects
Gustatory	tongue	taste
Olfactory	nose	smell
Photoreceptors	eye	light
Phonoreceptors	ear	sound, balance
Thermoreceptors	skin	heat, cold

Information from receptors travels to the brain or spinal cord through **sensory nerves**; orders come back to muscles and glands (**effectors**) through **motor nerves**.

Key terms

- **Stimulus:** a change in the surroundings that an organism responds to.
- **Effector:** the muscle or gland that carries out the response.
- **Impulse:** the electrical signal that travels along a neuron.
- **Neurotransmitter:** chemical released at a nerve ending that carries the signal across a gap.
- **Endocrine gland:** a ductless gland that pours hormones directly into blood.

Neuron (nerve cell)

The structural and functional unit of the nervous system. **Dendrites** (branched tips of the cell body) pick up information; it travels as an electrical impulse from the dendrite to the **cell body (cyton)**, then along the long **axon** to its nerve ending.

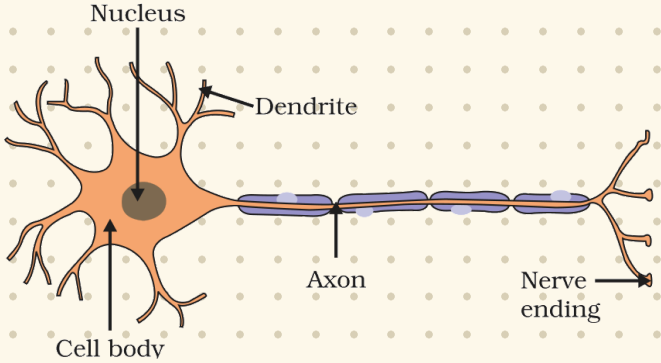


Fig. 6.1 (a) Structure of a neuron

Synapse: the tiny gap between the nerve ending of one neuron and the dendrite of the next. At the end of the axon the electrical impulse releases **chemicals (neurotransmitters)**, which cross the gap and start a fresh electrical impulse in the next neuron. The narrow gap is the **synaptic cleft**; a common neurotransmitter is **acetylcholine**.

Dendrite → Cell body → Axon → Nerve ending → Synapse → Next neuron

Neuromuscular junction: the point where the nerve ending of a motor neuron meets a **muscle fibre**. Chemicals released into the gap act on the muscle fibre, which contracts, so the body part moves. Nerve endings pass impulses to gland cells in the same way. The many **mitochondria** near the junction supply energy for this transfer.

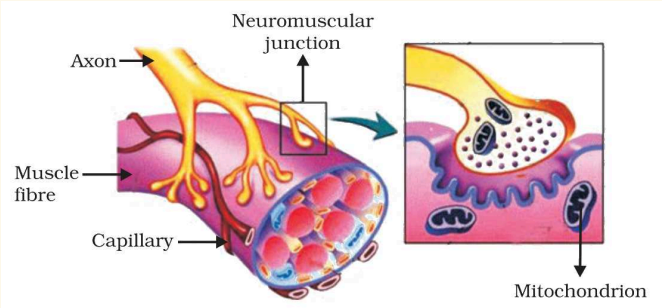
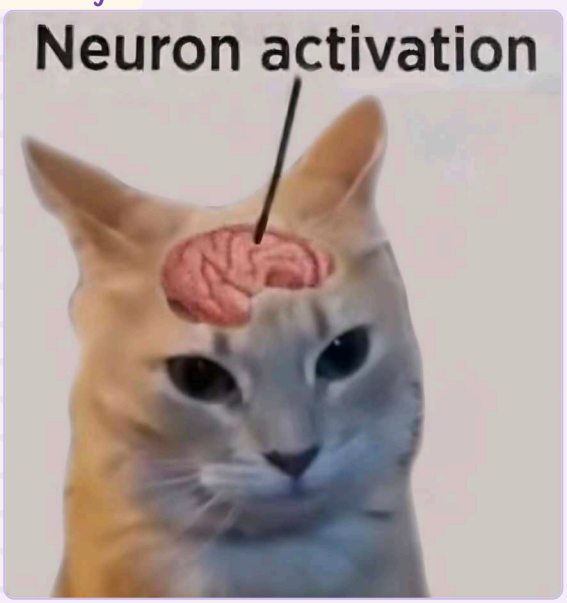


Fig. 6.1 (b) Neuromuscular junction

Meme samjho



Garam chai ka cup chhua aur dimaag mein bulb jal gaya: yahi hai impulse ka safar!

Asli baat: impulse hamesha ek hi direction mein jaata hai: dendrite ⟶ cell body ⟶ axon ⟶ synapse. Ulta kabhi nahi.

Activity 6.1: Does a blocked nose change taste?

Done: Put some sugar in the mouth and taste it; then press the nose shut and taste sugar again; eat lunch with the nose blocked. **Observed:** With the nose blocked, sugar and food taste much **less distinct**, just like during a cold. **Conclusion:** flavour is sensed by **taste (tongue) and smell (nose) receptors together**; with the nose blocked, food seems flat.

Reflex action

Reflex action: a sudden, automatic and quick response to a stimulus, done without thinking. Examples: pulling the hand away from a flame, knee-jerk, sneezing, coughing, blinking when something comes near the eye, the **pupil becoming smaller in bright light.**

Thinking involves the brain and takes time. In an emergency there is no time to think, so the connection between the detecting nerve and the acting nerve is made **directly in the spinal cord.** The information also reaches the brain, but the response has already been given.

Reflex arc: the pathway followed by nerve impulses in a reflex action. Reflex arcs evolved because the thinking process of the brain is not fast enough.

Receptor → Sensory neuron → Relay neuron (spinal cord) → Motor neuron → Effector (muscle)

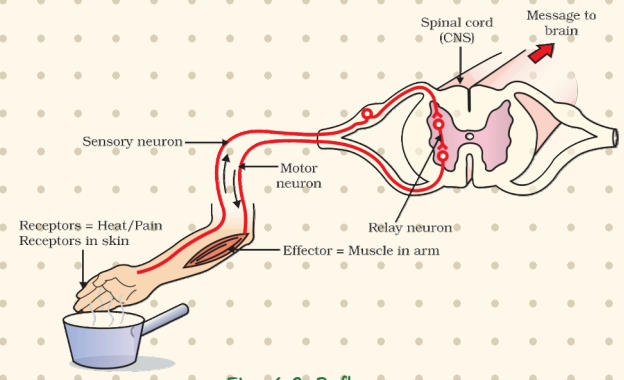


Fig. 6.2 Reflex arc

Exam Alert

- → **Reflex = spinal cord ka kaam**, brain ko khabar baad mein milti hai.
- → Reflex arc ke diagram mein **arrows** (direction of impulse) zaroor lagao: receptor se effector tak.
- → Neuron ka diagram: dendrite, cell body, nucleus, axon, nerve ending: 5 labels pakke.

Quick fact

Information in a nerve travels as an **electrical impulse**, but it crosses every synapse as a **chemical.** So nerve communication is electrical plus chemical.

Human nervous system

Part	Made of	Job
Central (CNS)	brain and spinal cord	receives information, decides, sends orders
Peripheral (PNS)	cranial nerves (from brain) and spinal nerves (from spinal cord)	carry messages between CNS and body parts

Action	Controlled by	Examples
Voluntary	fore-brain	writing, talking, walking
Involuntary	mid-brain and hind-brain	heartbeat, breathing, salivation, vomiting
Reflex	spinal cord	knee-jerk, blinking

Human brain

The brain is the main **coordinating centre** of the body. It has three major parts:

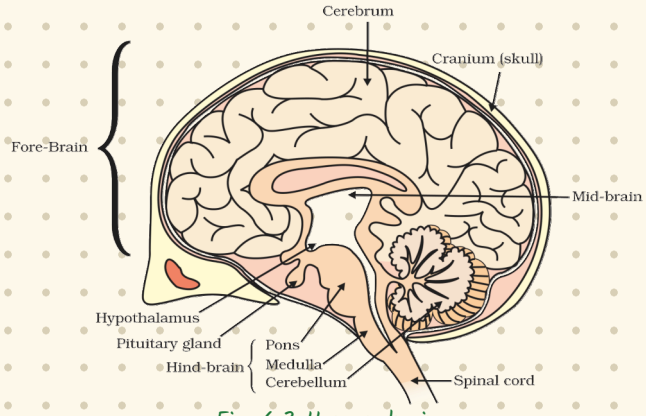
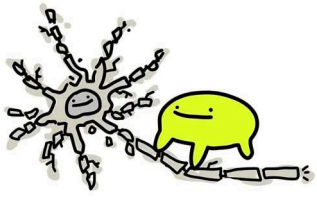


Fig. 6.3 Human brain

1. **Fore-brain:** the main thinking part. Sensory areas receive impulses (separate areas for hearing, smell, sight); association areas interpret them using stored memory; motor areas control voluntary muscles. It has the **centre for hunger**. Cerebrum is its largest part; the **hypothalamus** controls body temperature, hunger and thirst.
2. **Mid-brain:** controls involuntary actions such as reflex movements of the head, neck and eyes.
3. **Hind-brain: medulla** controls blood pressure, salivation and vomiting; **pons** regulates breathing; **cerebellum** keeps the precision of voluntary actions, posture and balance: riding a bicycle, picking up a pencil.

Meme samjho



Cerebellum ne kaha: "Le bhai, ek braincell rakh le, cycle chalate waqt balance mein kaam aayega."

Asli baat: posture aur balance ***cerebellum*** (hind-brain) sambhalta hai; sochne ka kaam fore-brain ka hai.

Yaad karo: hind-brain

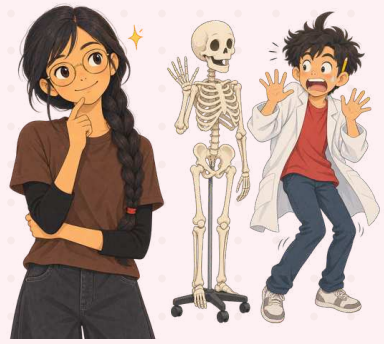
"Mere Paas Cycle": Medulla (BP, vomiting), Pons (breathing), Cerebellum (balance).

How are these tissues protected? The brain sits inside a bony box, the **skull (cranium)**, and floats in a fluid-filled balloon

(**cerebrospinal fluid**) that absorbs shocks. The spinal cord is protected by the **vertebral column** (backbone).

How does nervous tissue cause action? When a nerve impulse reaches a muscle, special proteins in the muscle cells change their **shape and arrangement**, so the cell becomes shorter and the part moves. Muscles moved by will are voluntary; heart and gut muscles are involuntary.

Paheli-Bujho



Bujho: Paheli, garam tawa chhua toh haath pehle hata ya dard pehle hua? Main toh soch ke hataya tha!

Paheli: Sochne ka time hi nahi tha! **Spinal cord** ne reflex arc se haath hata diya, brain ko khabar baad mein mili.

Bujho: Matlab mera dimaag late-comer hai?

Paheli: Emergency mein haan! Isliye reflex itna fast hota hai.

Limitations of electrical impulses: (1) they reach only cells **connected by nervous tissue**, not every cell; (2) a cell needs time to **reset** after an impulse, so it cannot send impulses continuously. Hence animals also use chemical communication (hormones), and plants use only that.

NCERT in-text quick answers

→ **Reflex action vs walking:** a reflex is sudden and automatic, controlled by the spinal cord; walking is voluntary, controlled by the fore-brain (balance by the cerebellum).

→ **Spinal cord injury:** sensory impulses to the brain and motor impulses to muscles below the injury are both disrupted, along with reflexes handled there.

→ **Why chemical communication?** Impulses reach only cells joined by nerves and need reset time; hormones reach every cell through blood.

2. Coordination in Plants

Plants have **neither a nervous system nor muscles**, yet they respond to stimuli. Plant movements are of two kinds:

(a) Immediate response (not due to growth)

Leaves of the sensitive plant (**touch-me-not, Mimosa**) fold and droop the moment they are touched. Information passes from cell to cell by **electrical-chemical** means (no special tissue). Cells change shape by changing the amount of water in them: they swell or shrink, and the leaf folds.



Fig. 6.4 The sensitive plant: (a) before and (b) after touching

Nastic movements: such non-growth movements, whose direction does **not** depend on the direction of the stimulus.

(b) Movement due to growth (tropic movements)

Slow, directional growth of a plant part towards or away from a stimulus. **Tendrils** of a pea plant coil around a support: auxin collects on the side **away from the support**, so that side grows faster than the side touching the support, and the tendril coils round it.

Tropism	Stimulus	Example
Photo-	light	shoot towards light (+), root away (−)
Geo-	gravity	root down (+), shoot up (−)
Hydro-	water	roots grow towards water (+)
Chemo-	chemicals	pollen tube grows towards ovule
Thigmo-	touch	tendrils coil round a support

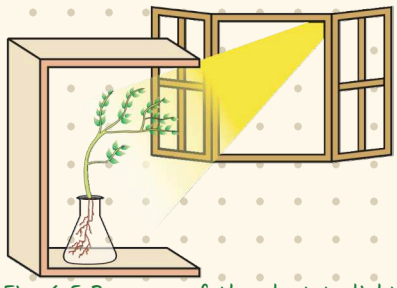


Fig. 6.5 Response of the plant to light

Activity 6.2: Response of plant to light

Done: Freshly germinated bean seeds are kept on a wire mesh over a water-filled flask inside a box open on one side towards a window. After 2–3 days the flask is turned so the shoots face away from light. **Observed:** Shoots bend **towards light**, roots away. After turning, the **new parts** again bend towards light; old parts do not change direction. **Conclusion:** shoots are positively and roots negatively phototropic; tropic movement is by **growth**, so only growing parts respond.

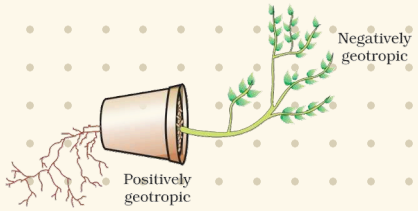


Fig. 6.6 Plant showing geotropism

Experiment for hydrotropism: grow a seedling in dry soil with a porous clay pot filled with water buried on one side. After a few days, dig out gently: the **roots have bent towards the pot**.

Nastic movement	Tropic movement
not directional	directional (towards/away)
not due to growth: cells gain or lose water	due to unequal growth of cells
fast, seen at once	slow, seen after hours or days
folding of Mimosa leaves	shoot bending towards light

Why are plant responses slow? Plants have no nerves or muscles; information passes from cell to cell by chemicals,

and most movements need growth, which takes hours or days. Only a few movements, like folding of Mimosa leaves or the snapping of a Venus flytrap, are quick because they depend on **water movement** in cells.

Exam Alert

- → Shoot light ki taraf (+), root light se door (−). **Gravity ke liye ulta**: root neeche (+), shoot upar (−).
- → Mimosa ka sharmana = **paani ka khel**, growth nahi. Isliye turant hota hai aur wapas bhi khul jaata hai.
- → Activity 6.2 mein "sirf **naye parts** mudte hain" likhna: yahi growth movement ka proof hai.

Plant hormones (phytohormones)

Chemicals that coordinate growth, development and responses to the environment.

- → **Auxin:** made at the **shoot tip**; makes cells grow longer. When light comes from one side, auxin diffuses to the shady side, cells there grow longer, and the shoot bends towards light.
- → **Gibberellins:** help the growth of the stem.
- → **Cytokinins:** promote **cell division**; present in plenty in fruits and seeds.
- → **Abscisic acid: inhibits growth**; causes wilting of leaves (a "stop" signal).

Yaad karo: plant hormones

"Aao Ghar Chalein, Abhi ruko!": Auxin, Gibberellin, Cytokinin = growth badhao; Abscisic acid = ruk jao!

Paheli-Bujho



Bujho: Paheli, mere kamre ka money plant khidki ki taraf jhuk gaya. Kya use bahar jaana hai?

Paheli: Nahi Bujho, ye **phototropism** hai. Light ek side se aati hai toh **auxin** chhaon wali side chala jaata hai.

Bujho: Phir chhaon wali side kya karti hai?

Paheli: Wahan ke cells zyada lambe hote hain, isliye stem roshni ki taraf mud jaata hai. Nerve nahi, sirf hormone ka khel!

3. Hormones in Animals

Hormones: chemical messengers secreted by **endocrine (ductless) glands** directly into the blood, which carries them to target organs where they act.

Adrenaline (fight or flight): when an animal is scared, the **adrenal glands** release adrenaline into the blood. The heart beats faster to supply more oxygen to muscles; blood to the digestive system and skin is reduced and diverted to skeletal muscles; diaphragm and rib muscles contract, so breathing becomes faster. The body is ready to **fight or run**.

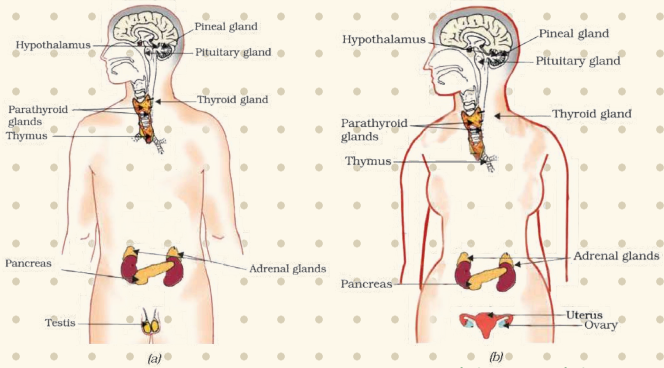


Fig. 6.7 Endocrine glands in human beings: (a) male (b) female

Activity 6.3: Endocrine glands

Done: Study Fig. 6.7, identify the endocrine glands and read about them. **Observed:** Pineal, hypothalamus and pituitary in the brain; thyroid and parathyroid in the neck; thymus in the chest; pancreas and adrenals in the abdomen; testes (males), ovaries (females). **Conclusion:** endocrine glands are spread all over the body and each secretes **specific hormones**.

Thyroxine and iodine: the thyroid needs **iodine** to make thyroxine, which regulates carbohydrate, protein and fat metabolism for balanced growth. Lack of iodine causes goitre (swollen neck); hence **iodised salt** is advised.

Growth hormone: from the **pituitary**; regulates growth and development. Too little in childhood causes **dwarfism**, too much causes **gigantism**. When its level is low, the **hypothalamus** releases a growth hormone releasing factor, which makes the pituitary release growth hormone.

Sex hormones: at puberty, **testosterone** (males) and **oestrogen** (females) bring the changes of the body. **Insulin:** made by the **pancreas**; regulates blood sugar. Too little insulin raises blood sugar and causes **diabetes**; such patients are given insulin injections.

Other glands: pineal (melatonin, sleep-wake cycle), parathyroid (parathormone, blood calcium), thymus (thymosin, immunity in children).

Hormone	Gland	Function
Growth hormone	pituitary	growth of all organs
Thyroxine	thyroid	metabolism for growth
Insulin	pancreas	blood sugar level
Testosterone	testes	male organs, beard, deep voice
Oestrogen	ovaries	female organs, menstrual cycle
Adrenaline	adrenal	prepares body for emergency
Releasing hormones	hypothalamus	make pituitary release hormones

Paheli-Bujho



Bujho: Paheli, exam hall mein mera dil dhak-dhak kyun karta hai? Kya main bimar hoon?

Paheli: Bimar nahi! Darr ke time **adrenal gland** adrenaline chhodta hai: dil tez, saans tez, body ready for fight or flight.

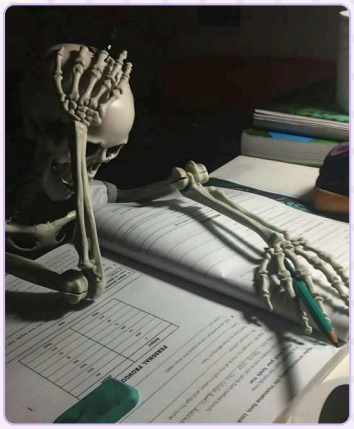
Bujho: Toh main exam se bhaag sakta hoon?

Paheli: Bilkul nahi! Bas padh ke jaao, darr apne aap kam ho jayega.

Hormone problem	Disorder
thyroxine low (lack of iodine)	goitre: swollen neck
growth hormone	less: dwarfism; more: gigantism
insulin low	diabetes (high blood sugar)

Feedback mechanism: the timing and amount of hormone released are regulated by feedback. Example: when blood sugar rises, it is detected by cells of the pancreas, which secrete **more insulin**; when it falls, less insulin is secreted.

Meme samjho



Raat ke 2 baje, hormones ki table ratte-ratte: "bas ek baar aur padh leta hoon..."

Asli baat: gland + hormone + disorder ke 3 pairs pakke yaad karo: iodine kam = goitre, insulin kam = diabetes, growth hormone kam = dwarfism.

Nervous system	Endocrine system
electrical impulses (chemicals at synapses)	chemicals (hormones)
travel along nerve fibres	travel through blood
very fast, short-lived	slow, long-lasting
reach only connected cells	reach all cells; act on targets

Exam Alert

- → **Iodine kam = goitre; insulin kam = diabetes; growth hormone kam = dwarfism.** Teeno board mein pakke!
- → Endocrine glands ko **ductless** kehte hain: hormone seedha blood mein jaata hai.
- → Hormone ka question ho toh **gland ka naam** bhi likho: aadhe marks wahi ke hote hain.

Quick fact

The **pituitary** is often called the master gland because its hormones control several other glands, but it works under orders from the **hypothalamus**.

1 minute recap

Stimulus → Receptor → Neuron + synapse → Reflex arc / brain → Plant tropisms + auxin → Hormones + feedback

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM Board fav Which of the following is a plant hormone?
(a) Insulin (b) Thyroxin
(c) Oestrogen (d) Cytokinin
Ans: (d) Cytokinin; the other three are animal hormones.

Q2. IM Board fav The gap between two neurons is called a:
(a) dendrite (b) axon
(c) synapse (d) impulse
Ans: (c) synapse.

Q3. IM The brain is responsible for:
(a) thinking (b) regulating heartbeat
(c) balancing the body (d) all of these
Ans: (d) all of these.

Q4. IM Which part of the brain maintains posture and balance of the body?
(a) cerebrum (b) cerebellum
(c) medulla (d) pons
Ans: (b) Cerebellum (hind-brain).

Q5. IM Name the hormone which helps a plant shoot bend towards light.
Ans: Auxin; it makes the cells on the shaded side grow longer.

Q6. IM Which gland secretes growth hormone, and what does its deficiency cause in childhood?
Ans: Pituitary gland; dwarfism.

Q7. IM **Assertion (A):** A person with a blocked nose cannot taste food properly. **Reason (R):** Flavour is detected by both gustatory and olfactory receptors.
Ans: (a) Both A and R are true and R is the correct explanation of A.

Q8. IM Board fav **Assertion (A):** Tendrils coil around a support. **Reason (R):** Cells of the tendril away from the support grow faster.
Ans: (a) Both A and R are true and R is the correct explanation of A.

Q9. IM Which of these is an example of a nastic movement?
(a) shoot bending to light (b) Mimosa leaves folding
(c) root growing down (d) pollen tube to ovule
Ans: (b) folding of the leaves of Mimosa on touch.

Q10. 1M Which endocrine gland is present in males but not in females?
(a) testes (b) pituitary
(c) thyroid (d) adrenal
Ans: (a) Testes.

Q11. 1M Name the receptors that detect (a) light (b) smell.
Ans: (a) Photoreceptors (eye) (b) olfactory receptors (nose).

Q12. 1M Which part of the hind-brain regulates breathing, and which controls blood pressure?
Ans: Pons regulates breathing; medulla controls blood pressure (also salivation and vomiting).

Short answer (2-3 marks)

Q13. 2M Board fav What is a synapse? How does a nerve impulse cross it?
Ans: A synapse is the gap between the nerve ending of one neuron and the dendrite of the next. The electrical impulse at the axon end releases chemicals (neurotransmitters such as acetylcholine), which diffuse across the gap and start a new electrical impulse in the dendrite of the next neuron.

Q14. 3M Board fav What is a reflex action? Why have reflex arcs evolved? Give two examples.
Ans: A sudden, automatic and quick response to a stimulus, without thinking, controlled by the spinal cord. Thinking by the brain is slow, so in emergencies a direct connection between sensory and motor nerves in the spinal cord saves time. Examples: pulling the hand from a flame, knee-jerk, sneezing.

Q15. 3M Board fav Name the three main parts of the brain and give one function of each.
Ans: Fore-brain: thinking, voluntary actions, centre for hunger. Mid-brain: reflex movements of head, neck and eyes. Hind-brain: medulla controls blood pressure and vomiting, pons breathing, cerebellum balance and posture.

Q16. 2M How are the brain and the spinal cord protected?
Ans: The brain is enclosed in the bony skull (cranium) and surrounded by cerebrospinal fluid that absorbs shocks. The spinal cord is enclosed in the vertebral column (backbone).

Q17. 2M What are the limitations of using electrical impulses for communication in the body?
Ans: (i) They reach only those cells connected by nervous tissue, not every cell. (ii) After generating an impulse a cell needs time to reset, so it cannot send impulses continuously. So chemical communication (hormones) is also used.

Q18. 3M Board fav How does the sensitive plant (Mimosa) respond to touch? How is it different from the movement of a shoot towards light?
Ans: On touch, electrical-chemical signals pass from cell to cell; cells lose water and shrink, so the leaves fold and droop. It is a fast, non-directional, non-growth (nastic) movement, whereas a shoot bending towards light is a slow, directional growth movement (tropism).

Q19. 2M Name the plant hormones: (a) promotes cell division (b) inhibits growth and wilts leaves (c) helps stem growth.
Ans: (a) Cytokinin (b) abscisic acid (c) gibberellin.

Q20. 2M Board fav Why is the use of iodised salt advised? Which disorder is caused by its deficiency?
Ans: Iodine is needed by the thyroid gland to make thyroxine, which regulates carbohydrate, protein and fat metabolism for balanced growth. Its deficiency causes goitre (swollen neck).

Q21. 2M How does the feedback mechanism regulate insulin secretion?
Ans: When blood sugar rises, cells of the pancreas detect it and secrete more insulin, which lowers sugar. When sugar falls, insulin secretion is reduced. So the amount and timing of the hormone are controlled automatically.

Q22. 3M Give three differences between nervous and hormonal coordination.
Ans: Nervous: electrical impulses, travel along nerves, very fast and short-lived, reach only connected cells. Hormonal: chemicals, travel through blood, slow but long-lasting, reach all cells and act on target organs.

Q23. 3M Why is a reflex arc formed in the spinal cord and not in the brain? Which signals are lost in a spinal cord injury?
Ans: The brain is far away and thinking takes time; a connection in the spinal cord gives a quick response in an emergency. In a spinal cord injury, sensory signals from the body to the brain and motor signals from the brain to muscles below the injury are disrupted.

Q24. 2M Describe an experiment to demonstrate hydrotropism.
Ans: Grow a seedling in dry soil in a tray and bury a porous clay pot filled with water on one side. Keep it for a few days and then dig out the seedling gently. The roots are seen to bend towards the pot, showing that roots grow towards water (positive hydrotropism).

Q25. 2M How do muscle cells move when a nerve impulse reaches them?
Ans: Muscle cells contain special proteins. When an impulse arrives, these proteins change their shape and arrangement, so the muscle cell becomes shorter (contracts) and the body part moves.

Q26. 3M Name the gland and hormone: (a) regulates blood calcium (b) controls the sleep-wake cycle (c) prepares the body for emergencies.
Ans: (a) Parathyroid: parathormone (b) pineal: melatonin (c) adrenal: adrenaline.

Q27. 2M Give two functions of the hypothalamus.
Ans: It controls body temperature, hunger and thirst, and releases hormones (releasing factors) that make the pituitary secrete its hormones.

Long answer (5 marks)

Q28. 5M Board fav (a) Draw the structure of a neuron and label dendrite, cell body, axon and nerve ending. (b) Describe the path of a nerve impulse. (c) What is a neuromuscular junction?

Ans: (a) Diagram as Fig. 6.1 (a). (b) Dendrites receive the stimulus and set off an electrical impulse; it travels to the cell body, along the axon to the nerve ending, where chemicals are released into the synapse; they start a new impulse in the next neuron. (c) The junction between the nerve ending of a motor neuron and a muscle fibre; chemicals released there make the muscle contract.

Q29. 5M Board fav (a) Draw a labelled reflex arc. (b) Trace the path of a reflex action when you touch a hot object. (c) Why does the response happen before the brain feels the pain?

Ans: (a) Diagram as Fig. 6.2: receptor in skin, sensory neuron, relay neuron in spinal cord, motor neuron, effector muscle. (b) Heat receptors in skin ⟶ sensory neuron ⟶ relay neuron in spinal cord ⟶ motor neuron ⟶ arm muscle contracts and the hand is pulled away. (c) The spinal cord gives the response directly; the information reaches the brain later, so pain is felt after the hand moves.

Q30. 5M Board fav (a) Describe an activity to show that a shoot bends towards light and a root away from it. (b) Explain the role of auxin. (c) Name the movement of a root towards water and of a pollen tube towards the ovule.

Ans: (a) Activity 6.2: germinated bean seeds on a wire mesh over a flask in a box open towards light; shoots bend to light, roots away; on turning the flask the new parts bend again. (b) Auxin, made at the shoot tip, diffuses to the shaded side; cells there grow longer, so the shoot bends towards light. (c) Hydrotropism; chemotropism.

Q31. 5M Board fav (a) What are hormones? Why are endocrine glands called ductless? (b) Name the gland, hormone and function: (i) pancreas (ii) thyroid (iii) testes (iv) ovaries. (c) What happens if insulin is not secreted in the proper amount?

Ans: (a) Hormones are chemical messengers secreted by endocrine glands directly into the blood, which carries them to target organs. These glands have no ducts; they release hormones straight into blood. (b) (i) Pancreas: insulin, regulates blood sugar (ii) thyroid: thyroxine, regulates carbohydrate, protein and fat metabolism (iii) testes: testosterone, male changes at puberty (iv) ovaries: oestrogen, female changes and menstrual cycle. (c) Blood sugar rises and the person suffers from diabetes; such patients are treated with insulin injections.

Case-based (4 marks)

Q32. 4M Board fav Rohan, aged 14, is much shorter than his classmates. The doctor found a low level of a hormone in his blood. (a) Name the hormone and the gland. (b) Name the disorder. (c) Which part of the brain controls the release of this hormone? (d) What would excess of it in childhood cause?

Ans: (a) Growth hormone; pituitary gland. (b) Dwarfism. (c) Hypothalamus (releases growth hormone releasing factor). (d) Gigantism.

Q33. 4M Meena saw a snake on her way home. Her heart started beating faster and she breathed rapidly. (a) Name the hormone and gland. (b) Why does the heart beat faster? (c) Where is blood diverted? (d) Why is it called a fight or flight hormone?

Ans: (a) Adrenaline; adrenal glands. (b) To supply more oxygen to the muscles. (c) From the digestive system and skin to skeletal muscles. (d) It prepares the body to fight or run away in an emergency.

Q34. 4M A farmer noticed that the roots of plants near a leaking water pipe had grown towards the pipe, while the shoots of plants near a wall had bent towards an open gap in it. (a) Name the movement of roots towards water. (b) Name the movement of shoots towards light. (c) Which hormone is responsible for the shoot bending? (d) Is this response fast or slow, and why?

Ans: (a) Hydrotropism (positive). (b) Phototropism (positive). (c) Auxin, which collects on the shaded side and makes those cells grow longer. (d) Slow: it is a growth movement, which takes hours or days.

More board favourites

Q35. 2M Board fav Why does a nerve impulse travel only in one direction across a synapse?

Ans: The chemical (neurotransmitter) is released only at the axon ending of one neuron, and the receptors for it are present only on the dendrites of the next neuron. So the signal can pass only from axon to dendrite.

Q36. 3M Board fav How do plant cells communicate without nerves? Why are plant responses usually slow?

Ans: Plants have no nervous or muscular tissue. Their cells use electrical-chemical signals and plant hormones; a response often needs growth (cells growing longer on one side), which takes hours or days.

Q37. 2M Board fav How is an involuntary action different from a reflex action? Give one example of each.

Ans: Involuntary actions (heartbeat, breathing, salivation) go on all the time, controlled by the mid-brain and hind-brain. A reflex action is a sudden response to a stimulus through a reflex arc, mostly in the spinal cord, e.g. pulling the hand away from a flame.

Exam Alert: diagrams to practise

Neuron, reflex arc, human brain, neuromuscular junction aur phototropism set-up: ye 5 diagrams roz ek baar banao. Board mein 3 marks pakke! Diagram pencil se banao, labels ek taraf seedhi lines se, aur heading neeche likho.

How do Organisms Reproduce?



Most asked: modes of asexual reproduction with examples, L.S. of flower + fertilisation, human reproductive systems, contraception

Every kind of organism we see around us exists because its members **reproduce**. Reproduction is not needed to keep one organism alive, but it keeps the **species** going.

1. Do Organisms Create Exact Copies of Themselves?

Reproduction: the process by which living organisms produce new individuals of their own kind. It is essential for the continuity of the species, not for the survival of an individual. Organisms reproduce in two ways: **asexual** and **sexual**.

The **DNA** in the chromosomes of the nucleus carries information for making proteins: the blueprint of body design. The basic event in reproduction is DNA copying (replication). The cell makes two copies of its DNA, builds extra cellular apparatus and divides into two cells, each with one copy.

DNA copying is **never absolutely accurate**. These small errors give rise to variations, so daughter cells are similar but not identical to the parent.

Importance of variation

- → Organisms occupy well-defined places (**niches**) in an ecosystem using their ability to reproduce.
- → If a niche changes drastically, a population may be wiped out; individuals with **useful variations** survive.
- → Example: if water temperature rises suddenly, most bacteria die, but a few **heat-resistant variants** survive and grow in number.
- → Variation is the basis of **evolution**; consistent DNA copying keeps the body design suited to the niche, so reproduction keeps populations **stable**.
- → Variation is useful for the **survival of the species** over time, not necessarily for the individual.

Quick fact

DNA = deoxyribonucleic acid, found in the **chromosomes** of the nucleus. Even a small change in it can change the body design slightly.

Paheli-Bujho



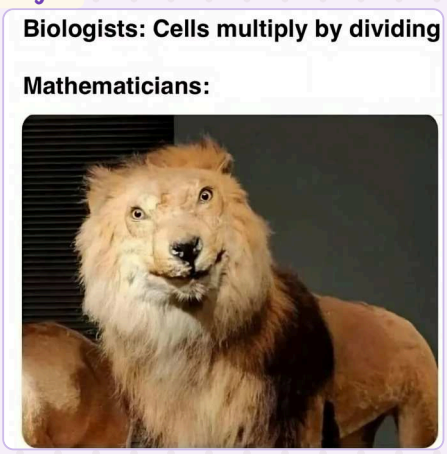
Bujho: Paheli, DNA ki photocopy bilkul same hoti hai na?

Paheli: Nahi! Copying mein chhoti galtiyaan hoti hain, isliye variation aata hai.

Bujho: Matlab galti achhi hai?

Paheli: Thodi si haan! Isi se species bachti hai.

Meme samjho



Biology mein cell **divide** hoke **multiply** hoti hai!

Asli baat: DNA copy karke cell do mein divide hoti hai: reproduction ka basic event.

2. Modes of Reproduction Used by Single Organisms

Asexual reproduction: a new individual is produced from a single parent without fusion of gametes; offspring are almost exact copies of the parent.

Asexual reproduction	Sexual reproduction
one parent	two parents (male and female)
no gametes formed	male and female gametes fuse
offspring nearly identical	offspring show variations
fast; simple organisms	slower; complex organisms
fission, budding, spores	flowering plants, humans

1. Fission (unicellular organisms)

Binary fission: the cell divides into **two equal halves**. *Amoeba* can split in any plane; *Leishmania* (causes **kala-azar**) has a whip-like structure at one end, so it splits lengthwise, in a definite plane only.

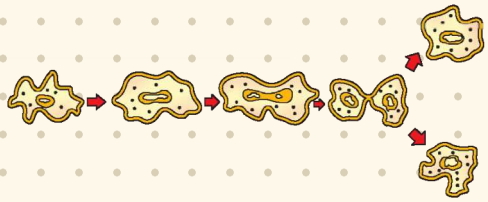


Fig. 7.1(a) Binary fission in Amoeba

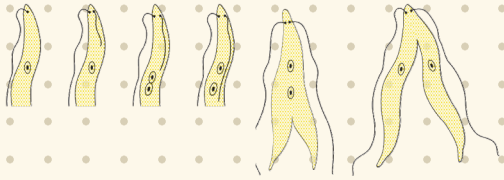


Fig. 7.1(b) Binary fission in Leishmania

Multiple fission: the cell divides into **many daughter cells** at once, as in the malarial parasite *Plasmodium*.

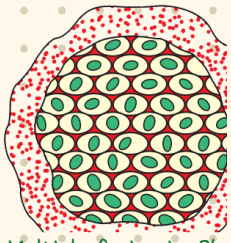


Fig. 7.2 Multiple fission in Plasmodium

Activity 7.3: Binary fission in Amoeba

Done: Observe a permanent slide of *Amoeba*, then a slide showing binary fission, and compare. **Observed:** The nucleus divides first, then the cytoplasm pinches into two. **Conclusion:** One parent gives two daughter cells.

Activity 7.1: Budding in yeast

Done: Dissolve 10 g sugar in 100 mL water, add a pinch of yeast to 20 mL of it, plug and keep warm; after 1–2 hours observe a drop under a microscope. **Observed:** Yeast cells show small outgrowths (**buds**), some forming chains. **Conclusion:** Yeast reproduces asexually by budding.

2. Fragmentation

A simple multicellular organism breaks into pieces on maturing and each fragment grows into a new individual, e.g. *Spirogyra*.

Activity 7.4: Fragmentation in Spirogyra

Done: Place one or two dark-green filaments from a pond on a slide in a drop of glycerine; cover and observe under a microscope. **Observed:** All cells of the filament look alike; no different tissues. **Conclusion:** Any piece can grow into a new filament, so *Spirogyra* reproduces by fragmentation.

Most multicellular organisms **cannot** simply divide cell by cell, because their cells are organised into tissues and organs placed at definite positions. In them, reproduction is the job of a specific cell type.

3. Regeneration

The ability of a **fully differentiated** organism to give rise to new individuals from its body parts. If *Planaria* or *Hydra* is cut into pieces, each piece grows into a complete organism. Specialised cells multiply into a mass of cells, which changes into various cell types and tissues.

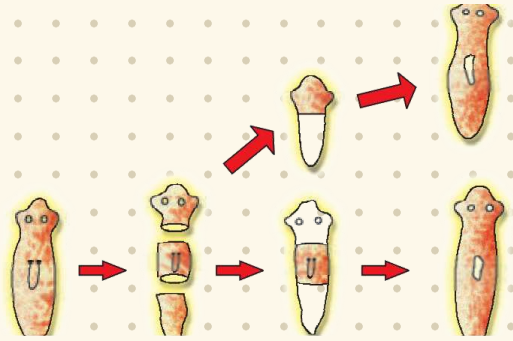


Fig. 7.3 Regeneration in Planaria

Regeneration is not the same as reproduction: most organisms do not normally depend on being cut into pieces.

4. Budding

A small **outgrowth (bud)** forms on the parent due to repeated cell division at one specific site. It grows into a tiny individual and detaches when mature, as in *Hydra* and yeast.

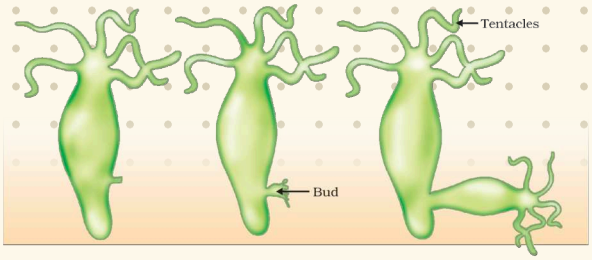


Fig. 7.4 Budding in Hydra

Quick fact

Hydra is special: it reproduces by **budding** and can also **regenerate** a whole body from a small piece.

Yaad karo: kaun kya karta hai

Amoeba aur Leishmania = binary fission; Plasmodium = multiple; Spirogyra = fragmentation; Planaria = regeneration; Hydra aur yeast = budding; Rhizopus = spores. Hydra regeneration bhi karta hai!

5. Vegetative propagation

New plants grow from vegetative parts (root, stem, leaf) under suitable conditions.

Part	Examples
Root	sweet potato, dahlia
Stem	potato (eyes), ginger, onion, money plant
Leaf	*Bryophyllum* (buds on leaf margin)
Artificial	layering, grafting: sugarcane, rose, grapes

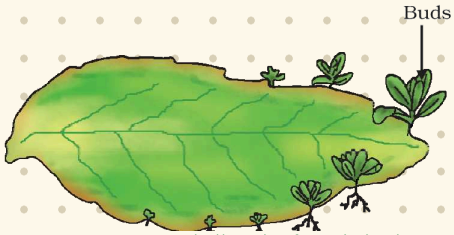


Fig. 7.5 Bryophyllum leaf with buds

Advantages: plants bear flowers and fruits **earlier** than those from seeds; plants that have lost the capacity to produce seeds (banana, orange, rose, jasmine) can be grown; all new plants are **genetically similar** to the parent.

Activity 7.5: Potato pieces

Done: Place potato pieces, some with buds (eyes) and some without, on wet cotton for a few days. **Observed:** Only pieces with buds give green shoots and roots. **Conclusion:** New plants arise from buds on the stem (tuber).

Activity 7.6: Money plant

Done: Dip cuttings with at least one leaf and cuttings from between two leaves in water. **Observed:** Only cuttings with a leaf (node) grow fresh leaves. **Conclusion:** Buds at the nodes give rise to new plants.

Tissue culture: cells from the **growing tip** of a plant are placed in an artificial medium, where they divide rapidly into a mass of

cells called callus. The callus is moved to a medium with hormones for growth and differentiation, and the plantlets are planted in soil. Many **disease-free** plants grow from one parent in a short time; widely used for ornamental plants.

Paheli-Bujho



Bujho: Kele mein beej kahan gaye?
Paheli: Kela beej nahi banata, isliye vegetative propagation se ugta hai.
Bujho: Aur aloo ki aankhen?
Paheli: Wo buds hain!

6. Spore formation

The bread mould *Rhizopus* has thread-like **hyphae** (not reproductive) and tiny blob-on-a-stick structures called sporangia, which contain **spores**. Spores have thick walls that protect them until they reach a moist surface and grow.

Activity 7.2: Bread mould

Done: Wet a slice of bread and keep it in a cool, moist, dark place; observe its surface with a magnifying glass for a week. **Observed:** White cotton-like growth spreads; black dots (sporangia) on fine threads. **Conclusion:** *Rhizopus* reproduces by spore formation.

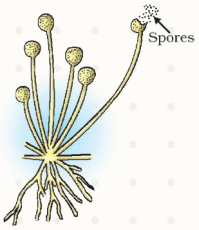


Fig. 7.6 *Rhizopus*

Exam Alert

- → "Example do" wale 1 mark questions: Amoeba, Leishmania, Plasmodium, Spirogyra, Planaria, Hydra, Rhizopus, Bryophyllum: har ek ka mode yaad rakho.
- → Regeneration ko reproduction mat likhna: ye dono alag processes hain.
- → Tissue culture ka keyword: **callus**; spores ka keyword: **thick protective wall.**

3. Sexual Reproduction

Why sexual reproduction? Combining DNA from **two individuals** creates new combinations, so variations arise much faster than by copying errors alone.

If each parent gave a full set of DNA, the amount would **double every generation**. So germ cells (gametes) are made by a special division called meiosis, with only **half** the chromosomes and DNA; fusion of two gametes restores the normal amount.

One gamete is **small and motile** (male); the other is **large, non-motile** and stores food (female egg).

Sexual reproduction in flowering plants

- → **Sepals** (green, protect the bud; together the **calyx**)
- → **Petals** (coloured, attract insects; together the **corolla**)
- → **Stamen** (male): filament + anther; anther makes **pollen**
- → **Pistil/carpel** (female): sticky **stigma**, long **style**, swollen **ovary** with ovules, each with an egg

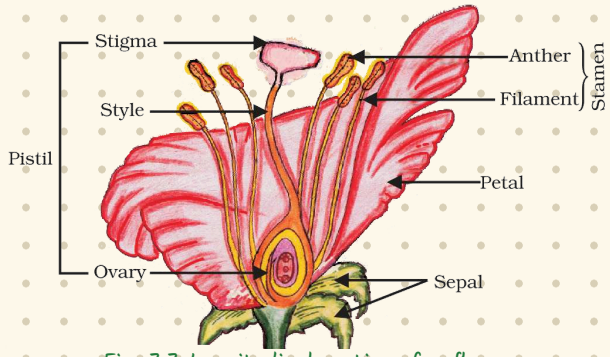
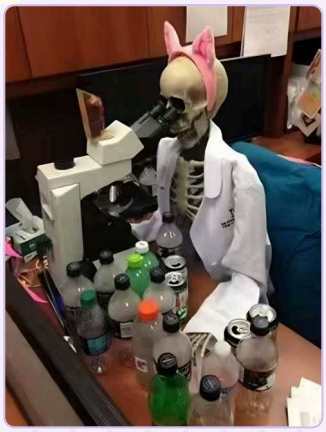


Fig. 7.7 Longitudinal section of a flower

Meme samjho



Asexual variation ke intezaar mein skeleton ban gaye!
Asli baat: sexual reproduction mein do parents ka DNA milta hai, variation tezi se aata hai.

Unisexual flower	Bisexual flower
either stamens or pistil	both stamens and pistil
papaya, watermelon	*Hibiscus*, mustard

Pollination: transfer of pollen from the **anther to the stigma**. Self-pollination: within the same flower; cross-pollination: to another flower, by wind, water or animals.

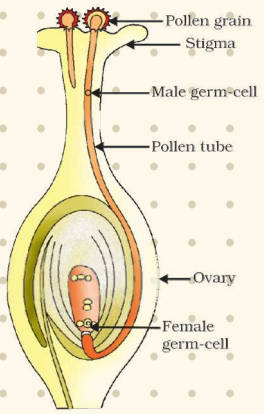


Fig. 7.8 Germination of pollen on stigma

Fertilisation: a pollen tube grows from the pollen grain through the style to the ovule; the male germ cell fuses with the egg to form the **zygote**.

After fertilisation	Becomes
zygote	embryo
ovule	seed
ovary	fruit
petals, sepals, stamens, style, stigma	shrivel and fall off

Germination: the seed holds the future plant (**embryo**), which grows into a seedling in suitable conditions. Parts: **plumule** (future shoot), **radicle** (future root), **cotyledons** (food store).

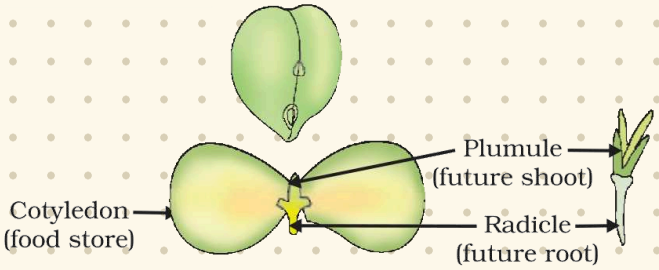


Fig. 7.9 Germination

Activity 7.7: Parts of a seed

Done: Soak gram (chana) seeds overnight, keep them under a wet cloth for a day, then cut them open. **Observed:** Each seed shows two cotyledons, a plumule and a radicle. **Conclusion:** The seed contains the embryo (future plant) and its stored food.

Exam Alert

- → Flower L.S. ka diagram 3-5 marks mein aata hai: stigma, style, ovary, ovule, anther, filament label zaroor karo.
- → Ovule ⟶ seed aur ovary ⟶ fruit: inko kabhi ulta mat likhna, marks katenge!
- → Fertilisation ke answer mein **pollen tube** ka naam zaroor likho, examiner yahi dhoondhta hai.

4. Reproduction in Human Beings

Puberty: the period of adolescence when reproductive tissues mature and sex hormones are made. Common changes: thick hair in armpits and genital area, oily skin and pimples, awareness of our own and others' bodies. Girls: breasts enlarge, **menstruation** begins. Boys: facial hair appears, voice begins to crack.

Reproductive tissues mature only as general body growth slows down, because till then the body's resources go into growth. Sexual maturity does **not** mean the body or mind is ready for sex or children, whatever the pressure from friends, family or society. Changes like new hair growth are signals of sexual maturity.

Sperm (male gamete)	Egg (female gamete)
made in testes	made in ovaries
tiny, with a long tail	large, no tail
motile (swims)	non-motile
millions released	one matures per month
no stored food	stores food

Quick fact

Puberty changes are brought about by **sex hormones**: testosterone in boys and oestrogen in girls. In girls they usually begin a little earlier than in boys.

Male reproductive system

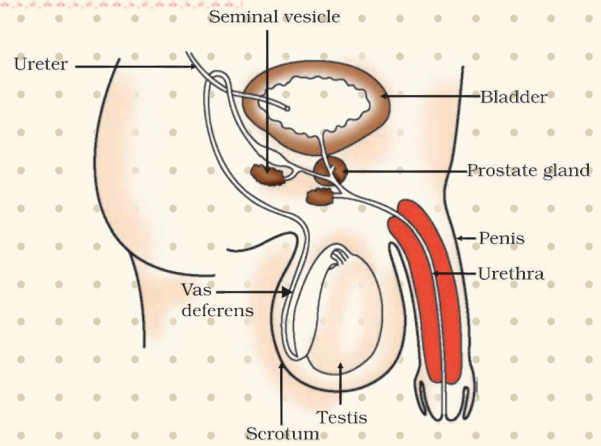


Fig. 7.10 Human male reproductive system

- → **Testes** make **sperms** and the hormone **testosterone**.
- → Testes lie in the scrotum outside the abdomen, because sperm formation needs a **lower temperature** than the normal temperature inside the abdomen.
- → Sperms travel through the **vas deferens**, which joins a tube from the bladder; the **urethra** is the common passage for sperms and urine.
- → **Seminal vesicles** and the **prostate gland** add secretions that nourish sperms and make their transport easy.
- → Each sperm is a tiny cell with a **long tail** that helps it swim towards the egg in the oviduct.

Female reproductive system

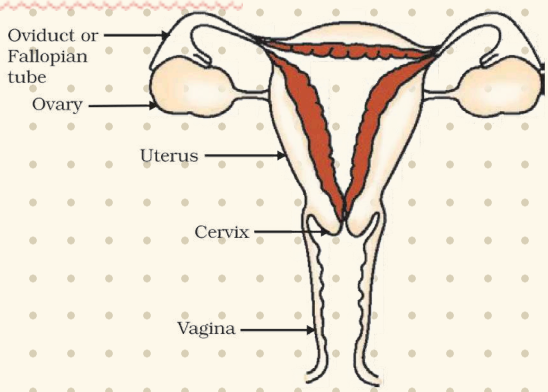


Fig. 7.11 Human female reproductive system

- → **Ovaries** make **eggs**, **oestrogen** and **progesterone**.
- → A girl is born with thousands of immature eggs; from puberty one egg matures each month.
- → The egg travels through the thin **oviduct (fallopian tube)** to the elastic, bag-like **uterus**, which opens into the **vagina** through the **cervix**.

Fertilisation: sperms enter through the vagina and reach the oviduct, where one sperm fuses with the egg. The zygote divides into an **embryo**, which is implanted in the thick, soft lining of the uterus.

Placenta: a **disc-like tissue** embedded in the uterine wall. On the embryo's side it has finger-like villi with a large surface area for glucose and oxygen to pass from mother's blood to the embryo; wastes pass back the same way. The baby develops in about **nine months** and is born by rhythmic contractions of the muscles of the uterus.

If the egg is not fertilised: the egg lives for about one day. The thick, spongy, blood-rich lining of the uterus slowly breaks and comes out through the vagina as blood and mucus: menstruation, roughly every month, lasting about **2 to 8 days**.

Ovary (egg) → Oviduct (fertilisation) → Uterus (implantation) → Placenta (nutrition) → Birth (~9 months)

Paheli-Bujho



Bujho: Testes body ke bahar kyun hote hain?

Paheli: Sperm banne ke liye body se kam temperature chahiye.

Bujho: Aur baby ko khana kaise milta hai?

Paheli: Placenta ke villi se: glucose aur oxygen.

5. Reproductive Health

Sexually transmitted diseases (STDs): bacterial: **gonorrhoea, syphilis**; viral: **warts, HIV-AIDS**. Using a covering (condom) on the penis helps prevent many of them.

Pregnancy makes major demands on a woman's **body and mind**; if she is not ready, her health suffers. So contraception (preventing pregnancy) is used. The methods are compared in the table.

Method	How it works
Mechanical barrier	condom, cervical covering: sperms cannot reach the egg
Hormonal (oral pills)	change hormonal balance so eggs are not released; side effects possible
Loop / copper-T	placed in the uterus; may cause irritation
Surgical	vasectomy (vas deferens blocked), tubectomy (fallopian tube blocked); infection risk if not done properly

Surgery can also remove unwanted pregnancies, but its misuse for female foeticide is illegal. **Prenatal sex determination is prohibited by law**; a healthy society needs a balanced sex ratio.

Exam Alert

- Human diagrams mein **oviduct, uterus, cervix, vas deferens, urethra, seminal vesicle, prostate** labels sabse zyada poochhe jaate hain.
- "Why contraception?" ke answer mein woman ki health aur STD prevention dono likho.
- Female foeticide illegal hai: ye line case-based questions mein value point hai.

Key terms

- **Callus:** mass of undifferentiated cells in tissue culture.
- **Gametes:** germ cells with half the chromosomes.
- **Zygote:** cell formed when two gametes fuse.
- **Villi:** finger-like projections of the placenta.

1 minute recap

DNA copying → Variation → Asexual modes → Pollination → Fertilisation → Human systems → Reproductive health

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM Board fav Multiple fission is seen in:
(a) *Amoeba* (b) *Leishmania* (c) *Plasmodium* (d) yeast
Ans: (c) *Plasmodium*.

Q2. IM In a flower, the part that develops into the fruit after fertilisation is the:
(a) ovule (b) ovary (c) stigma (d) anther
Ans: (b) ovary.

Q3. IM Board fav Which of these is a sexually transmitted disease caused by a virus?
(a) gonorrhoea (b) syphilis (c) malaria (d) HIV-AIDS
Ans: (d) HIV-AIDS (gonorrhoea and syphilis are bacterial).

Q4. IM Name the structure through which the embryo gets nourishment from the mother's blood.
Ans: Placenta (through its villi).

Q5. IM Board fav Where does fertilisation take place in human females?
Ans: In the oviduct (fallopian tube).

Q6. IM Board fav **Assertion (A):** Banana is grown by vegetative propagation. **Reason (R):** Banana plants have lost the capacity to produce seeds.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q7. IM Name the mass of cells formed in tissue culture.
Ans: Callus.

Q8. IM Which hormone is produced by the testes?
Ans: Testosterone.

Q9. IM Fill in the blanks: (a) *Bryophyllum* is propagated by _____. (b) *Leishmania* causes _____.
Ans: (a) buds on the leaf margin. (b) kala-azar.

Q10. IM Board fav **Assertion (A):** Gametes have half the number of chromosomes. **Reason (R):** Fusion of two gametes restores the normal number of chromosomes in the zygote.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q11. 1M Which of these is the correct pair?
(a) Vasectomy: fallopian tube blocked
(b) Oral pills: barrier method
(c) Copper-T: placed in the uterus
(d) Condom: hormonal method
Ans: (c) Copper-T: placed in the uterus.

Q12. 1M Name the parts of a seed that develop into the future shoot and the future root.
Ans: Plumule: future shoot; radicle: future root.

Q13. 1M Board fav A human body cell has 46 chromosomes. How many chromosomes does a human egg or sperm have?
Ans: 23 (gametes have half the number, made by meiosis).

Q14. 1M Name the hormones produced by the human ovaries.
Ans: Oestrogen and progesterone.

Short answer (2-3 marks)

Q15. 3M Board fav Why is DNA copying an essential part of reproduction? Why is variation useful for a species?
Ans: DNA carries the information for making proteins (the body design), so each new cell must get a copy of it. Copying is not perfectly accurate, which creates variations; if the niche changes (e.g. water heats up), individuals with useful variations (heat-resistant bacteria) survive, so the species is not wiped out.

Q16. 2M Board fav How does binary fission differ from multiple fission? Give one example of each.
Ans: Binary fission: one cell divides into two equal daughter cells (*Amoeba*, *Leishmania*). Multiple fission: one cell divides into many daughter cells at the same time (*Plasmodium*).

Q17. 3M Why can *Spirogyra* reproduce by fragmentation while complex multicellular organisms cannot?
Ans: *Spirogyra* has a simple body whose cells are all alike, so any fragment can grow into a new individual. In complex organisms cells are organised into tissues and organs at definite positions, so a piece cannot rebuild the whole body; reproduction needs specialised cells.

Q18. 2M What is regeneration? Why is it not reproduction?
Ans: Regeneration is the ability of a fully differentiated organism to form a whole new individual from its body parts, e.g. *Planaria*, *Hydra*. It is not reproduction, because organisms do not normally depend on being cut into pieces.

Q19. 3M Board fav Give two advantages of vegetative propagation. Why is it used for banana and rose?
Ans: Plants bear flowers and fruits earlier; all plants are genetically similar to the parent (desirable traits kept). Banana, rose, orange and jasmine cannot produce seeds, so vegetative propagation is the only way to grow them.

Q20. 3M Board fav Distinguish between self- and cross-pollination. Name what the ovule and ovary become.
Ans: Self-pollination: pollen reaches the stigma of the same flower. Cross-pollination: pollen reaches the stigma of another flower, by wind, water or animals. After fertilisation the ovule becomes the seed and the ovary becomes the fruit.

Q21. 3M Board fav Why are testes outside the abdomen? State the role of seminal vesicles and prostate gland.
Ans: Sperm formation needs a temperature lower than body temperature, so testes lie in the scrotum. Seminal vesicles and prostate add secretions that make a fluid which nourishes sperms and makes their transport easier.

Q22. 2M What happens when the egg is not fertilised?
Ans: The egg lives for about one day. The thick, blood-rich lining of the uterus is not needed, so it breaks down and comes out through the vagina as blood and mucus: menstruation (every month, 2-8 days).

Q23. 3M Board fav List four methods of contraception. Why is prenatal sex determination banned?
Ans: Condoms (barrier), oral pills (hormonal), copper-T or loop, surgery (vasectomy/tubectomy). It is banned because it is misused for female foeticide, which reduces the female sex ratio; a healthy society needs a balanced sex ratio.

Q24. 3M What is tissue culture? Give two advantages.
Ans: Cells from the growing tip of a plant are grown in an artificial nutrient medium to form a callus, which is moved to a hormone medium to form plantlets that are planted in soil. Advantages: many plants from one parent in a short time; plants are disease-free; widely used for ornamental plants.

Q25. 2M Board fav Why do gametes have half the number of chromosomes? Name the process that makes them.
Ans: If each gamete had the full set, the zygote would have double the DNA and the amount would keep doubling every generation. Meiosis makes gametes with half the chromosomes, so fusion restores the normal number and keeps the DNA of the species stable.

Q26. 2M Why is sexual reproduction more advantageous than asexual reproduction for a species?
Ans: It combines DNA from two parents, so new combinations and variations arise much faster. Variations help some individuals survive when the environment changes, and they are the basis of evolution.

Q27. 2M Differentiate between pollination and fertilisation.
Ans: Pollination is the transfer of pollen grains from the anther to the stigma. Fertilisation is the fusion of the male germ cell (carried by the pollen tube) with the egg in the ovule to form a zygote. Pollination comes first.

Q28. 2M Distinguish between unisexual and bisexual flowers with two examples of each.
Ans: Unisexual flowers have either stamens or pistil (papaya, watermelon). Bisexual flowers have both stamens and pistil (*Hibiscus*, mustard).

Long answer (5 marks)

Q29. 5M Board fav (a) Describe any five modes of asexual reproduction with one example each. (b) Why is regeneration not counted as a mode of reproduction?

Ans: Binary fission: cell splits into two (*Amoeba*). Multiple fission: many daughter cells at once (*Plasmodium*). Fragmentation: body breaks into pieces, each grows (*Spirogyra*). Budding: outgrowth from repeated cell division at one site detaches when mature (*Hydra*, yeast). Spore formation: sporangia release thick-walled spores (*Rhizopus*). Vegetative propagation: new plants from root/stem/leaf (potato, *Bryophyllum*). All of these need only one parent and give offspring almost identical to it. (b) Organisms do not normally depend on being cut up to reproduce, so regeneration is not a mode of reproduction.

Q30. 5M Board fav (a) Draw a labelled diagram of the longitudinal section of a flower. (b) Describe pollination and fertilisation. (c) What changes follow fertilisation?

Ans: (a) Label: sepal, petal, stamen (anther, filament), pistil (stigma, style, ovary), ovule. (b) Pollination is transfer of pollen from anther to stigma (self or cross). On the stigma the pollen grain grows a pollen tube through the style to the ovule; the male germ cell fuses with the egg to form the zygote: fertilisation. (c) Zygote forms the embryo, ovule becomes seed, ovary becomes fruit; petals, sepals, stamens, style and stigma shrivel and fall off. The seed later germinates: plumule forms the shoot, radicle the root, and cotyledons supply food.

Q31. 5M Board fav (a) Draw the human female reproductive system and label ovary, oviduct, uterus, cervix, vagina. (b) How does the embryo get nourishment? (c) How long does the baby take to develop and how is it born?

Ans: (a) Labelled diagram as in Fig. 7.11. (b) The embryo is implanted in the uterine wall; the placenta, a disc-like tissue, has villi that give a large surface area for glucose and oxygen to pass from the mother's blood to the embryo, and wastes pass back. (c) About nine months; the baby is born by rhythmic contractions of the muscles of the uterus. If the egg is not fertilised, the lining breaks down and comes out as menstruation.

Q32. 5M (a) Describe the male reproductive system with the role of each part. (b) List the changes at puberty. (c) Why are testes located in the scrotum?

Ans: (a) Testes make sperms and testosterone; scrotum keeps them at lower temperature; vas deferens carries sperms; urethra is the common passage for sperms and urine; seminal vesicles and prostate add nourishing fluid; each sperm has a long tail to swim to the egg. (b) Both: hair in armpits and genital area, oily skin, pimples. Girls: breasts enlarge, menstruation starts. Boys: facial hair appears and the voice begins to crack. (c) Sperm formation needs a temperature lower than body temperature.

Q33. 5M Board fav (a) What are sexually transmitted diseases? Name two bacterial and two viral STDs. (b) Describe three methods of contraception with one limitation of each. (c) Why is prenatal sex determination banned?

Ans: (a) Diseases spread through sexual contact. Bacterial: gonorrhoea, syphilis; viral: warts, HIV-AIDS. (b) Condom (mechanical barrier; also prevents many STDs, must be used every time); oral pills (hormonal; may have side effects); copper-T/loop in the uterus (may cause irritation); surgery: vasectomy or tubectomy (infection if not done properly). (c) They are misused to remove female foetuses, which disturbs the sex ratio; a healthy society needs a balanced sex ratio.

Case-based (4 marks)

Q34. 4M Board fav A student kept a moist slice of bread in a dark, warm place. After a few days a white cotton-like growth with black dots appeared. (a) Name the organism. (b) What are the black dots and what do they contain? (c) How do spores survive harsh conditions? (d) Name the mode of reproduction.

Ans: (a) *Rhizopus* (bread mould). (b) Sporangia; they contain spores. (c) Spores have thick protective walls until they reach a moist surface. (d) Spore formation (asexual).

Q35. 4M Board fav Riya's biology teacher showed a chart of a pregnant woman's uterus with a disc-like tissue joining the embryo to the uterine wall. (a) Name the tissue. (b) How is it suited for its function? (c) Name two substances that pass from mother to embryo. (d) How are its wastes removed?

Ans: (a) Placenta. (b) It has finger-like villi on the embryo's side that give a large surface area for exchange. (c) Glucose and oxygen. (d) Wastes pass from the embryo to the mother's blood through the placenta.

Q36. 4M A farmer wants many disease-free orchid plants quickly from one good plant, and also wants to grow more rose plants that have stopped forming seeds. (a) Which method should he use for the orchids? (b) Name the mass of cells formed in it. (c) How can roses be grown? (d) Give one advantage of plants grown this way.

Ans: (a) Tissue culture. (b) Callus. (c) By vegetative propagation, e.g. layering or grafting (stem cuttings). (d) They are genetically similar to the parent and flower and fruit earlier than plants from seeds.

Exam Alert: diagrams to practise

- → Fig. 7.4 budding in *Hydra*, Fig. 7.1 binary fission in *Amoeba*, Fig. 7.6 *Rhizopus*.
- → Fig. 7.7 L.S. of flower and Fig. 7.8 pollen tube on stigma.
- → Fig. 7.10 and 7.11 human reproductive systems: pencil se banao, labels seedhi line se, right side mein.
- → Asexual vs sexual ke 3 differences aur contraception ke 4 methods (har ek ki ek limitation ke saath): ye 3 marks pakke hain.
- → Placenta aur villi ka function case-based questions mein baar baar aata hai.
- → Puberty ka reason: body growth slow hone par hi reproductive tissues mature hote hain.

HEREDITY



Most asked: monohybrid cross with Punnett square (3:1, 1:2:1), dihybrid 9:3:3:1, Mendel's laws, sex determination in humans

Every child looks like its parents, yet no two children are exactly alike. How are features passed on from parents, and why do small differences appear in every generation?

1. Heredity and Variation

Heredity: the passing of traits (characteristics) from parents to offspring through genes. **Variation:** the differences between individuals of the same species.

Gene: a section of **DNA** that carries the information for making one protein; genes are the units of heredity.

Inherited traits	Acquired traits
come from parents through DNA	developed during life (environment, practice)
eye colour, earlobe type, blood group	a scar, learning to swim, gym muscles
passed on to offspring	not in the DNA, so not passed on

Paheli-Bujho



Bujho: Papa ke gym wale biceps mujhe milenge?
Paheli: Nahi! Wo acquired trait hai, DNA mein nahi jaata.
Bujho: Aur papa ki bhoori aankhen?
Paheli: Wo inherited hai, genes se aa sakti hain.

Accumulation of variation during reproduction

Each generation inherits the **basic body design** plus small changes from its parents, and adds new changes of its own, so differences pile up over generations.

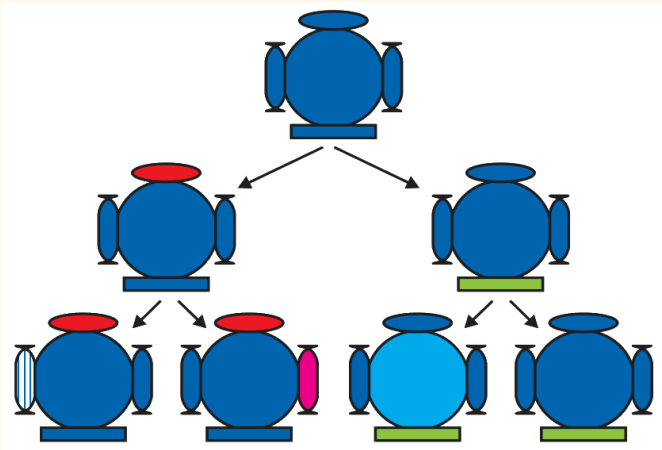


Fig. 8.1 Creation of diversity over succeeding generations

Reading Fig. 8.1: the original organism (top) gives two individuals of the second generation with the same body design but subtle differences. Each again gives two, so the four in the third generation differ from one another: some differences are **inherited**, others are **newly created**.

Fig. 8.1 shows a single individual reproducing, as in asexual reproduction: if one bacterium divides and the two daughters divide again, the four bacteria are very similar, with only minor differences from small inaccuracies in DNA copying.

Asexual reproduction	Sexual reproduction
offspring almost identical	offspring show more variation
only minor variation (DNA copying errors)	DNA from two parents mixes: much greater variation

Why variation matters: not all variants survive equally. Bacteria that can withstand heat survive a heat wave better. This selection of variants by the environment is the basis of **evolution**.

Quick fact

Trait A is in 10% and trait B in 60% of an asexually reproducing population. Trait B arose earlier: an older variation has had more generations to spread.

Activity 8.1: Earlobes in the class

Done: List classmates with free and attached earlobes, calculate the percentage of each, and compare every student with their parents. **Observed:** Both types occur; free earlobes are usually more common, and children often share the earlobe type of a parent. **Conclusion:** Earlobe type is an **inherited trait** controlled by genes from both parents (free earlobe behaves as dominant).



Fig. 8.2 (a) free (b) attached

2. Mendel and the Rules of Inheritance

Gregor Johann Mendel (1822–1884), the Father of Genetics, studied science and maths in Vienna and grew garden peas in his monastery. He was the first to **count** the offspring showing each trait in every generation.

Why pea plants? clear **contrasting characters**, a short life cycle, many seeds, and easy self- or cross-pollination.

Character	Dominant	Recessive
Seed shape	round	wrinkled
Seed colour	yellow	green
Flower colour	violet	white
Plant height	tall	short (dwarf)

Paheli-Bujho



Bujho: Mendel ne pea kyun liye?
Paheli: Clear traits, chhota life cycle, bahut seeds!
Bujho: F1 mein short kahan chhupa?
Paheli: Tt mein t hidden tha, F2 mein wapas aaya.

Monohybrid cross (one trait): tall × short

Parents: pure tall (TT) × pure short (tt). **F1:** all tall (Tt); no medium-height plants, so traits do not blend.

F2 (F1 selfed): tall and short appear again in the ratio 3 : 1. Shortness was present in F1 but hidden. Mendel concluded that every trait is controlled by **two copies of a factor** (now called genes), one from each parent.

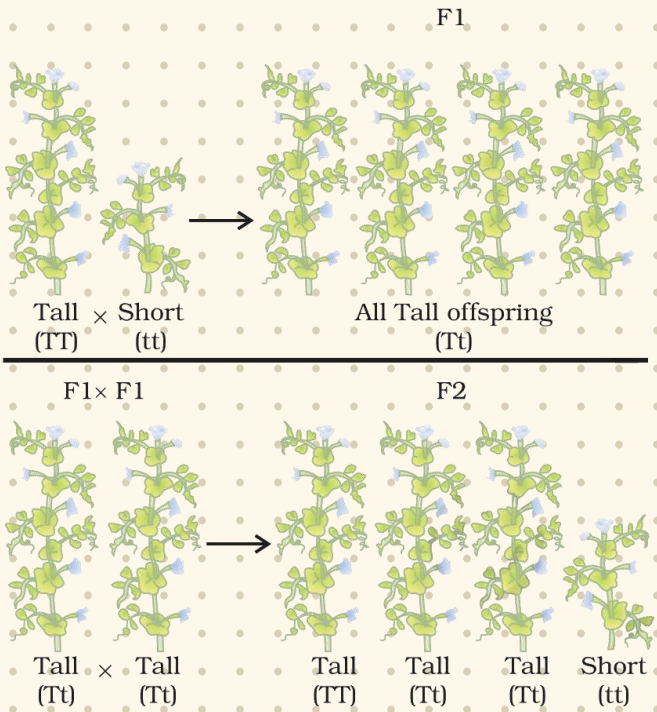


Fig. 8.3 Inheritance of a single trait (height) over two generations

Gametes	T	t
T	TT (tall)	Tt (tall)
t	Tt (tall)	tt (short)

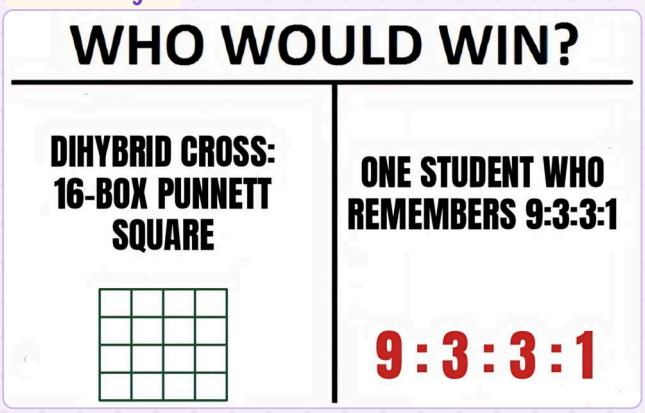
F2 ratios: phenotype 3 tall : 1 short; genotype 1 TT : 2 Tt : 1 tt.

Monohybrid cross: a cross between plants differing in **one pair** of contrasting traits. **Dominant trait:** expressed even when only one copy is present (T). **Recessive trait:** expressed only when both copies are recessive (tt).

Key terms

- → **Allele:** different forms of one gene (T and t).
- → **Genotype:** gene combination (TT, Tt, tt).
- → **Phenotype:** visible trait (tall, short).
- → **Homozygous:** both copies same (TT or tt), pure line.
- → **Heterozygous:** two different copies (Tt), hybrid.

Meme samjho



Ratio ratta ek taraf, 16 boxes wala Punnett square doosri taraf!
Asli baat: dihybrid F2 = 9 : 3 : 3 : 1; board mein Punnett square bana ke dikhao.

Activity 8.2: Checking 1 : 2 : 1 (monohybrid cross)

Done: Prove that F2 tall plants are of two kinds (TT and Tt); work out the dominant trait in Fig. 8.4, the **monohybrid cross for flower colour: violet (VV) × white (vv).** **Observed:** Selfing F2 plants: TT gives only tall, Tt gives tall and short (3 : 1), tt gives only short, i.e. 1 : 2 : 1.

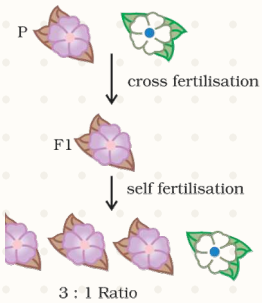


Fig. 8.4

Flower cross: F1 all violet (Vv); F2 3 violet : 1 white (1 VV : 2 Vv : 1 vv). **Conclusion:** Violet flower colour is dominant over white.

Test cross

Crossing a plant showing the **dominant trait** (genotype unknown: TT or Tt) with a pure recessive plant (tt). The offspring reveal the hidden genotype.

Cross	Offspring	Result
TT × tt	all Tt (tall)	pure tall (TT)
Tt × tt	1 tall : 1 short	hybrid tall (Tt)

Exam Alert

- → Test cross mein hamesha **recessive (tt)** se cross karo: saare tall = TT; aadhe short = Tt (1 : 1).
- → Phenotype 3 : 1 aur genotype 1 : 2 : 1 alag hain, answer mein dono ratio likho.

Quick fact

Father blood group A, mother O, daughter O: we **cannot** tell which trait is dominant. The father could be AO and pass on O, and one family is too small a sample; we need data from many families.

How to write any cross in the board exam

1. Write the parents with genotypes (TT × tt).
2. Write the gametes each parent makes (T; t).
3. Draw the Punnett square and fill each box.
4. Write F1, then self F1 and repeat for F2.
5. End with both ratios: phenotype and genotype.

Dihybrid cross (two traits)

Parents: round green (RRyy) × wrinkled yellow (rrYY). **F1:** all **round yellow** (RrYy), so round and yellow are dominant. Each F1 plant makes four kinds of gametes in equal numbers: **RY, Ry, rY, ry.**

F2 by selfing F1 (RrYy × RrYy):

♀ / ♂	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

F2 seeds (SS6)	Number	Ratio
round yellow	315	9
round green	108	3
wrinkled yellow	101	3
wrinkled green	32	1

Round yellow and wrinkled green are new combinations not seen in the parents, so seed shape and seed colour are inherited independently. RRYY × rryy gives the same 9 : 3 : 3 : 1. Tall round × short wrinkled behaves the same way: new types (tall wrinkled, short round) appear in F2.

Point	Monohybrid	Dihybrid
traits studied	one pair	two pairs
F2 phenotype ratio	3 : 1	9 : 3 : 3 : 1
law shown	dominance, segregation	independent assortment

Mendel's laws of inheritance

1. **Law of Dominance:** of two different alleles, the dominant one expresses itself and hides the recessive one.
2. **Law of Segregation:** the two copies of a gene separate during gamete formation; each gamete gets only one copy.
3. **Law of Independent Assortment:** genes for different traits are sorted into gametes independently of each other (seen in the dihybrid cross).

Yaad karo: 9-3-3-1

Dihybrid F2 ke 16 boxes: 9 mein dono dominant, 3 + 3 mein ek-ek dominant, aur sirf 1 box wrinkled green (rryy). Monohybrid = 3 : 1, dihybrid = 9 : 3 : 3 : 1.

3. How do Traits get Expressed?

DNA in the cell is the information source for making proteins; a gene gives the instructions for one protein.

Example, plant height: a gene makes an **enzyme**, the enzyme makes a **growth hormone**. An efficient enzyme means more hormone and a tall plant; an altered gene gives a less efficient enzyme, less hormone and a short plant. So genes control traits through proteins.

Gene (DNA) → Enzyme → Hormone → Height

Chromosomes: each gene set exists as separate, independent pieces of DNA called chromosomes. Every body cell has **two copies** of each chromosome: one from each parent.

Germ cells take only one chromosome from each pair. When two germ cells fuse, the normal number is restored, which keeps the **DNA of the species stable** and ensures an equal genetic contribution from both parents. Because chromosomes sort separately, traits like R and y are inherited independently of each other.

Exam Alert

- → Gene ⟶ enzyme ⟶ hormone ⟶ height ka flow chart 3 marks mein pakka aata hai.
- → "Equal contribution of both parents" ka answer: germ cells mein har pair ka sirf ek chromosome.

4. Sex Determination

Sex determination: the process by which the sex of a new individual is decided.

- → **By environment:** in some reptiles, the **temperature** at which fertilised eggs are kept decides male or female.
- → **Sex change:** snails can change sex, so their sex is not genetically fixed.
- → **Humans:** sex is largely genetic: 22 pairs of **autosomes** plus one pair of **sex chromosomes**. Women are **XX**; men are **XY** (Y is short).

Gametes	X (egg)	X (egg)
X (sperm)	XX girl	XX girl
Y (sperm)	XY boy	XY boy

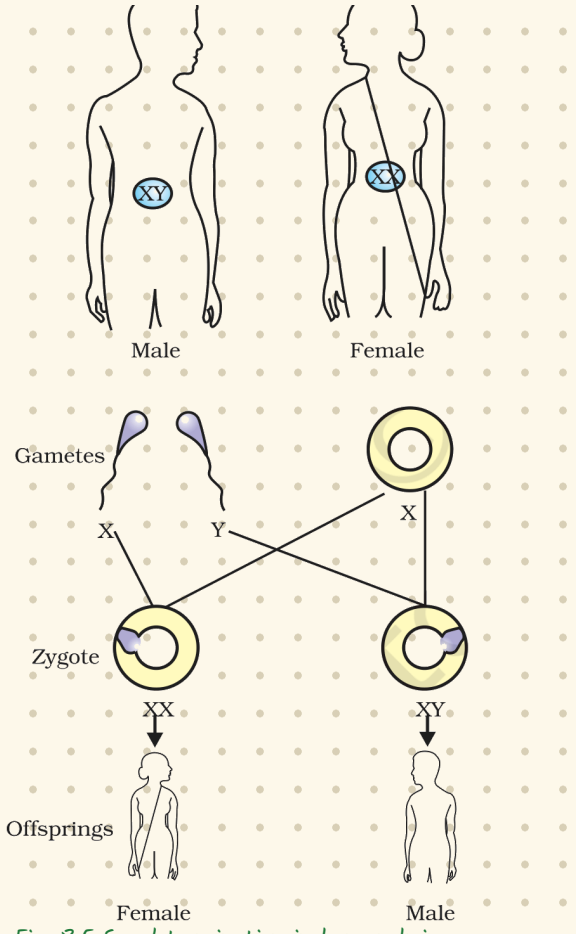


Fig. 8.5 Sex determination in human beings

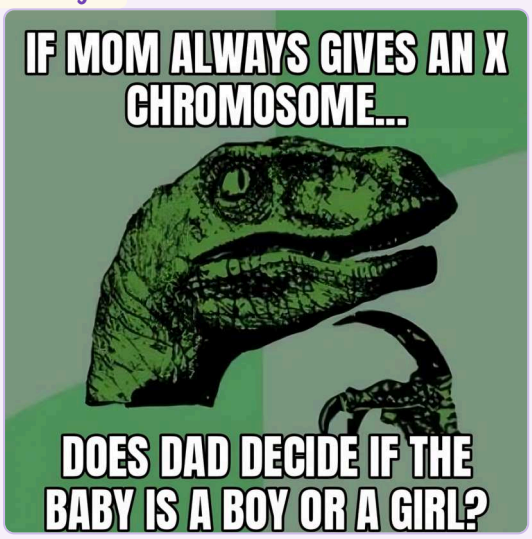
The mother always gives **X**. The father gives X or Y: X + X = girl, X + Y = boy. So the father's sperm decides the sex of the child, and the chance is 50 : 50.

Paheli-Bujho



Bujho: Beti hui toh blame mummy pe?
Paheli: Galat! Mummy sirf X deti hai, papa ka X ya Y decide karta hai.
Bujho: Toh blame kyun? Scam!
Paheli: Haan, biology padho!

Meme samjho



Maa toh hamesha X hi deti hai... phir beta hoga ya beti, ye decide kaun karta hai?

Asli baat: pita ka sperm X ya Y laata hai: X mila toh ***XX = girl***, Y mila toh ***XY = boy***. Chance 50-50, aur decide pita ka chromosome karta hai.

1 minute recap

Variation → Mendel → 3 : 1 → 9 : 3 : 3 : 1 → Gene ⟶ trait → XX / XY

Most Important Questions

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. 1M Board fav A cross between pure tall (TT) and pure short (tt) pea plants gives F1 plants that are:
(a) all tall (b) all short (c) medium height (d) 3 tall : 1 short
Ans: (a) all tall (Tt).

Q2. 1M Board fav F2 phenotypic ratio of a dihybrid cross:
(a) 3 : 1 (b) 1 : 2 : 1 (c) 9 : 3 : 3 : 1 (d) 1 : 1
Ans: (c) 9 : 3 : 3 : 1.

Q3. 1M A human male has which sex chromosomes?
Ans: XY (one X and a short Y).

Q4. 1M Which of these is an acquired trait?
(a) eye colour (b) a scar (c) blood group (d) earlobe type
Ans: (b) a scar from an injury (not in DNA, not passed on).

Q5. 1M Board fav **Assertion (A):** The sex of a child is decided by the father. **Reason (R):** The father produces sperms carrying either X or Y chromosome, while all eggs carry X.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q6. 1M In the F2 generation of a monohybrid cross, what fraction of plants is homozygous recessive?
Ans: One-fourth (1 tt out of 4).

Q7. 1M Name the genetic material that carries the information for making proteins.
Ans: DNA (deoxyribonucleic acid); a gene is a section of it.

Q8. 1M How many kinds of gametes does RrYy make?
Ans: Four: RY, Ry, rY, ry, in equal numbers.

Q9. 1M Who is known as the Father of Genetics?
Ans: Gregor Johann Mendel.

Q10. 1M How many pairs of autosomes are present in a human body cell?
Ans: 22 pairs (plus one pair of sex chromosomes).

Q11. 1M Board fav **Assertion (A):** Acquired traits are not passed on to the next generation. **Reason (R):** Acquired traits do not bring any change in the DNA of germ cells.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q12. 1M Which Mendel law does the dihybrid cross show?
Ans: Law of independent assortment.

Q13. 1M What is the genotype of a pure dwarf pea plant?
Ans: tt (homozygous recessive).

Q14. 1M Board fav Which chromosome decides that a child will be a boy?
Ans: The Y chromosome from the father's sperm.

Q15. 1M Write the F2 genotypic ratio of a monohybrid cross.
Ans: 1 TT : 2 Tt : 1 tt.

Short answer (2-3 marks)

Q16. 2M Board fav Differentiate between inherited and acquired traits with one example each.
Ans: Inherited traits come from parents through DNA and are passed on (eye colour, blood group). Acquired traits develop during life due to environment or practice; they are not in DNA and are not passed on (a scar, learning to swim).

Q17. 3M Board fav Why is variation greater in sexual than in asexual reproduction? How does variation help a species?
Ans: In sexual reproduction DNA from two parents combines, creating new combinations; in asexual reproduction only small DNA-copying errors arise. Variation helps some individuals survive when the environment changes (e.g. heat-resistant bacteria in a heat wave), so the species survives.

Q18. 3M Why did Mendel choose garden pea plants? Name four pairs of contrasting characters.
Ans: Clear contrasting characters, short life cycle, many seeds, easy self- and cross-pollination. Pairs: round/wrinkled seed, yellow/green seed, violet/white flower, tall/short plant.

Q19. 3M Board fav How do Mendel's experiments show that traits may be dominant or recessive?
Ans: In TT × tt, all F1 plants were tall: tallness expressed, shortness hidden, so tall is dominant. In F2, short plants (tt) reappeared in 1 out of 4, showing the recessive trait was present but hidden in F1.

Q20. 2M Board fav A man with blood group A marries a woman with blood group O; their daughter has blood group O. Can you tell which trait is dominant?
Ans: No. The father could be AO and pass on O, so the data fit either case; also one family is too small a sample. Data from many families are needed.

Q21. 3M What is a test cross? A tall plant crossed with a short plant gives half tall and half short plants. What is the genotype of the tall parent?
Ans: Crossing a plant with the dominant trait (unknown genotype) with a pure recessive (tt) plant to find its genotype. 1 tall : 1 short means the tall parent was Tt (hybrid).

Q22. 3M Board fav Explain with an example how genes control the traits of an organism.
Ans: A gene carries information for a protein. For plant height, a gene makes an enzyme that makes a growth hormone. An efficient enzyme gives more hormone and a tall plant; an altered gene gives a less efficient enzyme, less hormone and a short plant.

Q23. 2M Board fav How is the equal genetic contribution of male and female parents ensured in the progeny?
Ans: Each body cell has two copies of every chromosome. Germ cells get only one chromosome of each pair; when a male and a female gamete fuse, the zygote gets one set from each parent, restoring the normal number with equal contribution.

Q24. 2M How is sex determined in reptiles and snails?
Ans: In some reptiles, the temperature at which the fertilised eggs are incubated decides the sex. Snails can change sex, so their sex is not genetically determined.

Q25. 3M Board fav Tall plants with violet flowers were crossed with short plants with white flowers. All progeny had violet flowers, but almost half were short. What is the genotype of the tall parent: TTWW, TTww, TtWW or TtWw?
Ans: TtWW. All violet means the tall parent is WW (no white appears). Half short means it is Tt, since Tt × tt gives 1 tall : 1 short.

Q26. 2M Children with light-coloured eyes are likely to have light-eyed parents. Can we say whether light eye colour is dominant or recessive?
Ans: No. This only shows that eye colour is inherited. To find dominance we need crosses between light-eyed and dark-eyed parents and their offspring over many families, as Mendel did with large numbers.

Q27. 3M How would you find the dominant coat colour in dogs?
Ans: Cross pure black and pure white dogs (pure lines over generations). Note the coat colour of all F1 pups: the colour shown is dominant. Then cross F1 dogs among themselves; F2 should show about 3 dominant : 1 recessive.

Q28. 3M Define (a) allele (b) genotype (c) phenotype, with one example each.
Ans: (a) Different forms of one gene: T and t. (b) The gene combination of an organism: TT, Tt or tt. (c) The visible trait: tall or short.

Q29. 2M Distinguish between homozygous and heterozygous with examples. Which of them breeds true?
Ans: Homozygous: both copies of a gene are the same (TT or tt). Heterozygous: two different copies (Tt). Homozygous plants are pure lines and breed true; heterozygous (hybrid) plants give both types in the next generation.

Q30. 2M State the law of dominance with an example.
Ans: Of two different alleles of a gene, the dominant one expresses itself and hides the recessive one. Example: Tt pea plants are tall, because T (tall) is dominant over t (short).

Q31. 2M In Mendel's cross TT × tt, why were no plants of medium height obtained in F1? What does this show?
Ans: F1 plants (Tt) were all tall because T is dominant and fully hides t. There was no mixing of traits, which shows that traits do not blend; each gene keeps its identity and t reappears unchanged in F2.

Q32. 2M All variations in a species do not have equal chances of survival. Justify.
Ans: Only variations that suit the changed environment help survival. In a heat wave, heat-resistant bacteria survive while others die. The environment thus selects some variants, which is the basis of evolution.

Q33. 2M Board fav How can you find out whether a tall pea plant is TT or Tt?
Ans: By a test cross with a short (tt) plant, or by selfing it. If all offspring are tall, it is TT; if about half (test cross) or one-fourth (selfing) are short, it is Tt.

Q34. 2M Why do offspring produced by asexual reproduction look almost identical to the parent?
Ans: They get a copy of the DNA of a single parent; only small errors in DNA copying cause minor differences, and no new combinations of DNA are formed.

Long answer (5 marks)

Q35. 5M Board fav (a) A pure tall pea plant (TT) is crossed with a pure short plant (tt). Show F1 and F2 with a Punnett square. (b) Write the phenotypic and genotypic ratios of F2. (c) State the law of segregation.
Ans: (a) Gametes T and t; F1 all Tt (tall). F1 selfed: Tt × Tt gives TT, Tt, Tt, tt. (b) Phenotype 3 tall : 1 short; genotype 1 TT : 2 Tt : 1 tt. (c) The two copies of a gene separate during gamete formation, so each gamete gets only one copy.

Q36. 5M Board fav (a) Explain the dihybrid cross between round green (RRyy) and wrinkled yellow (rrYY) pea plants up to F2. (b) How does it show independent inheritance?
Ans: (a) F1: all round yellow (RrYy). Gametes RY, Ry, rY, ry; the 16-box Punnett square gives 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green. (b) New combinations (round yellow, wrinkled green) not present in the parents appear, so the genes for shape and colour are sorted into gametes independently (law of independent assortment).

Q37. 5M Board fav (a) What is sex determination? (b) Explain with a flow chart how the sex of a child is determined in human beings. (c) Why is it wrong to blame the mother for the birth of a girl?
Ans: (a) The process by which the sex of a new individual is decided. (b) Mother (XX) gives only X eggs; father (XY) gives X or Y sperms. X egg + X sperm = XX girl; X egg + Y sperm = XY boy; chance 50 : 50. (c) Sex depends on the sperm (X or Y), i.e. on the father; the mother always gives X.

Q38. 5M (a) What are chromosomes? How many copies of each are present in a body cell? (b) How do germ cells keep the DNA of a species stable? (c) Why are traits acquired during life not inherited?
Ans: (a) Separate independent pieces of DNA carrying sets of genes; two copies of each, one from the father and one from the mother. (b) Germ cells receive only one chromosome of each pair; when two fuse, the normal number is restored, so the DNA amount stays constant from generation to generation. (c) Acquired traits (a scar, trained muscles) change only body cells, not the DNA of germ cells, so they cannot be passed on.

Q39. 5M Board fav (a) Distinguish between monohybrid and dihybrid crosses (three points). (b) Write the F2 phenotypic ratio of each. (c) Name the law shown by each cross.
Ans: (a) Monohybrid: one pair of contrasting traits (tall × short), F1 gametes 2 kinds, 4-box Punnett square. Dihybrid: two pairs (seed shape + colour), F1 gametes 4 kinds, 16-box Punnett square. (b) Monohybrid 3 : 1; dihybrid 9 : 3 : 3 : 1. (c) Monohybrid shows dominance and segregation; dihybrid shows independent assortment.

Case-based (4 marks)

Q40. 4M Board fav Tall pea plants with violet flowers (TTVV) were crossed with short plants with white flowers (ttvv). (a) Give the F1 phenotype. (b) Which traits are dominant? (c) Write the gametes of F1. (d) What is the F2 phenotypic ratio?
Ans: (a) All tall with violet flowers (TtVv). (b) Tall and violet. (c) TV, Tv, tV, tv. (d) 9 tall violet : 3 tall white : 3 short violet : 1 short white.

Q41. 4M Board fav Riya listed the earlobes of her classmates: 70% had free earlobes and 30% attached. She found that most children shared the earlobe type of one of their parents. (a) Is earlobe type inherited or acquired? (b) Which type appears dominant? (c) What carries this trait from parents to children? (d) Could two parents with free earlobes have a child with attached earlobes? Explain.
Ans: (a) Inherited. (b) Free earlobe (more common, behaves as dominant). (c) Genes (sections of DNA) in the chromosomes of the germ cells. (d) Yes, if both parents are hybrid (Ff), the child can get f from each and be ff (attached), like tt in Mendel's cross.

Q42. 4M Board fav A couple has two daughters. Some relatives blame the mother for not having a son. (a) Which chromosomes does the mother carry and give? (b) Which chromosomes does the father give? (c) Who decides the sex of the child? (d) What is the chance of a boy in each pregnancy?
Ans: (a) Mother is XX and gives only X. (b) Father is XY and gives either X or Y. (c) The father: an X sperm gives a girl (XX), a Y sperm gives a boy (XY). (d) 50% (1 in 2), every time.

More board favourites

Q43. 2M Board fav How does the creation of variations in a species promote survival?
Ans: Variations make some individuals better suited to a change in the environment (heat, drought, disease). These individuals survive and reproduce, so the species does not die out completely.

Q44. 2M What is a gene? Where are genes located in a cell?
Ans: A gene is a section of DNA that carries the information for one protein, and so controls one trait. Genes are located on the chromosomes in the nucleus.

Q45. 2M Board fav Give two examples each of inherited traits and acquired traits in humans.
Ans: Inherited: eye colour, type of earlobe, blood group. Acquired: a scar from an injury, a language learnt, muscles built by exercise. Only inherited traits pass to the next generation.

Q46. 2M A plant with genotype Tt is self-pollinated. (a) What fraction of the offspring will be short? (b) What fraction will be tall but not pure (Tt)?
Ans: (a) 1/4 (tt). (b) 2/4 = 1/2 (Tt).
Ratio TT : Tt : tt = 1 : 2 : 1.

Q47. 2M Board fav Why does sexual reproduction produce more variation than asexual reproduction?
Ans: Two parents contribute DNA, and each germ cell carries a random mix of chromosomes; their fusion makes new combinations in every offspring. In asexual reproduction only small DNA-copying errors cause variation.

Yaad karo: heredity ke 4 numbers
3 : 1 (monohybrid phenotype), **1 : 2 : 1** (monohybrid genotype), **9 : 3 : 3 : 1** (dihybrid), **50%** (beta ya beti ka chance). Inhe ek baar likh lo, MCQ pakka!

LIGHT: REFLECTION AND REFRACTION



Most asked: mirror and lens ray diagrams, mirror/lens formula numericals, refractive index, power of a lens, uses of mirrors

We see objects because light from them reaches our eyes. Light travels in **straight lines**; this chapter is about light bouncing off mirrors (**reflection**) and bending through glass, water and lenses (**refraction**).

1. Reflection of Light

Reflection: the bouncing back of light into the same medium after striking a polished surface such as a mirror.

Laws of reflection

1. The angle of incidence is equal to the angle of reflection ($\angle i = \angle r$).
2. The incident ray, the normal to the mirror at the point of incidence and the reflected ray all lie in the **same plane**.

These laws hold for all surfaces, including spherical mirrors.

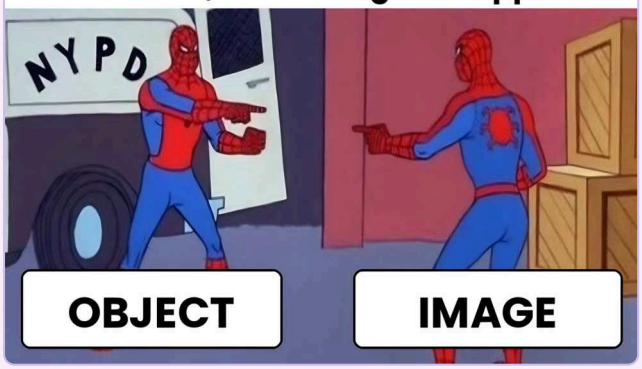
Real image	Virtual image
Rays actually meet after reflection/refraction	Rays only appear to meet (when extended backwards)
Can be obtained on a screen	Cannot be obtained on a screen
Always inverted	Always erect
e.g. image on a cinema screen	e.g. image in a plane mirror

Image formed by a plane mirror

- ✓ always **virtual and erect**
- ✓ **same size** as the object (m = +1)
- ✓ as far **behind** the mirror as the object is in front of it
- ✓ **laterally inverted** (left looks right)

Meme samjho

Plane mirror: same size, same distance, but left-right swapped



Mirror bole: "jo dikhta hai wahi milega", bas left-right ulta karke!

Asli baat: plane mirror ki image virtual, erect, same size aur laterally inverted hoti hai. AMBULANCE isi liye ulta likha jaata hai.

2. Spherical Mirrors

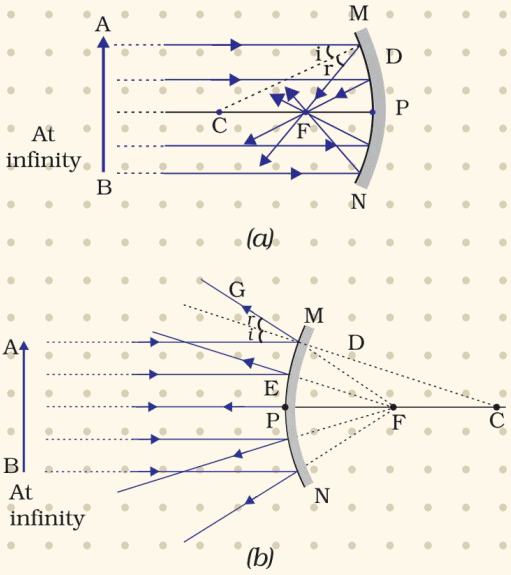


Fig. 9.2 (a) Concave mirror (b) Convex mirror

A mirror whose reflecting surface is part of a hollow sphere is a **spherical mirror**.

- → **Concave mirror:** reflecting surface curved **inwards** (faces the centre of the sphere); a converging mirror.
- → **Convex mirror:** reflecting surface curved **outwards**; a diverging mirror.

Activity 9.1: Images in a shining spoon

Done: Look at your face in the inner surface of a shining spoon, move it away, then repeat with the back surface. **Observed:** Inner side: erect and enlarged when very close, inverted farther away. Back side: always erect and smaller. **Conclusion:** The two curved surfaces behave like a concave and a convex mirror.

Terms for spherical mirrors

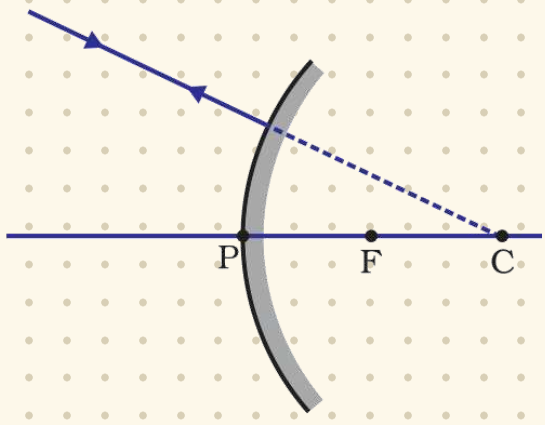


Fig. 9.5 Convex mirror: P, F and C

Key terms

- **Pole (P):** centre of the reflecting surface.
- **Centre of curvature (C):** centre of the sphere of which the mirror is a part (in front of a concave mirror, behind a convex mirror).
- **Radius of curvature (R):** radius of that sphere, distance PC.
- **Principal axis:** straight line through P and C; normal to the mirror at P.
- **Principal focus (F):** point where rays parallel to the principal axis meet after reflection (concave) or appear to come from (convex).
- **Focal length (f):** distance PF. **Aperture:** diameter of the reflecting surface.

$$R = 2f$$ (for mirrors of small aperture)

So the principal focus lies midway between P and C.

Activity 9.2: Concave mirror burns paper

Done: Point a concave mirror at the Sun and hold a sheet of paper at the bright, sharp spot for a few minutes. **Observed:** The paper starts smoking and then catches fire. **Conclusion:** A concave mirror converges sunlight at its focus; the bright spot is the image of the Sun and its distance from the mirror is the approximate focal length.

3. Rules for Drawing Ray Diagrams

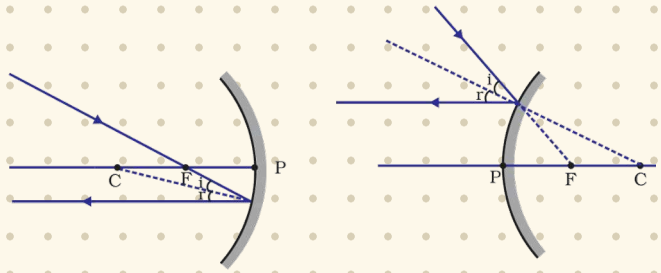


Fig. 9.4 Ray through F (concave) and towards F (convex)

1. A ray **parallel** to the principal axis passes **through F** after reflection (concave), or appears to diverge from F (convex).
2. A ray **through F** (concave), or directed towards F (convex), becomes **parallel** to the principal axis.
3. A ray **through C** (concave), or directed towards C (convex), is reflected back **along the same path.**
4. A ray falling obliquely at P is reflected making an **equal angle** with the principal axis.

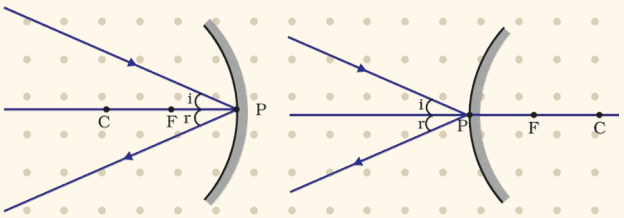


Fig. 9.6 Ray incident obliquely at the pole

Exam Alert

- Image wahi banta hai jahan kam se kam **do reflected rays milein** (real) ya peeche badhaane par milti dikhein (virtual, **dotted lines**).
- Ray diagram mein arrow lagana mat bhoolna, bina arrow ke marks katte hain.

4. Image Formation by a Concave Mirror

Object at	Image at	Size	Nature
Infinity	At F	Point-sized	Real, inverted
Beyond C	Between F and C	Diminished	Real, inverted
At C	At C	Same size	Real, inverted
Between C and F	Beyond C	Enlarged	Real, inverted
At F	Infinity	Highly enlarged	Real, inverted
Between P and F	Behind mirror	Enlarged	Virtual, erect

Activity 9.3 + 9.4: Candle before a concave mirror

Done: Place a lit candle at different distances from a concave mirror, catch the image on a screen where possible and draw ray diagrams. **Observed:** Position, size and nature of the image change (table above). **Conclusion:** The nature of the image depends on where the object is placed.

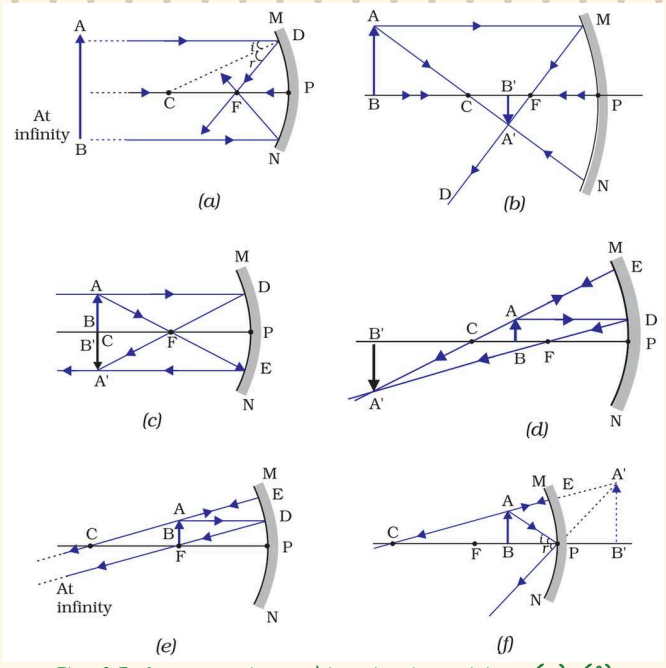


Fig. 9.7 Concave mirror: object in six positions (a)-(f)

Uses of concave mirrors

1. Torches, searchlights and vehicle headlights: bulb kept at F gives a powerful **parallel beam.**
2. Shaving and make-up mirrors: face within F gives an **enlarged, erect** image.
3. Dentists use them to see large images of teeth.
4. Large concave mirrors concentrate sunlight in **solar furnaces.**

Meme samjho

Object rakha F par... image kahan hai? At INFINITY!



Object jitna mirror ke paas, image utni door aur badi. F ke andar aaya toh image mirror ke peeche!

Asli baat: object P aur F ke beech ho toh concave mirror virtual, erect aur enlarged image deta hai; object F par ho toh image infinity par.

5. Image Formation by a Convex Mirror

Activity 9.5: Pencil in front of a convex mirror

Done: Hold a pencil upright before a convex mirror and move it away. **Observed:** The image is always erect and diminished, and shrinks as the pencil moves away. **Conclusion:** A convex mirror always forms a virtual, erect, diminished image.

Object at	Image at	Size	Nature
Infinity	At F, behind mirror	Point-sized	Virtual, erect
Between infinity and P	Between P and F, behind	Diminished	Virtual, erect

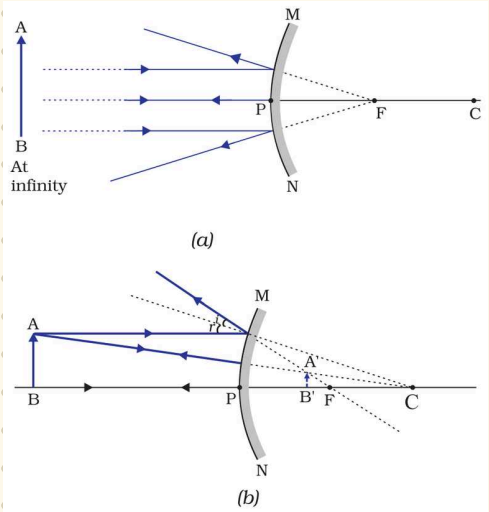


Fig. 9.8 Convex mirror: object (a) at infinity (b) nearby

Activity 9.6: Full-length image

Done: Observe the image of a distant tall tree or building in a plane mirror, a concave mirror and a convex mirror. **Observed:** Only the convex mirror shows the full-length image. **Conclusion:** A convex mirror gives the widest field of view, so it shows the whole tree.

Uses of convex mirrors: rear-view (wing) mirrors of vehicles, because they always give an erect image and have a wider field of view (they are curved outwards); also street and blind-turn mirrors and shop security mirrors.

Concave mirror	Convex mirror
Reflecting surface curved inwards	Reflecting surface curved outwards
Converging	Diverging
Real or virtual image (depends on object position)	Always virtual, erect, diminished
f negative	f positive
Headlights, shaving mirror, solar furnace	Rear-view mirror, shop security mirror

Exam Alert: ray diagrams

→ Pencil aur scale se banao; real rays **solid**, virtual extensions **dotted**.

→ Har ray par **arrow**, aur P, F, C (lens mein O, F_1, $2F_1$) zaroor label karo.

→ Answer mein image ki **position, size, nature** teeno likho: 3 marks ka pura package.

Yaad karo: kaunsa mirror kahan

"Concave = Close-up": shaving, dentist, headlight, solar furnace. "Convex = Car ke piche dekhna": rear-view, blind turn, shop security.

6. Sign Convention and Mirror Formula

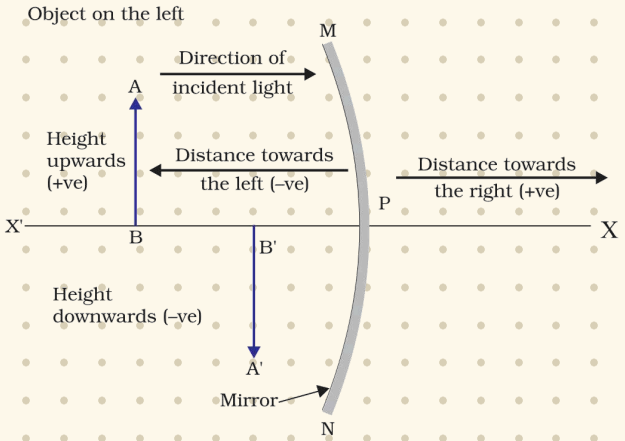


Fig. 9.9 New Cartesian sign convention for spherical mirrors

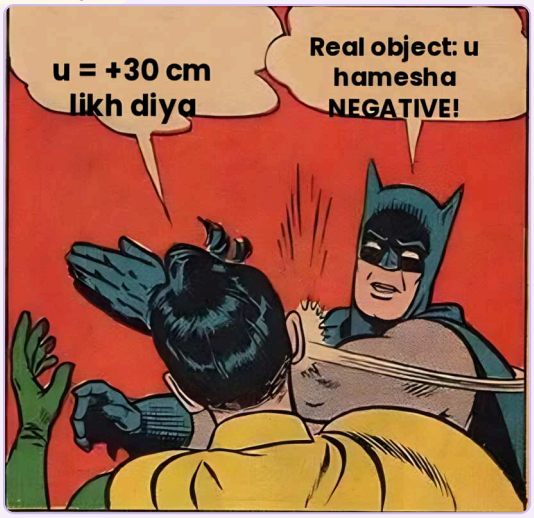
New Cartesian sign convention

1. The object is always placed to the **left** of the mirror (light falls from the left).
2. All distances are measured from the **pole P**.
3. Distances in the direction of incident light (to the right) are **positive**; against it (to the left) **negative**.
4. Heights **above** the principal axis are positive; below it negative.

Result: u is always negative; f is negative for a concave mirror and positive for a convex mirror.

	u	v (real image)	v (virtual image)	f
Concave mirror	–	– (in front)	+ (behind)	–
Convex mirror	–	never	+ (behind)	+
Convex lens	–	+ (other side)	– (same side)	+
Concave lens	–	never	– (same side)	–

Meme samjho



Board copy mein u = +30 cm likha? Examiner ka thappad pakka! Object hamesha left mein, toh u hamesha minus.

Asli baat: pehle har value sign ke saath likho (u hamesha −), phir formula lagao. Real image ki height negative, virtual erect ki positive.

Paheli-Bujho



Bujho: Concave mirror ka f negative? Length negative kaise?

Paheli: Length nahi, direction hai. Focus incident light ki ulti taraf hai, toh minus.

Bujho: Toh minus = ulti direction. Samjha!

Mirror formula and magnification

$$1/v + 1/u = 1/f \quad | \quad m = h'/h = -v/u$$

u = object distance, v = image distance, f = focal length, h = object height, h′ = image height. **Magnification (m)** tells how many times the image is enlarged relative to the object.

- ✓ m **negative**: real and inverted; m **positive**: virtual and erect
- ✓ |m| > 1 enlarged; |m| < 1 diminished; |m| = 1 same size

Quick fact

For a mirror, a **negative v** means a real image in front of the mirror; a **positive v** means a virtual image behind it. The sign itself tells the nature.

Yaad karo: numerical in 3 steps

"Sign, Formula, Nature": (1) given values sign ke saath likho, (2) formula mein daalo, (3) answer ka sign padh ke nature batao.

Solved numericals

Example 1. A convex mirror used as a rear-view mirror has R = 3.00 m. A bus is 5.00 m from it. Find the position, nature and size of the image.

Given: u = −5.00 m, f = +R/2 = +1.50 m. **Solution:** 1/v = 1/f − 1/u = 1/1.50 + 1/5.00 = 1.30/1.50, so **v = +1.15 m** (behind the mirror). m = −v/u = −1.15/(−5.00) = **+0.23**: virtual, erect, about one-fourth the size.

Example 2. A 4.0 cm tall object is 25.0 cm in front of a concave mirror of focal length 15.0 cm. Where should a screen be placed for a sharp image? Find the image size.

Given: u = −25.0 cm, f = −15.0 cm. **Solution:** 1/v = −1/15 + 1/25 = −2/75, so **v = −37.5 cm**: screen 37.5 cm in front of the mirror. h′ = −vh/u = −(−37.5)(4.0)/(−25.0) = **−6.0 cm**: real, inverted, enlarged.

Example 3. A concave mirror produces a real image three times the size of an object placed 10 cm in front of it. Where is the image?

Real image, so m = −3. m = −v/u gives v = 3u = 3(−10) = **−30 cm**: the image is 30 cm in front of the mirror.

7. Refraction of Light

Refraction: the bending of light when it passes obliquely from one transparent medium into another. It happens because light travels at different speeds in different media.

Everyday examples: a pencil looks bent in water; a tank or pond looks shallower; letters look raised under a glass slab; lemons in a glass of water look bigger.

Activity 9.7: Coin in a bucket

Done: Try to pick up a coin from a water-filled bucket in one go. **Observed:** We miss it. **Conclusion:** Light bends at the water surface, so the coin appears raised.

Activity 9.8: The coin reappears

Done: Move back till a coin in a bowl just disappears from view; a friend pours water gently into the bowl. **Observed:** The coin becomes visible again. **Conclusion:** Refraction bends light from the coin into our eye.

Activity 9.9: A line under a glass slab

Done: Place a glass slab over a straight ink line, first at an angle, then perpendicular to the line. **Observed:** At an angle the line looks bent at the edges; perpendicular to it, it does not. **Conclusion:** Light bends only when it falls obliquely, not along the normal.

Refraction through a rectangular glass slab

Activity 9.10: Tracing a ray through a glass slab

Done: Trace the slab ABCD, fix pins E and F on one side, find pins G and H in line with their images from the other side, and join the points. **Observed:** The ray bends **towards** the normal on entering glass and **away** from it on leaving; the emergent ray is **parallel** to the incident ray. **Conclusion:** The bending at the two parallel faces is equal and opposite, so ∠i = ∠e; the ray is only shifted sideways (**lateral displacement**).

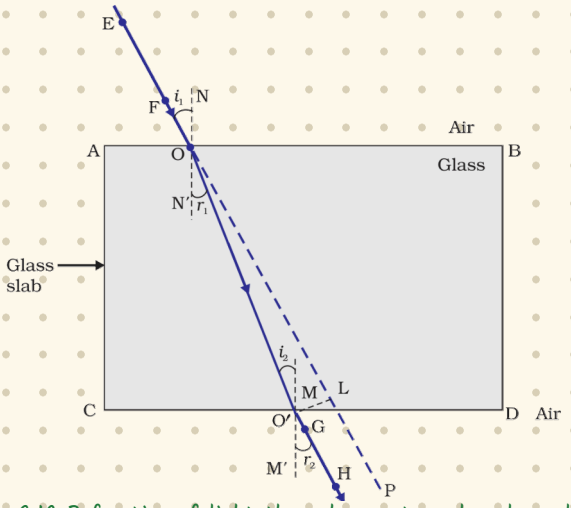


Fig. 9.10 Refraction of light through a rectangular glass slab

→ Rarer to denser (air to glass): light **slows down** and bends towards the normal.

→ Denser to rarer: speeds up, bends away from the normal.

→ A ray along the normal ($\angle i = 0$) goes **straight**, with no bending at all.

Paheli-Bujho



Bujho: Pencil paani mein toot gayi! Ab exam kisse likhunga?

Paheli: Toot-ti nahi, refraction hai. Light paani se hawa mein normal se door mudti hai.

Bujho: Paani ne light ko moda, pencil ko nahi!

Laws of refraction

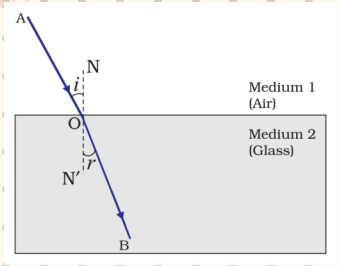


Fig. 9.11 Refraction from air into glass

1. The incident ray, the refracted ray and the normal at the point of incidence all lie in the **same plane**.
2. **Snell's law:** for light of a given colour and a given pair of media, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is **constant**.

$$\sin i / \sin r = \text{constant} = n_{21}$$

This constant is the **refractive index** of the second medium with respect to the first.

8. Refractive Index

$$n_{21} = v_1/v_2 \quad | \quad n_{12} = v_2/v_1 = 1/n_{21} \quad | \quad n_m = c/v$$

Absolute refractive index of a medium = speed of light in vacuum ($c = 3 \times 10^8$ m s^{-1}) / speed of light in that medium. Of two media, the one with the larger refractive index is **optically denser** (light travels slower in it).

Exam Alert

→ Optical density ≠ mass density! Kerosene paani par tairta hai, phir bhi optically **denser** hai (n = 1.44 > 1.33).

→ Refractive index ki koi **unit nahi** hoti (ratio hai).

Example: n of glass = 1.50, so speed of light in glass = $c/n = 3 \times 10^8/1.50 = \mathbf{2 \times 10^8}$ **m s^{-1}**; in water (n = 1.33) it is 2.25×10^8 m s^{-1}.

Medium	n	Medium	n
Air	1.0003	Crown glass	1.52
Ice	1.31	Canada balsam	1.53
Water	1.33	Rock salt	1.54
Alcohol	1.36	Carbon disulphide	1.63
Kerosene	1.44	Dense flint glass	1.65
Fused quartz	1.46	Ruby	1.71
Turpentine oil	1.47	Sapphire	1.77
Benzene	1.50	Diamond	2.42

Quick fact

Diamond has the highest refractive index in the table (2.42), so light travels **slowest** in it:
$v = c/2.42 \approx 1.24 \times 10^8$ m s^{-1}.

9. Spherical Lenses

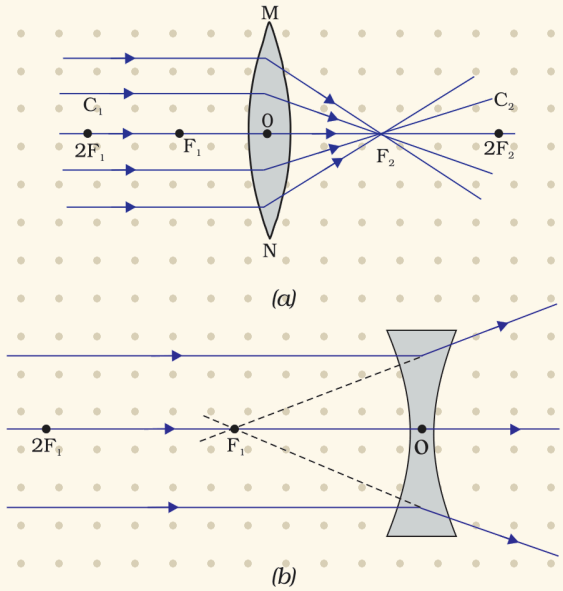


Fig. 9.12 (a) Convex lens converges (b) concave lens diverges

A transparent material bound by two surfaces, at least one of them spherical, is a **lens**.

→ **Convex lens:** thicker at the middle than at the edges; converges light (converging lens).

→ **Concave lens:** thicker at the edges; diverges light (diverging lens).

Key terms

→ **Centres of curvature:** two, C_1 and C_2; **principal foci:** two, F_1 and F_2.

→ **Optical centre (O):** central point of the lens; a ray through it goes **undeviated**.

→ **Principal axis:** line through both centres of curvature. **Focal length:** distance OF.

→ **Aperture:** effective diameter of the lens.

Activity 9.11: Convex lens burns paper

Done: Hold a convex lens towards the Sun and focus the light on a sheet of paper till a sharp bright spot forms. **Observed:** The paper begins to smoke and catches fire. **Conclusion:** A convex lens converges sunlight to its principal focus; the distance of the spot from the lens is roughly its focal length.

Rules for ray diagrams (lenses)

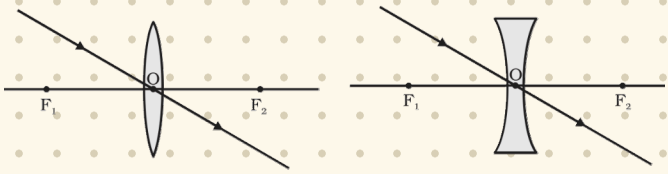


Fig. 9.15 Ray through the optical centre goes undeviated

1. A ray **parallel** to the principal axis passes through F_2 (convex) or appears to diverge from F_1 (concave).
2. A ray through F_1 (convex) or directed towards F_2 (concave) emerges **parallel** to the principal axis.
3. A ray through the **optical centre O** passes straight, undeviated.

10. Image Formation by Lenses

Activity 9.12: Candle in front of a convex lens

Done: Place a candle at different distances from a convex lens and try to catch its image on a screen. **Observed:** A sharp image forms on the screen in all positions except at F_1 and inside F_1. **Conclusion:** Nature, size and position of the image depend on the object position (table).

Object at	Image at	Size	Nature
Infinity	At F_2	Point-sized	Real, inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real, inverted
At $2F_1$	At $2F_2$	Same size	Real, inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real, inverted
At F_1	At infinity	Infinitely large	Real, inverted
Between F_1 and O	Same side as object	Enlarged	Virtual, erect

Quick fact

→ Object between F_1 and O: the convex lens works as a **magnifying glass** (enlarged, erect, virtual image). For reading small print a lens of f = 5 cm is better than one of f = 50 cm (smaller f, more power, larger image).

→ Cover **half of a convex lens** with black paper: the image is still **complete**, only dimmer, because the uncovered half still receives rays from every point of the object.

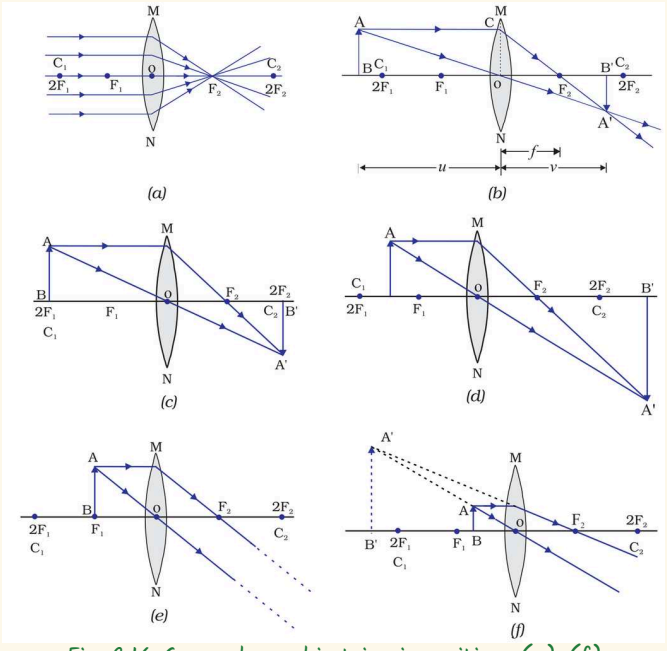


Fig. 9.16 Convex lens: object in six positions (a)-(f)

Activity 9.13: Candle in front of a concave lens

Done: Look at a candle through a concave lens and move the candle away. **Observed:** No image on a screen; the image is always erect, diminished and between F_1 and O on the candle's side. **Conclusion:** A concave lens always gives a virtual, erect, diminished image.

Object at	Image at	Size	Nature
Infinity	At F_1	Point-sized	Virtual, erect
Between infinity and O	Between F_1 and O	Diminished	Virtual, erect

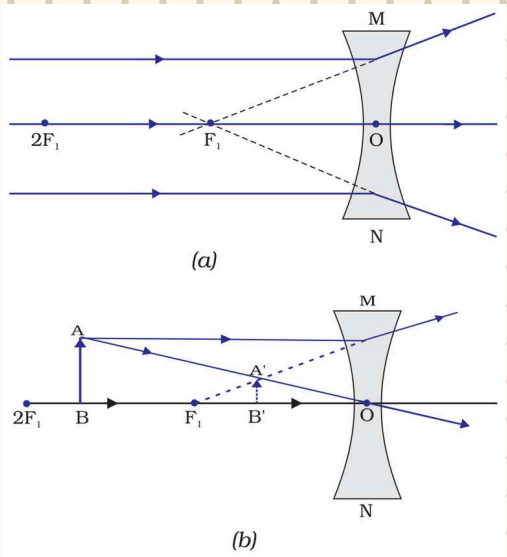


Fig. 9.17 Concave lens: object (a) at infinity (b) nearby

Convex lens	Concave lens
Thicker at the middle	Thicker at the edges
Converging lens	Diverging lens
Real or virtual image	Always virtual, erect, diminished
f and P positive	f and P negative
Magnifying glass, camera, correcting hypermetropia	Correcting myopia, peep-hole of doors

11. Lens Formula and Magnification

Sign convention is the same, but distances are measured from the **optical centre O**. f of a convex lens is positive; of a concave lens negative.

$$1/v - 1/u = 1/f \quad | \quad m = h'/h = v/u$$

Yaad karo: mirror vs lens

Mirror mein PLUS, lens mein MINUS. Mirror: $1/v + 1/u = 1/f$, $m = -v/u$. Lens: $1/v - 1/u = 1/f$, $m = +v/u$. Bas yeh ulta yaad rakho, formula kabhi mix nahi hoga.

Example 4. A concave lens has a focal length of 15 cm. At what distance should an object be placed so that it forms an image 10 cm from the lens? Find m.

Given: f = −15 cm, v = −10 cm (virtual, same side). **Solution:** $1/u = 1/v - 1/f = -1/10 + 1/15 = -1/30$, so **u = −30 cm**. $m = v/u = (-10)/(-30) = +0.33$: virtual, erect, one-third the size.

Example 5. A 2.0 cm tall object is placed 15 cm from a convex lens of focal length 10 cm. Find the position, nature and size of the image.

Given: u = −15 cm, f = +10 cm. **Solution:** $1/v = 1/f + 1/u = 1/10 - 1/15 = 1/30$, so **v = +30 cm** (other side). $m = v/u = 30/(-15) = -2$, h′ = **−4.0 cm**: real, inverted, twice the size.

12. Power of a Lens

Power is the degree of convergence or divergence of light by a lens: the reciprocal of the focal length in metres.

$$P = 1/f \text{ (f in metres)} \quad | \quad P = P_1 + P_2 + P_3 + \ldots$$

→ SI unit: **dioptre (D)**; 1 D is the power of a lens of focal length 1 m ($1\ D = 1\ m^{-1}$).

→ Power: convex lens **positive**, concave lens **negative**.

→ Lenses in contact: powers simply add (used by opticians to test eyes).

Example: a concave lens of f = 2 m has $P = 1/(-2) = -0.5$ D. A lens of power +2.0 D has $f = 1/2.0 = +0.5$ m: a **convex lens of focal length 50 cm**.

Quick formula sheet	Formula	Remember
Mirror	$1/v + 1/u = 1/f$ $m = -v/u$	$f = R/2$
Lens	$1/v - 1/u = 1/f$ $m = v/u$	f: convex +, concave −
Refractive index	$n_{21} = \sin i/\sin r = v_1/v_2$	$n = c/v$
Power	$P = 1/f$ (f in m)	$P = P_1 + P_2$

NCERT in-text questions: quick answers

→ **Principal focus of a concave mirror:** point on the axis where rays parallel to the principal axis meet after reflection.

→ **R = 20 cm, f = ?** f = R/2 = 10 cm.

→ **Mirror giving an erect, enlarged image:** concave mirror, object between P and F.

→ **Light from air into water bends** towards the normal (water is optically denser).

→ **Highest n in the table:** diamond (2.42), so light is slowest in it; lowest: air.

→ **n of glass = 1.5 means:** light travels 1.5 times slower in glass than in vacuum.

1 minute recap

Reflection: ∠i = ∠r → Mirrors: R = 2f → $1/v + 1/u = 1/f$ → Refraction: n = c/v → Lens: $1/v - 1/u = 1/f$ → P = 1/f (D)

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM The focal length of a concave mirror of radius of curvature 32 cm is:
(a) 32 cm (b) 64 cm (c) −16 cm (d) 8 cm
Ans: (c) −16 cm (f = R/2, negative for a concave mirror).

Q2. IM Board fav No matter how far you stand from a mirror, your image appears erect. The mirror is likely to be:
(a) only plane (b) only concave
(c) only convex (d) either plane or convex
Ans: (d) either plane or convex.

Q3. IM Board fav Where should an object be placed in front of a convex lens to get a real image of the same size as the object?
(a) at F (b) at 2F (c) at infinity (d) between O and F
Ans: (b) At $2F_1$, the image forms at $2F_2$.

Q4. IM A spherical mirror and a thin spherical lens each have a focal length of −15 cm. They are likely to be:
(a) both concave (b) both convex
(c) mirror concave, lens convex
(d) mirror convex, lens concave
Ans: (a) both concave.

Q5. IM Board fav Which lens should be used to read small letters in a dictionary?
(a) convex, f = 50 cm (b) concave, f = 50 cm
(c) convex, f = 5 cm (d) concave, f = 5 cm
Ans: (c) A convex lens of focal length 5 cm (smaller f gives larger magnification).

Q6. IM The magnification produced by a plane mirror is +1. What does this mean?
Ans: The image is the same size as the object (|m| = 1) and is virtual and erect (m positive).

Q7. IM Board fav **Assertion (A):** A ray passing through the optical centre of a lens goes undeviated. **Reason (R):** Near the optical centre the lens behaves like a thin glass slab with parallel faces.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q8. IM Find the power of a concave lens of focal length 2 m.
Ans: P = 1/f = 1/(−2 m) = −0.5 D.

Q9. IM Which mirror can give a real, inverted image of the same size as the object?
Ans: A concave mirror, with the object at C (the image also forms at C).

Q10. 1M Define 1 dioptre.
Ans: 1 dioptre is the power of a lens whose focal length is 1 metre ($1\,D = 1\,m^{-1}$).

Q11. 1M Board fav **Assertion (A):** Convex mirrors are used as rear-view mirrors. **Reason (R):** A convex mirror always forms a real, enlarged image.
Ans: (c) A is true but R is false: a convex mirror always forms a virtual, erect and diminished image, and it gives a wider field of view.

Short answer (2–3 marks)

Q12. 2M State the two laws of reflection. List the properties of the image formed by a plane mirror.
Ans: (i) $\angle i = \angle r$. (ii) Incident ray, reflected ray and the normal at the point of incidence lie in the same plane. Plane mirror image: virtual, erect, same size, as far behind the mirror as the object is in front, laterally inverted.

Q13. 3M Board fav Name the mirror used in (a) headlights of cars (b) rear-view mirrors of vehicles (c) solar furnaces. Give reasons.
Ans: (a) Concave: bulb at its focus gives a powerful parallel beam. (b) Convex: always an erect, diminished image and a wider field of view. (c) Concave: converges sunlight at its focus to produce high temperature.

Q14. 3M Board fav Draw a ray diagram for an object placed between F and C of a concave mirror. State the position and nature of the image.
Ans: Use one ray parallel to the axis (reflects through F) and one ray through C (reflects back on itself); they meet beyond C. Image: beyond C, real, inverted and enlarged (Fig. 9.7 d).

Q15. 3M Board fav An object 5.0 cm tall is placed 20 cm in front of a convex mirror of radius of curvature 30 cm. Find the position, nature and size of the image.
Ans: $u = -20$ cm, $f = +15$ cm. $1/v = 1/f - 1/u = 1/15 + 1/20 = 7/60$, $v = +8.57$ cm (behind the mirror). $m = -v/u = 8.57/20 = +0.43$, $h' = 0.43 \times 5.0 = +2.14$ cm. The image is virtual, erect and diminished.

Q16. 2M Board fav Light enters from air to glass of refractive index 1.50. Find the speed of light in glass ($c = 3 \times 10^8$ m s^{-1}).
Ans: $n = c/v$, so $v = c/n = 3 \times 10^8/1.50 = 2 \times 10^8$ m s^{-1}.

Q17. 2M The refractive index of diamond is 2.42. What is the meaning of this statement?
Ans: The speed of light in vacuum is 2.42 times its speed in diamond ($v = c/2.42 \approx 1.24 \times 10^8$ m s^{-1}).

Q18. 2M Why does a ray of light bend when it travels from one medium into another? Which medium is optically denser: kerosene ($n = 1.44$) or water ($n = 1.33$)?
Ans: Light has different speeds in different media; the change in speed at the boundary bends an oblique ray. Kerosene is optically denser (larger n), so light travels slower in it.

Q19. 2M Board fav An object is placed 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image.
Ans: $u = -10$ cm, $f = +15$ cm. $1/v = 1/15 + 1/10 = 5/30$, $v = +6$ cm. Virtual, erect, behind the mirror; $m = +0.6$ (diminished).

Q20. 2M Board fav One half of a convex lens is covered with black paper. Will it produce a complete image? Verify.
Ans: Yes, a complete image forms but it is less bright (dimmer), because every point of the object still sends rays through the uncovered half. Verify with a candle and screen.

Q21. 3M Board fav An object is placed 50 cm from a concave lens of focal length 20 cm. Find v, m and the power of the lens.
Ans: $u = -50$ cm, $f = -20$ cm. $1/v = 1/f + 1/u = -1/20 - 1/50 = -7/100$, $v = -14.3$ cm. $m = v/u = +0.29$.
$P = 1/(-0.20\ m) = -5$ D.

Q22. 2M Board fav A lens has a power of −2.0 D. Find its focal length and name the lens.
Ans: $f = 1/P = 1/(-2.0) = -0.5$ m $= -50$ cm: a concave (diverging) lens.

Q23. 2M Why is AMBULANCE written in reverse on the front of the vehicle?
Ans: The driver ahead sees it in the rear-view mirror. Mirror images are laterally inverted, so the reversed word reads correctly in the mirror.

Q24. 2M A ray of light travelling in air enters obliquely into water. Does it bend towards or away from the normal? Why?
Ans: Towards the normal: water is optically denser than air, so light slows down on entering water.

Q25. 3M Board fav An object 7.0 cm tall is placed 27 cm in front of a concave mirror of focal length 18 cm. At what distance should a screen be placed to get a sharp image? Find the size and nature.
Ans: $u = -27$ cm, $f = -18$ cm. $1/v = 1/f - 1/u = -1/18 + 1/27 = -1/54$, $v = -54$ cm: screen 54 cm in front of the mirror. $h' = -vh/u = -(-54)(7.0)/(-27) = -14$ cm: real, inverted, enlarged.

Q26. 2M Board fav The image formed by a concave mirror is observed to be virtual, erect and larger than the object. Where should the object be placed? Also state the image position.
Ans: Between the pole P and the focus F. The image forms behind the mirror.

Q27. 2M Board fav You are given kerosene, turpentine and water. In which does light travel fastest? Use n: kerosene 1.44, turpentine 1.47, water 1.33.
Ans: $v = c/n$, so light is fastest in the medium with the smallest n: water (1.33).

Q28. 3M Board fav A convex lens forms a real and inverted image of a needle at 50 cm from it. The image is the same size as the needle. Where is the needle placed? Find the power of the lens.
Ans: Same-size real image: object and image both at 2F, so $u = -50$ cm and $2f = 50$ cm, $f = +25$ cm $= 0.25$ m. $P = 1/0.25 = +4$ D.

Q29. 2M Why do we prefer a convex mirror as a rear-view mirror in vehicles? Why can it not be used for shaving?
Ans: It always gives an erect, diminished image and a wider field of view, so the driver sees more traffic behind. For shaving an enlarged image is needed, which a convex mirror never gives (a concave mirror is used).

Q30. 2M What is meant by "the refractive index of glass with respect to water is 1.13"? If n of water is 1.33, find n of glass.
Ans: Light travels 1.13 times faster in water than in glass ($n_{gw} = v_w/v_g$). $n_g = n_{gw} \times n_w = 1.13 \times 1.33 \approx 1.50$.

Q31. 3M Board fav Draw a ray diagram to show the image of an object placed between F_1 and O of a convex lens. Where is this used?
Ans: Ray parallel to the axis refracts through F_2; ray through O goes straight. The refracted rays diverge, so extend them backwards (dotted): they meet on the object's side, giving a virtual, erect, enlarged image (Fig. 9.16 f). Used as a magnifying glass.

Long answer (5 marks)

Q32. 5M Board fav (a) A 2.0 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 10 cm, at 15 cm from it. Find the nature, position and size of the image. (b) Draw the ray diagram. (c) What is its magnification?
Ans: (a) u = −15 cm, f = +10 cm. 1/v = 1/10 − 1/15 = 1/30, v = +30 cm on the other side of the lens. h′ = h(v/u) = 2.0 × 30/(−15) = −4.0 cm: real, inverted, enlarged. (b) Object between F_1 and $2F_1$: parallel ray refracts through F_2, ray through O goes straight; they meet beyond $2F_2$ (Fig. 9.16 d). (c) m = v/u = −2.

Q33. 5M Board fav (a) State the laws of refraction. (b) Define absolute refractive index. (c) Trace the path of a ray through a rectangular glass slab and show that the emergent ray is parallel to the incident ray. (d) Speed of light in water is 2.25×10^8 m s^{-1}; find n of water.
Ans: (a) Incident ray, refracted ray and normal lie in one plane; sin i/sin r = constant (Snell's law). (b) n = speed of light in vacuum / speed in the medium. (c) Ray bends towards the normal at the first face and away at the second; since the faces are parallel, the bending is equal and opposite, so ∠i = ∠e and the emergent ray is parallel but laterally displaced (Fig. 9.10). (d) $n = 3 \times 10^8/2.25 \times 10^8 = 1.33$.

Q34. 5M (a) Define power of a lens and its SI unit. (b) A doctor prescribes a lens of power +1.5 D. Find its focal length and name the lens. (c) Two lenses of power +3.5 D and −2.5 D are in contact. Find the net power and focal length.
Ans: (a) P = 1/f (f in metres); dioptre (D), 1 D = 1 m^{-1}. (b) f = 1/1.5 = +0.67 m = 66.7 cm: a convex (converging) lens. (c) P = 3.5 − 2.5 = +1.0 D; f = 1/1.0 = +1 m (converging).

Q35. 5M Board fav (a) Draw ray diagrams to show image formation by a concave mirror when the object is (i) beyond C (ii) between P and F. (b) State the nature and size of the image in each case. (c) Give two uses of concave mirrors.
Ans: (a) (i) Parallel ray reflects through F, ray through C returns on itself; they meet between F and C (Fig. 9.7 b). (ii) Rays diverge after reflection; extended behind the mirror (dotted) they meet behind the mirror (Fig. 9.7 f). (b) (i) Real, inverted, diminished, between F and C. (ii) Virtual, erect, enlarged, behind the mirror. (c) Shaving/make-up mirrors; reflectors in torches and headlights (also dentist mirrors, solar furnaces).

Case-based (4 marks)

Q36. 4M Board fav A student places a candle 30 cm in front of a concave mirror of focal length 15 cm and moves a screen to get a sharp image. (a) Where is the object relative to C? (b) Find v. (c) State the nature and size of the image. (d) Where should the candle be placed to get a virtual image?
Ans: (a) At C (R = 2f = 30 cm). (b) u = −30, f = −15: 1/v = −1/15 + 1/30 = −1/30, v = −30 cm (at C). (c) Real, inverted, same size (m = −1). (d) Between P and F, i.e. less than 15 cm from the mirror.

Q37. 4M Ravi uses a convex mirror on a sharp hill turn and his sister uses a concave mirror for make-up. (a) Why is a convex mirror used on the turn? (b) Where must her face be for an enlarged, erect image? (c) A convex mirror forms an image 1/3 the size of a car 30 m away. Find v. (d) Is this image real or virtual?
Ans: (a) It always gives an erect, diminished image and has a wider field of view, so vehicles round the turn are seen. (b) Between the pole P and the focus F. (c) m = −v/u = +1/3, so v = −u/3 = −(−30)/3 = +10 m (behind the mirror). (d) Virtual (v positive, m positive).

Q38. 4M Board fav In a lab, Meera got a sharp image of a distant tree on a screen kept 20 cm from a lens. (a) Name the lens and find its focal length. (b) Find its power. (c) She now places a candle 30 cm from the same lens; find v. (d) State the nature of this image. (e) What happens to the image if she covers the lower half of the lens?
Ans: (a) Convex lens; a distant object images at F, so f = +20 cm. (b) P = 1/0.20 m = +5 D. (c) u = −30 cm: 1/v = 1/f + 1/u = 1/20 − 1/30 = 1/60, v = +60 cm. (d) Real, inverted and enlarged (m = v/u = −2). (e) The full image still forms, only dimmer, because the upper half still refracts rays from every point of the candle.

Q39. 4M A torch maker wants a strong parallel beam, and a hotel wants a mirror at a blind corridor turn. (a) Which mirror does each need? (b) Where is the torch bulb placed? (c) State the nature of the image seen in the corridor mirror.
Ans: (a) Torch: concave mirror; corridor: convex mirror. (b) At the principal focus F of the concave mirror, so the reflected rays go out parallel. (c) Virtual, erect and diminished, with a wide field of view.

The Human Eye and the Colourful World



Most asked: myopia and hypermetropia with correction, dispersion through a prism, twinkling of stars, blue sky and red sunset

The eye works like a camera with a **natural convex lens**. Here we study how it focuses, its defects and their correction, and the effects of light in nature: **rainbow, twinkling stars, blue sky**.

1. The Human Eye

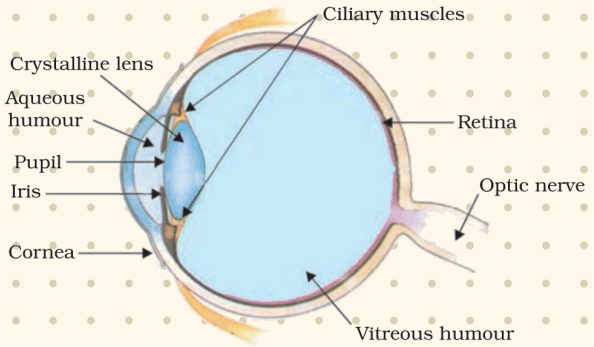


Fig. 10.1 The human eye

The eyeball is nearly spherical, about **2.3 cm** in diameter. Light enters through a thin transparent membrane, the **cornea**; most of the refraction of light happens at its outer surface.

- → **Iris:** dark muscular diaphragm behind the cornea that controls the size of the pupil.
- → **Pupil:** regulates the amount of light entering the eye: **small in bright light**, large in dim light.
- → **Eye lens:** crystalline convex lens that makes the **fine adjustment of focal length**; forms a real, inverted image on the retina.
- → **Retina:** delicate membrane full of light-sensitive cells; they generate electrical signals.
- → **Optic nerve:** carries these signals to the **brain**, which interprets them so we see objects as they are.

Yaad karo: path of light

"Chacha Pappu Laddoo Rakh Ke Office Bhaage": Cornea → Pupil → Lens → Retina → Optic nerve → Brain.

2. Power of Accommodation

The **ciliary muscles** change the curvature, and hence the focal length, of the eye lens.

Looking at	Ciliary muscles	Eye lens	Focal length
Distant object	Relax	Thin	Increases
Nearby object	Contract	Thick	Decreases

Accommodation: the ability of the eye lens to adjust its focal length. The **near point** (least distance of distinct vision) of a normal young adult is about 25 cm; the **far point** is infinity. So a normal eye sees clearly from 25 cm to infinity.

Exam Alert

- → Object door ho ya paas, **image distance (lens se retina) SAME** rehta hai; sirf focal length badalti hai.
- → Focal length ek minimum se kam nahi ho sakti, isliye **25 cm se paas** saaf nahi dikhta.

Cataract: in old age the crystalline lens may become **milky and cloudy**, causing partial or complete loss of vision; vision is restored by **cataract surgery**, in which the cloudy lens is replaced by an artificial lens.

Quick fact

Coming from bright sunlight into a dark cinema hall we cannot see clearly for a while: the pupil, which had become small, takes some time to **widen** and let in more light. The iris controls this change.

Eye donation

- → Anyone of any age can donate, even people who wear spectacles or have had cataract surgery.
- → Eyes must be removed within **4 to 6 hours** of death.
- → One donated pair helps up to **four corneal-blind people**.

3. Defects of Vision and their Correction

1. Myopia (near-sightedness)

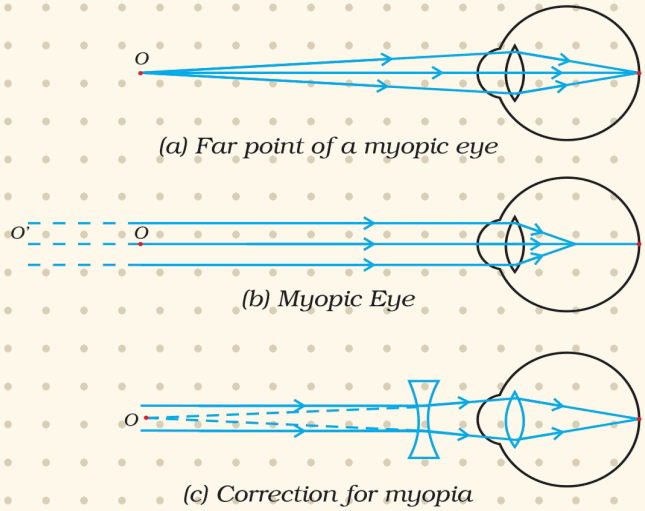


Fig. 10.2 Myopia: (a) far point (b) myopic eye (c) correction

- → Nearby objects are seen clearly, **distant objects are blurred**; the far point comes nearer than infinity.
- → Image of a distant object forms in front of the retina.
- → **Causes:** (i) excessive curvature of the eye lens (ii) elongation of the eyeball.
- → **Correction:** a **concave (diverging) lens** of suitable power brings the image back onto the retina.

Meme samjho



Concave lagaun ya convex? Bas yaad rakho: door wala nahi dikhta toh concave, paas wala nahi dikhta toh convex.

Asli baat: myopia mein image retina ke aage banti hai, concave lens rays ko thoda phaila kar retina par laata hai.

2. Hypermetropia (far-sightedness)

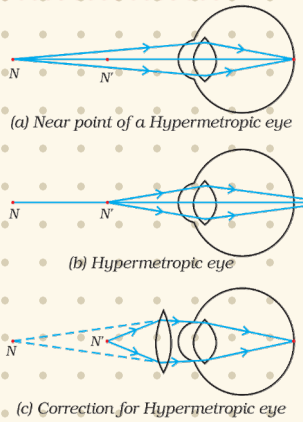


Fig. 10.3 Hypermetropia and its correction

- → Distant objects are seen clearly, **nearby objects are blurred**; the near point is farther than 25 cm.
- → Image of a nearby object forms behind the retina.
- → **Causes:** (i) focal length of the eye lens is too long (ii) the eyeball has become too small.
- → **Correction:** a **convex (converging) lens** of suitable power.

Paheli-Bujho



Bujho: Board nahi dikhta, toh convex chashma lagaunga!

Paheli: Ulta! Door nahi dikhta = myopia. Image retina ke aage, toh concave lens.

Bujho: Paas nahi dikhe toh convex, door nahi dikhe toh concave. Done!

3. Presbyopia

With age the power of accommodation decreases and the near point recedes, so nearby objects are hard to see without glasses. **Cause:** gradual weakening of the ciliary muscles and reduced flexibility of the eye lens. **Correction:** convex lens. A person with both myopia and hypermetropia uses **bi-focal lenses**: upper part **concave** (distant vision), lower part **convex** (reading). Today defects can also be corrected with contact lenses or surgery.

Defect	Cannot see	Image forms	Lens
Myopia	Distant objects	In front of retina	Concave
Hypermetropia	Nearby objects	Behind retina	Convex
Presbyopia	Nearby (old age)	Behind retina	Convex / bi-focal

Example 1. The far point of a myopic person is 80 cm in front of the eye. Find the nature and power of the lens needed.

The lens must form the image of a distant object ($u = -\infty$) at the far point, $v = -80$ cm. $1/f = 1/v - 1/u = -1/80$, so $f = -80$ cm $= -0.8$ m. $P = 1/f = -1.25$ D: a concave lens.

Example 2. The near point of a hypermetropic eye is 1 m. What lens lets the person read at 25 cm?

Object at $u = -25$ cm must image at the near point $v = -100$ cm. $1/f = 1/v - 1/u = -1/100 + 1/25 = 3/100$, $f = +33.3$ cm. $P = 1/0.333$ m $= +3$ D: a convex lens.

4. Refraction through a Glass Prism

A **prism** has two triangular bases and three rectangular lateral surfaces. The angle between its two refracting faces is the **angle of the prism** ($\angle A$).

Activity 10.1: Path of light through a glass prism

Done: Trace a glass prism, fix pins P and Q on a line inclined to face AB, locate pins R and S in line with their images through face AC and join the points. **Observed:** The ray bends towards the normal on entering (air to glass) and away from the normal on leaving (glass to air). The emergent ray is **not parallel** to the incident ray.

Conclusion: Because the faces are not parallel, the emergent ray is bent at an angle to the incident ray: the **angle of deviation** ($\angle D$).

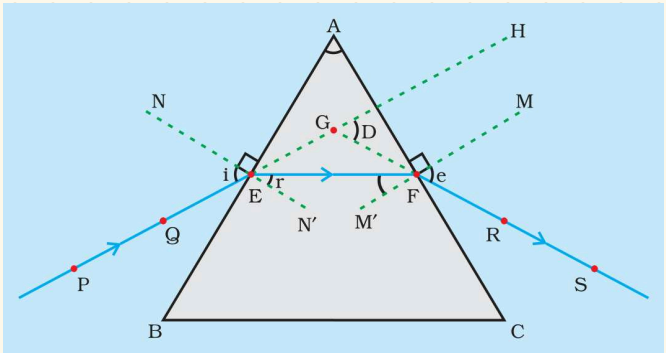


Fig. 10.4 Refraction of light through a triangular glass prism

Key terms (Fig. 10.4)

- → PE incident ray, EF refracted ray, FS emergent ray.
- → $\angle i$, $\angle r$, $\angle e$: angles of incidence, refraction and emergence.
- → $\angle A$ angle of the prism, $\angle D$ angle of deviation.

5. Dispersion of White Light

Activity 10.2: Dispersion of white light by a prism

Done: Pass a narrow beam of sunlight from a hole in a cardboard sheet through a glass prism and turn the prism until the light falls on a screen. **Observed:** A band of **seven colours** appears: violet, indigo, blue, green, yellow, orange, red (**VIBGYOR**). **Conclusion:** Each colour bends through a different angle: **red bends the least, violet the most.**

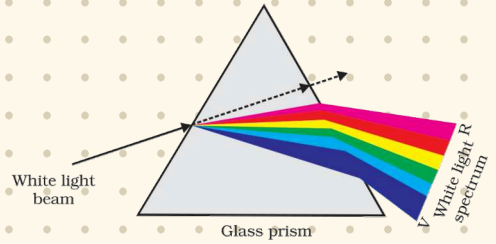


Fig. 10.5 Dispersion of white light by a glass prism

- **Spectrum:** the band of coloured components obtained when a beam of light is split.
- **Dispersion:** the splitting of white light into its seven component colours by a prism.
- **White light:** any light that gives a spectrum similar to that of sunlight, like the light from the Sun itself.

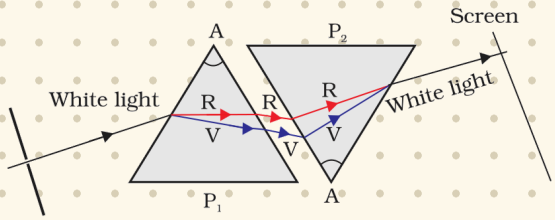


Fig. 10.6 Recombination of the spectrum of white light

Isaac Newton first obtained the spectrum of sunlight with a glass prism. Placing a second identical prism **upside down**, he got white light back from the spectrum, showing that sunlight is made up of seven colours.

Yaad karo: VIBGYOR

VIBGYOR: Violet sabse zyada mudta hai, Red sabse kam. Red ki wavelength sabse badi, isliye usko mudne mein sabse zyada aalas!

6. Rainbow

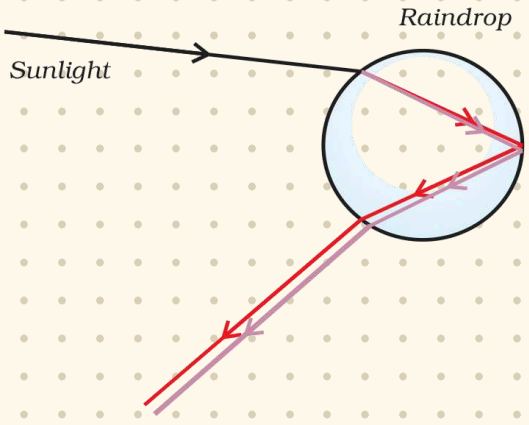


Fig. 10.7 Rainbow formation in a raindrop

A **rainbow** is a natural spectrum seen in the sky after a rain shower, always in the direction opposite to the Sun.

Tiny water droplets act like small prisms:

Refraction + dispersion (entering) → Internal reflection → Refraction (leaving) → Colours reach the eye

A rainbow can also be seen on a sunny day through a waterfall or a fountain, with the Sun behind us.

7. Atmospheric Refraction

Refraction of light by the Earth's atmosphere, whose layers have different optical densities, is **atmospheric refraction**. Objects seen through the hot air above a fire seem to **waver** for the same reason.

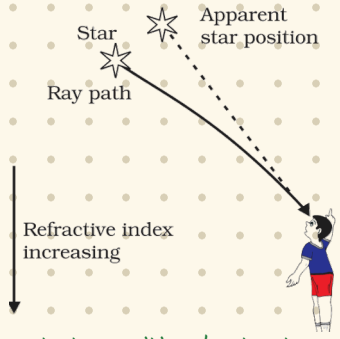


Fig. 10.8 Apparent star position due to atmospheric refraction

Twinkling of stars: starlight bends towards the normal as it enters the atmosphere, so a star appears **slightly higher** than its actual position. As atmospheric conditions keep changing, the apparent position and the amount of light entering the eye flicker, so the star twinkles.

Planets do not twinkle: they are much closer and appear as **extended sources** (a collection of point sources); the variations average out to zero.

Paheli-Bujho



Bujho: Taare timtimate hain kyunki humein aankh maarte hain!

Paheli: Nahi! Hawa ki badalti layers starlight ko baar-baar modti hain, toh chamak hilti hai.

Bujho: Aur planets bade disc jaise hain, toh unka flicker cancel. Samajh gaya!

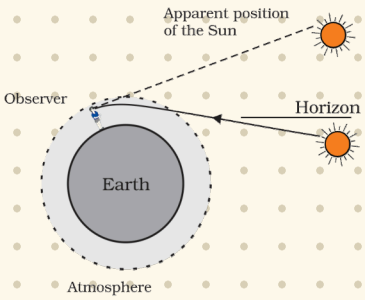


Fig. 10.9 Advance sunrise and delayed sunset

Advance sunrise and delayed sunset: because of atmospheric refraction the Sun is visible about **2 minutes before** actual sunrise and about **2 minutes after** actual sunset. The same effect makes the Sun's disc look flattened at sunrise and sunset.

8. Scattering of Light

Tyndall effect: scattering of light by colloidal particles (smoke, dust, water droplets) makes the path of a beam visible, e.g. sunlight entering a smoky room through a small hole or a dense forest canopy. Very fine particles scatter mainly **blue** light; larger ones scatter longer wavelengths; still larger ones look **white**.

Meme samjho



Aasmaan blue samundar ki wajah se nahi hai, hawa ke chhote molecules blue ko sabse zyada bikherte hain!

Asli baat: air molecules light ki wavelength se chhote hain, isliye short wavelength (blue) zyada scatter hoti hai.

Why is the sky blue? Air molecules and fine particles are smaller than the wavelength of visible light, so they scatter **blue light** (shorter wavelength) much more strongly than red (red has a wavelength about **1.8 times** that of blue).

Without an atmosphere: there would be no scattering, so the sky would look **dark**, as it does to astronauts flying at very high altitudes in space.

Reddish Sun at sunrise and sunset: near the horizon sunlight crosses a much **thicker layer of air**, so most of the blue is scattered away and mainly red reaches us. At noon the Sun is overhead, so its light crosses less air and the Sun looks white.

Danger signals are red: red light is scattered the least by fog or smoke, so it can be seen in the same colour from far away.

1 minute recap

Eye: real image on retina → Myopia: concave →
Hypermetropia: convex → Prism: VIBGYOR →
Twinkling: refraction → Blue sky: scattering

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. 1M Board fav The human eye can focus on objects at different distances by adjusting the focal length of the eye lens. This is due to:
(a) presbyopia (b) myopia
(c) accommodation (d) hypermetropia
Ans: (c) accommodation.

Q2. 1M The human eye forms the image of an object at its:
(a) cornea (b) iris (c) pupil (d) retina
Ans: (d) retina.

Q3. 1M Board fav The least distance of distinct vision for a young adult with normal vision is about:
(a) 25 m (b) 2.5 cm (c) 25 cm (d) 2.5 m
Ans: (c) 25 cm.

Q4. 1M The change in focal length of the eye lens is caused by the action of the:
(a) pupil (b) retina
(c) ciliary muscles (d) iris
Ans: (c) ciliary muscles.

Q5. 1M Which colour of white light deviates the most on passing through a prism?
Ans: Violet (it has the shortest wavelength); red deviates the least.

Q6. 1M Board fav **Assertion (A):** The sky appears blue in the daytime. **Reason (R):** Blue light has a shorter wavelength and is scattered more by the fine particles of air.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q7. 1M **Assertion (A):** A rainbow is always formed opposite to the Sun. **Reason (R):** Light is reflected internally inside the raindrops.
Ans: (a) Both A and R are true, and R is the correct explanation of A (the light turns back towards the observer after internal reflection).

Q8. 1M Why does the Sun appear reddish at sunrise?
Ans: Near the horizon sunlight passes through a thicker layer of air; most blue light is scattered away and mainly red reaches our eyes.

Short answer (2–3 marks)

Q9. 2M Board fav What is meant by power of accommodation of the eye? What are the near point and far point of a normal eye?
Ans: The ability of the eye lens to adjust its focal length to see near and far objects clearly. Near point: about 25 cm. Far point: infinity.

Q10. 2M Board fav Name the part of the eye that (a) controls the size of the pupil (b) changes the focal length of the eye lens (c) carries signals to the brain.
Ans: (a) Iris (b) ciliary muscles (c) optic nerve.

Q11. 3M Board fav A student cannot read the blackboard from the last bench. Name the defect, give two causes and say how it is corrected.
Ans: Myopia (near-sightedness). Causes: excessive curvature of the eye lens; elongation of the eyeball. Corrected with a concave lens of suitable power, which forms the image of distant objects at the far point.

Q12. 2M Board fav The far point of a myopic eye is 1.2 m. Find the nature and power of the correcting lens.
Ans: $u = -\infty$, $v = -1.2$ m. $1/f = 1/v - 1/u = -1/1.2$, $f = -1.2$ m. $P = 1/f = -0.83$ D: a concave lens.

Q13. 2M Why do older people often need bi-focal lenses? What does each part do?
Ans: In old age the ciliary muscles weaken and the eye lens loses flexibility (presbyopia); often myopia is also present. Upper part: concave, for distant vision. Lower part: convex, for reading.

Q14. 2M Why is a normal eye not able to see clearly the objects placed closer than 25 cm?
Ans: The focal length of the eye lens cannot be decreased below a minimum limit, so the image of a very close object cannot be focused on the retina.

Q15. 2M Board fav What happens to the image distance in the eye when we increase the distance of an object from the eye?
Ans: Nothing: the image distance (lens to retina) stays the same. The ciliary muscles relax and the eye lens becomes thinner, increasing its focal length.

Q16. 3M Board fav Draw a labelled diagram of refraction through a glass prism and mark the angle of deviation.
Ans: Draw the prism ABC; incident ray PE bends towards the normal at E (refracted ray EF), and bends away from the normal at F (emergent ray FS). Mark ∠i, ∠r, ∠e, ∠A at the top vertex, and ∠D between the extended incident ray and the emergent ray (Fig. 10.4).

Q17. 3M Board fav Why do stars twinkle but planets do not?
Ans: Starlight undergoes continuous atmospheric refraction; a star is a point source, so its apparent position and brightness flicker. Planets are much closer and act as extended sources; the variations from their many points average out, so they do not twinkle.

Q18. 2M Explain why the sky appears dark to an astronaut at very high altitude.
Ans: At such heights there is almost no atmosphere, so sunlight is not scattered; no scattered light reaches the eye and the sky looks dark.

Q19. 2M Board fav Why are danger signal lights red in colour?
Ans: Red light has the longest wavelength and is scattered the least by fog, smoke and dust, so it can be seen in the same colour from a long distance.

Q20. 2M What is the Tyndall effect? Give two examples from daily life.
Ans: Scattering of light by colloidal particles that makes the path of the beam visible. Examples: a beam of sunlight in a smoky or dusty room entering through a small hole; sunlight passing through a dense forest canopy.

Q21. 2M Board fav A person needs a lens of power −5.5 D for distant vision. Find its focal length and name the defect it corrects.
Ans: f = 1/P = 1/(−5.5) = −0.182 m = −18.2 cm: a concave lens, which corrects myopia.

Long answer (5 marks)

Q22. 5M Board fav (a) Name the defect of vision in which a person can see nearby objects clearly but not distant ones. (b) Give two causes. (c) Draw ray diagrams of (i) the defective eye (ii) its correction. (d) A person with this defect has a far point of 80 cm; find the power of the lens needed.
Ans: (a) Myopia. (b) Excessive curvature of the eye lens; elongation of the eyeball. (c) (i) Rays from a distant object meet in front of the retina. (ii) A concave lens diverges the rays so that they appear to come from the far point and meet on the retina (Fig. 10.2). (d) v = −80 cm, u = −∞: f = −80 cm = −0.8 m, P = −1.25 D (concave).

Q23. 5M Board fav (a) What is dispersion of white light? What is its cause? (b) Describe Newton's experiment showing that white light is made of seven colours. (c) Explain with a diagram how a rainbow is formed.
Ans: (a) Splitting of white light into its seven colours; each colour bends through a different angle in the prism (red least, violet most). (b) A second identical prism placed upside down in the path of the spectrum recombines the colours into white light. (c) Raindrops act as tiny prisms: sunlight is refracted and dispersed on entering, reflected internally and refracted again on leaving, so the colours reach the observer, who has the Sun behind (Fig. 10.7).

Q24. 5M (a) Name the defect in which a person cannot see nearby objects clearly. Give its causes. (b) The near point of such an eye is 1 m; find the power of the lens needed to read at 25 cm. (c) Name the lens.
Ans: (a) Hypermetropia: focal length of the eye lens too long; eyeball too small. (b) u = −25 cm, v = −100 cm: 1/f = −1/100 + 1/25 = 3/100, f = +33.3 cm = 0.333 m, P = +3 D. (c) Convex (converging) lens.

Case-based (4 marks)

Q25. 4M Board fav Rohan, 50, holds his newspaper at arm's length to read but sees road signs clearly. His son, 15, sits in the front row because the board looks blurred. (a) Name the defect of each. (b) Which lens does each need? (c) Why does Rohan's defect arise? (d) Which lens can correct both defects in one pair of spectacles?
Ans: (a) Rohan: presbyopia; son: myopia. (b) Rohan: convex lens; son: concave lens. (c) Ciliary muscles weaken and the eye lens loses flexibility with age, so the near point recedes. (d) A bi-focal lens: upper concave part for distance, lower convex part for reading.

Q26. 4M During a school trip, students saw a rainbow in the spray of a waterfall in the morning. (a) In which direction was the Sun? (b) Name the three processes in a drop. (c) Which colour was at the top of the rainbow? (d) Could they see it facing the Sun?
Ans: (a) Behind the students (rainbow opposite to the Sun). (b) Refraction with dispersion, internal reflection, refraction. (c) Red (violet at the bottom). (d) No: a rainbow is seen only with the Sun behind the observer.

ELECTRICITY



Most asked: Ohm's law, series/parallel numericals, $R = \rho l/A$, Joule's heating, power and kW h

Fans, lights, chargers and geysers all run on **electricity**. Here we learn what makes charges flow, what controls the current, how resistors combine, and how current gives heat and power.

1. Electric Charge and Current

Charge: the property of matter due to which it shows electric effects. A body is negatively charged by an excess of electrons and positively charged by a deficiency. SI unit: **coulomb (C)**. Charge on an electron, $e = -1.6 \times 10^{-19}$ C. Like charges repel, unlike charges attract.

$$Q = n\,e \quad \text{(charge is quantised)}$$

Key terms

- → **Current (I):** charge flowing per second; $I = Q/t$.
- → **Potential difference (V):** work per unit charge; $V = W/Q$.
- → **Resistance (R):** opposes flow of charge; $R = V/I$, in ohm.
- → **Resistivity (ρ):** R of unit length and unit area; Ω m.
- → **Power (P):** rate of using electric energy; $P = VI$.

1 C is the charge of about 6.25×10^{18} electrons (NCERT: nearly 6×10^{18}). **Conductors** (copper, aluminium) let charges flow easily; **insulators** (glass, rubber, plastic) do not.

Electric current (I): the rate of flow of charge through any cross-section of a conductor. In metal wires the moving charges are electrons.

$$I = Q/t \quad \text{or} \quad Q = I\,t$$

SI unit: ampere (A). **1 A** = flow of 1 coulomb of charge per second. 1 mA = 10^{-3} A, 1 µA = 10^{-6} A.

Electric circuit: a continuous and closed path of electric current. A switch makes it a **closed circuit** (ON) or an **open circuit** (OFF).

Conventional current flows from the **positive to the negative terminal** outside the cell, opposite to the flow of electrons (electrons were unknown when the convention was made).

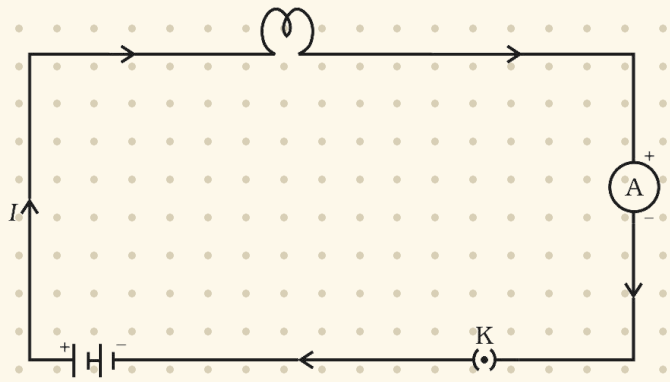


Fig. 11.1: Cell, bulb, ammeter and plug key

Ammeter: measures current; always connected in **series**; very **low resistance**, so it does not change the current it measures.

Solved: current and charge

Given: I = 0.5 A, t = 10 min = 600 s
Formula: $Q = I\,t$
Solution: $Q = 0.5 \times 600 = \mathbf{300\ C}$. Electrons in 1 C: $n = Q/e = 1/(1.6 \times 10^{-19}) = \mathbf{6.25 \times 10^{18}}$.

Solved: heat from charge (NCERT)

Given: Q = 96000 C, V = 50 V, t = 1 h
Formula: $H = V\,Q$
Solution: $H = 50 \times 96000 = \mathbf{4.8 \times 10^{6}}$ J. (Heat depends on V and Q only; the time is not needed.)

2. Electric Potential and Potential Difference

Water flows only when there is a pressure difference; charges flow only when there is an **electric pressure difference**, the potential difference, made by a cell through chemical action.

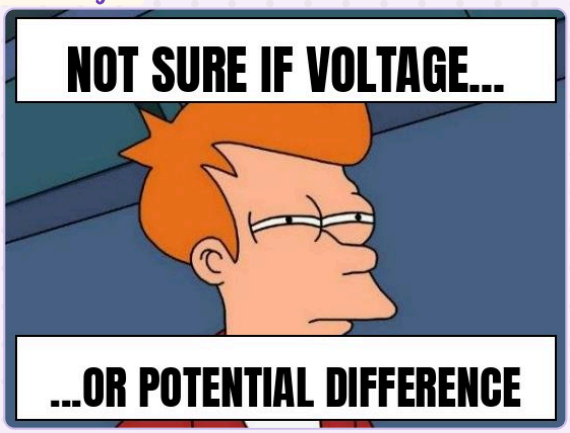
Electric potential at a point: work done to bring a unit positive charge from infinity to that point.

Potential difference (V): the work done to move a unit charge from one point to the other in a current-carrying circuit.

$$V = W/Q \qquad 1\,V = 1\,J/1\,C$$

SI unit: volt (V). **1 volt**: 1 joule of work is done to move 1 coulomb of charge from one point to the other. **Voltmeter:** measures p.d.; always connected in **parallel**; very **high resistance**.

Meme samjho



Voltage bolein ya potential difference? Dono ek hi cheez hain, unit bhi ek: volt!

Asli baat: potential difference (voltage) = work done per unit charge, $V = W/Q$; measured by a voltmeter in parallel.

Solved: work done

Given: Q = 2 C, V = 12 V
Formula: $W = V\,Q$
Solution: $W = 12 \times 2 = \mathbf{24\ J}$. (A 6 V battery gives 6 J to every coulomb.)

Yaad karo: meter connection

"A-S, V-P": Ammeter Series mein (Amma saath mein), Voltmeter Parallel mein (Voltu paas mein khada).

3. Circuit Diagram: Symbols

A **circuit diagram** shows each component by its standard symbol (NCERT Table 11.1). These are asked directly.

Component	Symbol	Component	Symbol
Electric cell		Wire joint	
Battery		Wires crossing	
Key (open)		Electric bulb	
Key (closed)		Resistor	
Ammeter		Rheostat	
Voltmeter			

Exam Alert

→ Cell ke symbol mein **lambi patli line = positive (+)**, chhoti moti line = negative (−).

→ Circuit draw karte waqt ammeter series mein aur voltmeter resistor ke across lagao; current ka arrow + se − ki taraf dikhao.

4. Ohm's Law

Ohm's law (Georg Simon Ohm, 1827): the potential difference across the ends of a metallic wire is directly proportional to the current through it, provided its temperature remains the same.

$$V \propto I \Rightarrow V/I = R \Rightarrow \mathbf{V = IR}$$

Activity 11.1: Verifying Ohm's law

Done: A 0.5 m nichrome wire XY was joined to an ammeter (series), a voltmeter (across XY) and 1, 2, 3, then 4 cells; V and I were noted each time. **Observed:** V/I came out nearly the same every time; the V-I graph is a **straight line through the origin**. **Conclusion:** V ∝ I for a given wire at constant temperature (Ohm's law).

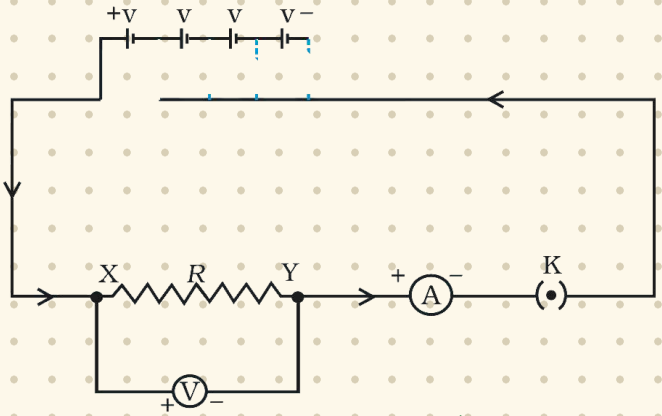


Fig. 11.2: Circuit to verify Ohm's law

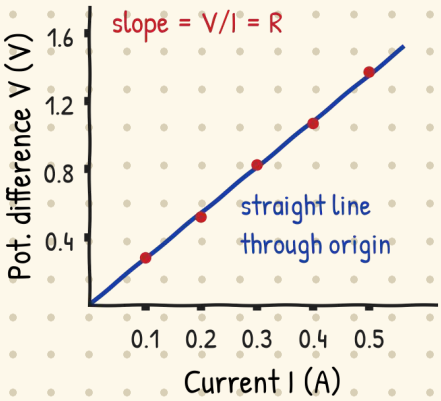


Fig. 11.3: V–I graph for a nichrome wire

Resistance (R): the property of a conductor to **oppose the flow of charges**. SI unit: **ohm (Ω)**. **1 ohm**: a p.d. of 1 V across a conductor drives a current of 1 A (1 Ω = 1 V/1 A). Since I = V/R, current is inversely proportional to resistance.

Why resistance? atoms attract the moving electrons and slow them down. Low resistance: good conductor; appreciable resistance: **resistor**; very high: insulator. A **rheostat** changes the current without changing the source voltage.

Activity 11.2: Different components

Done: Four cells and an ammeter were joined in series leaving a gap XY. A nichrome wire, a torch bulb and a 10 W bulb were put in the gap one by one. **Observed:** The ammeter reading was different each time. **Conclusion:** Different components offer different resistance.

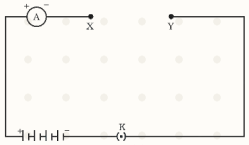


Fig. 11.4

Quick fact

Slope of the V-I graph = R: the **steeper** the line, the higher the resistance. A graph that bends means the temperature (and so R) is changing, e.g. a bulb filament.

Solved: Ohm's law

Given: V = 220 V; bulb R = 1200 Ω, heater R = 100 Ω
Formula: I = V / R
Solution: bulb I = 220/1200 = **0.18 A**; heater I = 220/100 = **2.2 A**. A heater taking 4 A at 60 V has R = 15 Ω, so at 120 V it takes 120/15 = **8 A**.

Solved: bulb filament

Given: a bulb draws I = 0.5 A from V = 220 V
Formula: R = V / I
Solution: Bulb filament: R = 220/0.5 = **440 Ω**. A filament this hot must not melt, hence tungsten.

5. Factors on which Resistance Depends

Resistance of a uniform metallic conductor is **(i)** directly proportional to its **length** (R ∝ l), **(ii)** inversely proportional to its **area of cross-section** (R ∝ 1/A), and **(iii)** depends on the **nature of the material** (and its temperature).

$$R \propto l/A \Rightarrow \mathbf{R = \rho l / A}$$

Resistivity (ρ): the resistance of a conductor of unit length and unit area of cross-section. SI unit: **Ω m**. It is a property of

the material: it does not change with length or thickness, but changes with temperature.

Activity 11.3: Length, thickness and material

Done: A nichrome wire was put in gap XY; then one of double length, then a thicker one of the same length, then a copper wire of the same size. **Observed:** Double length: current halved. Thicker wire: current increased. Copper: much more current. **Conclusion:** R increases with length, decreases with area of cross-section and depends on the material.

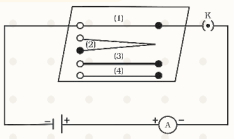


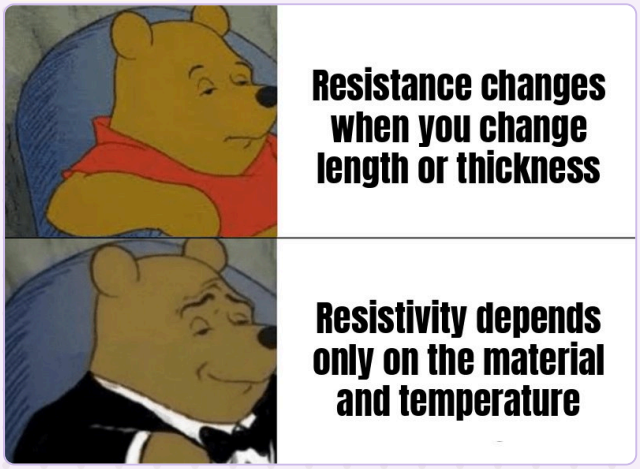
Fig. 11.5

Exam Alert: two V–I graphs

→ Do resistors ke V–I graph diye hon (V on y-axis): jiska **slope zyada**, uska R zyada. Agar I y-axis par ho, toh ulta: steeper line = **kam R**.

→ Graph ke axes, units aur origin se guzarti seedhi line zaroor dikhana.

Meme samjho



Wire lambi-moti karne se resistance badalta hai, par resistivity? Woh toh nawab hai, sirf material aur temperature ki sunta hai!

Asli baat: $R = \rho l/A$ depends on length and area; ρ depends only on the material and temperature.

Material	ρ (Ω m) at 20°C	Material	ρ (Ω m)
Silver	1.60×10^{-8}	Constantan	49×10^{-6}
Copper	1.62×10^{-8}	Manganin	44×10^{-6}
Aluminium	2.63×10^{-8}	Nichrome	100×10^{-6}
Tungsten	5.20×10^{-8}	Glass	$10^{10} - 10^{14}$
Iron	10.0×10^{-8}	Hard rubber	$10^{13} - 10^{16}$
Mercury	94.0×10^{-8}	Ebonite	$10^{15} - 10^{17}$

→ Metals and alloys: 10^{-8} to 10^{-6} Ω m; insulators: 10^{12} to 10^{17} Ω m (values need not be memorised).

→ An **alloy** has higher resistivity than its constituent metals and **does not oxidise (burn) easily** at high temperature, so **nichrome** is used in irons, toasters and heaters.

→ **Tungsten**: bulb filaments. **Copper, aluminium** (lowest ρ): transmission and household wires.

Solved: resistivity

Given: $l = 1$ m, $d = 0.3$ mm $= 3 \times 10^{-4}$ m, $R = 26$ Ω
Formula: $\rho = R A / l$, $A = \pi d^2/4$
Solution: $A = 3.14 \times 9 \times 10^{-8}/4 = 7.07 \times 10^{-8}$ m^2; $\rho = 26 \times 7.07 \times 10^{-8}/1 =$ **1.84×10^{-6} Ω m** (manganese). Diameter doubled: area $\times 4$, so $R = 26/4 =$ **6.5 Ω**.

Solved: wire stretched to double length

Given: $R = 10$ Ω, $l \to 2l$ (volume same, so $A \to A/2$)
Formula: $R = \rho l / A$
Solution: $R' = \rho(2l)/(A/2) = 4\rho l/A =$ **$4R = 40$ Ω**. A wire cut into two equal halves: each half has **$R/2$**.

Exam Alert

→ ρ material ki property hai: wire kaato, khincho ya moti karo, **ρ same rahega**; sirf temperature se badlega.

→ Numericals mein mm ko m mein, cm^2 ko m^2 mein badalna mat bhoolna.

6. Resistors in Series

Resistors joined **end to end**, so that the same current flows through each, are in **series**.

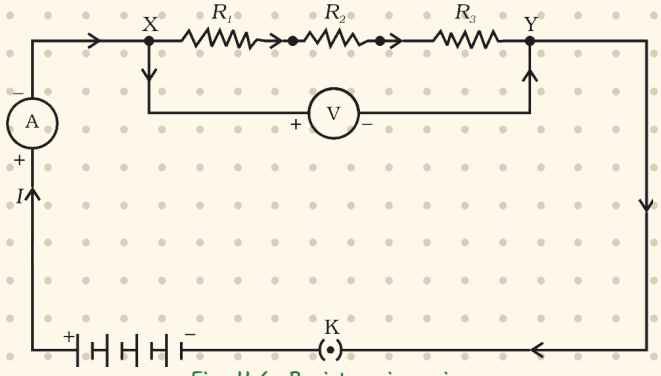


Fig. 11.6: Resistors in series

Activity 11.4: Current in series

Done: Resistors of 1, 2 and 3 Ω were joined in series with a 6 V battery, an ammeter and a key; the ammeter was then moved between the resistors. **Observed:** The ammeter reading stayed the same everywhere. **Conclusion:** In series, the **current is the same** in every part of the circuit.

Activity 11.5: Voltage in series

Done: A voltmeter was put across the whole combination (V), then across each resistor (V_1, V_2, V_3). **Observed:** $V = V_1 + V_2 + V_3$. **Conclusion:** In series, the total p.d. is the **sum** of the p.d.s across the resistors.

Derivation: let R_s be the equivalent resistance and I the common current. $V = V_1 + V_2 + V_3$, with $V = IR_s$, $V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$:

$$IR_s = IR_1 + IR_2 + IR_3 \Rightarrow \mathbf{R_s = R_1 + R_2 + R_3}$$

The equivalent resistance in series is greater than the largest individual resistance.

Solved: series circuit

Given: bulb 20 Ω and conductor 4 Ω in series, $V = 6$ V
Formula: $R_s = R_1 + R_2$; $I = V/R_s$
Solution: $R_s = 24$ Ω; $I = 6/24 =$ **0.25 A**;
$V_1 = 0.25 \times 20 =$ **5 V**, $V_2 = 0.25 \times 4 =$ **1 V**; total 6 V.

7. Resistors in Parallel

Resistors joined between the **same two points**, so that each has the same potential difference across it, are in **parallel**.

Activity 11.6: Current in parallel

Done: Three resistors were joined in parallel between X and Y with a battery, key and ammeter. Total current I was noted, then the current in each branch; a voltmeter was put across the combination. **Observed:** $I = I_1 + I_2 + I_3$; the same V across each resistor. **Conclusion:** In parallel, total current = sum of branch currents; p.d. is the same across each.

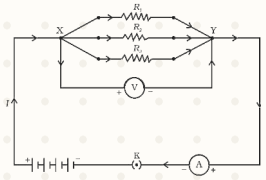


Fig. 11.7

Derivation: V is common; $I = I_1 + I_2 + I_3$ gives $V/R_p = V/R_1 + V/R_2 + V/R_3$:

$$1/R_p = 1/R_1 + 1/R_2 + 1/R_3$$

R_p is less than the smallest individual resistance. For two resistors: $R_p = R_1R_2/(R_1 + R_2)$. So 1 Ω in parallel with 10^6 Ω gives just under 1 Ω.

Solved: parallel circuit

Given: 5 Ω, 10 Ω, 30 Ω in parallel, V = 12 V
Formula: $I = V/R$; $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$
Solution: I_1 = 2.4 A, I_2 = 1.2 A, I_3 = 0.4 A, total **4 A**; $1/R_p = 10/30$, R_p = **3 Ω** (= 12 V/4 A).

Solved: appliances on 220 V

Given: lamp 100 Ω, toaster 50 Ω, filter 500 Ω in parallel on 220 V
Formula: $1/R_p = \Sigma 1/R$; $I = V/R_p$
Solution: $1/R_p = 1/100 + 1/50 + 1/500 = 16/500$, R_p = **31.25 Ω** (an iron of this R draws the same current); $I = 220/31.25$ = **7.04 A**.

Meme samjho



Gru ka plan: bulbs series mein jodo... R badha, current ghata, bulbs dhime!

Asli baat: in series $R_s = R_1 + R_2 + R_3$ increases, so $I = V/R_s$ decreases.

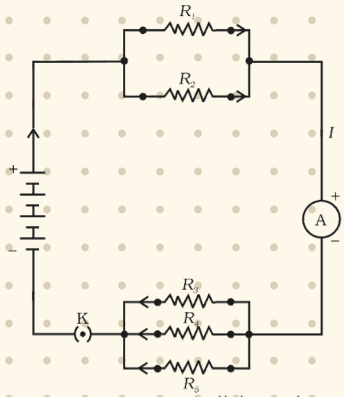


Fig. 11.12: Series + parallel combination

Solved: combination (NCERT Ex. 11.9)

Given: R_1 = 10 Ω ‖ R_2 = 40 Ω, in series with R_3 = 30 Ω, R_4 = 20 Ω, R_5 = 60 Ω in parallel; V = 12 V
Formula: reduce each parallel group, then add in series
Solution: $1/R' = 1/10 + 1/40 = 5/40$, R' = 8 Ω; $1/R'' = 1/30 + 1/20 + 1/60 = 6/60$, R'' = 10 Ω. R = 8 + 10 = **18 Ω**; I = 12/18 = **0.67 A**.

Series	Parallel
Same current through each	Same p.d. across each
$V = V_1 + V_2 + V_3$	$I = I_1 + I_2 + I_3$
$R_s = R_1 + R_2 + R_3$ (increases)	$1/R_p = \Sigma 1/R$ (decreases)
One device fails, all stop	One fails, others keep working

Paheli-Bujho



Bujho: Diwali wali fairy lights mein ek bulb fuse hua aur poori ladi band! Light hi chali gayi kya?
Paheli: Nahi re, wo bulbs series mein hain. Ek toota toh circuit hi open, current kahin nahi jaayega.
Bujho: Toh ghar mein ek bulb fuse hone pe fan kyun chalta rehta hai?
Paheli: Kyunki ghar ki wiring parallel hai. Har appliance ka apna raasta, poore 220 V ke saath.
Bujho: Aur taar pe baithi chidiya ko current kyun nahi lagta?
Paheli: Uske dono pair ek hi taar pe hain, beech mein potential difference almost zero. V nahi toh I nahi!

Why houses use parallel wiring: (1) every appliance gets the full 220 V; (2) each draws the current it needs (a bulb and a heater need very different currents); (3) each has its own switch and if one fails the rest still work; (4) the total resistance is lower.

8. Heating Effect of Electric Current

A source keeps spending energy to maintain current. Part may do useful work (a fan rotating); the rest becomes heat. In a **purely resistive** circuit all the energy is dissipated as heat: the heating effect of electric current.

Derivation: current I flows through R for time t, carrying charge Q across p.d. V. Work done = VQ, so power P = VQ/t = VI and energy in time t = VIt, all converted to heat:

$$H = VIt \quad \text{with } V = IR: \quad H = I^2Rt = V^2t/R$$

Joule's law of heating: the heat produced in a resistor is directly proportional to **(i)** the square of the current ($H \propto I^2$), **(ii)** the resistance ($H \propto R$) and **(iii)** the time for which current flows ($H \propto t$).

Yaad karo: heat ka hisaab

Current **double** karo toh heat **4 guna** (I^2)! R double ya time double karo toh heat sirf **double**.

Solved: Joule heating

Given: H = 100 J each second (t = 1 s), R = 4 Ω
Formula: $H = I^2Rt$, $V = IR$
Solution: $I^2 = 100/(4 \times 1) = 25$, I = 5 A; V = 5 × 4 = **20 V**. An iron of 20 Ω taking 5 A for 30 s: H = 25 × 20 × 30 = **15000 J**.

Applications: electric iron, toaster, oven, kettle and room heater. Heating is often wasteful too: it can damage components.

Electric bulb: the filament is **tungsten** (melting point 3380°C), thermally isolated by insulating supports; the bulb is filled with inactive **nitrogen and argon** to prolong the filament's life. Most of the power becomes heat, only a little light.

Electric fuse: a safety device in **series** with the appliance; a wire of a metal or alloy of suitable **low melting point** (aluminium, copper, iron, lead; commonly tin-lead alloy) in a porcelain cartridge. Too much current heats it, it melts and breaks the circuit. Ratings: 1, 2, 3, 5, 10 A. A 1 kW iron at 220 V draws 4.54 A, so a **5 A fuse** is used.

Quick fact

The heater element glows but its cord does not: nichrome has a high R, so I^2R heat there is large; the copper cord has tiny R, so the same current makes very little heat in it.

9. Electric Power

Electric power (P): the rate at which electric energy is consumed or dissipated in a circuit.

$$P = VI = I^2R = V^2/R$$

SI unit: watt (W). **1 W** = power used by a device carrying 1 A at 1 V (1 W = 1 V A). 1 kW = 1000 W, 1 MW = 10^6 W, 1 hp ≈ 746 W.

Electrical energy = power × time (E = Pt). Commercial unit: the **kilowatt hour (kW h)**, one "unit" on the electricity bill.

$$1\text{ kW h} = 1000\text{ W} \times 3600\text{ s} = 3.6 \times 10^6\text{ J}$$

More to know: electrons are **not consumed** in a circuit; we pay for the **energy** used to move them through our appliances.

Paheli-Bujho



Bujho: Mera bill 200 unit aaya, matlab mere ghar ki power 200 kW h hai!
Paheli: Galat! kW h power ki nahi, energy ki unit hai. Power watt mein hoti hai, energy = power × time.
Bujho: 1 unit = kitne joule?
Paheli: 1 kW h = 1000 W × 3600 s = 3.6×10^6 J. Aur bill electrons ka nahi, energy ka aata hai.

Solved: electricity bill

Given: fridge P = 400 W = 0.4 kW, 8 h a day, 30 days, Rs 3.00 per kW h
Formula: E = P × t; cost = units × rate
Solution: E = 0.4 × 8 × 30 = 96 kW h; cost = 96 × 3 = **Rs 288**.

Solved: bulb on half voltage

Given: bulb 100 W, 220 V; used on 110 V
Formula: $R = V^2/P$ (R stays fixed)
Solution: R = 220 × 220/100 = **484 Ω**; on 110 V, P = $110^2/484$ = **25 W**. Since $P \propto V^2$, half the voltage gives one-fourth the power.

Solved: energy: TV or toaster? (NCERT)

Given: TV 250 W for 1 h; toaster 1200 W for 10 min
Formula: E = P × t
Solution: TV: 250 × 3600 = **9×10^5 J**; toaster: 1200 × 600 = **7.2×10^5 J**. The TV uses more energy.

Solved: motor: power and energy

Given: I = 5 A, V = 220 V, t = 2 h
Formula: P = VI; E = P t
Solution: P = 220 × 5 = **1100 W**; E = 1.1 kW × 2 h = **2.2 kW h** = **7.92×10^6 J**.

Solved: two lamps in parallel (NCERT)

Given: lamps 100 W and 60 W, in parallel on 220 V
Formula: I = P/V for each; total $I = I_1 + I_2$
Solution: I = 100/220 + 60/220 = 160/220 = **0.73 A**.

Board favourite numericals (quick results)

1. Wire R cut into **n equal parts**, all joined in parallel: R′ = **R/n^2** (5 parts: R/25).
2. Hot plate, two 24 Ω coils on 220 V: one coil 220/24 = **9.17 A**; series (48 Ω) **4.58 A**; parallel (12 Ω) **18.33 A**.
3. 4, 8, 12, 24 Ω: highest = series = **48 Ω**; lowest = parallel, $1/R_p = 12/24$, **2 Ω**.
4. Same material, length l/2 and area 2A: R′ = ρ(l/2)/(2A) = **R/4** (NCERT Ex. 11.6 type).
5. Two identical wires on the same V: series 2R, parallel R/2; H = V^2t/R gives $H_s : H_p$ = **1 : 4**.
6. 2 Ω, 3 Ω, 6 Ω: 4 Ω = 2 + (3 ∥ 6); all three in parallel = 1 Ω.
7. Parallel count: 10 W lamps on 220 V with 5 A allowed: 5 ÷ (10/220) = **110**; 176 Ω resistors for 5 A: **4**.

Exam Alert

→ Har numerical: **Given, Formula, Solution**, aur last mein **unit ke saath boxed answer** (step marking isi par).

→ Pehle units SI mein (min → s, W → kW for bills). **kW h = energy, W = power**: ye galti mat karna.

Formula sheet

Quantity	Formula	Unit
Charge, current	$Q = ne$; $I = Q/t$	C, A
Potential difference	$V = W/Q$	V
Ohm's law	$V = IR$	Ω
Resistance of wire	$R = \rho l/A$	ρ in Ω m
Series	$R_s = R_1 + R_2 + R_3$	Ω
Parallel	$1/R_p = 1/R_1 + 1/R_2 + 1/R_3$	Ω
Joule's law	$H = I^2Rt = VIt = V^2t/R$	J
Power	$P = VI = I^2R = V^2/R$	W
Energy	$E = Pt$; 1 kW h = 3.6×10^6 J	J, kW h

NCERT in-text questions (quick answers)

→ **What does an electric circuit mean?** A continuous and closed path of electric current.

→ **Name a device that helps maintain a p.d. across a conductor.** A cell or battery.

→ **A wire of the same material, which is thicker or thinner, has less R?** The thicker one ($R \propto 1/A$).

→ **Why are coils of toasters and irons made of an alloy?** High resistivity and no oxidation at high temperature.

→ **Why is a series arrangement not used for domestic circuits?** Appliances need different currents, all would get less than 220 V and one fault would switch all off.

→ **How does the resistance of a wire vary with its area of cross-section?** Inversely: $R \propto 1/A$.

Common mistakes

→ **R_p nikal ke $1/R_p$ hi likh diya**: last step mein ulta karna mat bhoolo.

→ Power ke liye galat formula: same V par devices ho (parallel) toh **V^2/R**, same I ho (series) toh **I^2R** use karo.

→ Fuse hamesha **series** mein aur **live wire** mein lagta hai.

1 minute recap

$Q = It$ → $V = W/Q$ → $V = IR$ → $R = \rho l/A$ → Series / parallel → $H = I^2Rt$ → $P = VI$, 1 kW h = 3.6 MJ

Most Important Questions

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM Board fav The SI unit of resistivity is:
(a) Ω (b) Ω/m (c) Ω m (d) Ω m^2
Ans: (c) Ω m

Q2. IM Board fav A wire of resistance R is cut into five equal parts and these are joined in parallel. If the equivalent resistance is R′, then R/R′ is:
(a) 1/25 (b) 1/5 (c) 5 (d) 25
Ans: (d) 25 (each part R/5; five in parallel give R/25).

Q3. IM Board fav Which of these does NOT represent electric power?
(a) I^2R (b) IR^2 (c) VI (d) V^2/R
Ans: (b) IR^2. Power = I^2R = VI = V^2/R.

Q4. IM Board fav An electric bulb is rated 220 V and 100 W. When it is operated on 110 V, the power consumed will be:
(a) 100 W (b) 75 W (c) 50 W (d) 25 W
Ans: (d) 25 W ($P \propto V^2$; R = 484 Ω stays the same).

Q5. IM How is a voltmeter connected in a circuit to measure the p.d. between two points?
Ans: In **parallel** across the two points.

Q6. IM 1 kW h = ____ J
Ans: 3.6×10^6 J

Q7. IM Board fav **Assertion (A):** Alloys are used in electric heating devices. **Reason (R):** Alloys have high resistivity and do not oxidise (burn) readily at high temperature.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q8. IM **Assertion (A):** In series, the current through each resistor is the same. **Reason (R):** The equivalent resistance in series is less than the smallest resistance.
Ans: (c) A is true but R is false: R_s is more than the largest R.

Q9. IM The resistance of a conductor is directly proportional to its:
(a) area of cross-section (b) length
(c) density (d) melting point
Ans: (b) length

Q10. IM Electric potential difference is measured by a ____ connected in ____.
Ans: Voltmeter; parallel.

Q11. IM Board fav What is the resistance of a 100 W, 220 V bulb?
Ans: $R = V^2/P = 48400/100 = 484$ Ω.

Q12. IM Two wires of the same material and length; A is twice as thick (diameter) as B. Which has more resistance and how much more?
Ans: B: area of A is 4 times, so $R_B = 4R_A$.

Q13. IM Which of the following is used for heating elements of electric irons?
(a) copper (b) tungsten (c) nichrome (d) silver
Ans: (c) nichrome (high resistivity, does not oxidise).

Q14. IM The direction of conventional current is:
(a) along the flow of electrons
(b) opposite to the flow of electrons
Ans: (b) opposite to electron flow (+ to − outside the cell).

Q15. IM Name the SI unit of electric charge and state how many electrons make 1 C.
Ans: Coulomb (C); about 6.25×10^{18} electrons.

Q16. 1M Two resistors of 4 Ω and 12 Ω are connected in parallel. Their equivalent resistance is:
(a) 3 Ω (b) 8 Ω (c) 16 Ω (d) 48 Ω
Ans: (a) 3 Ω ($4 \times 12/16$).

Q17. 1M In an electric circuit, an ammeter has ____ resistance and is connected in ____.
Ans: Very low; series.

Q18. 1M **Assertion (A):** A fuse wire is always connected in parallel with the appliance. **Reason (R):** A fuse wire has a low melting point.
Ans: (d) A is false but R is true; a fuse goes in series.

Short answer (2–3 marks)

Q19. 2M Board fav Define 1 volt. How much work is done in moving a charge of 2 C across two points having a potential difference of 12 V?
Ans: 1 V: when 1 J of work is done to move 1 C of charge from one point to the other. $W = VQ = 12 \times 2 =$ **24 J**.

Q20. 3M Board fav State Ohm's law. Draw the V–I graph for a metallic conductor. What does its slope give?
Ans: At constant temperature, the p.d. across a metallic wire is directly proportional to the current through it ($V = IR$). The graph is a straight line through the origin; slope (V/I) = resistance R.

Q21. 3M On what factors does the resistance of a conductor depend? Why are copper and aluminium wires used for transmission of electricity?
Ans: On length ($R \propto l$), area of cross-section ($R \propto 1/A$), nature of material and temperature: $R = \rho l/A$. Copper and aluminium have very low resistivity, so very little energy is lost as heat in the wires.

Q22. 2M Why is tungsten used almost exclusively for filaments of electric lamps? Why are bulbs filled with nitrogen and argon?
Ans: Tungsten has a very high melting point (3380°C) and glows at high temperature without melting. Inactive N_2/Ar prevent oxidation of the hot filament and prolong its life.

Q23. 3M Board fav How can three resistors of 2 Ω, 3 Ω and 6 Ω be connected to give a total resistance of (a) 4 Ω (b) 1 Ω?
Ans: (a) 3 Ω and 6 Ω in parallel (= 2 Ω) in series with 2 Ω: total 4 Ω. (b) All three in parallel: $1/R = 1/2 + 1/3 + 1/6 = 1$, $R = 1$ Ω.

Q24. 3M Board fav What are the advantages of connecting electrical devices in parallel with the battery instead of in series?
Ans: Each device gets the full voltage; each draws the current it needs; each can be switched on/off independently; if one fails the others work; total resistance is less.

Q25. 3M Board fav An electric iron consumes 840 W at the highest setting and 360 W at the lowest, on 220 V. Find the current and resistance in each case.
Ans: Max: $I = P/V = 840/220 =$ **3.82 A**, $R = V/I =$ **57.6 Ω**. Min: $I = 360/220 =$ **1.64 A**, $R =$ **134.4 Ω**.

Q26. 3M Compare the power used in a 2 Ω resistor in each case: (i) a 6 V battery in series with 1 Ω and 2 Ω (ii) a 4 V battery in parallel with 12 Ω and 2 Ω.
Ans: (i) $I = 6/3 = 2$ A, $P = I^2R = 4 \times 2 =$ **8 W**. (ii) V across 2 Ω = 4 V, $P = V^2/R = 16/2 =$ **8 W**. Same power in both.

Q27. 3M Why does the cord of an electric heater not glow while the heating element does? What is an electric fuse and how is it connected?
Ans: Element (nichrome) has high resistance, so much I^2Rt heat; the copper cord has very low resistance, so little heat. A fuse is a low-melting-point wire in series with the appliance; excess current melts it and breaks the circuit.

Q28. 2M Why is the series arrangement not used for domestic circuits? Why are fuse wires connected in the live wire?
Ans: In series, appliances share the voltage, need different currents and all go off if one fails. The fuse is in the live wire so that when it melts the appliance is cut off from the high potential.

Q29. 2M Board fav How many 176 Ω resistors (in parallel) are required to carry 5 A on a 220 V line?
Ans: Each draws $220/176 = 1.25$ A, so $5/1.25 =$ **4 resistors** in parallel ($R_p = 176/4 = 44$ Ω $= 220/5$).

Q30. 3M Board fav Show how you would connect three resistors, each of 6 Ω, so that the combination has a resistance of (i) 9 Ω (ii) 4 Ω.
Ans: (i) Two in parallel (3 Ω) in series with the third: $3 + 6 =$ **9 Ω**. (ii) Two in series (12 Ω) in parallel with the third: $12 \times 6/18 =$ **4 Ω**.

Q31. 3M Board fav Draw a schematic diagram of a circuit consisting of a battery of three 2 V cells, a 5 Ω resistor, an 8 Ω resistor, a 12 Ω resistor and a plug key, all in series. Find the current and the p.d. across the 12 Ω resistor.
Ans: Draw cells, key and the three resistors in one loop with an ammeter in series and a voltmeter across 12 Ω. $R = 25$ Ω, $V = 6$ V, $I = 6/25 =$ **0.24 A**; $V_{12} = 0.24 \times 12 =$ **2.88 V**.

Q32. 2M Why do we use copper and aluminium wires for transmission, but nichrome for heaters? Give one reason each.
Ans: Copper/aluminium have very low resistivity, so little energy is wasted as heat in long wires. Nichrome has high resistivity and does not oxidise at high temperature, so it produces much heat safely.

Q33. 2M Board fav A piece of wire of resistance 20 Ω is drawn out so that its length is increased to twice its original length. Calculate the resistance of the wire in the new situation.
Ans: Volume constant: $l \to 2l$, $A \to A/2$. $R' = \rho(2l)/(A/2) = 4R =$ **80 Ω**.

Q34. 2M Why is an ammeter connected in series and why should it have very low resistance?
Ans: In series, the whole current of the circuit passes through it. Its resistance must be very low so that it does not change the current it measures.

Q35. 2M Which uses more energy: a 250 W TV set in 1 hour or a 1200 W toaster in 10 minutes?
Ans: TV: $250 \times 3600 = 9 \times 10^5$ J. Toaster: $1200 \times 600 = 7.2 \times 10^5$ J. The **TV** uses more energy.

Q36. 3M What is meant by electric power? Express 1 kW h in joules. A bulb draws 0.5 A from a 220 V line: find its power.
Ans: Rate at which electric energy is consumed. 1 kW h = $1000 \text{ W} \times 3600 \text{ s} = \mathbf{3.6 \times 10^6}$ J. $P = VI = 220 \times 0.5 = \mathbf{110}$ W.

Q37. 2M Board fav Two conducting wires of the same material, equal length and equal diameter are connected first in series and then in parallel across the same p.d. Find the ratio of heat produced.
Ans: Series $2R$, parallel $R/2$; $H = V^2t/R$, so $H_s : H_p = (1/2R) : (2/R) = \mathbf{1:4}$.

Q38. 2M Why is the series arrangement used in decorative fairy lights but not for house wiring? What happens if one bulb of such a string fuses?
Ans: Series is cheap and divides the mains voltage among many tiny bulbs. If one bulb fuses, the circuit breaks and the whole string goes off; in houses each appliance needs full 220 V and independent switching, so parallel is used.

Q39. 2M A bird sits on a single high-voltage wire and is not harmed. Why? What would happen if it touched two wires at once?
Ans: Both feet are on the same wire, so there is almost no potential difference across its body and no current flows. Touching two wires puts a large p.d. across it: current flows and it gets a fatal shock.

Long answer (5 marks)

Q40. 5M Board fav (a) Derive the expression for the equivalent resistance of three resistors in series. (b) A 20 Ω bulb and a 4 Ω conductor are in series with a 6 V battery. Find the total resistance, the current and the p.d. across each.
Ans: (a) Same I through each; $V = V_1 + V_2 + V_3$; $IR_s = IR_1 + IR_2 + IR_3$, so $R_s = R_1 + R_2 + R_3$. (b) $R_s = \mathbf{24\ \Omega}$; $I = 6/24 = \mathbf{0.25}$ A; $V_{bulb} = 0.25 \times 20 = \mathbf{5}$ V; $V_{conductor} = \mathbf{1}$ V.

Q41. 5M Board fav (a) Derive $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$. (b) A hot plate has two coils of 24 Ω each, on 220 V. Find the current when they are used (i) separately (ii) in series (iii) in parallel.
Ans: (a) Same V across each; $I = I_1 + I_2 + I_3$; $V/R_p = V/R_1 + V/R_2 + V/R_3$; dividing by V gives the result. (b) (i) $220/24 = \mathbf{9.17}$ A (ii) $220/48 = \mathbf{4.58}$ A (iii) $220/12 = \mathbf{18.33}$ A.

Q42. 5M (a) State Joule's law of heating and derive $H = I^2Rt$. (b) A 400 W refrigerator runs 8 hours a day. Find the cost of energy for 30 days at Rs 3.00 per kW h. (c) Why is the commercial unit kW h and not joule?
Ans: (a) $H \propto I^2$, R, t. Work to move Q across V is $VQ = VIt$; with $V = IR$, $H = I^2Rt$. (b) $E = 0.4 \times 8 \times 30 = 96$ kW h; cost = **Rs 288**. (c) The joule is too small for household energy use (1 kW h = 3.6×10^6 J).

Q43. 5M Board fav (a) A wire of length 1 m and diameter 0.3 mm has resistance 26 Ω. Find its resistivity. (b) What is the resistance if its diameter is doubled? (c) Why does resistivity not change in (b)?
Ans: (a) $A = \pi d^2/4 = 7.07 \times 10^{-8}$ m^2; $\rho = RA/l = \mathbf{1.84 \times 10^{-6}\ \Omega}$ m. (b) Area ×4, so $R = 26/4 = \mathbf{6.5\ \Omega}$. (c) Resistivity depends only on the material and temperature, not on the dimensions.

Case-based (4 marks)

Q44. 4M Board fav Riya joined resistors of 1 Ω, 2 Ω and 3 Ω in series with a 6 V battery and an ammeter. She moved the ammeter to different places in the circuit. (a) What does she observe? (b) Find the current. (c) Find the p.d. across the 3 Ω resistor. (d) If the 2 Ω resistor burns out, what happens to the other two?
Ans: (a) The ammeter reading is the same everywhere: current is the same in series. (b) $R = 6\ \Omega$, $I = 6/6 = \mathbf{1}$ A. (c) $V = IR = 1 \times 3 = \mathbf{3}$ V. (d) The circuit breaks, so no current flows through the others.

Q45. 4M A house has a 5 A fuse on its 220 V line. The family wants to run a 1 kW iron. (a) What current does the iron draw? (b) Is the 5 A fuse suitable? (c) What will happen if a 2 kW heater is also run on the same line? (d) Why is a fuse wire made of a low melting point alloy?
Ans: (a) $I = 1000/220 = \mathbf{4.54}$ A. (b) Yes, 4.54 A is below 5 A. (c) Total 3 kW draws 13.6 A, above 5 A: the fuse melts and the circuit breaks (overloading). (d) So that excess current heats and melts it quickly before wiring/appliances are damaged.

Q46. 4M Board fav Aman measured V and I for a nichrome wire: (0.2 A, 1.6 V), (0.4 A, 3.4 V), (0.6 A, 5.1 V), (0.8 A, 6.7 V). (a) What will the V–I graph look like? (b) Find the resistance from the data. (c) Which law does this verify and under what condition? (d) If a thicker nichrome wire of the same length is used, how will the graph change?
Ans: (a) A straight line passing through the origin. (b) $R = V/I \approx 8.4\ \Omega$ (about 8–8.5 Ω from each pair). (c) Ohm's law, provided the temperature of the wire stays the same. (d) Thicker wire has less resistance, so the line will be **less steep** (smaller slope V/I).

Q47. 4M Two bulbs, 60 W and 100 W, both rated 220 V, are connected in parallel to the 220 V mains. (a) Which bulb has more resistance? (b) Find the total current drawn. (c) Which glows brighter? (d) How many units are used if both run for 5 hours?
Ans: (a) $R = V^2/P$: the 60 W bulb (806.7 Ω) has more resistance than the 100 W bulb (484 Ω). (b) $I = 160/220 = \mathbf{0.73}$ A. (c) The 100 W bulb (more power at the same V). (d) $E = 0.16$ kW × 5 h = **0.8 kW h** (0.8 unit).

Yaad karo: power ka funda
Same 220 V par **kam R = zyada power** ($P = V^2/R$): isliye heater ka R bulb se kam hota hai. Series mein ulta: same I, toh **zyada R wala zyada garam** ($P = I^2R$).

Magnetic Effects of Electric Current



Most asked: properties of field lines, right-hand thumb rule, solenoid vs bar magnet, Fleming's left-hand rule, domestic circuits and safety

In the last chapter we saw the **heating effect** of current. Now the second one: a current-carrying wire acts like a **magnet**.

1. Magnetic Effect of Current

Oersted's discovery (1820): Hans Christian Oersted noticed by accident that a compass needle got deflected when an electric current passed through a nearby metallic wire. This proved that **electricity and magnetism are related**. The unit of magnetic field strength, the **oersted**, is named after him.

So a **moving charge (current) produces a magnetic field**, and a magnetic field in turn exerts a **force on a moving charge**. This chapter studies both effects, the rules that give their directions, and how electricity is used safely at home.

Activity 12.1: Compass near a current-carrying wire

Done: A small compass was kept near a thick copper wire XY joined to a cell, a key and a resistor; the key was plugged in. **Observed:** The compass needle got **deflected**. **Conclusion:** An electric current produces a **magnetic field** around the conductor (magnetic effect of current).

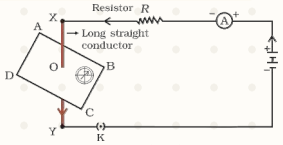


Fig. 12.1

Quick fact

Current wali wire = ek chhota sa magnet. Poora chapter isi ek line par chalta hai: current → magnetic field → force.

Key terms

- **Magnetic field:** region around a magnet where its force is felt; unit tesla (T).
- **Field line:** path along which a free north pole would move.
- **Solenoid:** closely wound coil of insulated copper wire.
- **Electromagnet:** soft iron core magnetised by a current-carrying solenoid.
- **Earthing:** joining an appliance's metal body to earth.

2. Magnetic Field and Field Lines

A compass needle is a **small bar magnet**. Its north-pointing end is the **north-seeking (N) pole**, the other end the **south-seeking (S) pole**. Like poles repel, unlike poles attract.

Magnetic field: the region around a magnet in which its force can be detected. It has both magnitude and direction. SI unit: **tesla (T)**.

Activity 12.2: Iron filings around a bar magnet

Done: Iron filings were sprinkled on a sheet placed over a bar magnet and the sheet was tapped gently. **Observed:** The filings arranged themselves in a pattern of curved lines from pole to pole. **Conclusion:** The magnet exerts a force in the region around it; the lines along which the filings align are **magnetic field lines**.

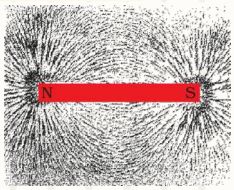


Fig. 12.2

Activity 12.3: Drawing field lines with a compass

Done: A compass was placed near the N pole, both needle ends marked; it was moved so its S end sat on the last mark, and repeated up to the S pole. The points were joined by a smooth curve. **Observed:** Curves run from N to S; the needle deflects more near the poles. **Conclusion:** These are the field lines of the bar magnet; the field is **strongest near the poles**.



Fig. 12.3

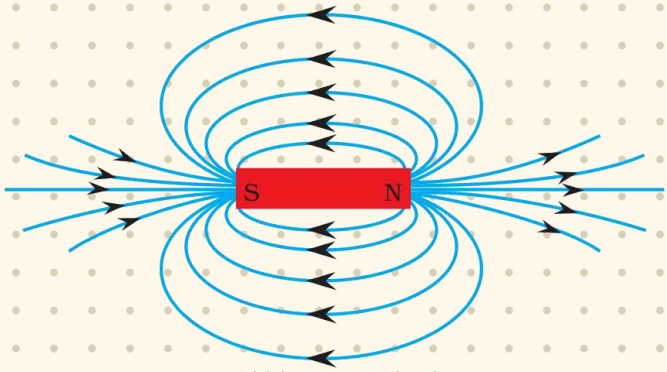


Fig. 12.4: Field lines around a bar magnet

Properties of magnetic field lines

Direction of field: the direction in which the **north pole of a compass** moves in it; the **tangent** at any point of a field line gives the field direction there.

- Outside the magnet they go from **N to S**; inside the magnet from **S to N**. So they are closed continuous curves.
- Where lines are **crowded** the field is **stronger** (near the poles); far apart means weaker.
- **No two field lines ever intersect**: a compass at the crossing point would have to point in two directions at once.
- Parallel, equally spaced lines mean a **uniform** field.

Meme samjho



Do field lines ke milne ka intezaar... skeleton ban gaye, par milan nahi hua!

Asli baat: field lines never intersect, because a compass cannot point in two directions at one point.

Exam Alert

→ "Why don't field lines intersect?" 2 marks pakka: likho **"two directions at one point"**.

→ Bar magnet ka diagram banao toh lines par **arrows N se S (bahar)** dikhana mat bhoolna.

3. Field due to a Current-carrying Conductor

(a) Straight conductor

The pattern of the field depends on the shape of the conductor. For a long straight wire it is a set of concentric circles centred on the wire, in a plane **perpendicular** to it.

Activity 12.4: Deflection reverses with current

Done: A straight wire was held over and parallel to a compass needle and connected to cells and a key; then the cells were reversed. **Observed:** Current north to south: the N pole moved **east**. Reversed current: it moved **west**. **Conclusion:** Reversing the current **reverses the direction** of the magnetic field.

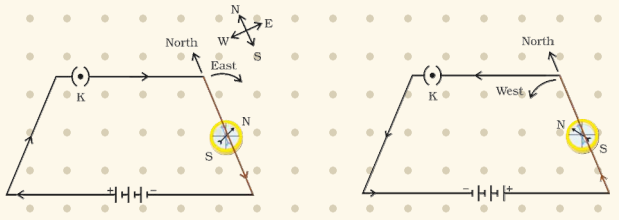


Fig. 12.5: Compass below a current-carrying wire

Activity 12.5: Field lines of a straight wire

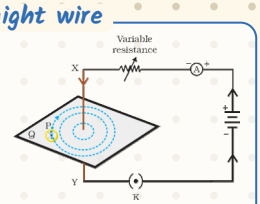


Fig. 12.6

Done: A thick copper wire was passed vertically through a cardboard, joined to a battery, rheostat, ammeter and key; iron filings were sprinkled and tapped; a compass was placed at P, then farther at Q. **Observed:** Filings form **concentric circles**. More current: more deflection; farther point: less deflection. **Conclusion:** The field **increases with current** and **decreases with distance** from the wire.

$$B \propto I \quad \text{and} \quad B \propto 1/r$$

Right-hand thumb rule

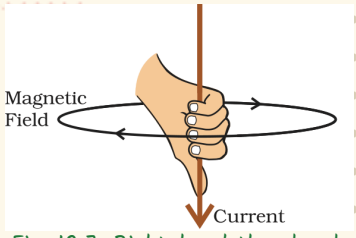


Fig. 12.7: Right-hand thumb rule

Hold the current-carrying straight conductor in your **right hand** so that the thumb points in the direction of current. Then the direction in which your **fingers curl** around the wire gives the direction of the **magnetic field lines**. Also called **Maxwell's corkscrew rule**.

Meme samjho



Field nikaalna hai? Right hand uthao, thumb current ki taraf, ungliyan bata dengi field!

Asli baat: right-hand thumb rule: thumb = current, curled fingers = field.

Solved: power line (NCERT Ex. 12.1)

Given: a horizontal power line carries current from **east to west**

Rule: right-hand thumb rule, thumb towards west

Solution: Field lines are circles in a vertical plane: **clockwise** seen from the east end, anticlockwise from the west end. Field is towards the **south below** the wire and towards the **north above** it.

Quick fact

Reverse the current and the curl of your fingers reverses too, so the field reverses: the compass needle under the wire now turns the other way (Activity 12.4).

(b) Circular loop

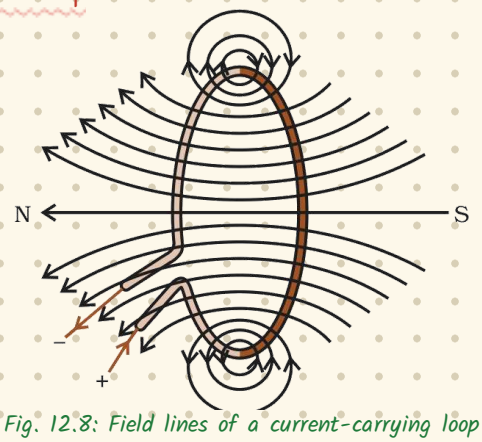


Fig. 12.8: Field lines of a current-carrying loop

→ Near the wire the lines are nearly concentric circles; at the **centre** of the loop the arcs look like **straight lines**.

→ Every part of the wire adds field in the **same direction** inside the loop.

→ Field at the centre **increases** with current and number of turns, and is **weaker for a larger radius**.

→ For a coil of **n turns**, the field is **n times** that of a single turn (currents in all turns are in the same direction).

Activity 12.6: Field of a circular coil

Done: A coil of many turns was passed through a cardboard, joined to a battery, key and rheostat; iron filings were sprinkled and tapped. **Observed:** Circles around each side of the coil, nearly straight lines at its centre. **Conclusion:** A coil's field is nearly straight and uniform at its centre.

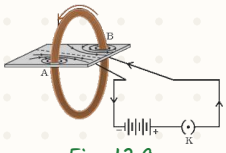


Fig. 12.9

(c) Solenoid

Solenoid: a coil of many circular turns of **insulated copper wire** wrapped closely in the shape of a cylinder.

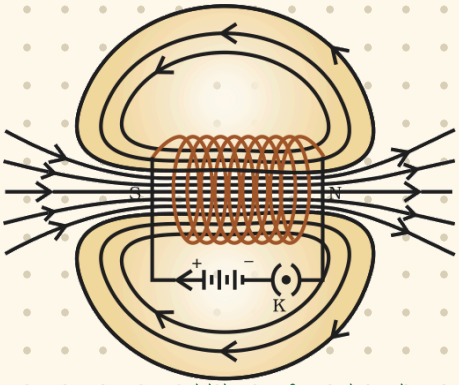


Fig. 12.10: Field lines of a solenoid

→ Its field pattern is just like that of a bar magnet: one end acts as N pole, the other as S pole.

→ Inside, the lines are **parallel straight lines**: the field is **uniform** (same at all points) inside.

→ Strength increases with the **number of turns**, the **current** and a **soft iron core**; reversing the current swaps the poles.

Conductor	Field pattern	Field increases with
Straight wire	Concentric circles	More I, nearer to the wire
Circular loop	Circles near wire, straight at centre	More I, more turns, smaller radius
Solenoid	Like a bar magnet; uniform inside	More I, more turns, soft iron core

Electromagnet

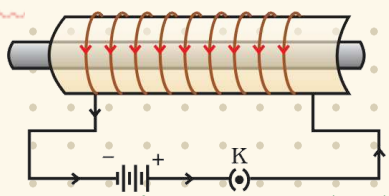


Fig. 12.11: Soft iron core in a solenoid

Electromagnet: a piece of **soft iron** placed inside a current-carrying solenoid gets strongly magnetised. Its magnetism lasts only while the current flows. Uses: electric bells, electric locks, cranes lifting scrap iron, relays, **MRI machines** and **magnetic separators**.

Paheli-Bujho



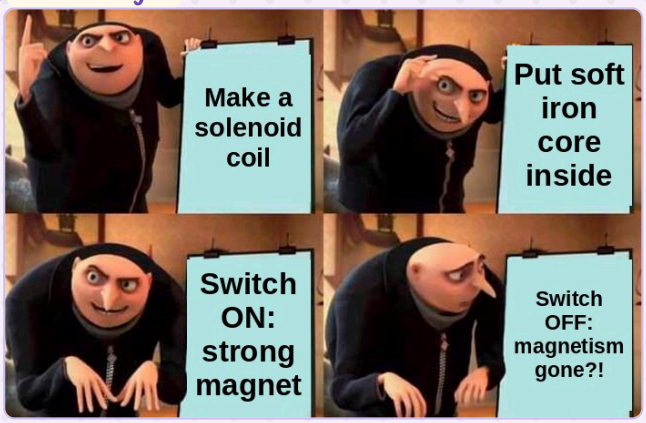
Bujho: Solenoid aur bar magnet mein koi farak hi nahi? Toh magnet hi le aate!

Paheli: Field pattern same hai, par solenoid ka magnetism current se on-off hota hai, aur current ya turns badha kar strength bhi badha sakte ho.

Bujho: Matlab solenoid ek remote-control wala magnet hai!

Electromagnet	Permanent magnet
Solenoid with a soft iron core	Steel or alloys like alnico
Temporary: only while current flows	Keeps its magnetism
Strength can be changed	Strength is fixed
Polarity can be reversed	Polarity is fixed

Meme samjho



Gru ka plan perfect tha, bas switch off karte hi magnet gayab! Yahi to electromagnet hai.

Asli baat: soft iron core is magnetised only while current flows; that is why electromagnets are used in cranes and bells.

4. Force on a Conductor in a Magnetic Field

A current exerts a force on a magnet, so **Andre Marie Ampere** (1775–1836) suggested that the magnet must exert an equal and opposite force on the conductor.

Activity 12.7: Rod in a horseshoe magnet

Done: A small aluminium rod AB was hung between the poles of a horseshoe magnet (field upwards) and current passed from B to A; then the current, then the poles, were reversed. **Observed:** The rod moved **left**; with reversed current or poles it moved **right**. **Conclusion:** A force acts on a current-carrying conductor in a field; its direction depends on both the **current** and the **field**.

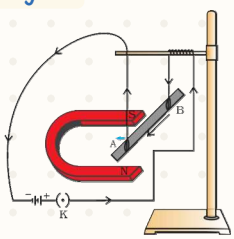


Fig. 12.12

Size of the force: largest when the current is at right angles to the field, zero when parallel. It increases with more current, a stronger magnet and a longer rod.

Fleming's left-hand rule

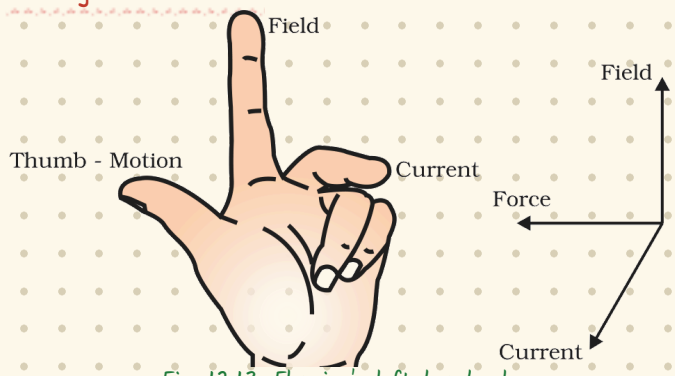


Fig. 12.13: Fleming's left-hand rule

Stretch the **thumb, forefinger and middle finger** of your **left hand** mutually perpendicular. If the **forefinger** points along the **field** and the **middle finger** along the **current**, the **thumb** gives the direction of force (motion) on the conductor.

> Fore = Field, Middle = Current, Thumb = Force

Fleming's right-hand rule (induced current): the same three fingers of the **right hand**; forefinger = field, thumb = motion, middle finger = **induced current**.

Devices using a current-carrying conductor in a field: electric motor, electric generator, loudspeakers, microphones and measuring instruments.

Exam Alert

→ Electron ya negative charge ka sawaal aaye toh pehle **current ki direction ulti** karo, phir left hand lagao.

→ Field = right hand (thumb rule); force = left hand (Fleming). Dono mix mat karna!

Solved: electron beam (NCERT Ex. 12.2)

Given: an electron enters a magnetic field at right angles (Fig. 12.14)

Rule: current is opposite to electron motion; then Fleming's left-hand rule

Solution: The force on the electron is **into the page**.

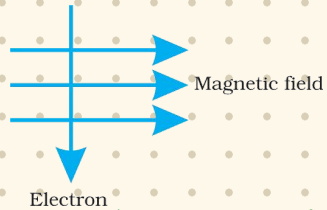


Fig. 12.14: Electron entering a field

Solved: alpha particle (NCERT Ex. 12.3)

Given: a positive alpha particle moving **west** is deflected **north**

Rule: middle finger (current) west, thumb (force) north

Solution: The forefinger (field) points **upward**. A magnetic force only turns a moving charge: its velocity and momentum change, its mass and speed do not.

Quick fact: magnetism in medicine

Weak ion currents in **nerve cells** make tiny magnetic fields; the **heart** and **brain** make significant ones. **MRI** (Magnetic Resonance Imaging) uses fields inside the body to image organs. A **galvanometer** detects current: its pointer deflects either side of zero.

Right-hand thumb rule	Fleming's left-hand rule
Gives direction of magnetic field	Gives direction of force (motion)
Thumb = current	Thumb = force
Curled fingers = field	Forefinger = field, middle = current
Straight wire, loop, solenoid	Conductor placed in a field (motor)

5. Direct and Alternating Current

DC (direct current): flows in **one direction only** (cells, batteries). **AC (alternating current): reverses its direction periodically.** In India AC has a frequency of **50 Hz**, so it changes direction every 1/100 s.

Advantage of AC over DC: AC can be transmitted over **long distances without much loss of energy.**

DC	AC
Flows in one direction	Reverses direction periodically
Cells, batteries	Power stations, mains (220 V, 50 Hz)
Large loss over long distances	Sent far with little loss

6. Domestic Electric Circuits

Power reaches homes through the **mains** (overhead poles or underground cables), then the **main fuse**, the **electricity meter** and the **main switch**, to the house wiring.

Wire	Insulation	Role
Live (phase)	Red (new: brown)	At 220 V, brings current
Neutral	Black (new: blue)	At zero potential, return path
Earth	Green	Safety, to a metal plate deep in the earth

Potential difference between live and neutral: **220 V**; AC frequency **50 Hz**.

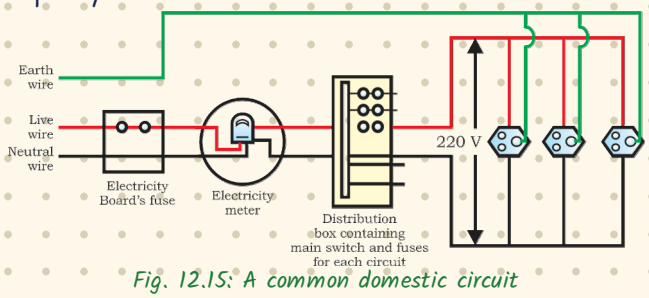


Fig. 12.15: A common domestic circuit

→ Two separate circuits: a **15 A** circuit for high-power appliances (geysers, air coolers, ovens) and a **5 A** circuit for bulbs and fans.

→ All appliances are connected in parallel: each gets the same 220 V and has its own switch.

→ Each circuit has its own **fuse**, so a fault blows only that fuse.

Earthing: the metal body of appliances (electric press, toaster, refrigerator) is joined to the earth wire, a **low-**

resistance path for leakage current. The body stays at earth potential, so the user does not get a **severe shock**.

Electric fuse: a thin wire of low melting point in **series** with the circuit. An unduly high current heats it (Joule heating), it melts and breaks the circuit, protecting the wiring and appliances. An **MCB** (miniature circuit breaker) trips instead and is simply reset.

Short-circuiting: live and neutral wires come in **direct contact** (damaged insulation or a faulty appliance); the resistance becomes nearly zero, so the current **increases heavily**.

Overloading: excess current due to **too many appliances on one socket**, an accidental **hike in supply voltage**, or a short circuit.

Safety measures

- ✓ Fuse or MCB in every circuit
- ✓ Earthing of metallic appliances
- ✓ Correct current rating (5 A / 15 A) for each circuit
- ✓ Never plug many appliances into one socket

Paheli-Bujho



Bujho: Short circuit aur overloading ek hi cheez hai na? Dono mein fuse udta hai.
Paheli: Nahi! Short circuit mein live aur neutral seedhe touch hote hain. Overloading mein ek socket par bahut appliances ya voltage hike se current badh jaata hai.
Bujho: Matlab short circuit overloading ka ek reason hai. Ab samajh aaya!

Solved: overloading (NCERT Ex. 12.4)

Given: oven 2 kW on a 220 V circuit rated 5 A
Rule: $I = P / V$
Solution: $I = 2000/220 = 9.1$ A, more than 5 A: the circuit is overloaded, the fuse blows (or the MCB trips). Use the 15 A circuit.

Exam Alert

- → Colour code: **Red/Brown = Live, Black/Blue = Neutral, Green = Earth.** "Green wire is live": False!
- → Fuse ki rating appliance ke current se thodi zyada ho: 1 kW iron (4.5 A) ke liye 5 A fuse.

Diagram tips

- → Field lines par **arrows** zaroor; solenoid ke ends par **N aur S** likho.
- → Domestic circuit mein **fuse live wire mein**, earth wire green, appliances parallel.

NCERT in-text questions (quick answers)

- → **Why does a compass needle get deflected near a bar magnet?** The magnet's field exerts force on the needle's poles.
- → **Draw field lines around a bar magnet.** Closed curves from N to S outside, S to N inside, crowded near poles.
- → **Which particles moving in a field experience a force?** Charged particles, such as electrons, protons and alpha particles.
- → **Name two safety measures in domestic circuits.** Fuse/MCB and earthing.
- → **Why are two field lines never found to cross?** A compass at that point would point in two directions.
- → **What precaution should be taken to avoid overloading?** Do not connect too many appliances to one socket.

1 minute recap

Oersted → Field lines (never cross) → Thumb rule → Loop, solenoid → Electromagnet → Fleming LHR → AC vs DC → Fuse, earthing

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. IM Board fav The magnetic field inside a long straight solenoid carrying current:
(a) is zero (b) decreases towards its end
(c) increases towards its end (d) is the same at all points
Ans: (d) is the same at all points (uniform field).

Q2. IM Board fav Which property of a proton can change while it moves freely in a magnetic field?
(a) mass (b) speed (c) velocity (d) none
Ans: (c) velocity (and momentum); mass and speed stay the same.

Q3. IM Inside a bar magnet, the magnetic field lines go from:
(a) N pole to S pole (b) S pole to N pole
(c) both ways (d) there are no lines inside
Ans: (b) S pole to N pole (outside, N to S).

Q4. IM Board fav The most important safety method used for protecting home appliances from short-circuiting or overloading is:
(a) earthing (b) use of a stabiliser
(c) use of fuse (d) use of electric meter
Ans: (c) use of fuse (or MCB).

Q5. IM Name the rule used to find the direction of the magnetic field around a straight current-carrying conductor.
Ans: Right-hand thumb rule (Maxwell's corkscrew rule).

Q6. IM What is the frequency of AC in India? After how much time does it change direction?
Ans: 50 Hz; it changes direction every 1/100 s.

Q7. IM Board fav **Assertion (A):** Magnetic field lines never intersect. **Reason (R):** At the point of intersection a compass needle would point in two directions, which is not possible.
Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q8. 1M **Assertion (A):** The green wire in house wiring is the live wire. **Reason (R):** The live wire is at a potential of 220 V.
Ans: (d) A is false but R is true; green is the earth wire.

Q9. 1M Name the SI unit of magnetic field.
Ans: Tesla (T).

Q10. 1M What is the potential difference between the live and neutral wires in Indian homes?
Ans: 220 V (AC, 50 Hz).

Q11. 1M A current-carrying rod is placed parallel to a magnetic field. What is the force on it?
Ans: Zero (force is maximum when the current is perpendicular to the field).

Q12. 1M The strength of the magnetic field at the centre of a circular loop decreases when:
(a) current is increased (b) radius is increased
(c) number of turns is increased (d) iron core is used
Ans: (b) the radius of the loop is increased.

Q13. 1M In Fleming's left-hand rule, the middle finger indicates the direction of:
(a) magnetic field (b) current (c) force (d) motion
Ans: (b) current.

Q14. 1M Board fav Which of the following correctly describes the magnetic field near a long straight wire?
(a) straight lines perpendicular to the wire
(b) straight lines parallel to the wire
(c) radial lines from the wire (d) concentric circles centred on the wire
Ans: (d) The field consists of concentric circles centred on the wire.

Short answer (2–3 marks)

Q15. 2M Board fav Why does a compass needle get deflected when brought near a bar magnet?
Ans: A compass needle is a small magnet. The magnetic field of the bar magnet exerts forces on its poles (like poles repel, unlike attract), which turn the needle.

Q16. 3M Board fav List the properties of magnetic field lines. Why don't two field lines ever intersect?
Ans: Go from N to S outside and S to N inside (closed curves); crowded where the field is strong; tangent gives direction; never intersect, because a compass at the crossing would point in two directions at once, which is impossible.

Q17. 3M Draw the field lines around a straight current-carrying conductor. How does the field change if (a) the current is increased (b) the point is moved away?
Ans: Concentric circles centred on the wire in a perpendicular plane, direction by the right-hand thumb rule. (a) Field increases. (b) Field decreases ($B \propto I$, $B \propto 1/r$).

Q18. 2M Board fav Consider a circular loop of wire lying in the plane of the table carrying a clockwise current. Find the direction of the field inside and outside the loop.
Ans: By the right-hand thumb rule: inside the loop the field is **into the table** (downward); outside the loop it is **out of the table** (upward).

Q19. 3M Board fav What is a solenoid? Draw its field lines. How is its field like that of a bar magnet?
Ans: A coil of many circular turns of insulated copper wire wound closely in a cylinder. Its field lines are like a bar magnet's: one end behaves as N, the other as S; inside, the lines are parallel (uniform field).

Q20. 2M State two differences between an electromagnet and a permanent magnet.
Ans: Electromagnet: temporary (only while current flows), strength and polarity can be changed. Permanent magnet: keeps its magnetism, strength and polarity fixed.

Q21. 3M Board fav State Fleming's left-hand rule. In Activity 12.7, how will the displacement of rod AB change if (a) the current is increased (b) a stronger magnet is used (c) the length of AB is increased?
Ans: Forefinger = field, middle finger = current (mutually perpendicular): thumb gives force. In all three cases the force, and so the displacement, **increases**.

Q22. 3M Board fav An electron beam moves horizontally from the back wall towards the front wall and is deflected to your right by a magnetic field. Find the field direction.
Ans: Current is opposite to the electrons: front to back. Thumb (force) to the right, middle finger (current) towards the back gives forefinger (field) **vertically downward**.

Q23. 2M Name four devices that use a current-carrying conductor in a magnetic field. Name two organs of the body where magnetic fields are significant.
Ans: Electric motor, electric generator, loudspeaker, microphone (and measuring instruments). Heart and brain.

Q24. 2M Board fav Why is it necessary to earth metallic electric appliances? What is the colour of the earth wire?
Ans: If the live wire touches the metal body due to leakage, current flows to the earth through the low-resistance earth wire, so the body stays at zero potential and the user gets no severe shock. Green.

Q25. 3M Board fav What is short-circuiting? What is overloading? Name a device that protects against both.
Ans: Short circuit: live and neutral wires touch, current rises heavily. Overloading: too many appliances on one socket or a voltage hike draws excess current. Fuse or MCB.

Q26. 2M Write one advantage of AC over DC. Name two sources of DC.
Ans: AC can be transmitted over long distances without much loss of energy. DC sources: dry cell, battery (storage cell).

Q27. 2M Why is a fuse wire always connected in the live wire and not in the neutral wire?

Ans: If the fuse is in the live wire, it disconnects the appliance from the 220 V supply when it melts. In the neutral wire the appliance would stay connected to the live wire and could give a shock.

Q28. 2M A magnetic compass is placed at the centre of a current-carrying circular loop. What will be the effect of increasing the current and the number of turns?

Ans: The magnetic field at the centre increases, so the compass needle deflects more (field ∝ current × number of turns).

Q29. 2M A current-carrying solenoid is suspended freely. In which direction will it come to rest, and why?

Ans: It comes to rest in the north-south direction, like a bar magnet, because its field pattern is like that of a bar magnet: one end acts as N pole, the other as S pole.

Q30. 2M What is an MCB? Give one advantage it has over a fuse wire.

Ans: A miniature circuit breaker: a switch that trips automatically and breaks the circuit when the current exceeds a safe value. Advantage: it is simply reset after the fault is removed; a fuse wire has to be replaced.

Q31. 2M A circular loop in the plane of the paper carries an anticlockwise current. Find the field direction at its centre.

Ans: By the right-hand thumb rule the field at the centre is **perpendicular to the paper, coming out** (towards the viewer).

Long answer (5 marks)

Q32. 5M Board fav (a) Describe an activity to show that a magnetic field exists around a current-carrying straight conductor and that its field lines are concentric circles. (b) How can its direction be found? (c) What happens to the field if the current is reversed?

Ans: (a) Pass a thick wire vertically through a cardboard, connect a battery, rheostat, ammeter and key; sprinkle iron filings and tap: they form concentric circles; a compass placed at a point shows the direction. (b) Right-hand thumb rule: thumb along current, curled fingers give the field. (c) The field direction reverses.

Q33. 5M Board fav (a) Draw a schematic diagram of a common domestic circuit with fuse, meter, main switch and earth wire. (b) Why are appliances connected in parallel? (c) Why are 15 A and 5 A circuits kept separate? (d) What is the function of a fuse?

Ans: (a) Mains → main fuse → meter → main switch → distribution box, with live, neutral and earth wires, each circuit with its own fuse, appliances in parallel. (b) Each gets 220 V, has its own switch and works independently. (c) High-power appliances draw large currents (15 A); bulbs and fans need less (5 A). (d) It melts when current is too high and breaks the circuit, protecting wiring and appliances.

Q34. 5M (a) What is an electromagnet? Draw a labelled diagram. (b) List three factors on which the strength of the field of a solenoid depends. (c) Give two uses of electromagnets.

Ans: (a) Soft iron core inside a current-carrying solenoid; magnetised only while current flows (diagram: coil on soft iron rod with battery and key). (b) Number of turns, current, nature of the core (soft iron). (c) Electric bells, cranes lifting scrap iron, MRI machines, magnetic separators.

Case-based (4 marks)

Q35. 4M Board fav Riya hung an aluminium rod between the poles of a horseshoe magnet and passed current through it. The rod moved to the left. (a) Name the rule that gives the direction of the force. (b) What happens if the current is reversed? (c) When is the force maximum? (d) Name one device that works on this principle.

Ans: (a) Fleming's left-hand rule. (b) The rod moves to the right (force reverses). (c) When the current is perpendicular to the field. (d) Electric motor (or loudspeaker).

Q36. 4M A family runs a 2 kW geyser, a 1 kW iron and four 60 W bulbs on a 220 V line that has a 5 A fuse. (a) Find the current drawn by the geyser alone. (b) Can all appliances run together on this line? (c) What will happen? (d) Suggest a safe arrangement.

Ans: (a) $I = 2000/220 = 9.1$ A. (b) No: total 3.24 kW draws 14.7 A, far above 5 A. (c) Overloading: the fuse melts (or MCB trips) and the circuit breaks. (d) Put the geyser and iron on a 15 A circuit and the bulbs on the 5 A circuit.

Q37. 4M Board fav A student drew the field lines of a bar magnet and of a current-carrying solenoid side by side. (a) What similarity will she see? (b) What is special about the field inside the solenoid? (c) How can the poles of the solenoid be reversed? (d) Why is soft iron used as the core?

Ans: (a) Both patterns are identical: one end acts as N pole and the other as S pole. (b) Inside, the field lines are parallel straight lines, so the field is uniform. (c) By reversing the direction of current. (d) Soft iron is magnetised strongly and loses its magnetism as soon as the current stops.

Q38. 4M Board fav Aman passed a thick copper wire vertically through a horizontal cardboard and sent current upward through it. He placed a compass at P, 2 cm from the wire, and then at Q, 6 cm away. (a) What is the shape of the field lines? (b) Clockwise or anticlockwise, seen from above? (c) Where is the deflection more, P or Q? (d) What happens if the current is doubled?

Ans: (a) Concentric circles centred on the wire. (b) Anticlockwise (right-hand thumb rule, thumb upward). (c) At P, since the field decreases with distance. (d) The field becomes stronger, so the deflection increases.

Yaad karo: kaunsa haath?

Right hand = field (thumb rule); **Left** hand = force on current. Yaad rakho: "Left se Laat (force)"!

Our Environment



Most asked: food chain and 10 percent law, biological magnification, ozone depletion, biodegradable vs non-biodegradable

Plants, animals, microbes, air, water and soil are all linked. This chapter shows how **energy flows** through living things and how our activities, like using CFCs and throwing garbage, change the environment.

1. Ecosystem and its Components

Environment: all the physical and biological surroundings in which organisms live. **Ecosystem**: all the interacting organisms of an area together with the non-living components of the environment (term coined by A.G. Tansley, 1935).

Biotic components	Abiotic components
Living organisms: plants, animals, microbes	Temperature, rainfall, wind, soil, minerals
Producers, consumers, decomposers	Physical and chemical factors

→ **Natural ecosystems**: forests, ponds, lakes, grasslands.
→ **Human-made (artificial) ecosystems**: gardens, aquariums, crop fields. Crop fields are agro-ecosystems, the largest human-made ecosystems.

Activity 13.1: An aquarium

Done: Set up a jar with water, an aerator, fish food and a few aquatic plants and animals. **Observed:** It is nearly self-sustaining but still needs cleaning. **Conclusion:** It is a human-made ecosystem; in ponds and lakes, decomposers do the cleaning, so they never need it.

Producers, consumers and decomposers

Producers (autotrophs): green plants and some bacteria (blue-green algae) that make food from simple inorganic substances by photosynthesis using sunlight. **Consumers (heterotrophs)**: organisms that depend on producers for food, directly or indirectly.

Consumer type	Food	Examples
Herbivores	Plants	Cow, deer, goat
Carnivores	Other animals	Lion, tiger, frog
Omnivores	Plants and animals	Humans, crow
Parasites	Live on a host	Lice, Cuscuta, tapeworm
Saprophytes	Dead matter	Fungi, some bacteria

Decomposers (bacteria and fungi) break down dead remains and wastes of organisms into simple inorganic substances that return to the soil and are used again by plants. Without them, dead bodies and garbage would pile up and the soil would not be replenished.

Yaad karo: decomposers

Decomposers = nature ke safai karamchari. Ye na hon toh nutrients wapas soil mein nahi jaayenge. Producers hamesha food chain ki pehli kadi (first trophic level) hote hain!

2. Food Chain and Trophic Levels

Activity 13.2: Who eats whom

Done: Write the organisms found in a pond or aquarium in order of who eats whom. **Observed:** Algae ⟶ small fish ⟶ big fish; a predator eats the others. **Conclusion:** All groups depend on each other; producers support all the others.

Food chain: a series of organisms in which each one feeds on the one before it; energy passes along the chain as food.

Grass → Grasshopper → Frog → Snake → Eagle

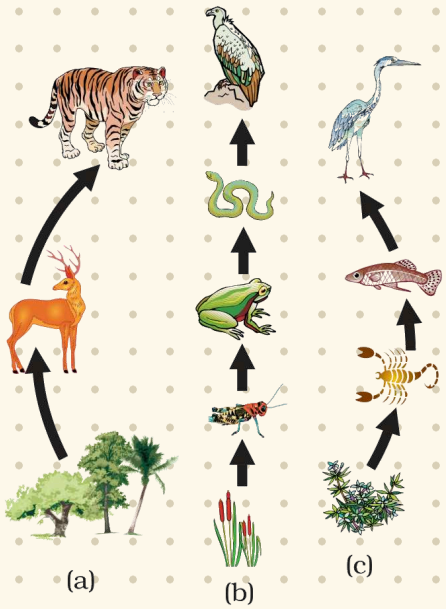


Fig. 13.1 Food chains: (a) forest (b) grassland (c) pond

Trophic level: each step or level of a food chain where transfer of energy occurs.

Trophic level	Organisms	Example
1st	Producers (autotrophs)	Grass, plants
2nd	Herbivores (primary consumers)	Grasshopper, deer
3rd	Small carnivores (secondary consumers)	Frog
4th	Large carnivores (tertiary consumers)	Snake, eagle

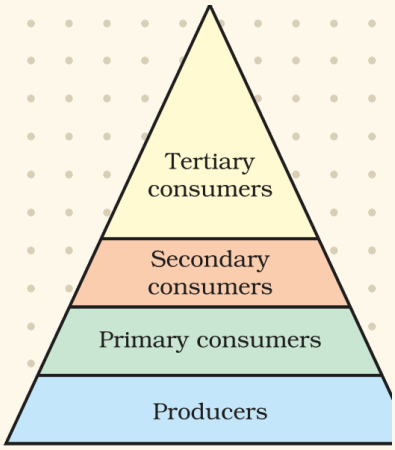


Fig. 13.2 Trophic levels

Energy flow and the 10 percent law

- Green plants capture only about 1% of the sunlight falling on their leaves and convert it into food energy.
- When a consumer eats, part of the food energy is used for digestion, work, growth and reproduction, and much is lost as heat.
- **10 percent law**: only about 10% of the energy of one trophic level reaches the next.
- So food chains have only 3 or 4 steps: little energy is left after the fourth level to support another one.
- Producers are the most numerous; the number of organisms falls at each higher level.
- Energy is lost mainly through respiration at every level, as heat to the surroundings.

Quick fact

10,000 J grass ⟶ 1,000 J deer ⟶ 100 J lion. Har step pe 90% energy gayab! Isliye jungle mein ghaas sabse zyada aur sher sabse kam hote hain.

3. Food Web and Energy Flow

Food web: a network of many food chains linked together. Each organism is generally eaten by two or more kinds of organisms, so the links branch, and one organism can occupy more than one trophic level.


Fig. 13.3 A food web

- **Unidirectional flow**: energy captured by autotrophs does not go back to the Sun, and energy passed to herbivores does not come back to plants. It moves only forward, from one level to the next.
- The energy available keeps decreasing at each trophic level.
- **Conservation of energy**: energy only changes form (sunlight ⟶ chemical energy ⟶ heat); it is not created or destroyed.

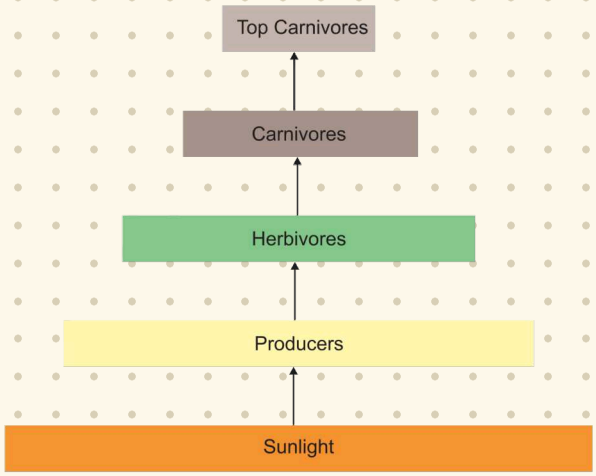


Fig. 13.4 Energy flow in an ecosystem

Ecological pyramids: pyramids of numbers and biomass may be upright or inverted (one tree, many insects), but the pyramid of energy is always upright.

Why food chains matter: they move energy and nutrients and keep nature in balance. If all organisms of one trophic level are killed, e.g. all herbivores, plants grow unchecked while the carnivores starve. Every level depends on the one below it.

Exam Alert

- Food chain ka arrow khaane wale ki taraf: Grass ⟶ Deer, yaani energy grass se deer mein gayi.
- Pyramid of energy hamesha seedha (upright). Numbers aur biomass ulta bhi ho sakta hai.
- 10% law ka numerical aata hai: har level pe energy ko 10 se divide karo.

4. Biological Magnification

Pesticides and other chemicals used to protect crops are washed into the soil and water bodies. From there they are absorbed by plants and aquatic organisms, and enter the food chain.

Biological magnification: these chemicals are non-biodegradable, so they accumulate and their concentration increases at each higher trophic level. Humans occupy the top level in most food chains, so the maximum concentration collects in our bodies. This is why grains, vegetables, fruits and meat carry pesticide residues that washing cannot fully remove.

Water (low DDT) → Plankton → Small fish → Big fish → Birds / humans (highest)

Activity 13.3: Pesticides in food

Done: Find out and debate why some ready-made foods were banned for high pesticide levels. **Observed:** Pesticides reach us through many foods from treated crops. **Conclusion:** Wash and peel food well, and use fewer pesticides.

Paheli-Bujho



Bujho: Paheli, maine kabhi pesticide khaya hi nahi, phir mere body mein kaise aaya?

Paheli: Pesticide paani aur mitti se crops aur machhli mein jaata hai, aur har trophic level pe badhta jaata hai.

Bujho: Aur hum food chain mein sabse upar... toh sabse zyada humare andar!

Paheli: Bilkul. Isi ko **biological magnification** kehte hain.

5. Ozone Layer and its Depletion

Ozone (O_3): a molecule of three oxygen atoms. It is a deadly poison at ground level, but in the upper atmosphere it shields the Earth from the harmful ultraviolet (UV) radiation of the Sun, which causes skin cancer in humans and harms plants and animals.

Formation: UV radiation splits oxygen molecules into free oxygen atoms, which combine with O_2 to form ozone.

$$O_2 \xrightarrow{UV} O + O$$

$$O + O_2 \longrightarrow O_3 \text{ (ozone)}$$

Depletion: the amount of ozone fell sharply from the 1980s, mainly due to chlorofluorocarbons (CFCs) used as refrigerants and in fire extinguishers and aerosol sprays. In 1987 the UNEP got an agreement to freeze CFC production at 1986 levels; now refrigerators must be CFC-free.

Activity 13.4: Ozone research

Done: Find out which chemicals deplete the ozone layer and whether the rules on them work. **Observed:** CFCs deplete it; since 1987 their use has fallen and the ozone hole is slowly healing. **Conclusion:** Global rules on CFCs protect the ozone layer.

6. Managing the Garbage we Produce

Biodegradable	Non-biodegradable
Broken down by biological processes (bacteria, saprophytes)	Not broken down by biological processes
Peels, paper, cotton, wood, leather	Plastics, polythene, glass, DDT, metal cans
Return nutrients to the soil	Persist for long and harm the ecosystem

Reason: enzymes are specific in their action, so microbes have no enzyme to break down most plastics (just as we cannot get energy by eating coal). Even biodegradable waste in excess smells foul and breeds flies, so it must be segregated and composted.

Activity 13.5: Burying household waste

Done: Bury a day's kitchen waste, paper, medicine strips, old clothes and broken footwear under 15 cm of soil; keep it moist and check every 15 days. **Observed:** Peels, food and paper rot fastest; plastic, polythene and bubble packs stay unchanged. **Conclusion:** Biodegradable waste is broken down by microbes; non-biodegradable waste persists.

Activity	What to find out	Conclusion
13.6	Biodegradable plastics	Plastics last centuries; new ones need testing
13.7	Waste from home and class	Mostly biodegradable but mixed: segregate
13.8	Sewage and factory waste	Some reaches rivers untreated: treat it
13.9	E-waste and recycling	Lead, mercury: use e-waste centres

Disposable cups in trains: returnable plastic glasses gave way to disposable plastic cups (hygienic, but huge waste). Clay kulhads on a large scale would remove fertile top-soil, so now biodegradable paper cups are used.

Meme samjho



Ek cup chuno: kulhad mitti mein wapas mil jaata hai, plastic cup 500 saal tak wahin pada rehta hai!

Asli baat: Biodegradable materials return to nature; non-biodegradable plastics persist and pollute.

Waste management: segregation, collection, transport, treatment, recycling and safe disposal. Our part: cloth bags, composting kitchen waste, switching off fans and lights, and the 3 Rs: Reduce, Reuse, Recycle.

Yaad karo: dustbin colours

Green bin = gila (biodegradable), Blue bin = sukha (plastic, paper, metal). Chips ka packet green mein nahi, blue mein!

1 minute recap

Ecosystem → Producers, consumers, decomposers → Food chain, trophic levels → 10% law → Food web → Biomagnification → Ozone and CFCs → Garbage, 3 Rs

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Objective (1 mark)

Q1. 1M Board fav Which of the following groups contains only biodegradable items?

(a) grass, flowers, polythene
(b) grass, wood, plastic
(c) grass, flowers, leather
(d) grass, plastic, glass

Ans: (c) Grass, flowers and leather.

Q2. 1M Board fav Which of the following constitute a food chain?

(a) grass, wheat, mango
(b) grass, goat, wheat
(c) grass, goat, human
(d) goat, cow, elephant

Ans: (c) Grass, goat and human.

Q3. 1M If 10,000 J of solar energy is available to producers, how much energy will reach a secondary consumer?

Ans: About 100 J: producers trap it, 10% (1,000 J) reaches herbivores and 10% of that (100 J) reaches secondary consumers.

Q4. 1M Board fav Name the chemical mainly responsible for depletion of the ozone layer.

Ans: Chlorofluorocarbons (CFCs).

Q5. 1M Which trophic level has the maximum concentration of a non-biodegradable pesticide?

Ans: The highest (top) trophic level, e.g. humans or large carnivores.

Q6. 1M Why is an aquarium called an artificial ecosystem?

Ans: It is made by humans and needs regular cleaning and feeding, unlike a natural pond.

Q7. 1M Board fav **Assertion (A):** Food chains generally have only three or four steps. **Reason (R):** Only about 10% of energy passes to the next trophic level.

Ans: (a) Both A and R are true, and R is the correct explanation of A.

Q8. 1M **Assertion (A):** The flow of energy in an ecosystem is unidirectional. **Reason (R):** Energy captured by autotrophs does not revert to the Sun.

Ans: (a) Both A and R are true, and R is the correct explanation of A.

Short answer (2–3 marks)

Q9. 3M Board fav What is an ecosystem? Name its biotic and abiotic components. Why does a pond not need cleaning like an aquarium?

Ans: All the interacting organisms of an area with the non-living components. Biotic: plants, animals, microbes. Abiotic: temperature, rainfall, wind, soil, minerals. In a pond, decomposers break down dead matter and wastes, so it cleans itself.

Q10. 2M Board fav What is the role of decomposers in an ecosystem?

Ans: They break down dead remains and wastes into simple inorganic substances that return to the soil and are used again by plants. They recycle nutrients and keep the environment clean.

Q11. 3M Board fav What are trophic levels? Give an example of a food chain and state its trophic levels.

Ans: Each step of a food chain where transfer of energy occurs. Grass (1st, producer) ⟶ grasshopper (2nd, herbivore) ⟶ frog (3rd, small carnivore) ⟶ snake (4th, large carnivore).

Q12. 2M Board fav Why is the flow of energy in an ecosystem unidirectional?

Ans: Energy captured by autotrophs does not go back to the Sun, and energy passed to herbivores does not come back to autotrophs. At each level it is used or lost as heat, so it moves only forward and is not available to the previous level.

Q13. 3M Board fav What will happen if we kill all the organisms in one trophic level?

Ans: The balance is upset. If all herbivores are killed, producers grow unchecked and carnivores starve and die out. If carnivores are removed, herbivores increase and overgraze, damaging plants.

Q14. 3M Board fav What is biological magnification? Will its level differ at different trophic levels?

Ans: The increase in concentration of non-biodegradable chemicals (like pesticides) at each higher trophic level. Yes: it is lowest at producers and highest at the top level (humans).

Q15. 3M Board fav What is ozone and how is it formed? Why is damage to the ozone layer a cause for concern?

Ans: O_3, a molecule of three oxygen atoms. UV splits O_2 into O atoms, which join O_2 to form O_3. The ozone layer absorbs harmful UV rays; its depletion lets them reach us, causing skin cancer, eye damage and harm to plants.

Q16. 3M Board fav Why are some substances biodegradable and some non-biodegradable? Give two ways in which non-biodegradable substances affect the environment.

Ans: Microbes break substances down with enzymes, which are specific; there are no enzymes for plastics, glass, DDT. Effects: they persist and pollute soil and water; they enter food chains and undergo biological magnification; plastics choke drains and harm animals.

Q17. 2M Why are crop fields called artificial ecosystems?

Ans: They are created and maintained by humans: crops are chosen, sown, watered, fertilised and protected from pests by people. They are agro-ecosystems.

Q18. 2M Give two ways in which we can reduce the problem of waste disposal.

Ans: Segregate waste and compost the biodegradable part; reduce use of plastic by carrying cloth bags; reuse and recycle paper, glass and metal (3 Rs).

Q19. 2M Why did the Indian Railways replace kulhads with paper cups?

Ans: Disposable plastic cups created huge non-biodegradable waste. Kulhads on a large scale would remove fertile top-soil. Paper cups are biodegradable and do not use up soil.

Q20. 2M Distinguish between a food chain and a food web (any two points).

Ans: Food chain: a single straight sequence; each organism has one food source; removing one link breaks it. Food web: many interlinked chains; organisms have several food sources; it is more stable.

Q21. 2M Why is the pyramid of energy always upright?

Ans: Energy decreases at every higher trophic level (only about 10% passes on), so each level always has less energy than the one below it.

Long answer (5 marks)

Q22. 5M Board fav (a) What is a food chain? Draw a four-step food chain. (b) State the 10 percent law. (c) Why do food chains have only three or four steps? (d) Why are producers the most numerous?

Ans: (a) A series of organisms in which each eats the one before it, transferring energy. Grass ⟶ grasshopper ⟶ frog ⟶ snake. (b) Only about 10% of the energy at one trophic level is passed on to the next. (c) Energy keeps decreasing (90% lost at each step), so after 3–4 levels too little is left to support another level. (d) They hold the maximum energy, which can support only fewer organisms at each higher level.

Q23. 5M Board fav (a) Differentiate between a food chain and a food web. (b) What is biological magnification? Explain with an example. (c) How can it be reduced?

Ans: (a) A food chain is a single straight sequence of who eats whom; a food web is a network of interconnected food chains, giving organisms more than one food source. (b) Non-biodegradable chemicals like DDT accumulate and their concentration rises at each trophic level: water (low) ⟶ plankton ⟶ small fish ⟶ big fish ⟶ birds/humans (highest). (c) Use fewer chemical pesticides, prefer biological pest control and organic farming, and wash and peel food.

Q24. 5M (a) How is ozone formed in the upper atmosphere? (b) What is the cause of its depletion and its harmful effects? (c) What steps have been taken to protect it?

Ans: (a) $O_2 \xrightarrow{UV} O + O$; $O + O_2 \longrightarrow O_3$. (b) CFCs used in refrigerants, fire extinguishers and aerosol sprays; more UV reaches the Earth, causing skin cancer, cataracts and damage to crops and aquatic life. (c) In 1987 UNEP froze CFC production at 1986 levels; CFC-free refrigerators are now compulsory, and CFC use has fallen.

Case-based (4 marks)

Q25. 4M Board fav In a field, DDT was sprayed on crops. Its concentration was measured in different organisms: water 0.02 ppm, plankton 0.04 ppm, small fish 0.5 ppm, big fish 2 ppm and fish-eating birds 25 ppm. (a) Name this phenomenon. (b) Why is the concentration highest in birds? (c) Why do these chemicals not get broken down? (d) Which organism of this chain would show the effects first if humans ate the fish?

Ans: (a) Biological magnification. (b) Birds are at the highest trophic level and eat many fish, so the chemical accumulates in them. (c) They are non-biodegradable: no microbe has enzymes to break them down. (d) Humans, being at the top, would collect the highest concentration and be affected most.

Q26. 4M Board fav A student wrote: Grass (10,000 J) ⟶ Deer ⟶ Lion. (a) How much energy will the deer and the lion get? (b) Name the law used. (c) What happens to the remaining energy at each step? (d) Is the pyramid of energy ever inverted?

Ans: (a) Deer 1,000 J, lion 100 J. (b) 10 percent law. (c) It is used in life processes (digestion, work, growth) and lost as heat. (d) No, the pyramid of energy is always upright.

Exam Alert

- → Food chain hamesha producer se shuru karo aur arrows energy ki direction mein lagao.
- → Biological magnification ke answer mein "non-biodegradable" aur "highest at top trophic level" dono likho.
- → Ozone ke answer mein equation + CFC + UNEP 1987: teeno points = full marks.
- → Garbage wale answer mein example ke saath 3 Rs (Reduce, Reuse, Recycle) zaroor likho.

REVISION Exam se pehle wali raat

LAST-NIGHT REVISION

Most asked: har chapter ke reactions, formulas, one-liners + quick check (answers saath mein)

1. Chemistry: Reactions to Write

Ch 1: Chemical Reactions and Equations

$2Mg + O_2 \longrightarrow 2MgO$ (white, dazzling flame)

$Pb(NO_3)_2 + 2KI \longrightarrow PbI_2$ (yellow ppt) $+ 2KNO_3$

$CaO + H_2O \longrightarrow Ca(OH)_2 +$ heat (combination, exothermic)

$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ (whitewashing)

$2FeSO_4 \xrightarrow{\text{Heat}} Fe_2O_3 + SO_2 + SO_3$

$CaCO_3 \xrightarrow{\text{Heat}} CaO + CO_2$

$2Pb(NO_3)_2 \xrightarrow{\text{Heat}} 2PbO + 4NO_2 + O_2$

$2H_2O \xrightarrow{\text{Electricity}} 2H_2 + O_2$

$2AgCl \xrightarrow{\text{Sunlight}} 2Ag + Cl_2$

$Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (blue to light green)

$Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4$ (white ppt) $+ 2NaCl$

$CuO + H_2 \xrightarrow{\text{Heat}} Cu + H_2O$

Ch 2: Acids, Bases and Salts

$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (pop test)

$2NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$

$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$

$NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$

$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3$ (milky) $+ H_2O$

$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$ (milkiness goes)

$NaOH + HCl \longrightarrow NaCl + H_2O$ (neutralisation)

$CuO + 2HCl \longrightarrow CuCl_2 + H_2O$ (blue-green)

$2NaCl + 2H_2O \longrightarrow 2NaOH + Cl_2 + H_2$ (chlor-alkali)

$2Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(ClO)_2 + CaCl_2 + 2H_2O$ (bleaching powder)

$NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$

$2NaHCO_3 \xrightarrow{\text{Heat}} Na_2CO_3 + H_2O + CO_2$

$Na_2CO_3 + 10H_2O \longrightarrow Na_2CO_3.10H_2O$ (washing soda)

$NaHCO_3 + H^+ \longrightarrow CO_2 + H_2O + Na^+$ salt (baking powder)

$CaSO_4.2H_2O \xrightarrow{373\,K} CaSO_4.\frac{1}{2}H_2O + 1\frac{1}{2}H_2O$

$CaSO_4.\frac{1}{2}H_2O + 1\frac{1}{2}H_2O \longrightarrow CaSO_4.2H_2O$ (PoP sets)

Ch 3: Metals and Non-metals

$4Na + O_2 \longrightarrow 2Na_2O$

$2Na + 2H_2O \longrightarrow 2NaOH + H_2 +$ heat

$2K + 2H_2O \longrightarrow 2KOH + H_2 +$ heat

$Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2$

$Mg + 2H_2O \longrightarrow Mg(OH)_2 + H_2$ (hot water)

$2Al + 3H_2O \longrightarrow Al_2O_3 + 3H_2$ (steam)

$3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$ (steam)

$Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$

$Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$ (amphoteric)

$Mg + 2HNO_3 \longrightarrow Mg(NO_3)_2 + H_2$ (very dilute)

$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$ (roasting)

$ZnCO_3 \longrightarrow ZnO + CO_2$ (calcination)

$ZnO + C \longrightarrow Zn + CO$ (reduction)

$2HgS + 3O_2 \longrightarrow 2HgO + 2SO_2$

$2HgO \longrightarrow 2Hg + O_2$

$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$

$2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$

$Fe_2O_3 + 2Al \longrightarrow 2Fe + Al_2O_3 +$ heat (thermite)

$3MnO_2 + 4Al \longrightarrow 3Mn + 2Al_2O_3 +$ heat

Ch 4: Carbon and its Compounds

$C + O_2 \longrightarrow CO_2 +$ heat + light

$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O +$ heat + light

$CH_3CH_2OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O$

$CH_3CH_2OH \xrightarrow[\text{or acidified } K_2Cr_2O_7]{\text{Alk. } KMnO_4 + \text{heat}} CH_3COOH$

$R_2C{=}CR_2 + H_2 \xrightarrow{\text{Ni catalyst}} R_2CH{-}CHR_2$

$CH_4 + Cl_2 \xrightarrow{\text{Sunlight}} CH_3Cl + HCl$

$2Na + 2CH_3CH_2OH \longrightarrow 2CH_3CH_2ONa + H_2$

$CH_3CH_2OH \xrightarrow[443\,K]{\text{Hot conc. } H_2SO_4} CH_2{=}CH_2 + H_2O$

$CH_3COOH + C_2H_5OH \xrightarrow{\text{Acid}} CH_3COOC_2H_5 + H_2O$

$CH_3COOC_2H_5 + NaOH \longrightarrow C_2H_5OH + CH_3COONa$ (saponification)

$$2CH_3COOH + 2Na \longrightarrow 2CH_3COONa + H_2$$

$$NaOH + CH_3COOH \longrightarrow CH_3COONa + H_2O$$

$$2CH_3COOH + Na_2CO_3 \longrightarrow 2CH_3COONa + H_2O + CO_2$$

$$CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2$$

Common names and formulas

Common name	Chemical name	Formula
Baking soda	Sodium hydrogencarbonate	$NaHCO_3$
Washing soda	Sodium carbonate decahydrate	$Na_2CO_3.10H_2O$
Bleaching powder	Calcium hypochlorite	$Ca(ClO)_2$
Plaster of Paris	Calcium sulphate hemihydrate	$CaSO_4.½H_2O$
Gypsum	Calcium sulphate dihydrate	$CaSO_4.2H_2O$
Quick lime	Calcium oxide	CaO
Slaked lime	Calcium hydroxide	$Ca(OH)_2$
Limestone / marble	Calcium carbonate	$CaCO_3$
Blue vitriol	Copper sulphate pentahydrate	$CuSO_4.5H_2O$
Green vitriol	Ferrous sulphate heptahydrate	$FeSO_4.7H_2O$
Rust	Hydrated iron(III) oxide	$Fe_2O_3.xH_2O$
Vinegar	5–8% ethanoic acid in water	CH_3COOH

Tests for gases

Gas	Test	Result
H_2	burning matchstick near it	burns with a pop sound
O_2	glowing splinter	splinter rekindles; candle burns brighter
CO_2	pass through lime water	lime water turns milky
Cl_2	moist blue litmus	turns red, then bleached
SO_2	smell	smell of burning sulphur

Indicators

Indicator	In acid	In base
Blue litmus	red	no change
Red litmus	no change	blue
Phenolphthalein	colourless	pink
Methyl orange	red	yellow
Turmeric	yellow	reddish-brown
Onion / vanilla / clove (olfactory)	smell stays	smell goes (onion, vanilla)

Chemistry one-liners

- Acids give H^+ (H_3O^+) ions in water; bases give OH^- ions. Water-soluble bases = alkalis.
- Dilute an acid by adding **acid to water** slowly (highly exothermic), never water to acid.
- pH 7 neutral, below 7 acidic, above 7 basic. Tooth decay starts below pH 5.5; stomach pH about 2 (HCl).
- Metals: lustrous, malleable, ductile, good conductors (silver and copper best); exceptions: mercury liquid, sodium and potassium soft (cut with a knife), iodine (non-metal) lustrous, graphite conducts.
- Ionic compounds: high melting point, soluble in water, conduct in molten state or solution, not in solid state.
- Alloys: brass (Cu + Zn), bronze (Cu + Sn), solder (Pb + Sn), stainless steel (Fe + Ni + Cr), amalgam (Hg + another metal).
- Carbon forms covalent bonds by sharing electrons; catenation + tetravalency make millions of compounds.
- Diamond: each C bonded to 4 C (hardest); graphite: each C bonded to 3 C in layers (conducts); fullerene C-60 (football shape).
- Saturated hydrocarbons burn with a clean blue flame; unsaturated give a yellow sooty flame.
- Micelles: soap molecules with ionic (hydrophilic) end in water and carbon chain (hydrophobic) in oil; trap dirt. Detergents work in hard water too.

2. Biology: One-Liners

Ch 5: Life Processes

- Autotrophs make food: $6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$.
- Photosynthesis events: light absorbed by chlorophyll, water split (photolysis), CO_2 reduced to carbohydrate.
- Stomata: gaseous exchange + transpiration; guard cells swell (water in) = pore opens.
- Saliva: salivary amylase, starch to sugar. Stomach: HCl (acidic medium, kills germs), pepsin (protein), mucus (protects lining).
- Liver makes bile (stored in gall bladder): emulsifies fats, makes medium alkaline. Pancreas: trypsin (protein), lipase (fat).
- Villi in small intestine increase surface area for absorption.
- Glucose ⟶ pyruvate (cytoplasm). No O_2: ethanol + CO_2 (yeast) or lactic acid (muscle cramps). With O_2: $CO_2 + H_2O$ + energy (mitochondria).
- ATP = energy currency of the cell. Alveoli = site of gas exchange; haemoglobin carries O_2.
- Heart: 4 chambers in mammals/birds (no mixing of blood); fish 2 chambers (single circulation); amphibians/reptiles 3. Double circulation in humans.
- Arteries carry blood away from the heart (thick elastic walls); veins bring it back (valves). Exception: pulmonary artery deoxygenated, pulmonary vein oxygenated.
- Platelets = clotting. Lymph = carries digested fat, drains fluid back to blood.
- Xylem: water + minerals (root pressure, transpiration pull). Phloem: food (translocation, uses ATP).
- Nephron = filtration unit of kidney (glomerulus + Bowman's capsule); urea and uric acid removed. Dialysis = artificial kidney.

Ch 6: Control and Coordination

- Neuron: dendrite ⟶ cell body ⟶ axon ⟶ nerve ending; impulse crosses the synapse chemically.
- Reflex arc: receptor ⟶ sensory neuron ⟶ spinal cord (relay) ⟶ motor neuron ⟶ effector.
- Cerebrum (fore-brain) = thinking, voluntary actions; cerebellum (hind-brain) = balance, posture; medulla = BP, salivation, vomiting.
- Brain protected by skull + cerebrospinal fluid; spinal cord by vertebral column.
- Plant hormones: auxin (cell elongation, phototropism), gibberellin (stem growth), cytokinin (cell division), abscisic acid (inhibits growth, wilting).
- Touch-me-not (Mimosa) = movement by change in water amount, not growth. Tendrils coil by faster growth on the side away from support.
- Thyroxin needs iodine (deficiency = goitre); insulin from pancreas lowers blood sugar (deficiency = diabetes); adrenaline = emergency (fight or flight).
- Growth hormone (pituitary): less = dwarfism, more = gigantism. Testosterone (testes), oestrogen (ovaries) = puberty changes.
- Feedback mechanism controls the timing and amount of hormones.

Ch 7: How do Organisms Reproduce?

- DNA copying makes variations, the basis of evolution; stable DNA = stable body design.
- Fission: Amoeba (binary, any plane), Leishmania (binary, definite plane), Plasmodium (multiple). Fragmentation: Spirogyra. Regeneration: Planaria, Hydra. Budding: Hydra, yeast. Spores: Rhizopus.
- Vegetative propagation: Bryophyllum leaf buds; banana, rose, orange, jasmine, grapes have lost seed-forming ability.
- Flower: sepals (calyx), petals (corolla), stamens (male, pollen), carpel (female: stigma, style, ovary). Zygote ⟶ embryo, ovule ⟶ seed, ovary ⟶ fruit.
- Self vs cross pollination; pollen tube carries male germ cell to ovule.
- Human male: testes (sperm + testosterone), vas deferens, seminal vesicles + prostate (fluid). Female: ovary (egg), oviduct (fertilisation), uterus (implantation), placenta (nutrition).
- If egg not fertilised: lining sheds as menstruation (about every month). Contraception: barrier (condom also stops STDs), pills, Copper-T, surgery. Female foeticide is illegal.

Ch 8: Heredity

- Mendel: pea plants; tall (TT) x short (tt) ⟶ F1 all tall (Tt) ⟶ F2 tall : short = 3 : 1 (genotype 1 TT : 2 Tt : 1 tt).
- Dihybrid (round yellow x wrinkled green): F2 = 9 : 3 : 3 : 1 ⟶ traits inherited independently.
- Gene = section of DNA for a protein; one copy from each parent. Germ cells get only one set.
- Sex: XX female, XY male; father's sperm (X or Y) decides the sex of the child. Probability of a boy or girl = 50%.

Ch 13: Our Environment

- Biodegradable (broken down by microbes) vs non-biodegradable (plastic, DDT).
- Food chain: producers ⟶ herbivores ⟶ carnivores. Only **10%** energy passes to the next trophic level (10% law), so chains have 3-4 levels.
- Biological magnification: pesticides like DDT collect more at each level; maximum in top carnivores (humans).
- Decomposers (bacteria, fungi) recycle nutrients. Ozone (O_3) layer shields UV; damaged by CFCs (UNEP agreement 1987 to freeze CFC production at 1986 levels).

Diagrams to practise (board favourites)

Chapter	Diagrams
Ch 5	stomata (open/closed), human alimentary canal, respiratory system, heart (sectional view), nephron, excretory system
Ch 6	neuron, reflex arc, human brain, phototropism
Ch 7	binary fission, budding in Hydra, longitudinal section of a flower, germination of pollen on stigma, male and female reproductive systems
Ch 8	sex determination (XX/XY) chart, monohybrid cross
Ch 9-10	ray diagrams (6 cases each), refraction through glass slab, human eye, dispersion by prism, myopia/hypermetropia correction
Ch 11-12	circuit diagrams with symbols, field lines of bar magnet, straight wire, loop and solenoid, domestic circuit

Exam Alert: differences that come every year

- Aerobic vs anaerobic respiration; arteries vs veins; xylem vs phloem.
- Self vs cross pollination; asexual vs sexual reproduction; inherited vs acquired traits.
- Real vs virtual image; myopia vs hypermetropia; series vs parallel; AC vs DC.
- Metals vs non-metals; soaps vs detergents; saturated vs unsaturated hydrocarbons; biodegradable vs non-biodegradable.

3. Physics: Formulas + Rules

Ch 9: Light

Quantity	Formula
Focal length (mirror)	$f = R/2$
Mirror formula	$1/v + 1/u = 1/f$
Magnification (mirror)	$m = h'/h = -v/u$
Lens formula	$1/v - 1/u = 1/f$
Magnification (lens)	$m = h'/h = v/u$
Power of lens	$P = 1/f$ (f in metre), unit dioptre (D)
Lenses in contact	$P = P_1 + P_2 + \ldots$
Refractive index	$n_{21} = v_1/v_2$; $n_m = c/v$
Snell's law	$\sin i / \sin r$ = constant (n_{21})

→ **Sign convention:** distances measured from the pole/optical centre; along incident light = +ve, against = −ve; heights up +ve, down −ve.
→ Concave mirror and convex lens: f is −ve for mirror, +ve for lens. Convex mirror: f +ve; concave lens: f −ve.
→ m negative = real and inverted image; m positive = virtual and erect.
→ Convex mirror: always virtual, erect, diminished (rear-view mirror). Concave mirror: torch, shaving mirror, dentist, solar furnace.
→ Plane mirror: m = +1; light bends towards normal going from rarer to denser medium. Glass slab: emergent ray parallel to incident ray (lateral shift).

Ch 10: Human Eye

→ Least distance of distinct vision = 25 cm; far point = infinity; ciliary muscles change focal length (power of accommodation).
→ Myopia (near-sighted): image before retina, corrected by **concave** lens. Hypermetropia (far-sighted): image behind retina, **convex** lens. Presbyopia: bifocal lens.
→ Prism: dispersion of white light into VIBGYOR; violet bends most, red least. Rainbow = dispersion + refraction + internal reflection in droplets.
→ Atmospheric refraction: twinkling of stars, advance sunrise and delayed sunset (about 2 min). Tyndall effect: scattering by fine particles; sky blue (blue scattered most), sun reddish at sunrise/sunset, danger signals red.

Ch 11: Electricity

Quantity	Formula (SI unit)
Current	$I = Q/t$ (ampere, A)
Potential difference	$V = W/Q$ (volt, V)
Ohm's law	$V = IR$
Resistance	$R = \rho l/A$ (ohm, Ω); ρ in Ω m
Series	$R = R_1 + R_2 + R_3$; same I
Parallel	$1/R = 1/R_1 + 1/R_2 + 1/R_3$; same V
Heat (Joule)	$H = I^2Rt$
Power	$P = VI = I^2R = V^2/R$ (watt)
Energy	$E = Pt$; 1 kWh = 3.6×10^6 J
Charge of electron	1.6×10^{-19} C; 1 C = 6.25×10^{18} electrons

→ Ammeter in series (low resistance), voltmeter in parallel (high resistance).
→ Resistance depends on length, area, material, temperature; not on current or voltage.
→ Wire stretched to double length: R becomes 4 times. Nichrome (high ρ) for heaters; tungsten for bulb filament; fuse wire has low melting point.

Ch 12: Magnetic Effects

→ Field lines: N to S outside, S to N inside; never cross; closer lines = stronger field.
→ Right-hand thumb rule: thumb = current, curled fingers = field. Fleming's left-hand rule: Forefinger = Field, Middle = Current, thumb = force (motion).
→ Straight wire: concentric circles, field decreases with distance. Solenoid: uniform field inside (like a bar magnet) used for electromagnets.
→ Domestic circuit: live (red), neutral (black), earth (green); 220 V, 50 Hz. Fuse/MCB protects from short circuit and overloading; earthing protects from shocks via metal body.

Concave mirror: image formation

Object at	Image at	Nature, size
Infinity	F	real, inverted, highly diminished (point)
Beyond C	between F and C	real, inverted, diminished
At C	at C	real, inverted, same size
Between C and F	beyond C	real, inverted, enlarged
At F	infinity	real, inverted, highly enlarged
Between P and F	behind mirror	virtual, erect, enlarged

Convex lens: image formation

Object at	Image at	Nature, size
Infinity	F_2	real, inverted, point size
Beyond $2F_1$	between F_2 and $2F_2$	real, inverted, diminished
At $2F_1$	at $2F_2$	real, inverted, same size
Between F_1 and $2F_1$	beyond $2F_2$	real, inverted, enlarged
At F_1	infinity	real, inverted, highly enlarged
Between F_1 and O	same side as object	virtual, erect, enlarged

Concave lens and convex mirror: image always **virtual, erect, diminished**, between F and the lens/mirror (at F for an object at infinity).

4. Quick Check (answers in green)

Question	Answer
Chemical formula of bleaching powder (NCERT 2026-27)?	$Ca(ClO)_2$
pH of pure water?	7
Gas evolved when metal reacts with acid?	Hydrogen
Most reactive metal in the activity series of NCERT?	Potassium
Non-metal that is liquid at room temperature?	Bromine
Metal that melts on your palm?	Gallium (also caesium)
Functional group of ethanoic acid?	−COOH (carboxylic acid)
General formula of alkenes?	C_nH_{2n}

Question	Answer
Site of photosynthesis?	Chloroplast
Enzyme in saliva?	Salivary amylase
Functional unit of kidney?	Nephron
Hormone that needs iodine?	Thyroxin
Plant hormone that inhibits growth?	Abscisic acid
Reproduction in yeast?	Budding
Male gamete in plants is in?	Pollen grain
F2 ratio of a dihybrid cross?	9 : 3 : 3 : 1

Question	Answer
SI unit of power of lens?	Dioptre
Image by convex mirror is always?	Virtual, erect, diminished
Defect corrected by concave lens?	Myopia
Colour scattered most by air?	Blue
1 kWh = ? J	3.6×10^6 J
Resistivity unit?	Ω m
Rule for direction of force on a conductor?	Fleming's left-hand rule
Frequency of AC in India?	50 Hz

Question	Answer
Energy passed to next trophic level?	10%
Chemicals that deplete ozone?	CFCs
Lens used to correct hypermetropia?	Convex lens
Device that protects from short circuit?	Fuse / MCB
Pigment that absorbs light in leaves?	Chlorophyll
Part of brain for balance?	Cerebellum
Who decides sex of the child?	Father (X or Y sperm)
Acid in stomach?	Hydrochloric acid (HCl)

Most expected board questions (2026-27)

1. Balance + state symbols; identify type of reaction; oxidised / reduced (Ch 1).
2. pH scale; preparation and uses of baking soda, washing soda, bleaching powder, PoP (Ch 2).
3. Reactivity series + extraction of metals (roasting, calcination, electrolytic refining); corrosion (Ch 3).
4. Covalent bonding (electron dot), homologous series, ethanol and ethanoic acid reactions, soaps vs detergents (Ch 4).
5. Nutrition in human beings (alimentary canal diagram), double circulation, nephron (Ch 5).
6. Reflex arc, brain parts, plant hormones, endocrine glands table (Ch 6).
7. Flower LS, pollination, fertilisation; male/female reproductive systems; contraception (Ch 7).
8. Mendel's monohybrid and dihybrid cross; sex determination (Ch 8).
9. Ray diagrams + mirror/lens formula numericals; refraction, Snell's law, power of lens (Ch 9).
10. Myopia/hypermetropia with diagrams; prism dispersion; twinkling of stars; blue sky (Ch 10).
11. Ohm's law, series/parallel numericals, heating effect, power and energy (Ch 11).
12. Field lines, right-hand thumb rule, Fleming's left-hand rule, domestic circuit and fuse (Ch 12).
13. Food chain, 10% law, biological magnification, ozone depletion (Ch 13).

SI units at a glance

Quantity	SI unit
Charge	coulomb (C)
Current	ampere (A)
Potential difference	volt (V)
Resistance	ohm (Ω)
Resistivity	ohm metre (Ω m)
Power	watt (W)
Commercial energy	kilowatt hour (kWh)
Power of lens	dioptre (D)
Focal length	metre (m) in P = 1/f

Activities examiners love

Activity	What it shows
Burning Mg ribbon (1.1)	combination, MgO white ash
Iron nail in $CuSO_4$ (1.9)	displacement, blue to light green
Litmus / indicators (2.1)	acids turn blue litmus red
Zn + H_2SO_4, pop test (2.3)	metal + acid gives H_2
Conductivity of solutions (2.8)	acids give H^+ ions in water
Rusting test tubes (3.14)	air + water both needed
Iodine on leaf (5.1)	starch made in light, needs chlorophyll
Lime water + exhaled air (5.8)	CO_2 in exhaled air
Seedling in flask near window (6.2)	phototropism
Glass slab (9.7)	lateral shift, emergent ray parallel
Prism (10.1)	dispersion, angle of deviation

Exam-day checklist

- ✓ Admit card, 2 blue pens, pencil, eraser, scale.
- ✓ Last 10 min: check units, signs, state symbols.

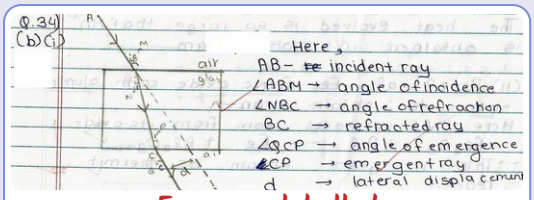
All the best!

Neend poori lo, answer neat likho, diagram pencil se. Nitesh Bhaiya ki taraf se: board exam bas ek aur test hai, aur tum ready ho!

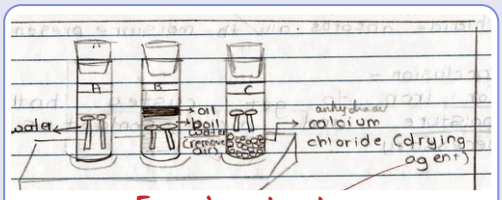
Topper Answer Sheets

dekho toppers kaise likhte hain!

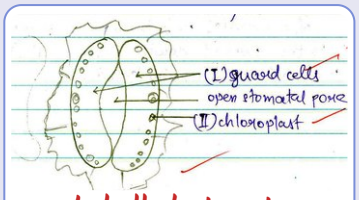
Real answers from CBSE Class 10 Science toppers, 2016 to 2025: diagrams, steps and boxed answers.



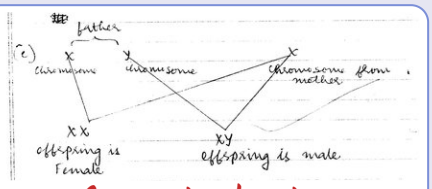
Every ray labelled
(CBSE 2025)



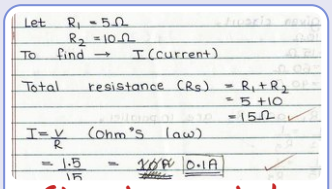
Experiment set-up
(CBSE 2025)



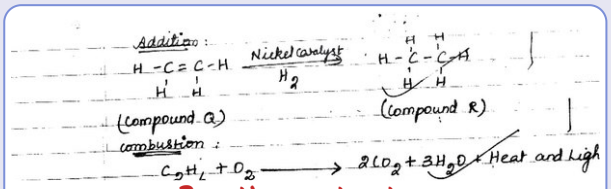
Labelled stomata
(CBSE 2024)



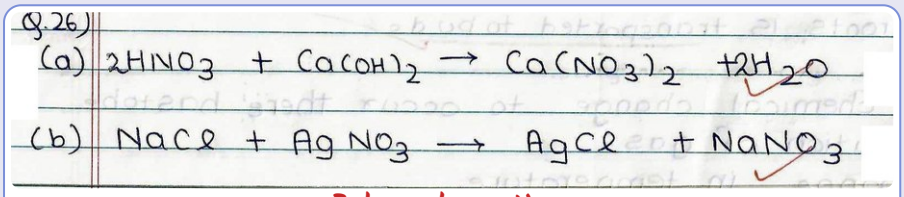
Cross, step by step
(CBSE 2023)



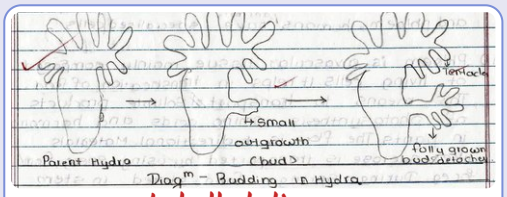
Steps in numericals
(CBSE 2025)



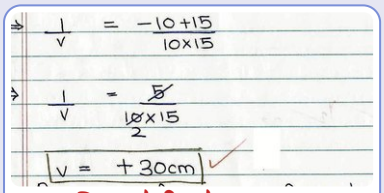
Reaction + structure
(CBSE 2016)



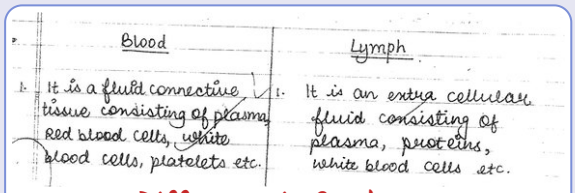
Balanced equations
(CBSE 2025)



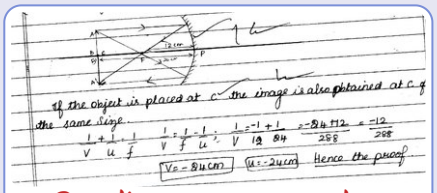
Labelled diagram
(CBSE 2025)



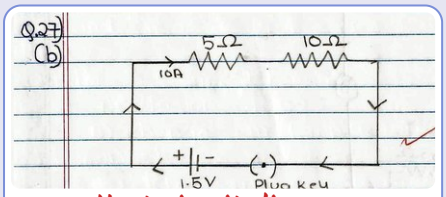
Boxed final answer
(CBSE 2025)



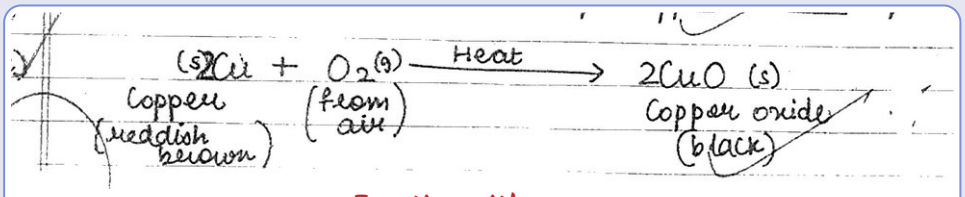
Difference in 2 columns
(CBSE 2019)



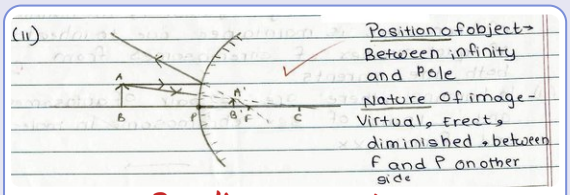
Ray diagram + answer box
(CBSE 2016)



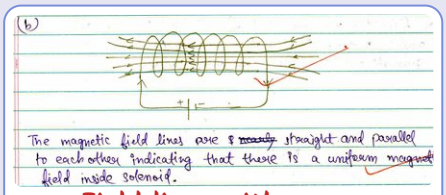
Neat circuit diagram
(CBSE 2025)



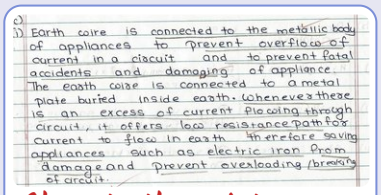
Equation with names
(CBSE 2019)



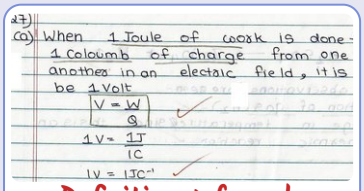
Ray diagram + nature
(CBSE 2025)



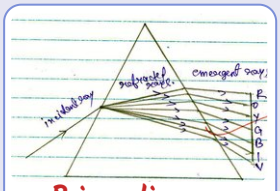
Field lines with arrows
(CBSE 2024)



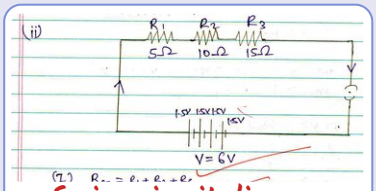
Clear, to-the-point answer
(CBSE 2025)



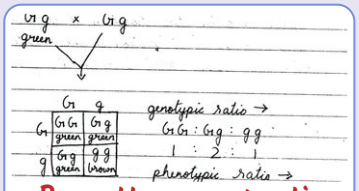
Definition + formula
(CBSE 2025)



Prism diagram
(CBSE 2024)



Series circuit diagram
(CBSE 2024)



Punnett square + ratio
(CBSE 2020)

#Toh yaad rakhna! Diagram pencil se, page ke left side mein, labels seedhi lines se. Keywords underline karo. Numericals mein Given, Formula, Solution, phir final answer box mein (unit ke saath). Equations hamesha balanced aur states ke saath likho.

Source: CBSE Toppers' Answer Books (cbse.gov.in)